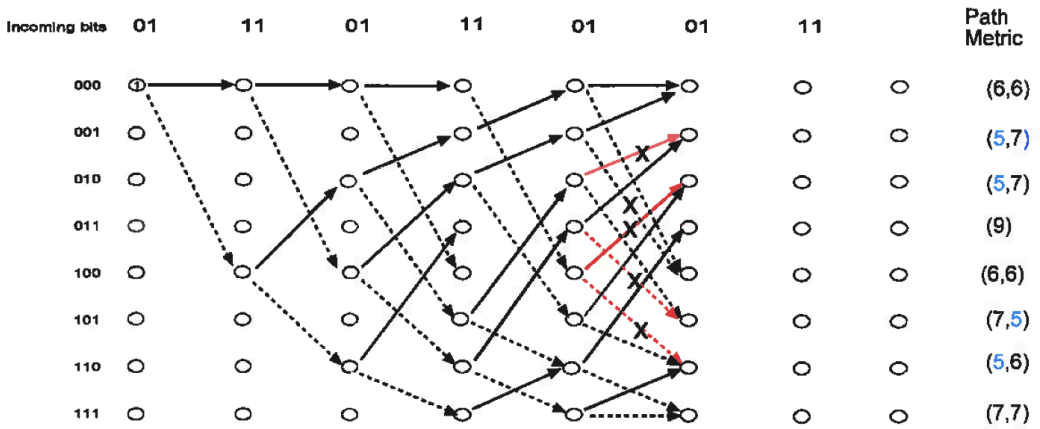
Viterbi Decoder Project

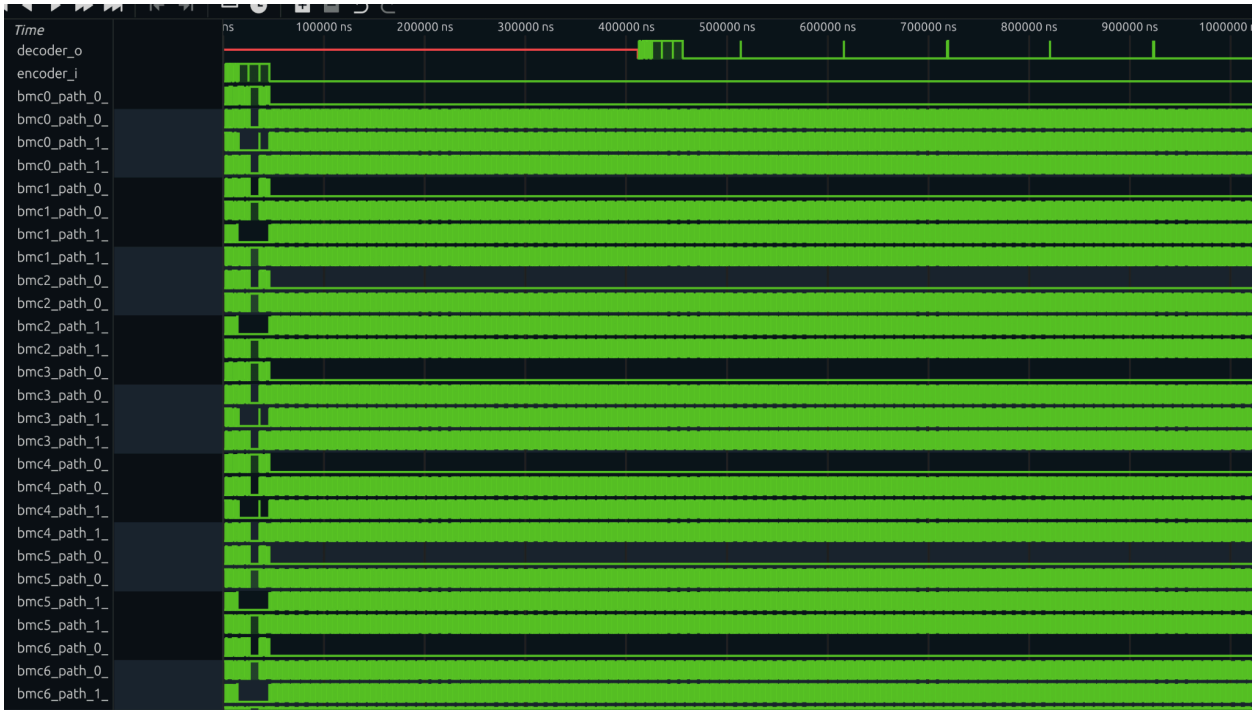
In this project, I implemented a Viterbi decoder from a given sequence of received bits with possible errors to decode the original sequence of bits by selecting the most likely path with the fewest errors. The decoder, encoder, and testbench was coded in System Verilog Code. On a web platform, test bench simulation was performed using Siemsen's Questa and synthesis was performed using Siemsen's Precision. The coding rate was ½ so for each input the convolutional encoder, 2 output bits are produced, and there are 8 possible states represented by 3 bits. Coding saves power because it can achieve the same error rate as raw transmission at a lower power despite any overheads. The decoder uses maximum likelihood principles to recover the original sequence by assuming the fewest amount of errors and that errors are generally spread randomly so it selects the path with minimum amount of errors. Always starting at state 0, at each cycle, the decoder receives 2 bits and compares the possible path destinations using a hamming distance metric. The possible destinations for each state are determined by the state machine of the convolutional encoder. As paths extend, the errors are accumulated and when there are two incoming paths to a single node, then the path with lower number of errors is selected as a survivor path and the other path is pruned. After the trellis or graph of nodes and states is filled up, the decoder traces back survivor paths to determine the original encoded sequence and decodes that encoded sequence to give the decoded output. To accomplish this, the architecture of the decoder has branch metric units(BMU)s, add compare select units(ACS), and traceback units(TBU). The branch metric unit does hard decoding since it uses hamming distance to compute distance for each possible path destination from the pair of received bits. The ACS units take in the branch metric unit results and update path cost and record the selection of which path is optional. The selections are written into memory which uses 4 rotating memory blocks. The pointer to memory is pipelined two cycles after which traceback module is activated tracing the survivor paths to recover the decoded value and writing the decoder output into display memory. Traceback is inherently sequential and cannot produce outputs at full throughput, so the design uses two traceback units operating in an interleaved fashion to maintain one decoded bit per cycle. The additional pipeline delay (Q3–Q5) aligns memory read latency and control signals so that the correct decoded bit from display memory is available at the output. To test the robustness of the design, errors were injected into the encoded sequence before feeding into the decoder. The MSB or the LSB or both bits were XORed with errors and the rate was also varied every 8 bits, every 16 bits or ever 32 bits. Moreover, varying number of consecutive pairs were also injected with errors and the rate and error injection was also randomized using a random function. After 5 consecutive pairs where errors were injected, the decoder breaks and does not correct all errors.



Simulation Waveform

The simulation waveform shows a correctly working viterbi decoder because the encoder input is reconstructed correctly in the decoder output.





Transcript

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000000 00          0         12
# error_counter,err_inj = 00000000 00          0         13
# error_counter,err_inj = 00000000 00          0         14
# error_counter,err_inj = 00000000 00          0         15
# error_counter,err_inj = 00000001 01          0         16
# error_counter,err_inj = 00000001 00          0         17
# error_counter,err_inj = 00000001 00          0         18
# error_counter,err_inj = 00000001 00          0         19
# error_counter,err_inj = 00000001 00          0         20
# error_counter,err_inj = 00000001 00          0         21
# error_counter,err_inj = 00000001 00          0         22
# error_counter,err_inj = 00000001 00          0         23
# error_counter,err_inj = 00000001 00          0         24
# error_counter,err_inj = 00000001 00          0         25
# error_counter,err_inj = 00000001 00          0         26
# error_counter,err_inj = 00000001 00          0         27
```

```
# error_counter,err_inj = 00000001 00          0          28
# error_counter,err_inj = 00000001 00          0          29
# error_counter,err_inj = 00000001 00          0          30
# error_counter,err_inj = 00000001 00          0          31
# error_counter,err_inj = 00000002 01          0          32
# error_counter,err_inj = 00000002 00          1          33
# error_counter,err_inj = 00000002 00          1          34
# error_counter,err_inj = 00000002 00          1          35
# error_counter,err_inj = 00000002 00          1          36
# error_counter,err_inj = 00000002 00          1          37
# error_counter,err_inj = 00000002 00          1          38
# error_counter,err_inj = 00000002 00          1          39
# error_counter,err_inj = 00000002 00          1          40
# error_counter,err_inj = 00000002 00          1          41
# error_counter,err_inj = 00000002 00          1          42
# error_counter,err_inj = 00000002 00          1          43
# error_counter,err_inj = 00000002 00          1          44
# error_counter,err_inj = 00000002 00          1          45
# error_counter,err_inj = 00000002 00          1          46
# error_counter,err_inj = 00000002 00          1          47
# error_counter,err_inj = 00000003 01          1          48
# error_counter,err_inj = 00000003 00          2          49
# error_counter,err_inj = 00000003 00          2          50
# error_counter,err_inj = 00000003 00          2          51
# error_counter,err_inj = 00000003 00          2          52
# error_counter,err_inj = 00000003 00          2          53
# error_counter,err_inj = 00000003 00          2          54
# error_counter,err_inj = 00000003 00          2          55
# error_counter,err_inj = 00000003 00          2          56
# error_counter,err_inj = 00000003 00          2          57
# error_counter,err_inj = 00000003 00          2          58
# error_counter,err_inj = 00000003 00          2          59
# error_counter,err_inj = 00000003 00          2          60
# error_counter,err_inj = 00000003 00          2          61
# error_counter,err_inj = 00000003 00          2          62
# error_counter,err_inj = 00000003 00          2          63
# error_counter,err_inj = 00000004 01          2          64
# error_counter,err_inj = 00000004 00          3          65
# error_counter,err_inj = 00000004 00          3          66
# error_counter,err_inj = 00000004 00          3          67
# error_counter,err_inj = 00000004 00          3          68
# error_counter,err_inj = 00000004 00          3          69
# error_counter,err_inj = 00000004 00          3          70
# error_counter,err_inj = 00000004 00          3          71
# error_counter,err_inj = 00000004 00          3          72
# error_counter,err_inj = 00000004 00          3          73
# error_counter,err_inj = 00000004 00          3          74
# error_counter,err_inj = 00000004 00          3          75
# error_counter,err_inj = 00000004 00          3          76
# error_counter,err_inj = 00000004 00          3          77
# error_counter,err_inj = 00000004 00          3          78
# error_counter,err_inj = 00000004 00          3          79
# error_counter,err_inj = 00000005 01          3          80
# error_counter,err_inj = 00000005 00          3          81
# error_counter,err_inj = 00000005 00          3          82
```

```
# error_counter,err_inj = 00000005 00           3          83
# error_counter,err_inj = 00000005 00           3          84
# error_counter,err_inj = 00000005 00           3          85
# error_counter,err_inj = 00000005 00           3          86
# error_counter,err_inj = 00000005 00           3          87
# error_counter,err_inj = 00000005 00           3          88
# error_counter,err_inj = 00000005 00           3          89
# error_counter,err_inj = 00000005 00           3          90
# error_counter,err_inj = 00000005 00           3          91
# error_counter,err_inj = 00000005 00           3          92
# error_counter,err_inj = 00000005 00           3          93
# error_counter,err_inj = 00000005 00           3          94
# error_counter,err_inj = 00000005 00           3          95
# error_counter,err_inj = 00000006 01           3          96
# error_counter,err_inj = 00000006 00           3          97
# error_counter,err_inj = 00000006 00           3          98
# error_counter,err_inj = 00000006 00           3          99
# error_counter,err_inj = 00000006 00           3         100
# error_counter,err_inj = 00000006 00           3         101
# error_counter,err_inj = 00000006 00           3         102
# error_counter,err_inj = 00000006 00           3         103
# error_counter,err_inj = 00000006 00           3         104
# error_counter,err_inj = 00000006 00           3         105
# error_counter,err_inj = 00000006 00           3         106
# error_counter,err_inj = 00000006 00           3         107
# error_counter,err_inj = 00000006 00           3         108
# error_counter,err_inj = 00000006 00           3         109
# error_counter,err_inj = 00000006 00           3         110
# error_counter,err_inj = 00000006 00           3         111
# error_counter,err_inj = 00000007 01           3         112
# error_counter,err_inj = 00000007 00           4         113
# error_counter,err_inj = 00000007 00           4         114
# error_counter,err_inj = 00000007 00           4         115
# error_counter,err_inj = 00000007 00           4         116
# error_counter,err_inj = 00000007 00           4         117
# error_counter,err_inj = 00000007 00           4         118
# error_counter,err_inj = 00000007 00           4         119
# error_counter,err_inj = 00000007 00           4         120
# error_counter,err_inj = 00000007 00           4         121
# error_counter,err_inj = 00000007 00           4         122
# error_counter,err_inj = 00000007 00           4         123
# error_counter,err_inj = 00000007 00           4         124
# error_counter,err_inj = 00000007 00           4         125
# error_counter,err_inj = 00000007 00           4         126
# error_counter,err_inj = 00000007 00           4         127
# error_counter,err_inj = 00000008 01           4         128
# error_counter,err_inj = 00000008 00           4         129
# error_counter,err_inj = 00000008 00           4         130
# error_counter,err_inj = 00000008 00           4         131
# error_counter,err_inj = 00000008 00           4         132
# error_counter,err_inj = 00000008 00           4         133
# error_counter,err_inj = 00000008 00           4         134
# error_counter,err_inj = 00000008 00           4         135
# error_counter,err_inj = 00000008 00           4         136
# error_counter,err_inj = 00000008 00           4         137
```

```
# error_counter,err_inj = 00000008 00          4         138
# error_counter,err_inj = 00000008 00          4         139
# error_counter,err_inj = 00000008 00          4         140
# error_counter,err_inj = 00000008 00          4         141
# error_counter,err_inj = 00000008 00          4         142
# error_counter,err_inj = 00000008 00          4         143
# error_counter,err_inj = 00000009 01          4         144
# error_counter,err_inj = 00000009 00          4         145
# error_counter,err_inj = 00000009 00          4         146
# error_counter,err_inj = 00000009 00          4         147
# error_counter,err_inj = 00000009 00          4         148
# error_counter,err_inj = 00000009 00          4         149
# error_counter,err_inj = 00000009 00          4         150
# error_counter,err_inj = 00000009 00          4         151
# error_counter,err_inj = 00000009 00          4         152
# error_counter,err_inj = 00000009 00          4         153
# error_counter,err_inj = 00000009 00          4         154
# error_counter,err_inj = 00000009 00          4         155
# error_counter,err_inj = 00000009 00          4         156
# error_counter,err_inj = 00000009 00          4         157
# error_counter,err_inj = 00000009 00          4         158
# error_counter,err_inj = 00000009 00          4         159
# error_counter,err_inj = 0000000a 01          4         160
# error_counter,err_inj = 0000000a 00          5         161
# error_counter,err_inj = 0000000a 00          5         162
# error_counter,err_inj = 0000000a 00          5         163
# error_counter,err_inj = 0000000a 00          5         164
# error_counter,err_inj = 0000000a 00          5         165
# error_counter,err_inj = 0000000a 00          5         166
# error_counter,err_inj = 0000000a 00          5         167
# error_counter,err_inj = 0000000a 00          5         168
# error_counter,err_inj = 0000000a 00          5         169
# error_counter,err_inj = 0000000a 00          5         170
# error_counter,err_inj = 0000000a 00          5         171
# error_counter,err_inj = 0000000a 00          5         172
# error_counter,err_inj = 0000000a 00          5         173
# error_counter,err_inj = 0000000a 00          5         174
# error_counter,err_inj = 0000000a 00          5         175
# error_counter,err_inj = 0000000b 01          5         176
# error_counter,err_inj = 0000000b 00          6         177
# error_counter,err_inj = 0000000b 00          6         178
# error_counter,err_inj = 0000000b 00          6         179
# error_counter,err_inj = 0000000b 00          6         180
# error_counter,err_inj = 0000000b 00          6         181
# error_counter,err_inj = 0000000b 00          6         182
# error_counter,err_inj = 0000000b 00          6         183
# error_counter,err_inj = 0000000b 00          6         184
# error_counter,err_inj = 0000000b 00          6         185
# error_counter,err_inj = 0000000b 00          6         186
# error_counter,err_inj = 0000000b 00          6         187
# error_counter,err_inj = 0000000b 00          6         188
# error_counter,err_inj = 0000000b 00          6         189
# error_counter,err_inj = 0000000b 00          6         190
# error_counter,err_inj = 0000000b 00          6         191
# error_counter,err_inj = 0000000c 01          6         192
```

```
# error_counter,err_inj = 0000000c 00          7         193
# error_counter,err_inj = 0000000c 00          7         194
# error_counter,err_inj = 0000000c 00          7         195
# error_counter,err_inj = 0000000c 00          7         196
# error_counter,err_inj = 0000000c 00          7         197
# error_counter,err_inj = 0000000c 00          7         198
# error_counter,err_inj = 0000000c 00          7         199
# error_counter,err_inj = 0000000c 00          7         200
# error_counter,err_inj = 0000000c 00          7         201
# error_counter,err_inj = 0000000c 00          7         202
# error_counter,err_inj = 0000000c 00          7         203
# error_counter,err_inj = 0000000c 00          7         204
# error_counter,err_inj = 0000000c 00          7         205
# error_counter,err_inj = 0000000c 00          7         206
# error_counter,err_inj = 0000000c 00          7         207
# error_counter,err_inj = 0000000d 01          7         208
# error_counter,err_inj = 0000000d 00          7         209
# error_counter,err_inj = 0000000d 00          7         210
# error_counter,err_inj = 0000000d 00          7         211
# error_counter,err_inj = 0000000d 00          7         212
# error_counter,err_inj = 0000000d 00          7         213
# error_counter,err_inj = 0000000d 00          7         214
# error_counter,err_inj = 0000000d 00          7         215
# error_counter,err_inj = 0000000d 00          7         216
# error_counter,err_inj = 0000000d 00          7         217
# error_counter,err_inj = 0000000d 00          7         218
# error_counter,err_inj = 0000000d 00          7         219
# error_counter,err_inj = 0000000d 00          7         220
# error_counter,err_inj = 0000000d 00          7         221
# error_counter,err_inj = 0000000d 00          7         222
# error_counter,err_inj = 0000000d 00          7         223
# error_counter,err_inj = 0000000e 01          7         224
# error_counter,err_inj = 0000000e 00          7         225
# error_counter,err_inj = 0000000e 00          7         226
# error_counter,err_inj = 0000000e 00          7         227
# error_counter,err_inj = 0000000e 00          7         228
# error_counter,err_inj = 0000000e 00          7         229
# error_counter,err_inj = 0000000e 00          7         230
# error_counter,err_inj = 0000000e 00          7         231
# error_counter,err_inj = 0000000e 00          7         232
# error_counter,err_inj = 0000000e 00          7         233
# error_counter,err_inj = 0000000e 00          7         234
# error_counter,err_inj = 0000000e 00          7         235
# error_counter,err_inj = 0000000e 00          7         236
# error_counter,err_inj = 0000000e 00          7         237
# error_counter,err_inj = 0000000e 00          7         238
# error_counter,err_inj = 0000000e 00          7         239
# error_counter,err_inj = 0000000f 01          7         240
# error_counter,err_inj = 0000000f 00          8         241
# error_counter,err_inj = 0000000f 00          8         242
# error_counter,err_inj = 0000000f 00          8         243
# error_counter,err_inj = 0000000f 00          8         244
# error_counter,err_inj = 0000000f 00          8         245
# error_counter,err_inj = 0000000f 00          8         246
# error_counter,err_inj = 0000000f 00          8         247
```

```
# error_counter,err_inj = 0000000f 00           8         248
# error_counter,err_inj = 0000000f 00           8         249
# error_counter,err_inj = 0000000f 00           8         250
# error_counter,err_inj = 0000000f 00           8         251
# error_counter,err_inj = 0000000f 00           8         252
# error_counter,err_inj = 0000000f 00           8         253
# error_counter,err_inj = 0000000f 00           8         254
# error_counter,err_inj = 0000000f 00           8         255
# word_count =       10440
# yaa! in = 0, out = 0, w_ct =          0, err = 00
# yaa! in = 0, out = 0, w_ct =          1, err = 00
# yaa! in = 0, out = 0, w_ct =          2, err = 00
# yaa! in = 0, out = 0, w_ct =          3, err = 00
# yaa! in = 0, out = 0, w_ct =          4, err = 00
# yaa! in = 0, out = 0, w_ct =          5, err = 00
# yaa! in = 0, out = 0, w_ct =          6, err = 00
# yaa! in = 0, out = 0, w_ct =          7, err = 00
# yaa! in = 0, out = 0, w_ct =          8, err = 00
# yaa! in = 0, out = 0, w_ct =          9, err = 00
# yaa! in = 0, out = 0, w_ct =         10, err = 00
# yaa! in = 1, out = 1, w_ct =         11, err = 00
# yaa! in = 0, out = 0, w_ct =         12, err = 00
# yaa! in = 0, out = 0, w_ct =         13, err = 00
# yaa! in = 1, out = 1, w_ct =         14, err = 00
# yaa! in = 1, out = 1, w_ct =         15, err = 00
# yaa! in = 0, out = 0, w_ct =         16, err = 00
# yaa! in = 0, out = 0, w_ct =         17, err = 00
# yaa! in = 0, out = 0, w_ct =         18, err = 00
# yaa! in = 1, out = 1, w_ct =         19, err = 00
# yaa! in = 1, out = 1, w_ct =         20, err = 00
# yaa! in = 1, out = 1, w_ct =         21, err = 00
# yaa! in = 0, out = 0, w_ct =         22, err = 00
# yaa! in = 0, out = 0, w_ct =         23, err = 00
# yaa! in = 0, out = 0, w_ct =         24, err = 00
# yaa! in = 0, out = 0, w_ct =         25, err = 00
# yaa! in = 1, out = 1, w_ct =         26, err = 00
# yaa! in = 1, out = 1, w_ct =         27, err = 00
# yaa! in = 1, out = 1, w_ct =         28, err = 00
# yaa! in = 1, out = 1, w_ct =         29, err = 00
# yaa! in = 0, out = 0, w_ct =         30, err = 00
# yaa! in = 0, out = 0, w_ct =         31, err = 00
# yaa! in = 0, out = 0, w_ct =         32, err = 00
# yaa! in = 0, out = 0, w_ct =         33, err = 00
# yaa! in = 0, out = 0, w_ct =         34, err = 00
# yaa! in = 1, out = 1, w_ct =         35, err = 00
# yaa! in = 1, out = 1, w_ct =         36, err = 00
# yaa! in = 1, out = 1, w_ct =         37, err = 00
# yaa! in = 1, out = 1, w_ct =         38, err = 00
# yaa! in = 1, out = 1, w_ct =         39, err = 00
# yaa! in = 0, out = 0, w_ct =         40, err = 00
# yaa! in = 1, out = 1, w_ct =         41, err = 00
# yaa! in = 0, out = 0, w_ct =         42, err = 00
# yaa! in = 1, out = 1, w_ct =         43, err = 00
# yaa! in = 0, out = 0, w_ct =         44, err = 00
# yaa! in = 1, out = 1, w_ct =         45, err = 00
```

```
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
```

```
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
```

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# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
```

```
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =           0
```

Resource Usage Report

```
# Info: ************************************************************
# Info: Device Utilization for 7A100TCSG324
# Info: ************************************************************
# Info: Resource                          Used    Avail   Utilization
# Info: ------------------------------------------------------------
# Info: IOs                               6       210       2.86%
# Info: Global Buffers                    1       32        3.12%
# Info: LUTs                              327     63400     0.52%
# Info: CLB Slices                        32      15850     0.20%
# Info: Dffs or Latches                   251     126800    0.20%
# Info: Block RAMs                        3       135       2.22%
# Info:    RAMB18E1                       6
# Info: DSP48E1s                          0       240       0.00%
# Info: ------------------------------------------------------------
# Info: ********************************************************
# Info: Library: work    Cell: decoder    View: INTERFACE
# Info: ********************************************************
# Info:  Number of ports :                            6
# Info:  Number of nets :                           739
# Info:  Number of instances :                      703
# Info:  Number of references to this view :          0
# Info: Total accumulated area :
# Info:  Number of Block RAMs :                       3
# Info:  Number of Dffs or Latches :                251
# Info:  Number of LUTs :                           327
# Info:  Number of Primitive LUTs :                 446
# Info:  Number of LUTs with LUTNM/HLUTNM :         238
# Info:  Number of MUX CARRYs :                      27
# Info:  Number of accumulated instances :          780
# Info: ****************************
```

Netlist Report

```
(* virtex_map_wide_clusters = "true" *)
module decoder ( clk, rst, enable, d_in, d_out ) ;

input clk ;
input rst ;
input enable ;
input [1:0]d_in ;
output d_out ;

wire [7:0]d_o_mem_A;
wire [7:0]d_o_mem_B;
wire [7:0]d_o_mem_C;
wire [7:0]d_o_mem_D;
wire wr_disp_mem_0, wr_disp_mem_1, d_in_disp_mem_0, d_in_disp_mem_1,
d_o_disp_mem_1, d_o_disp_mem_0, sclear_and_0_dup_125, nx8474z1,
sclear_and_1_dup_126, nx8475z1, sclear_and_2_dup_127, nx8476z1,
sclear_and_3_dup_128, nx8477z1, sclear_and_4_dup_129, nx8478z1,
sclear_and_5_dup_130, nx8479z1, sclear_and_6_dup_131, nx8480z1,
sclear_and_7_dup_132, nx8481z1, sclear_and_8_dup_133, nx51679z1,
sclear_and_9_dup_134, \dec_d(0) , nx63827z1, \dec_d(1) , nx6076z1,
\dec_d(2) , nx51675z1, \dec_d(3) , nx43890z1, \dec_d(4) , nx37092z1,
\dec_d(5) , nx29307z1, \dec_d(6) , nx21522z1, \dec_d(7) , nx13737z1,
\dec_d(8) , nx37206z1, \dec_d(9) , sload_mux_0_dup_119, nx60837z1,
```

```
sload_mux_1_dup_120, nx3086z1, sload_mux_2_dup_121, nx54665z1,
sload_mux_3_dup_122, nx46880z1, sload_mux_4_dup_123, nx40082z1,
sload_mux_5_dup_124, nx32297z1, sload_mux_6_dup_125, nx24512z1,
sload_mux_7_dup_126, nx16727z1, sload_mux_8_dup_127, nx34216z1,
sload_mux_9_dup_128;
wire clk_int;
wire rst_int, enable_int;
wire [1:0]d_in_int;
wire nx56042z3, nx63827z2, nx6076z2, nx51675z2, nx43890z2, nx37092z2,
nx29307z2, nx21522z2, nx13737z2, nx37206z2;
wire not_rst;
wire not_mem_bank_Q2_0, nx36344z1, nx37341z1, nx48298z1, nx48297z1,
nx65470z1, nx6235z1, nx5238z1, nx4241z1, nx3244z1, nx2247z1, nx1250z1,
nx253z1, nx64792z1, nx63795z1, nx62798z1, nx14654z1, nx13657z1,
nx12660z1, nx11663z1, nx10666z1, nx9669z1, nx8672z1, nx7675z1, nx6678z1,
nx5681z1, nx20648z1, nx19651z1, nx18654z1, nx17657z1, nx16660z1,
nx15663z1, nx14666z1, nx13669z1, nx54173z1, nx55170z1, nx56167z1,
nx57164z1, nx58161z1, nx59158z1, nx60155z1, nx61152z1, nx29067z1,
nx28070z1, nx27073z1, nx26076z1, nx25079z1, nx24082z1, nx23085z1,
nx22088z1, nx45754z1, nx46751z1, nx47748z1, nx48745z1, nx49742z1,
nx50739z1, nx51736z1, nx52733z1, nx29651z1, nx28654z1, nx27657z1,
nx26660z1, nx25663z1, nx24666z1, nx23669z1, nx22672z1, nx21675z1,
nx20678z1, nx52723z1, nx53720z1, nx54717z1, nx55714z1, nx56711z1,
nx57708z1, nx58705z1, nx59702z1, nx60699z1, nx61696z1, nx44304z1,
nx45301z1, nx46298z1, nx47295z1, nx48292z1, nx49289z1, nx50286z1,
nx51283z1, nx52280z1, nx53277z1, nx38070z1, nx37073z1, nx36076z1,
nx35079z1, nx34082z1, nx33085z1, nx32088z1, nx31091z1, nx30094z1,
nx29097z1, nx64075z1, nx65072z1, nx533z1, nx1530z1, nx2527z1, nx3524z1,
nx4521z1, nx5518z1, nx6515z1, nx7512z1, nx63z1, nx56042z2, nx34216z2,
nx16727z2, nx24512z2, nx32297z2, nx40082z2, nx46880z2, nx54665z2,
nx3086z2, nx53052z2, nx53052z1, nx37341z2, nx36344z2, nx58036z1,
nx57039z1, nx56042z1, nx55045z1, nx54048z1, nx53051z1, nx52054z1,
nx51057z1, nx24165z1, nx23168z1, nx23168z3, nx23168z2, nx22171z1,
nx21174z1, nx21174z2, nx20177z1, nx19180z1, nx18183z1, nx51057z2,
nx24165z2, nx24165z5, nx24165z8, nx24165z6, nx21174z3, nx24165z7,
nx22171z2, nx20177z2, nx19180z2, nx19180z3, nx20177z4, nx24165z4,
nx18183z4, nx18183z2, nx17186z1, nx51057z3, nx20177z3, nx24165z3,
nx20177z5, nx19030z1, nx18033z1, nx18033z6, nx18033z5, nx17036z1,
nx16039z1, nx16039z11, nx15042z1, nx14045z1, nx13048z1, nx16039z3,
nx18033z2, nx16039z6, nx16039z10, nx16039z8, nx16039z2, nx17036z2,
nx16039z9, nx15042z2, nx14045z2, nx16039z7, nx15042z4, nx16039z5,
nx13048z3, nx13048z2, nx12051z1, nx18033z3, nx15042z5, nx16039z4,
nx15042z3, nx51641z1, nx52638z1, nx52638z6, nx52638z5, nx53635z1,
nx54632z1, nx54632z2, nx55629z1, nx56626z1, nx57623z1, nx53635z4,
nx52638z2, nx53635z9, nx53635z15, nx53635z12, nx53635z14, nx53635z2,
nx53635z13, nx55629z2, nx56626z2, nx53635z11, nx53635z10, nx53635z7,
nx57623z3, nx57623z2, nx58620z1, nx52638z3, nx53635z8, nx53635z5,
nx53635z3, nx56776z1, nx57773z1, nx57773z2, nx58770z1, nx59767z1,
nx60764z1, nx61761z1, nx62758z1, nx57773z4, nx57773z11, nx57773z5,
nx61761z2, nx60764z2, nx57773z8, nx59767z2, nx58770z2, nx57773z9,
nx57773z7, nx57773z3, nx57773z6, nx57773z10, nx62758z2, nx63755z1,
nx63755z3, nx63755z5, nx63755z2, nx61761z3, nx61911z1, nx62908z1,
nx62908z2, nx63905z1, nx64902z1, nx363z1, nx1360z1, nx1360z5, nx1360z16,
nx1360z11, nx1360z14, nx1360z15, nx1360z13, nx62908z3, nx363z3,
nx1360z12, nx363z2, nx1360z6, nx1360z3, nx1360z2, nx2357z1, nx2357z2,
```

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nx1360z9, nx1360z10, nx3354z1, nx1360z7, nx1360z8, nx1360z4, nx2357z3,
nx18183z3, nx2357z4, nx1510z1, nx2507z1, nx2507z3, nx2507z2, nx3504z1,
nx42042z1, nx4501z1, nx4501z3, nx5498z1, nx6495z1, nx1510z2, nx1510z9,
nx1510z7, nx1510z3, nx1510z5, nx4501z2, nx3504z2, nx1510z6, nx5498z2,
nx5498z3, nx1510z4, nx6495z2, nx7492z1, nx7492z2, nx7492z5, nx1510z8,
nx8489z1, nx8489z2, nx8489z4, nx8489z3, nx7492z3, nx18033z4, nx7492z4,
nx6645z1, nx7642z1, nx7642z3, nx7642z2, nx8639z1, nx9636z1, nx10633z1,
nx11630z1, nx41045z1, nx12627z1, nx10633z6, nx7642z4, nx10633z9,
nx10633z13, nx10633z12, nx10633z10, nx9636z2, nx8639z2, nx10633z11,
nx10633z4, nx10633z2, nx11630z2, nx10633z8, nx12627z3, nx12627z2,
nx13624z1, nx10633z7, nx53635z6, nx10633z5, nx10633z14, nx52638z4,
nx10633z3, nx11780z1, nx1510z10, nx12777z1, nx12777z3, nx12777z2,
nx13774z1, nx14771z1, nx15768z1, nx16765z1, nx17762z1, nx11780z2,
nx11780z5, nx16765z2, nx15768z2, nx11780z6, nx14771z2, nx13774z2,
nx11780z7, nx15768z3, nx16765z3, nx11780z4, nx17762z4, nx18759z1,
nx17186z2, nx17186z3, nx18759z2, nx63755z4, nx16765z4, nx11780z3,
nx17762z3, nx17762z2, nx11312z1, nx11313z1, nx11315z1, nx11314z1,
nx64075z5, nx2527z2, nx60837z2, nx51679z2, nx8481z2, nx8480z2, nx8479z2,
nx8478z2, nx8477z2, nx8476z2, nx8475z2, nx8474z2, nx8473z1,
\rd_mem_counter_disp(0) , nx3524z2, nx3524z3, nx3524z4, nx1530z2,
nx1530z3, nx64075z2, nx64075z3, nx64075z4, nx62798z2;
wire [7:0]d_in_mem_C;
wire wr_mem_C, wr_mem_D, wr_mem_B;
wire [9:0]addr_mem_D;
wire [9:0]addr_mem_B;
wire [9:0]addr_mem_A;
wire [9:0]addr_mem_C;
wire wr_mem_A;
wire [1:0]mem_bank_Q1;
wire [1:0]mem_bank_Q2;
wire selection_tbu_1;
wire [7:0]d_in_1_tbu_1;
wire [7:0]d_in_0_tbu_1;
wire [7:0]d_in_1_tbu_0;
wire [7:0]d_in_0_tbu_0;
wire mem_bank_Q3;
wire [9:0]addr_disp_mem_1;
wire [9:0]addr_disp_mem_0;
wire mem_bank_Q5, d_out_1_0;
wire [7:0]\path_cost(0) ;
wire [7:0]\path_cost(1) ;
wire [7:0]\path_cost(2) ;
wire [7:0]\path_cost(3) ;
wire [7:0]\path_cost(4) ;
wire [7:0]\path_cost(5) ;
wire [7:0]\path_cost(6) ;
wire [7:0]\path_cost(7) ;
wire [7:0]selection;
wire \validity(2) , \validity(1) , \validity(4) , enable_tbu_0,
enable_tbu_1;
wire [9:0]wr_mem_counter;
wire [9:0]rd_mem_counter;
wire [9:0]wr_mem_counter_disp;
wire [1:0]mem_bank;
wire [97:0] \$dummy ;
```

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ram_dq_8_0 trelis_mem_A_mem (.wr_data1 ({d_in_mem_C[7],d_in_mem_C[6],
d_in_mem_C[5],d_in_mem_C[4],d_in_mem_C[3],d_in_mem_C[2],
d_in_mem_C[1],d_in_mem_C[0]}), .rd_data1 ({d_o_mem_A[7],
d_o_mem_A[6],d_o_mem_A[5],d_o_mem_A[4],d_o_mem_A[3],d_o_mem_A[2],
d_o_mem_A[1],d_o_mem_A[0]}), .addr1 ({addr_mem_A[9],addr_mem_A[8]
,addr_mem_A[7],addr_mem_A[6],addr_mem_A[5],addr_mem_A[4],
addr_mem_A[3],addr_mem_A[2],addr_mem_A[1],addr_mem_A[0]}), .wr_clk1 (
clk_int), .wr_ena1 (wr_mem_A)) ;
ram_dq_8_1 trelis_mem_B_mem (.wr_data1 ({d_in_mem_C[7],d_in_mem_C[6],
d_in_mem_C[5],d_in_mem_C[4],d_in_mem_C[3],d_in_mem_C[2],
d_in_mem_C[1],d_in_mem_C[0]}), .rd_data1 ({d_o_mem_B[7],
d_o_mem_B[6],d_o_mem_B[5],d_o_mem_B[4],d_o_mem_B[3],d_o_mem_B[2],
d_o_mem_B[1],d_o_mem_B[0]}), .addr1 ({addr_mem_B[9],addr_mem_B[8]
,addr_mem_B[7],addr_mem_B[6],addr_mem_B[5],addr_mem_B[4],
addr_mem_B[3],addr_mem_B[2],addr_mem_B[1],addr_mem_B[0]}), .wr_clk1 (
clk_int), .wr_ena1 (wr_mem_B)) ;
ram_dq_8_2 trelis_mem_C_mem (.wr_data1 ({d_in_mem_C[7],d_in_mem_C[6],
d_in_mem_C[5],d_in_mem_C[4],d_in_mem_C[3],d_in_mem_C[2],
d_in_mem_C[1],d_in_mem_C[0]}), .rd_data1 ({d_o_mem_C[7],
d_o_mem_C[6],d_o_mem_C[5],d_o_mem_C[4],d_o_mem_C[3],d_o_mem_C[2],
d_o_mem_C[1],d_o_mem_C[0]}), .addr1 ({addr_mem_C[9],addr_mem_C[8]
,addr_mem_C[7],addr_mem_C[6],addr_mem_C[5],addr_mem_C[4],
addr_mem_C[3],addr_mem_C[2],addr_mem_C[1],addr_mem_C[0]}), .wr_clk1 (
clk_int), .wr_ena1 (wr_mem_C)) ;
ram_dq_8_3 trelis_mem_D_mem (.wr_data1 ({d_in_mem_C[7],d_in_mem_C[6],
d_in_mem_C[5],d_in_mem_C[4],d_in_mem_C[3],d_in_mem_C[2],
d_in_mem_C[1],d_in_mem_C[0]}), .rd_data1 ({d_o_mem_D[7],
d_o_mem_D[6],d_o_mem_D[5],d_o_mem_D[4],d_o_mem_D[3],d_o_mem_D[2],
d_o_mem_D[1],d_o_mem_D[0]}), .addr1 ({addr_mem_D[9],addr_mem_D[8]
,addr_mem_D[7],addr_mem_D[6],addr_mem_D[5],addr_mem_D[4],
addr_mem_D[3],addr_mem_D[2],addr_mem_D[1],addr_mem_D[0]}), .wr_clk1 (
clk_int), .wr_ena1 (wr_mem_D)) ;
XORCY xorcy_0 (.O (sclear_and_0_dup_125), .CI (enable_int), .LI (nx8473z1)
) ;
XORCY xorcy_1 (.O (sclear_and_1_dup_126), .CI (nx8474z1), .LI (nx8474z2)) ;
XORCY xorcy_2 (.O (sclear_and_2_dup_127), .CI (nx8475z1), .LI (nx8475z2)) ;
XORCY xorcy_3 (.O (sclear_and_3_dup_128), .CI (nx8476z1), .LI (nx8476z2)) ;
XORCY xorcy_4 (.O (sclear_and_4_dup_129), .CI (nx8477z1), .LI (nx8477z2)) ;
XORCY xorcy_5 (.O (sclear_and_5_dup_130), .CI (nx8478z1), .LI (nx8478z2)) ;
XORCY xorcy_6 (.O (sclear_and_6_dup_131), .CI (nx8479z1), .LI (nx8479z2)) ;
XORCY xorcy_7 (.O (sclear_and_7_dup_132), .CI (nx8480z1), .LI (nx8480z2)) ;
XORCY xorcy_8 (.O (sclear_and_8_dup_133), .CI (nx8481z1), .LI (nx8481z2)) ;
XORCY xorcy_9 (.O (sclear_and_9_dup_134), .CI (nx51679z1), .LI (nx51679z2)
) ;
XORCY xorcy_0_1_10 (.O (\dec_d(0) ), .CI (nx63z1), .LI (nx56042z3)) ;
XORCY xorcy_1_1_12 (.O (\dec_d(1) ), .CI (nx63827z1), .LI (nx63827z2)) ;
XORCY xorcy_2_1_14 (.O (\dec_d(2) ), .CI (nx6076z1), .LI (nx6076z2)) ;
XORCY xorcy_3_1_16 (.O (\dec_d(3) ), .CI (nx51675z1), .LI (nx51675z2)) ;
XORCY xorcy_4_1_18 (.O (\dec_d(4) ), .CI (nx43890z1), .LI (nx43890z2)) ;
XORCY xorcy_5_1_20 (.O (\dec_d(5) ), .CI (nx37092z1), .LI (nx37092z2)) ;
XORCY xorcy_6_1_22 (.O (\dec_d(6) ), .CI (nx29307z1), .LI (nx29307z2)) ;
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XORCY xorcy_7_1_24 (.O (\dec_d(7) ), .CI (nx21522z1), .LI (nx21522z2)) ;
XORCY xorcy_8_1_26 (.O (\dec_d(8) ), .CI (nx13737z1), .LI (nx13737z2)) ;
XORCY xorcy_9_1_28 (.O (\dec_d(9) ), .CI (nx37206z1), .LI (nx37206z2)) ;
XORCY xorcy_0_1_39 (.O (sload_mux_0_dup_119), .CI (nx53052z1), .LI (
nx53052z2)) ;
XORCY xorcy_1_1_41 (.O (sload_mux_1_dup_120), .CI (nx60837z1), .LI (
nx60837z2)) ;
XORCY xorcy_2_1_43 (.O (sload_mux_2_dup_121), .CI (nx3086z1), .LI (nx3086z2)
) ;
XORCY xorcy_3_1_45 (.O (sload_mux_3_dup_122), .CI (nx54665z1), .LI (
nx54665z2)) ;
XORCY xorcy_4_1_47 (.O (sload_mux_4_dup_123), .CI (nx46880z1), .LI (
nx46880z2)) ;
XORCY xorcy_5_1_49 (.O (sload_mux_5_dup_124), .CI (nx40082z1), .LI (
nx40082z2)) ;
XORCY xorcy_6_1_51 (.O (sload_mux_6_dup_125), .CI (nx32297z1), .LI (
nx32297z2)) ;
XORCY xorcy_7_1_53 (.O (sload_mux_7_dup_126), .CI (nx24512z1), .LI (
nx24512z2)) ;
XORCY xorcy_8_1_55 (.O (sload_mux_8_dup_127), .CI (nx16727z1), .LI (
nx16727z2)) ;
XORCY xorcy_9_1_57 (.O (sload_mux_9_dup_128), .CI (nx34216z1), .LI (
nx34216z2)) ;
OBUF d_out_obuf (.O (d_out), .I (d_out_1_0)) ;
IBUF \d_in_ibuf(0) (.O (d_in_int[0]), .I (d_in[0])) ;
IBUF \d_in_ibuf(1) (.O (d_in_int[1]), .I (d_in[1])) ;
IBUF enable_ibuf (.O (enable_int), .I (enable)) ;
IBUF rst_ibuf (.O (rst_int), .I (rst)) ;
INV ix56042z26622 (.O (nx56042z3), .I (rd_mem_counter[0])) ;
INV ix63827z26622 (.O (nx63827z2), .I (rd_mem_counter[1])) ;
INV ix6076z26622 (.O (nx6076z2), .I (rd_mem_counter[2])) ;
INV ix51675z26622 (.O (nx51675z2), .I (rd_mem_counter[3])) ;
INV ix43890z26622 (.O (nx43890z2), .I (rd_mem_counter[4])) ;
INV ix37092z26622 (.O (nx37092z2), .I (rd_mem_counter[5])) ;
INV ix29307z26622 (.O (nx29307z2), .I (rd_mem_counter[6])) ;
INV ix21522z26622 (.O (nx21522z2), .I (rd_mem_counter[7])) ;
INV ix13737z26622 (.O (nx13737z2), .I (rd_mem_counter[8])) ;
INV ix37206z26622 (.O (nx37206z2), .I (rd_mem_counter[9])) ;
INV ix23921z1315 (.O (not_mem_bank_Q2_0), .I (mem_bank_Q2[0])) ;
LUT6 ix36344z45004 (.O (nx36344z1), .I0 (mem_bank[0]), .I1 (
wr_mem_counter[9]), .I2 (wr_mem_counter[8]), .I3 (wr_mem_counter[7]), .I4 (
wr_mem_counter[6]), .I5 (nx36344z2)) ;
defparam ix36344z45004.INIT = 64'h6AAAAAAAAAAAAAAA;
(* HLUTNM = "LUT62_3_85" *)
LUT3 ix37341z1434 (.O (nx37341z1), .I0 (mem_bank[0]), .I1 (nx37341z2), .I2 (
mem_bank[1])) ;
defparam ix37341z1434.INIT = 8'h78;
(* HLUTNM = "LUT62_3_81" *)
LUT3 ix48298z1562 (.O (nx48298z1), .I0 (mem_bank_Q2[0]), .I1 (mem_bank_Q2[1]
), .I2 (enable_tbu_1)) ;
defparam ix48298z1562.INIT = 8'hF8;
(* HLUTNM = "LUT62_3_80" *)
LUT3 ix48297z1558 (.O (nx48297z1), .I0 (mem_bank_Q2[0]), .I1 (mem_bank_Q2[1]
), .I2 (enable_tbu_0)) ;
defparam ix48297z1558.INIT = 8'hF4;
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(* HLUTNM = "LUT62_3_105" *)
LUT3 ix65470z1516 (.O (nx65470z1), .I0 (d_o_disp_mem_0), .I1 (d_o_disp_mem_1
), .I2 (mem_bank_Q5)) ;
defparam ix65470z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_99" *)
LUT3 ix6235z1516 (.O (nx6235z1), .I0 (\rd_mem_counter_disp(0) ), .I1 (
wr_mem_counter_disp[0]), .I2 (mem_bank_Q3)) ;
defparam ix6235z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_98" *)
LUT3 ix5238z1516 (.O (nx5238z1), .I0 (nx3524z2), .I1 (wr_mem_counter_disp[1]
), .I2 (mem_bank_Q3)) ;
defparam ix5238z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_98" *)
LUT3 ix4241z1516 (.O (nx4241z1), .I0 (nx3524z3), .I1 (wr_mem_counter_disp[2]
), .I2 (mem_bank_Q3)) ;
defparam ix4241z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_97" *)
LUT3 ix3244z1516 (.O (nx3244z1), .I0 (nx3524z4), .I1 (wr_mem_counter_disp[3]
), .I2 (mem_bank_Q3)) ;
defparam ix3244z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_96" *)
LUT3 ix2247z1516 (.O (nx2247z1), .I0 (nx1530z2), .I1 (wr_mem_counter_disp[4]
), .I2 (mem_bank_Q3)) ;
defparam ix2247z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_94" *)
LUT3 ix1250z1516 (.O (nx1250z1), .I0 (nx1530z3), .I1 (wr_mem_counter_disp[5]
), .I2 (mem_bank_Q3)) ;
defparam ix1250z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_97" *)
LUT3 ix253z1516 (.O (nx253z1), .I0 (nx64075z2), .I1 (wr_mem_counter_disp[6])
, .I2 (mem_bank_Q3)) ;
defparam ix253z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_103" *)
LUT3 ix64792z1516 (.O (nx64792z1), .I0 (nx64075z3), .I1 (
wr_mem_counter_disp[7]), .I2 (mem_bank_Q3)) ;
defparam ix64792z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_101" *)
LUT3 ix63795z1516 (.O (nx63795z1), .I0 (nx64075z4), .I1 (
wr_mem_counter_disp[8]), .I2 (mem_bank_Q3)) ;
defparam ix63795z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_101" *)
LUT3 ix62798z1516 (.O (nx62798z1), .I0 (nx62798z2), .I1 (
wr_mem_counter_disp[9]), .I2 (mem_bank_Q3)) ;
defparam ix62798z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_95" *)
LUT3 ix14654z1516 (.O (nx14654z1), .I0 (wr_mem_counter_disp[0]), .I1 (
\rd_mem_counter_disp(0) ), .I2 (mem_bank_Q3)) ;
defparam ix14654z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_95" *)
LUT3 ix13657z1516 (.O (nx13657z1), .I0 (wr_mem_counter_disp[1]), .I1 (
nx3524z2), .I2 (mem_bank_Q3)) ;
defparam ix13657z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_94" *)
LUT3 ix12660z1516 (.O (nx12660z1), .I0 (wr_mem_counter_disp[2]), .I1 (
nx3524z3), .I2 (mem_bank_Q3)) ;
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defparam ix12660z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_93" *)
LUT3 ix11663z1516 (.O (nx11663z1), .I0 (wr_mem_counter_disp[3]), .I1 (
nx3524z4), .I2 (mem_bank_Q3)) ;
defparam ix11663z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_93" *)
LUT3 ix10666z1516 (.O (nx10666z1), .I0 (wr_mem_counter_disp[4]), .I1 (
nx1530z2), .I2 (mem_bank_Q3)) ;
defparam ix10666z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_102" *)
LUT3 ix9669z1516 (.O (nx9669z1), .I0 (wr_mem_counter_disp[5]), .I1 (nx1530z3
), .I2 (mem_bank_Q3)) ;
defparam ix9669z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_104" *)
LUT3 ix8672z1516 (.O (nx8672z1), .I0 (wr_mem_counter_disp[6]), .I1 (
nx64075z2), .I2 (mem_bank_Q3)) ;
defparam ix8672z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_104" *)
LUT3 ix7675z1516 (.O (nx7675z1), .I0 (wr_mem_counter_disp[7]), .I1 (
nx64075z3), .I2 (mem_bank_Q3)) ;
defparam ix7675z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_103" *)
LUT3 ix6678z1516 (.O (nx6678z1), .I0 (wr_mem_counter_disp[8]), .I1 (
nx64075z4), .I2 (mem_bank_Q3)) ;
defparam ix6678z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_99" *)
LUT3 ix5681z1516 (.O (nx5681z1), .I0 (wr_mem_counter_disp[9]), .I1 (
nx62798z2), .I2 (mem_bank_Q3)) ;
defparam ix5681z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_63" *)
LUT3 ix20648z1516 (.O (nx20648z1), .I0 (d_o_mem_D[0]), .I1 (d_o_mem_B[0]), .I2 (
mem_bank_Q2[1])) ;
defparam ix20648z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_64" *)
LUT3 ix19651z1516 (.O (nx19651z1), .I0 (d_o_mem_D[1]), .I1 (d_o_mem_B[1]), .I2 (
mem_bank_Q2[1])) ;
defparam ix19651z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_65" *)
LUT3 ix18654z1516 (.O (nx18654z1), .I0 (d_o_mem_D[2]), .I1 (d_o_mem_B[2]), .I2 (
mem_bank_Q2[1])) ;
defparam ix18654z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_66" *)
LUT3 ix17657z1516 (.O (nx17657z1), .I0 (d_o_mem_D[3]), .I1 (d_o_mem_B[3]), .I2 (
mem_bank_Q2[1])) ;
defparam ix17657z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_67" *)
LUT3 ix16660z1516 (.O (nx16660z1), .I0 (d_o_mem_D[4]), .I1 (d_o_mem_B[4]), .I2 (
mem_bank_Q2[1])) ;
defparam ix16660z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_68" *)
LUT3 ix15663z1516 (.O (nx15663z1), .I0 (d_o_mem_D[5]), .I1 (d_o_mem_B[5]), .I2 (
mem_bank_Q2[1])) ;
defparam ix15663z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_69" *)
LUT3 ix14666z1516 (.O (nx14666z1), .I0 (d_o_mem_D[6]), .I1 (d_o_mem_B[6]), .I2 (
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mem_bank_Q2[1])) ;
defparam ix14666z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_70" *)
LUT3 ix13669z1516 (.O (nx13669z1), .I0 (d_o_mem_D[7]), .I1 (d_o_mem_B[7]), .I2 (
mem_bank_Q2[1])) ;
defparam ix13669z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_62" *)
LUT3 ix54173z1516 (.O (nx54173z1), .I0 (d_o_mem_C[0]), .I1 (d_o_mem_A[0]), .I2 (
mem_bank_Q2[1])) ;
defparam ix54173z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_61" *)
LUT3 ix55170z1516 (.O (nx55170z1), .I0 (d_o_mem_C[1]), .I1 (d_o_mem_A[1]), .I2 (
mem_bank_Q2[1])) ;
defparam ix55170z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_60" *)
LUT3 ix56167z1516 (.O (nx56167z1), .I0 (d_o_mem_C[2]), .I1 (d_o_mem_A[2]), .I2 (
mem_bank_Q2[1])) ;
defparam ix56167z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_59" *)
LUT3 ix57164z1516 (.O (nx57164z1), .I0 (d_o_mem_C[3]), .I1 (d_o_mem_A[3]), .I2 (
mem_bank_Q2[1])) ;
defparam ix57164z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_58" *)
LUT3 ix58161z1516 (.O (nx58161z1), .I0 (d_o_mem_C[4]), .I1 (d_o_mem_A[4]), .I2 (
mem_bank_Q2[1])) ;
defparam ix58161z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_57" *)
LUT3 ix59158z1516 (.O (nx59158z1), .I0 (d_o_mem_C[5]), .I1 (d_o_mem_A[5]), .I2 (
mem_bank_Q2[1])) ;
defparam ix59158z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_100" *)
LUT3 ix60155z1516 (.O (nx60155z1), .I0 (d_o_mem_C[6]), .I1 (d_o_mem_A[6]), .I2 (
mem_bank_Q2[1])) ;
defparam ix60155z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_100" *)
LUT3 ix61152z1516 (.O (nx61152z1), .I0 (d_o_mem_C[7]), .I1 (d_o_mem_A[7]), .I2 (
mem_bank_Q2[1])) ;
defparam ix61152z1516.INIT = 8'hCA;
(* HLUTNM = "LUT62_3_62" *)
LUT4 ix29067z53198 (.O (nx29067z1), .I0 (d_o_mem_A[0]), .I1 (d_o_mem_C[0]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix29067z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_61" *)
LUT4 ix28070z53198 (.O (nx28070z1), .I0 (d_o_mem_A[1]), .I1 (d_o_mem_C[1]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix28070z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_60" *)
LUT4 ix27073z53198 (.O (nx27073z1), .I0 (d_o_mem_A[2]), .I1 (d_o_mem_C[2]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix27073z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_59" *)
LUT4 ix26076z53198 (.O (nx26076z1), .I0 (d_o_mem_A[3]), .I1 (d_o_mem_C[3]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix26076z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_58" *)
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LUT4 ix25079z53198 (.O (nx25079z1), .I0 (d_o_mem_A[4]), .I1 (d_o_mem_C[4]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix25079z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_57" *)
LUT4 ix24082z53198 (.O (nx24082z1), .I0 (d_o_mem_A[5]), .I1 (d_o_mem_C[5]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix24082z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_81" *)
LUT4 ix23085z53198 (.O (nx23085z1), .I0 (d_o_mem_A[6]), .I1 (d_o_mem_C[6]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix23085z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_80" *)
LUT4 ix22088z53198 (.O (nx22088z1), .I0 (d_o_mem_A[7]), .I1 (d_o_mem_C[7]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix22088z53198.INIT = 16'hCAAC;
(* HLUTNM = "LUT62_3_63" *)
LUT4 ix45754z45548 (.O (nx45754z1), .I0 (d_o_mem_B[0]), .I1 (d_o_mem_D[0]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix45754z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_64" *)
LUT4 ix46751z45548 (.O (nx46751z1), .I0 (d_o_mem_B[1]), .I1 (d_o_mem_D[1]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix46751z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_65" *)
LUT4 ix47748z45548 (.O (nx47748z1), .I0 (d_o_mem_B[2]), .I1 (d_o_mem_D[2]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix47748z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_66" *)
LUT4 ix48745z45548 (.O (nx48745z1), .I0 (d_o_mem_B[3]), .I1 (d_o_mem_D[3]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix48745z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_67" *)
LUT4 ix49742z45548 (.O (nx49742z1), .I0 (d_o_mem_B[4]), .I1 (d_o_mem_D[4]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix49742z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_68" *)
LUT4 ix50739z45548 (.O (nx50739z1), .I0 (d_o_mem_B[5]), .I1 (d_o_mem_D[5]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix50739z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_69" *)
LUT4 ix51736z45548 (.O (nx51736z1), .I0 (d_o_mem_B[6]), .I1 (d_o_mem_D[6]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix51736z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_70" *)
LUT4 ix52733z45548 (.O (nx52733z1), .I0 (d_o_mem_B[7]), .I1 (d_o_mem_D[7]),
.I2 (mem_bank_Q2[1]), .I3 (mem_bank_Q2[0])) ;
defparam ix52733z45548.INIT = 16'hACCA;
(* HLUTNM = "LUT62_3_40" *)
LUT4 ix29651z62890 (.O (nx29651z1), .I0 (wr_mem_counter[0]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[0]), .I3 (mem_bank[0])) ;
defparam ix29651z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_39" *)
LUT4 ix28654z62890 (.O (nx28654z1), .I0 (wr_mem_counter[1]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[1]), .I3 (mem_bank[0])) ;
defparam ix28654z62890.INIT = 16'hF088;
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(* HLUTNM = "LUT62_3_38" *)
LUT4 ix27657z62890 (.O (nx27657z1), .I0 (wr_mem_counter[2]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[2]), .I3 (mem_bank[0])) ;
defparam ix27657z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_37" *)
LUT4 ix26660z62890 (.O (nx26660z1), .I0 (wr_mem_counter[3]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[3]), .I3 (mem_bank[0])) ;
defparam ix26660z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_36" *)
LUT4 ix25663z62890 (.O (nx25663z1), .I0 (wr_mem_counter[4]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[4]), .I3 (mem_bank[0])) ;
defparam ix25663z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_34" *)
LUT4 ix24666z62890 (.O (nx24666z1), .I0 (wr_mem_counter[5]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[5]), .I3 (mem_bank[0])) ;
defparam ix24666z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_33" *)
LUT4 ix23669z62890 (.O (nx23669z1), .I0 (wr_mem_counter[6]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[6]), .I3 (mem_bank[0])) ;
defparam ix23669z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_31" *)
LUT4 ix22672z62890 (.O (nx22672z1), .I0 (wr_mem_counter[7]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[7]), .I3 (mem_bank[0])) ;
defparam ix22672z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_30" *)
LUT4 ix21675z62890 (.O (nx21675z1), .I0 (wr_mem_counter[8]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[8]), .I3 (mem_bank[0])) ;
defparam ix21675z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_29" *)
LUT4 ix20678z62890 (.O (nx20678z1), .I0 (wr_mem_counter[9]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[9]), .I3 (mem_bank[0])) ;
defparam ix20678z62890.INIT = 16'hF088;
(* HLUTNM = "LUT62_3_41" *)
LUT4 ix52723z62822 (.O (nx52723z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[0]), .I2 (rd_mem_counter[0]), .I3 (mem_bank[0])) ;
defparam ix52723z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_42" *)
LUT4 ix53720z62822 (.O (nx53720z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[1]), .I2 (rd_mem_counter[1]), .I3 (mem_bank[0])) ;
defparam ix53720z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_43" *)
LUT4 ix54717z62822 (.O (nx54717z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[2]), .I2 (rd_mem_counter[2]), .I3 (mem_bank[0])) ;
defparam ix54717z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_44" *)
LUT4 ix55714z62822 (.O (nx55714z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[3]), .I2 (rd_mem_counter[3]), .I3 (mem_bank[0])) ;
defparam ix55714z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_45" *)
LUT4 ix56711z62822 (.O (nx56711z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[4]), .I2 (rd_mem_counter[4]), .I3 (mem_bank[0])) ;
defparam ix56711z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_46" *)
LUT4 ix57708z62822 (.O (nx57708z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[5]), .I2 (rd_mem_counter[5]), .I3 (mem_bank[0])) ;
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defparam ix57708z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_47" *)
LUT4 ix58705z62822 (.O (nx58705z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[6]), .I2 (rd_mem_counter[6]), .I3 (mem_bank[0])) ;
defparam ix58705z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_48" *)
LUT4 ix59702z62822 (.O (nx59702z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[7]), .I2 (rd_mem_counter[7]), .I3 (mem_bank[0])) ;
defparam ix59702z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_49" *)
LUT4 ix60699z62822 (.O (nx60699z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[8]), .I2 (rd_mem_counter[8]), .I3 (mem_bank[0])) ;
defparam ix60699z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_51" *)
LUT4 ix61696z62822 (.O (nx61696z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[9]), .I2 (rd_mem_counter[9]), .I3 (mem_bank[0])) ;
defparam ix61696z62822.INIT = 16'hF044;
(* HLUTNM = "LUT62_3_41" *)
LUT4 ix44304z18962 (.O (nx44304z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[0]), .I2 (rd_mem_counter[0]), .I3 (mem_bank[0])) ;
defparam ix44304z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_42" *)
LUT4 ix45301z18962 (.O (nx45301z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[1]), .I2 (rd_mem_counter[1]), .I3 (mem_bank[0])) ;
defparam ix45301z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_43" *)
LUT4 ix46298z18962 (.O (nx46298z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[2]), .I2 (rd_mem_counter[2]), .I3 (mem_bank[0])) ;
defparam ix46298z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_44" *)
LUT4 ix47295z18962 (.O (nx47295z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[3]), .I2 (rd_mem_counter[3]), .I3 (mem_bank[0])) ;
defparam ix47295z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_45" *)
LUT4 ix48292z18962 (.O (nx48292z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[4]), .I2 (rd_mem_counter[4]), .I3 (mem_bank[0])) ;
defparam ix48292z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_46" *)
LUT4 ix49289z18962 (.O (nx49289z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[5]), .I2 (rd_mem_counter[5]), .I3 (mem_bank[0])) ;
defparam ix49289z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_47" *)
LUT4 ix50286z18962 (.O (nx50286z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[6]), .I2 (rd_mem_counter[6]), .I3 (mem_bank[0])) ;
defparam ix50286z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_48" *)
LUT4 ix51283z18962 (.O (nx51283z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[7]), .I2 (rd_mem_counter[7]), .I3 (mem_bank[0])) ;
defparam ix51283z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_49" *)
LUT4 ix52280z18962 (.O (nx52280z1), .I0 (mem_bank[1]), .I1 (
wr_mem_counter[8]), .I2 (rd_mem_counter[8]), .I3 (mem_bank[0])) ;
defparam ix52280z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_51" *)
LUT4 ix53277z18962 (.O (nx53277z1), .I0 (mem_bank[1]), .I1 (
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wr_mem_counter[9]), .I2 (rd_mem_counter[9]), .I3 (mem_bank[0])) ;
defparam ix53277z18962.INIT = 16'h44F0;
(* HLUTNM = "LUT62_3_40" *)
LUT4 ix38070z36370 (.O (nx38070z1), .I0 (wr_mem_counter[0]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[0]), .I3 (mem_bank[0])) ;
defparam ix38070z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_39" *)
LUT4 ix37073z36370 (.O (nx37073z1), .I0 (wr_mem_counter[1]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[1]), .I3 (mem_bank[0])) ;
defparam ix37073z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_38" *)
LUT4 ix36076z36370 (.O (nx36076z1), .I0 (wr_mem_counter[2]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[2]), .I3 (mem_bank[0])) ;
defparam ix36076z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_37" *)
LUT4 ix35079z36370 (.O (nx35079z1), .I0 (wr_mem_counter[3]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[3]), .I3 (mem_bank[0])) ;
defparam ix35079z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_36" *)
LUT4 ix34082z36370 (.O (nx34082z1), .I0 (wr_mem_counter[4]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[4]), .I3 (mem_bank[0])) ;
defparam ix34082z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_34" *)
LUT4 ix33085z36370 (.O (nx33085z1), .I0 (wr_mem_counter[5]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[5]), .I3 (mem_bank[0])) ;
defparam ix33085z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_33" *)
LUT4 ix32088z36370 (.O (nx32088z1), .I0 (wr_mem_counter[6]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[6]), .I3 (mem_bank[0])) ;
defparam ix32088z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_31" *)
LUT4 ix31091z36370 (.O (nx31091z1), .I0 (wr_mem_counter[7]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[7]), .I3 (mem_bank[0])) ;
defparam ix31091z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_30" *)
LUT4 ix30094z36370 (.O (nx30094z1), .I0 (wr_mem_counter[8]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[8]), .I3 (mem_bank[0])) ;
defparam ix30094z36370.INIT = 16'h88F0;
(* HLUTNM = "LUT62_3_29" *)
LUT4 ix29097z36370 (.O (nx29097z1), .I0 (wr_mem_counter[9]), .I1 (
mem_bank[1]), .I2 (rd_mem_counter[9]), .I3 (mem_bank[0])) ;
defparam ix29097z36370.INIT = 16'h88F0;
LUT5 ix64075z34082 (.O (nx64075z1), .I0 (nx64075z2), .I1 (nx64075z3), .I2 (
nx64075z4), .I3 (nx64075z5), .I4 (nx62798z2)) ;
defparam ix64075z34082.INIT = 32'h7FFF8000;
LUT6 ix65072z1314 (.O (nx65072z1), .I0 (nx1530z2), .I1 (nx1530z3), .I2 (
nx64075z2), .I3 (nx64075z3), .I4 (nx64075z4), .I5 (nx2527z2)) ;
defparam ix65072z1314.INIT = 64'h7FFF8000FFFF0000;
(* HLUTNM = "LUT62_3_16" *)
LUT5 ix533z1058 (.O (nx533z1), .I0 (nx1530z2), .I1 (nx1530z3), .I2 (
nx64075z2), .I3 (nx64075z3), .I4 (nx2527z2)) ;
defparam ix533z1058.INIT = 32'h7F80FF00;
(* HLUTNM = "LUT62_3_16" *)
LUT4 ix1530z32274 (.O (nx1530z1), .I0 (nx1530z2), .I1 (nx1530z3), .I2 (
nx64075z2), .I3 (nx2527z2)) ;
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defparam ix1530z32274.INIT = 16'h78F0;
(* HLUTNM = "LUT62_3_96" *)
LUT3 ix2527z1434 (.O (nx2527z1), .I0 (nx1530z2), .I1 (nx2527z2), .I2 (
nx1530z3)) ;
defparam ix2527z1434.INIT = 8'h78;
LUT6 ix3524z1314 (.O (nx3524z1), .I0 (\rd_mem_counter_disp(0) ), .I1 (
enable_int), .I2 (nx3524z2), .I3 (nx3524z3), .I4 (nx3524z4), .I5 (
nx1530z2)) ;
defparam ix3524z1314.INIT = 64'h7FFFFFFF80000000;
(* HLUTNM = "LUT62_3_3" *)
LUT5 ix4521z34082 (.O (nx4521z1), .I0 (\rd_mem_counter_disp(0) ), .I1 (
enable_int), .I2 (nx3524z2), .I3 (nx3524z3), .I4 (nx3524z4)) ;
defparam ix4521z34082.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_71" *)
LUT4 ix5518z33954 (.O (nx5518z1), .I0 (\rd_mem_counter_disp(0) ), .I1 (
enable_int), .I2 (nx3524z2), .I3 (nx3524z3)) ;
defparam ix5518z33954.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_71" *)
LUT3 ix6515z1434 (.O (nx6515z1), .I0 (\rd_mem_counter_disp(0) ), .I1 (
enable_int), .I2 (nx3524z2)) ;
defparam ix6515z1434.INIT = 8'h78;
(* HLUTNM = "LUT62_3_92" *)
LUT2 ix7512z1320 (.O (nx7512z1), .I0 (\rd_mem_counter_disp(0) ), .I1 (
enable_int)) ;
defparam ix7512z1320.INIT = 4'h6;
GND ps_gnd (.G (nx63z1)) ;
VCC ps_vcc (.P (nx56042z2)) ;
LUT2 ix34216z1321 (.O (nx34216z2), .I0 (wr_mem_counter_disp[9]), .I1 (
rst_int)) ;
defparam ix34216z1321.INIT = 4'h7;
LUT2 ix16727z1321 (.O (nx16727z2), .I0 (wr_mem_counter_disp[8]), .I1 (
rst_int)) ;
defparam ix16727z1321.INIT = 4'h7;
LUT2 ix24512z1321 (.O (nx24512z2), .I0 (wr_mem_counter_disp[7]), .I1 (
rst_int)) ;
defparam ix24512z1321.INIT = 4'h7;
LUT2 ix32297z1321 (.O (nx32297z2), .I0 (wr_mem_counter_disp[6]), .I1 (
rst_int)) ;
defparam ix32297z1321.INIT = 4'h7;
LUT2 ix40082z1321 (.O (nx40082z2), .I0 (wr_mem_counter_disp[5]), .I1 (
rst_int)) ;
defparam ix40082z1321.INIT = 4'h7;
LUT2 ix46880z1321 (.O (nx46880z2), .I0 (wr_mem_counter_disp[4]), .I1 (
rst_int)) ;
defparam ix46880z1321.INIT = 4'h7;
LUT2 ix54665z1321 (.O (nx54665z2), .I0 (wr_mem_counter_disp[3]), .I1 (
rst_int)) ;
defparam ix54665z1321.INIT = 4'h7;
LUT2 ix3086z1321 (.O (nx3086z2), .I0 (wr_mem_counter_disp[2]), .I1 (rst_int)
) ;
defparam ix3086z1321.INIT = 4'h7;
LUT2 ix53052z1322 (.O (nx53052z2), .I0 (wr_mem_counter_disp[0]), .I1 (
rst_int)) ;
defparam ix53052z1322.INIT = 4'h7;
LUT2 ix53052z1321 (.O (nx53052z1), .I0 (rst_int), .I1 (enable_int)) ;
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defparam ix53052z1321.INIT = 4'h7;
LUT5 ix37341z1315 (.O (nx37341z2), .I0 (wr_mem_counter[9]), .I1 (
wr_mem_counter[8]), .I2 (wr_mem_counter[7]), .I3 (wr_mem_counter[6]), .I4 (
nx36344z2)) ;
defparam ix37341z1315.INIT = 32'h80000000;
LUT6 ix36344z1315 (.O (nx36344z2), .I0 (wr_mem_counter[5]), .I1 (
wr_mem_counter[4]), .I2 (wr_mem_counter[3]), .I3 (wr_mem_counter[2]), .I4 (
wr_mem_counter[1]), .I5 (wr_mem_counter[0])) ;
defparam ix36344z1315.INIT = 64'h8000000000000000;
(* HLUTNM = "LUT62_3_74" *)
LUT2 ix58036z1322 (.O (nx58036z1), .I0 (enable_int), .I1 (nx11780z2)) ;
defparam ix58036z1322.INIT = 4'h8;
(* HLUTNM = "LUT62_3_72" *)
LUT2 ix57039z1322 (.O (nx57039z1), .I0 (enable_int), .I1 (nx10633z6)) ;
defparam ix57039z1322.INIT = 4'h8;
LUT6 ix56042z42434 (.O (nx56042z1), .I0 (\validity(1) ), .I1 (\validity(4)
), .I2 (enable_int), .I3 (nx1510z3), .I4 (nx1510z7), .I5 (nx1510z9)) ;
defparam ix56042z42434.INIT = 64'hA0202020A0A0A0A0;
(* HLUTNM = "LUT62_3_78" *)
LUT2 ix55045z1322 (.O (nx55045z1), .I0 (enable_int), .I1 (nx1360z5)) ;
defparam ix55045z1322.INIT = 4'h8;
(* HLUTNM = "LUT62_3_77" *)
LUT2 ix54048z1318 (.O (nx54048z1), .I0 (nx57773z4), .I1 (enable_int)) ;
defparam ix54048z1318.INIT = 4'h4;
(* HLUTNM = "LUT62_3_76" *)
LUT2 ix53051z1322 (.O (nx53051z1), .I0 (enable_int), .I1 (nx53635z4)) ;
defparam ix53051z1322.INIT = 4'h8;
(* HLUTNM = "LUT62_3_73" *)
LUT2 ix52054z1322 (.O (nx52054z1), .I0 (enable_int), .I1 (nx16039z3)) ;
defparam ix52054z1322.INIT = 4'h8;
(* HLUTNM = "LUT62_3_75" *)
LUT2 ix51057z1322 (.O (nx51057z1), .I0 (enable_int), .I1 (nx51057z2)) ;
defparam ix51057z1322.INIT = 4'h8;
LUT6 ix24165z1314 (.O (nx24165z1), .I0 (\path_cost(7) [0]), .I1 (
\path_cost(6) [0]), .I2 (d_in_int[1]), .I3 (d_in_int[0]), .I4 (
nx41045z1), .I5 (nx24165z2)) ;
defparam ix24165z1314.INIT = 64'hA55A0000C33C0000;
LUT6 ix23168z1314 (.O (nx23168z1), .I0 (nx24165z2), .I1 (nx23168z2), .I2 (
\path_cost(7) [1]), .I3 (nx23168z3), .I4 (nx51057z2), .I5 (enable_int)
) ;
defparam ix23168z1314.INIT = 64'hF44F444400000000;
(* HLUTNM = "LUT62_3_84" *)
LUT3 ix23168z1528 (.O (nx23168z3), .I0 (\path_cost(7) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix23168z1528.INIT = 8'hD4;
(* HLUTNM = "LUT62_3_32" *)
LUT4 ix23168z7435 (.O (nx23168z2), .I0 (\path_cost(6) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0]), .I3 (\path_cost(6) [1])) ;
defparam ix23168z7435.INIT = 16'h17E8;
LUT6 ix22171z1314 (.O (nx22171z1), .I0 (nx22171z2), .I1 (nx24165z2), .I2 (
\path_cost(7) [2]), .I3 (nx20177z5), .I4 (nx51057z2), .I5 (enable_int)
) ;
defparam ix22171z1314.INIT = 64'h2FF2222200000000;
LUT5 ix21174z1314 (.O (nx21174z1), .I0 (nx21174z2), .I1 (nx51057z2), .I2 (
nx21174z3), .I3 (nx24165z2), .I4 (enable_int)) ;
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defparam ix21174z1314.INIT = 32'h88F80000;
LUT6 ix21174z28557 (.O (nx21174z2), .I0 (\path_cost(7) [3]), .I1 (
\path_cost(7) [2]), .I2 (\path_cost(7) [1]), .I3 (\path_cost(7) [0]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix21174z28557.INIT = 64'hAAAA6AAA6AAA6A6A;
LUT5 ix20177z1314 (.O (nx20177z1), .I0 (nx20177z2), .I1 (nx24165z2), .I2 (
nx20177z4), .I3 (nx51057z2), .I4 (enable_int)) ;
defparam ix20177z1314.INIT = 32'hF2220000;
LUT5 ix19180z1314 (.O (nx19180z1), .I0 (nx19180z2), .I1 (nx24165z2), .I2 (
nx19180z3), .I3 (nx51057z2), .I4 (enable_int)) ;
defparam ix19180z1314.INIT = 32'hF2220000;
(* HLUTNM = "LUT62_3_75" *)
LUT5 ix18183z1314 (.O (nx18183z1), .I0 (nx18183z2), .I1 (nx24165z2), .I2 (
nx18183z4), .I3 (nx51057z2), .I4 (enable_int)) ;
defparam ix18183z1314.INIT = 32'hF2220000;
LUT6 ix51057z2351 (.O (nx51057z2), .I0 (\validity(2) ), .I1 (\validity(1)
), .I2 (nx51057z3), .I3 (nx24165z3), .I4 (nx24165z4), .I5 (nx24165z5)
) ;
defparam ix51057z2351.INIT = 64'h0CCC0CCC0CCC040C;
LUT6 ix24165z17494 (.O (nx24165z2), .I0 (\validity(1) ), .I1 (nx51057z3), .I2 (
nx24165z3), .I3 (nx24165z4), .I4 (nx24165z5), .I5 (\validity(2) )) ;
defparam ix24165z17494.INIT = 64'h2A2A2A023F3F3F33;
LUT6 ix24165z30294 (.O (nx24165z5), .I0 (nx20177z4), .I1 (nx19180z3), .I2 (
nx19180z2), .I3 (nx20177z2), .I4 (nx24165z6), .I5 (nx24165z8)) ;
defparam ix24165z30294.INIT = 64'h00000000F3717130;
LUT6 ix24165z3505 (.O (nx24165z8), .I0 (nx1360z8), .I1 (nx20177z5), .I2 (
nx18183z3), .I3 (nx20177z3), .I4 (\path_cost(7) [6]), .I5 (
\path_cost(6) [6])) ;
defparam ix24165z3505.INIT = 64'h7000800007770888;
LUT6 ix24165z46626 (.O (nx24165z6), .I0 (nx22171z2), .I1 (nx24165z7), .I2 (
nx21174z3), .I3 (\path_cost(7) [3]), .I4 (\path_cost(7) [2]), .I5 (
nx20177z5)) ;
defparam ix24165z46626.INIT = 64'hFBB020F220F2B0FB;
LUT6 ix21174z1316 (.O (nx21174z3), .I0 (\path_cost(6) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0]), .I3 (\path_cost(6) [2]), .I4 (\path_cost(6) [1])
, .I5 (\path_cost(6) [3])) ;
defparam ix21174z1316.INIT = 64'h17FFFFFE8000000;
LUT6 ix24165z19207 (.O (nx24165z7), .I0 (\path_cost(7) [1]), .I1 (
\path_cost(7) [0]), .I2 (\path_cost(6) [0]), .I3 (\path_cost(6) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix24165z19207.INIT = 64'hEF8AF267F26745DF;
(* HLUTNM = "LUT62_3_15" *)
LUT5 ix22171z60707 (.O (nx22171z2), .I0 (\path_cost(6) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(6) [1]), .I4 (
\path_cost(6) [2])) ;
defparam ix22171z60707.INIT = 32'h17FFE800;
(* HLUTNM = "LUT62_3_13" *)
LUT4 ix20177z33955 (.O (nx20177z2), .I0 (\path_cost(6) [3]), .I1 (
\path_cost(6) [2]), .I2 (nx20177z3), .I3 (\path_cost(6) [4])) ;
defparam ix20177z33955.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_13" *)
LUT5 ix19180z34083 (.O (nx19180z2), .I0 (\path_cost(6) [4]), .I1 (
\path_cost(6) [3]), .I2 (\path_cost(6) [2]), .I3 (nx20177z3), .I4 (
\path_cost(6) [5])) ;
defparam ix19180z34083.INIT = 32'h7FFF8000;
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(* HLUTNM = "LUT62_3_14" *)
LUT5 ix19180z34084 (.O (nx19180z3), .I0 (\path_cost(7) [4]), .I1 (
\path_cost(7) [3]), .I2 (\path_cost(7) [2]), .I3 (nx20177z5), .I4 (
\path_cost(7) [5])) ;
defparam ix19180z34084.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_14" *)
LUT4 ix20177z33957 (.O (nx20177z4), .I0 (\path_cost(7) [3]), .I1 (
\path_cost(7) [2]), .I2 (nx20177z5), .I3 (\path_cost(7) [4])) ;
defparam ix20177z33957.INIT = 16'h7F80;
LUT6 ix24165z18793 (.O (nx24165z4), .I0 (\path_cost(7) [6]), .I1 (
\path_cost(6) [6]), .I2 (nx18183z3), .I3 (nx1360z8), .I4 (nx20177z5), .I5 (
nx20177z3)) ;
defparam ix24165z18793.INIT = 64'h2814141488444444;
LUT6 ix18183z45007 (.O (nx18183z4), .I0 (\path_cost(7) [6]), .I1 (
\path_cost(7) [1]), .I2 (\path_cost(7) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx1360z8)) ;
defparam ix18183z45007.INIT = 64'hAA6A6A66AAAAAAAA;
LUT6 ix18183z45005 (.O (nx18183z2), .I0 (\path_cost(6) [6]), .I1 (
\path_cost(6) [1]), .I2 (\path_cost(6) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx18183z3)) ;
defparam ix18183z45005.INIT = 64'h666A6AAAAAAAAAAA;
LUT5 ix17186z58690 (.O (nx17186z1), .I0 (\validity(2) ), .I1 (\validity(1)
), .I2 (nx17186z2), .I3 (nx24165z3), .I4 (nx51057z3)) ;
defparam ix17186z58690.INIT = 32'h0000E020;
LUT5 ix51057z1316 (.O (nx51057z3), .I0 (\path_cost(6) [6]), .I1 (nx18183z3)
, .I2 (nx20177z3), .I3 (\path_cost(6) [7]), .I4 (\validity(2) )) ;
defparam ix51057z1316.INIT = 32'h807F0000;
(* HLUTNM = "LUT62_3_15" *)
LUT4 ix20177z60708 (.O (nx20177z3), .I0 (\path_cost(6) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(6) [1])) ;
defparam ix20177z60708.INIT = 16'hE800;
(* HLUTNM = "LUT62_3_54" *)
LUT4 ix24165z33956 (.O (nx24165z3), .I0 (\path_cost(7) [6]), .I1 (nx1360z8)
, .I2 (nx20177z5), .I3 (\path_cost(7) [7])) ;
defparam ix24165z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_79" *)
LUT4 ix20177z12326 (.O (nx20177z5), .I0 (\path_cost(7) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(7) [1])) ;
defparam ix20177z12326.INIT = 16'h2B00;
LUT6 ix19030z1314 (.O (nx19030z1), .I0 (\path_cost(5) [0]), .I1 (
\path_cost(4) [0]), .I2 (d_in_int[1]), .I3 (d_in_int[0]), .I4 (
nx42042z1), .I5 (nx18033z2)) ;
defparam ix19030z1314.INIT = 64'h3CC300005AA50000;
LUT6 ix18033z1314 (.O (nx18033z1), .I0 (nx18033z2), .I1 (nx18033z5), .I2 (
nx18033z6), .I3 (\path_cost(4) [1]), .I4 (nx16039z3), .I5 (enable_int)
) ;
defparam ix18033z1314.INIT = 64'h4FF4444400000000;
(* HLUTNM = "LUT62_3_83" *)
LUT3 ix18033z1461 (.O (nx18033z6), .I0 (\path_cost(4) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix18033z1461.INIT = 8'h8E;
(* HLUTNM = "LUT62_3_26" *)
LUT4 ix18033z30388 (.O (nx18033z5), .I0 (\path_cost(5) [0]), .I1 (
d_in_int[0]), .I2 (d_in_int[1]), .I3 (\path_cost(5) [1])) ;
defparam ix18033z30388.INIT = 16'h718E;
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LUT6 ix17036z1314 (.O (nx17036z1), .I0 (nx17036z2), .I1 (nx16039z3), .I2 (
nx18033z2), .I3 (\path_cost(5) [2]), .I4 (nx15042z5), .I5 (enable_int)
) ;
defparam ix17036z1314.INIT = 64'h888F8F8800000000;
LUT5 ix16039z1314 (.O (nx16039z1), .I0 (nx16039z2), .I1 (nx16039z3), .I2 (
nx16039z11), .I3 (nx18033z2), .I4 (enable_int)) ;
defparam ix16039z1314.INIT = 32'h88F80000;
LUT6 ix16039z28630 (.O (nx16039z11), .I0 (\path_cost(5) [3]), .I1 (
\path_cost(5) [2]), .I2 (\path_cost(5) [1]), .I3 (\path_cost(5) [0]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix16039z28630.INIT = 64'h6AAA6A6AAAAA6AAA;
LUT5 ix15042z1314 (.O (nx15042z1), .I0 (nx15042z2), .I1 (nx16039z3), .I2 (
nx15042z4), .I3 (nx18033z2), .I4 (enable_int)) ;
defparam ix15042z1314.INIT = 32'h88F80000;
LUT5 ix14045z1314 (.O (nx14045z1), .I0 (nx14045z2), .I1 (nx18033z2), .I2 (
nx16039z7), .I3 (nx16039z3), .I4 (enable_int)) ;
defparam ix14045z1314.INIT = 32'hF2220000;
(* HLUTNM = "LUT62_3_73" *)
LUT5 ix13048z1314 (.O (nx13048z1), .I0 (nx13048z2), .I1 (nx18033z2), .I2 (
nx13048z3), .I3 (nx16039z3), .I4 (enable_int)) ;
defparam ix13048z1314.INIT = 32'hF2220000;
LUT6 ix16039z2352 (.O (nx16039z3), .I0 (\validity(1) ), .I1 (\validity(4)
), .I2 (nx18033z3), .I3 (nx16039z4), .I4 (nx16039z5), .I5 (nx16039z6)
) ;
defparam ix16039z2352.INIT = 64'h0CCC0CCC0CCC040C;
LUT6 ix18033z17494 (.O (nx18033z2), .I0 (\validity(4) ), .I1 (nx18033z3), .I2 (
nx16039z4), .I3 (nx16039z5), .I4 (nx16039z6), .I5 (\validity(1) )) ;
defparam ix18033z17494.INIT = 64'h2A2A2A023F3F3F33;
LUT6 ix16039z47130 (.O (nx16039z6), .I0 (nx15042z4), .I1 (nx16039z7), .I2 (
nx14045z2), .I3 (nx15042z2), .I4 (nx16039z8), .I5 (nx16039z10)) ;
defparam ix16039z47130.INIT = 64'h0000000030B2B2F3;
LUT6 ix16039z29995 (.O (nx16039z10), .I0 (nx18033z4), .I1 (nx15042z5), .I2 (
nx8489z4), .I3 (nx15042z3), .I4 (\path_cost(5) [6]), .I5 (
\path_cost(4) [6])) ;
defparam ix16039z29995.INIT = 64'h0888077780007000;
LUT6 ix16039z58919 (.O (nx16039z8), .I0 (nx16039z9), .I1 (nx17036z2), .I2 (
nx16039z2), .I3 (\path_cost(5) [3]), .I4 (\path_cost(5) [2]), .I5 (
nx15042z5)) ;
defparam ix16039z58919.INIT = 64'hFEE080F880F8E0FE;
LUT6 ix16039z1315 (.O (nx16039z2), .I0 (\path_cost(4) [0]), .I1 (d_in_int[0]
), .I2 (d_in_int[1]), .I3 (\path_cost(4) [2]), .I4 (\path_cost(4) [1])
, .I5 (\path_cost(4) [3])) ;
defparam ix16039z1315.INIT = 64'h4DFFFFFFB2000000;
(* HLUTNM = "LUT62_3_2" *)
LUT5 ix17036z46883 (.O (nx17036z2), .I0 (\path_cost(4) [0]), .I1 (
d_in_int[0]), .I2 (d_in_int[1]), .I3 (\path_cost(4) [1]), .I4 (
\path_cost(4) [2])) ;
defparam ix17036z46883.INIT = 32'h4DFFB200;
LUT6 ix16039z38147 (.O (nx16039z9), .I0 (\path_cost(5) [1]), .I1 (
\path_cost(5) [0]), .I2 (\path_cost(4) [0]), .I3 (\path_cost(4) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix16039z38147.INIT = 64'h8FD9FBA251F78FD9;
(* HLUTNM = "LUT62_3_10" *)
LUT4 ix15042z33955 (.O (nx15042z2), .I0 (\path_cost(4) [3]), .I1 (
\path_cost(4) [2]), .I2 (nx15042z3), .I3 (\path_cost(4) [4])) ;
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defparam ix15042z33955.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_8" *)
LUT5 ix14045z34083 (.O (nx14045z2), .I0 (\path_cost(5) [4]), .I1 (
\path_cost(5) [3]), .I2 (\path_cost(5) [2]), .I3 (nx15042z5), .I4 (
\path_cost(5) [5])) ;
defparam ix14045z34083.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_10" *)
LUT5 ix16039z34088 (.O (nx16039z7), .I0 (\path_cost(4) [4]), .I1 (
\path_cost(4) [3]), .I2 (\path_cost(4) [2]), .I3 (nx15042z3), .I4 (
\path_cost(4) [5])) ;
defparam ix16039z34088.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_8" *)
LUT4 ix15042z33957 (.O (nx15042z4), .I0 (\path_cost(5) [3]), .I1 (
\path_cost(5) [2]), .I2 (nx15042z5), .I3 (\path_cost(5) [4])) ;
defparam ix15042z33957.INIT = 16'h7F80;
LUT6 ix16039z10056 (.O (nx16039z5), .I0 (\path_cost(5) [6]), .I1 (
\path_cost(4) [6]), .I2 (nx18033z4), .I3 (nx8489z4), .I4 (nx15042z3), .I5 (
nx15042z5)) ;
defparam ix16039z10056.INIT = 64'h4812121288222222;
LUT6 ix13048z45006 (.O (nx13048z3), .I0 (\path_cost(4) [6]), .I1 (
\path_cost(4) [1]), .I2 (\path_cost(4) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx8489z4)) ;
defparam ix13048z45006.INIT = 64'h6AAA666AAAAAAAAA;
LUT6 ix13048z45005 (.O (nx13048z2), .I0 (\path_cost(5) [6]), .I1 (
\path_cost(5) [1]), .I2 (\path_cost(5) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx18033z4)) ;
defparam ix13048z45005.INIT = 64'h6A66AA6AAAAAAAAA;
LUT5 ix12051z1314 (.O (nx12051z1), .I0 (nx16039z4), .I1 (\validity(1) ), .I2 (
nx18033z3), .I3 (\validity(4) ), .I4 (nx17186z2)) ;
defparam ix12051z1314.INIT = 32'h0A0C0000;
LUT5 ix18033z1316 (.O (nx18033z3), .I0 (\path_cost(5) [6]), .I1 (nx18033z4)
, .I2 (nx15042z5), .I3 (\path_cost(5) [7]), .I4 (\validity(1) )) ;
defparam ix18033z1316.INIT = 32'h807F0000;
(* HLUTNM = "LUT62_3_26" *)
LUT4 ix15042z46886 (.O (nx15042z5), .I0 (\path_cost(5) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(5) [1])) ;
defparam ix15042z46886.INIT = 16'hB200;
(* HLUTNM = "LUT62_3_55" *)
LUT4 ix16039z33957 (.O (nx16039z4), .I0 (\path_cost(4) [6]), .I1 (nx8489z4)
, .I2 (nx15042z3), .I3 (\path_cost(4) [7])) ;
defparam ix16039z33957.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_25" *)
LUT4 ix15042z46884 (.O (nx15042z3), .I0 (\path_cost(4) [0]), .I1 (
d_in_int[0]), .I2 (d_in_int[1]), .I3 (\path_cost(4) [1])) ;
defparam ix15042z46884.INIT = 16'hB200;
LUT6 ix51641z1314 (.O (nx51641z1), .I0 (\path_cost(3) [0]), .I1 (
\path_cost(2) [0]), .I2 (d_in_int[1]), .I3 (d_in_int[0]), .I4 (
nx41045z1), .I5 (nx52638z2)) ;
defparam ix51641z1314.INIT = 64'h5AA500003CC30000;
LUT6 ix52638z1314 (.O (nx52638z1), .I0 (nx52638z2), .I1 (nx52638z5), .I2 (
nx52638z6), .I3 (\path_cost(3) [1]), .I4 (nx53635z4), .I5 (enable_int)
) ;
defparam ix52638z1314.INIT = 64'h4FF4444400000000;
(* HLUTNM = "LUT62_3_87" *)
LUT3 ix52638z1461 (.O (nx52638z6), .I0 (\path_cost(3) [0]), .I1 (d_in_int[1]
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), .I2 (d_in_int[0])) ;
defparam ix52638z1461.INIT = 8'h8E;
(* HLUTNM = "LUT62_3_50" *)
LUT4 ix52638z30388 (.O (nx52638z5), .I0 (\path_cost(2) [0]), .I1 (
d_in_int[0]), .I2 (d_in_int[1]), .I3 (\path_cost(2) [1])) ;
defparam ix52638z30388.INIT = 16'h718E;
LUT6 ix53635z1314 (.O (nx53635z1), .I0 (nx53635z2), .I1 (nx52638z2), .I2 (
\path_cost(3) [2]), .I3 (nx53635z3), .I4 (nx53635z4), .I5 (enable_int)
) ;
defparam ix53635z1314.INIT = 64'h2FF2222200000000;
(* HLUTNM = "LUT62_3_76" *)
LUT5 ix54632z1314 (.O (nx54632z1), .I0 (nx54632z2), .I1 (nx53635z4), .I2 (
nx53635z14), .I3 (nx52638z2), .I4 (enable_int)) ;
defparam ix54632z1314.INIT = 32'h88F80000;
LUT6 ix54632z28621 (.O (nx54632z2), .I0 (\path_cost(3) [3]), .I1 (
\path_cost(3) [2]), .I2 (\path_cost(3) [1]), .I3 (\path_cost(3) [0]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix54632z28621.INIT = 64'h6AAAAAA6A6A6AAA;
LUT5 ix55629z1314 (.O (nx55629z1), .I0 (nx55629z2), .I1 (nx52638z2), .I2 (
nx53635z10), .I3 (nx53635z4), .I4 (enable_int)) ;
defparam ix55629z1314.INIT = 32'hF2220000;
LUT5 ix56626z1314 (.O (nx56626z1), .I0 (nx56626z2), .I1 (nx52638z2), .I2 (
nx53635z11), .I3 (nx53635z4), .I4 (enable_int)) ;
defparam ix56626z1314.INIT = 32'hF2220000;
LUT5 ix57623z1314 (.O (nx57623z1), .I0 (nx57623z2), .I1 (nx52638z2), .I2 (
nx57623z3), .I3 (nx53635z4), .I4 (enable_int)) ;
defparam ix57623z1314.INIT = 32'hF2220000;
LUT6 ix53635z2353 (.O (nx53635z4), .I0 (\validity(2) ), .I1 (\validity(1)
), .I2 (nx52638z3), .I3 (nx53635z5), .I4 (nx53635z7), .I5 (nx53635z9)
) ;
defparam ix53635z2353.INIT = 64'h0CCC0CCC0CCC040C;
LUT6 ix52638z17494 (.O (nx52638z2), .I0 (\validity(1) ), .I1 (nx52638z3), .I2 (
nx53635z5), .I3 (nx53635z7), .I4 (nx53635z9), .I5 (\validity(2) )) ;
defparam ix52638z17494.INIT = 64'h2A2A2A023F3F3F33;
LUT6 ix53635z30298 (.O (nx53635z9), .I0 (nx53635z10), .I1 (nx53635z11), .I2 (
nx56626z2), .I3 (nx55629z2), .I4 (nx53635z12), .I5 (nx53635z15)) ;
defparam ix53635z30298.INIT = 64'h00000000F3717130;
LUT6 ix53635z3512 (.O (nx53635z15), .I0 (nx53635z6), .I1 (nx53635z3), .I2 (
nx52638z4), .I3 (nx53635z8), .I4 (\path_cost(3) [6]), .I5 (
\path_cost(2) [6])) ;
defparam ix53635z3512.INIT = 64'h7000800007770888;
LUT6 ix53635z54826 (.O (nx53635z12), .I0 (nx53635z13), .I1 (nx53635z2), .I2 (
nx53635z14), .I3 (\path_cost(3) [3]), .I4 (\path_cost(3) [2]), .I5 (
nx53635z3)) ;
defparam ix53635z54826.INIT = 64'hFDD040F440F4D0FD;
LUT6 ix53635z1327 (.O (nx53635z14), .I0 (\path_cost(2) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(2) [2]), .I4 (
\path_cost(2) [1]), .I5 (\path_cost(2) [3])) ;
defparam ix53635z1327.INIT = 64'h4DFFFFFFB2000000;
(* HLUTNM = "LUT62_3_18" *)
LUT5 ix53635z46883 (.O (nx53635z2), .I0 (\path_cost(2) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(2) [1]), .I4 (
\path_cost(2) [2])) ;
defparam ix53635z46883.INIT = 32'h4DFFB200;
LUT6 ix53635z63381 (.O (nx53635z13), .I0 (\path_cost(3) [1]), .I1 (
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\path_cost(3) [0]), .I2 (\path_cost(2) [0]), .I3 (\path_cost(2) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix53635z63381.INIT = 64'hF267EF8A45DFF267;
LUT4 ix55629z33955 (.O (nx55629z2), .I0 (\path_cost(2) [3]), .I1 (
\path_cost(2) [2]), .I2 (nx53635z8), .I3 (\path_cost(2) [4])) ;
defparam ix55629z33955.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_19" *)
LUT5 ix56626z34083 (.O (nx56626z2), .I0 (\path_cost(2) [4]), .I1 (
\path_cost(2) [3]), .I2 (\path_cost(2) [2]), .I3 (nx53635z8), .I4 (
\path_cost(2) [5])) ;
defparam ix56626z34083.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_17" *)
LUT5 ix53635z34092 (.O (nx53635z11), .I0 (\path_cost(3) [4]), .I1 (
\path_cost(3) [3]), .I2 (\path_cost(3) [2]), .I3 (nx53635z3), .I4 (
\path_cost(3) [5])) ;
defparam ix53635z34092.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_17" *)
LUT4 ix53635z33963 (.O (nx53635z10), .I0 (\path_cost(3) [3]), .I1 (
\path_cost(3) [2]), .I2 (nx53635z3), .I3 (\path_cost(3) [4])) ;
defparam ix53635z33963.INIT = 16'h7F80;
LUT6 ix53635z18796 (.O (nx53635z7), .I0 (\path_cost(3) [6]), .I1 (
\path_cost(2) [6]), .I2 (nx52638z4), .I3 (nx53635z6), .I4 (nx53635z3),
.I5 (nx53635z8)) ;
defparam ix53635z18796.INIT = 64'h2814141488444444;
LUT6 ix57623z45006 (.O (nx57623z3), .I0 (\path_cost(3) [6]), .I1 (
\path_cost(3) [1]), .I2 (\path_cost(3) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx53635z6)) ;
defparam ix57623z45006.INIT = 64'h6AAA666AAAAAAAAA;
LUT6 ix57623z45005 (.O (nx57623z2), .I0 (\path_cost(2) [6]), .I1 (
\path_cost(2) [1]), .I2 (\path_cost(2) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx52638z4)) ;
defparam ix57623z45005.INIT = 64'h6A66AA6AAAAAAAAA;
LUT5 ix58620z58690 (.O (nx58620z1), .I0 (\validity(2) ), .I1 (\validity(1)
), .I2 (nx17186z2), .I3 (nx53635z5), .I4 (nx52638z3)) ;
defparam ix58620z58690.INIT = 32'h0000E020;
LUT5 ix52638z1316 (.O (nx52638z3), .I0 (\path_cost(2) [6]), .I1 (nx52638z4)
, .I2 (nx53635z8), .I3 (\path_cost(2) [7]), .I4 (\validity(2) )) ;
defparam ix52638z1316.INIT = 32'h807F0000;
(* HLUTNM = "LUT62_3_50" *)
LUT4 ix53635z46889 (.O (nx53635z8), .I0 (\path_cost(2) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(2) [1])) ;
defparam ix53635z46889.INIT = 16'hB200;
(* HLUTNM = "LUT62_3_52" *)
LUT4 ix53635z33958 (.O (nx53635z5), .I0 (\path_cost(3) [6]), .I1 (nx53635z6)
, .I2 (nx53635z3), .I3 (\path_cost(3) [7])) ;
defparam ix53635z33958.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_82" *)
LUT4 ix53635z46884 (.O (nx53635z3), .I0 (\path_cost(3) [0]), .I1 (
d_in_int[0]), .I2 (d_in_int[1]), .I3 (\path_cost(3) [1])) ;
defparam ix53635z46884.INIT = 16'hB200;
LUT6 ix56776z13794 (.O (nx56776z1), .I0 (\path_cost(1) [0]), .I1 (
\path_cost(0) [0]), .I2 (enable_int), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx57773z4)) ;
defparam ix56776z13794.INIT = 64'hA05050A0C03030C0;
LUT6 ix57773z1314 (.O (nx57773z1), .I0 (\path_cost(1) [1]), .I1 (nx57773z2)
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, .I2 (\path_cost(0) [1]), .I3 (nx57773z3), .I4 (nx57773z4), .I5 (
enable_int)) ;
defparam ix57773z1314.INIT = 64'h6666F00F00000000;
(* HLUTNM = "LUT62_3_88" *)
LUT3 ix57773z1547 (.O (nx57773z2), .I0 (\path_cost(1) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix57773z1547.INIT = 8'hE8;
(* HLUTNM = "LUT62_3_77" *)
LUT5 ix58770z1314 (.O (nx58770z1), .I0 (nx58770z2), .I1 (\path_cost(1) [2])
, .I2 (nx61761z3), .I3 (nx57773z4), .I4 (enable_int)) ;
defparam ix58770z1314.INIT = 32'h3CAA0000;
LUT6 ix59767z1314 (.O (nx59767z1), .I0 (nx59767z2), .I1 (\path_cost(1) [2])
, .I2 (nx61761z3), .I3 (\path_cost(1) [3]), .I4 (nx57773z4), .I5 (
enable_int)) ;
defparam ix59767z1314.INIT = 64'h3FC0AAAA00000000;
(* HLUTNM = "LUT62_3_91" *)
LUT4 ix60764z45346 (.O (nx60764z1), .I0 (nx60764z2), .I1 (nx57773z7), .I2 (
nx57773z4), .I3 (enable_int)) ;
defparam ix60764z45346.INIT = 16'hAC00;
(* HLUTNM = "LUT62_3_92" *)
LUT4 ix61761z45346 (.O (nx61761z1), .I0 (nx61761z2), .I1 (nx57773z6), .I2 (
nx57773z4), .I3 (enable_int)) ;
defparam ix61761z45346.INIT = 16'hAC00;
LUT6 ix62758z19882 (.O (nx62758z1), .I0 (\path_cost(0) [6]), .I1 (enable_int
), .I2 (nx63755z4), .I3 (nx63755z5), .I4 (nx62758z2), .I5 (nx57773z4)
) ;
defparam ix62758z19882.INIT = 64'hCCCC000048884888;
LUT6 ix57773z1317 (.O (nx57773z4), .I0 (nx57773z5), .I1 (nx57773z11), .I2 (
nx62758z2), .I3 (nx63755z2), .I4 (nx63755z3), .I5 (\validity(1) )) ;
defparam ix57773z1317.INIT = 64'hABFF00AB00000000;
LUT6 ix57773z64637 (.O (nx57773z11), .I0 (nx57773z6), .I1 (nx57773z7), .I2 (
nx60764z2), .I3 (nx61761z2), .I4 (nx57773z8), .I5 (nx57773z10)) ;
defparam ix57773z64637.INIT = 64'h0000000007510F751;
LUT6 ix57773z1318 (.O (nx57773z5), .I0 (nx57773z6), .I1 (nx57773z7), .I2 (
nx57773z8), .I3 (nx60764z2), .I4 (nx61761z2), .I5 (nx57773z10)) ;
defparam ix57773z1318.INIT = 64'h80A8EAFE00000000;
(* HLUTNM = "LUT62_3_21" *)
LUT5 ix61761z34083 (.O (nx61761z2), .I0 (\path_cost(1) [4]), .I1 (
\path_cost(1) [3]), .I2 (\path_cost(1) [2]), .I3 (nx61761z3), .I4 (
\path_cost(1) [5])) ;
defparam ix61761z34083.INIT = 32'h7FFF8000;
LUT4 ix60764z33955 (.O (nx60764z2), .I0 (\path_cost(1) [3]), .I1 (
\path_cost(1) [2]), .I2 (nx61761z3), .I3 (\path_cost(1) [4])) ;
defparam ix60764z33955.INIT = 16'h7F80;
LUT6 ix57773z58919 (.O (nx57773z8), .I0 (nx57773z9), .I1 (nx58770z2), .I2 (
nx59767z2), .I3 (\path_cost(1) [3]), .I4 (\path_cost(1) [2]), .I5 (
nx61761z3)) ;
defparam ix57773z58919.INIT = 64'hFEE080F880F8E0FE;
LUT6 ix59767z1315 (.O (nx59767z2), .I0 (\path_cost(0) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0]), .I3 (\path_cost(0) [2]), .I4 (\path_cost(0) [1])
, .I5 (\path_cost(0) [3])) ;
defparam ix59767z1315.INIT = 64'hD4FFFFFF2B000000;
(* HLUTNM = "LUT62_3_1" *)
LUT5 ix58770z12323 (.O (nx58770z2), .I0 (\path_cost(0) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(0) [1]), .I4 (
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\path_cost(0) [2])) ;
defparam ix58770z12323.INIT = 32'hD4FF2B00;
LUT6 ix57773z22305 (.O (nx57773z9), .I0 (\path_cost(1) [1]), .I1 (
\path_cost(1) [0]), .I2 (\path_cost(0) [0]), .I3 (\path_cost(0) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix57773z22305.INIT = 64'hFBA28FD98FD951F7;
LUT5 ix57773z1448 (.O (nx57773z7), .I0 (\path_cost(0) [3]), .I1 (
\path_cost(0) [2]), .I2 (\path_cost(0) [1]), .I3 (nx57773z3), .I4 (
\path_cost(0) [4])) ;
defparam ix57773z1448.INIT = 32'hFF7F0080;
(* HLUTNM = "LUT62_3_87" *)
LUT3 ix57773z1528 (.O (nx57773z3), .I0 (\path_cost(0) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix57773z1528.INIT = 8'hD4;
(* HLUTNM = "LUT62_3_20" *)
LUT5 ix57773z34087 (.O (nx57773z6), .I0 (\path_cost(0) [4]), .I1 (
\path_cost(0) [3]), .I2 (\path_cost(0) [2]), .I3 (nx63755z5), .I4 (
\path_cost(0) [5])) ;
defparam ix57773z34087.INIT = 32'h7FFF8000;
LUT6 ix57773z45013 (.O (nx57773z10), .I0 (\path_cost(0) [6]), .I1 (
\path_cost(0) [1]), .I2 (\path_cost(0) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx63755z4)) ;
defparam ix57773z45013.INIT = 64'hAA6A6A66AAAAAAAA;
LUT6 ix62758z45005 (.O (nx62758z2), .I0 (\path_cost(1) [6]), .I1 (
\path_cost(1) [1]), .I2 (\path_cost(1) [0]), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx17762z3)) ;
defparam ix62758z45005.INIT = 64'h666A6AAAAAAAAAAA;
LUT4 ix63755z46370 (.O (nx63755z1), .I0 (nx63755z2), .I1 (\validity(1) ), .I2 (
nx63755z3), .I3 (nx17186z2)) ;
defparam ix63755z46370.INIT = 16'hB000;
(* HLUTNM = "LUT62_3_56" *)
LUT4 ix63755z33956 (.O (nx63755z3), .I0 (\path_cost(0) [6]), .I1 (nx63755z4)
, .I2 (nx63755z5), .I3 (\path_cost(0) [7])) ;
defparam ix63755z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_27" *)
LUT4 ix63755z12326 (.O (nx63755z5), .I0 (\path_cost(0) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(0) [1])) ;
defparam ix63755z12326.INIT = 16'h2B00;
(* HLUTNM = "LUT62_3_53" *)
LUT4 ix63755z33955 (.O (nx63755z2), .I0 (\path_cost(1) [6]), .I1 (nx17762z3)
, .I2 (nx61761z3), .I3 (\path_cost(1) [7])) ;
defparam ix63755z33955.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_28" *)
LUT4 ix61761z60708 (.O (nx61761z3), .I0 (\path_cost(1) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(1) [1])) ;
defparam ix61761z60708.INIT = 16'hE800;
LUT6 ix61911z1314 (.O (nx61911z1), .I0 (\path_cost(6) [0]), .I1 (
\path_cost(7) [0]), .I2 (\validity(1) ), .I3 (nx1360z5), .I4 (
nx1510z10), .I5 (enable_int)) ;
defparam ix61911z1314.INIT = 64'h5530AAC000000000;
(* HLUTNM = "LUT62_3_78" *)
LUT5 ix62908z1314 (.O (nx62908z1), .I0 (nx62908z2), .I1 (\path_cost(6) [1])
, .I2 (nx62908z3), .I3 (nx1360z5), .I4 (enable_int)) ;
defparam ix62908z1314.INIT = 32'hC3AA0000;
(* HLUTNM = "LUT62_3_22" *)
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LUT5 ix62908z1315 (.O (nx62908z2), .I0 (\path_cost(7) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0]), .I3 (\path_cost(7) [1]), .I4 (\validity(1) )) ;
defparam ix62908z1315.INIT = 32'h17E80000;
LUT6 ix63905z1314 (.O (nx63905z1), .I0 (\path_cost(7) [2]), .I1 (nx1360z4),
.I2 (\path_cost(6) [2]), .I3 (nx2357z4), .I4 (nx1360z5), .I5 (nx41045z1
)) ;
defparam ix63905z1314.INIT = 64'h0FF0666600000000;
LUT6 ix64902z26050 (.O (nx64902z1), .I0 (\path_cost(7) [3]), .I1 (
\path_cost(7) [2]), .I2 (nx41045z1), .I3 (nx1360z4), .I4 (nx1360z13), .I5 (
nx1360z5)) ;
defparam ix64902z26050.INIT = 64'hF0F0000060A060A0;
LUT6 ix363z1314 (.O (nx363z1), .I0 (\validity(2) ), .I1 (\validity(1) ), .I2 (
enable_int), .I3 (nx363z2), .I4 (nx363z3), .I5 (nx1360z5)) ;
defparam ix363z1314.INIT = 64'hE000E000E0E00000;
LUT6 ix1360z1314 (.O (nx1360z1), .I0 (\validity(2) ), .I1 (\validity(1) )
, .I2 (enable_int), .I3 (nx1360z2), .I4 (nx1360z3), .I5 (nx1360z5)) ;
defparam ix1360z1314.INIT = 64'hE000E000E0E00000;
LUT6 ix1360z1318 (.O (nx1360z5), .I0 (nx1360z6), .I1 (nx1360z3), .I2 (
nx1360z2), .I3 (nx1360z11), .I4 (nx1360z16), .I5 (\validity(2) )) ;
defparam ix1360z1318.INIT = 64'h5DDF555500000000;
LUT5 ix1360z12342 (.O (nx1360z11), .I0 (nx363z2), .I1 (nx1360z12), .I2 (
nx363z3), .I3 (nx1360z13), .I4 (nx1360z14)) ;
defparam ix1360z12342.INIT = 32'hAF2B2B0A;
LUT5 ix1360z12513 (.O (nx1360z14), .I0 (nx1360z15), .I1 (\path_cost(7) [2])
, .I2 (nx2357z4), .I3 (\path_cost(6) [2]), .I4 (nx1360z4)) ;
defparam ix1360z12513.INIT = 32'h8EE82BB2;
LUT6 ix1360z22311 (.O (nx1360z15), .I0 (\path_cost(7) [1]), .I1 (
\path_cost(7) [0]), .I2 (\path_cost(6) [0]), .I3 (\path_cost(6) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix1360z22311.INIT = 64'hFBA28FD98FD951F7;
LUT6 ix1360z28568 (.O (nx1360z13), .I0 (\path_cost(6) [3]), .I1 (
\path_cost(6) [2]), .I2 (\path_cost(6) [1]), .I3 (\path_cost(6) [0]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix1360z28568.INIT = 64'hAAAA6AAA6AAA6A6A;
(* HLUTNM = "LUT62_3_88" *)
LUT3 ix62908z1528 (.O (nx62908z3), .I0 (\path_cost(6) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix62908z1528.INIT = 8'hD4;
LUT4 ix363z33956 (.O (nx363z3), .I0 (\path_cost(7) [3]), .I1 (
\path_cost(7) [2]), .I2 (nx1360z4), .I3 (\path_cost(7) [4])) ;
defparam ix363z33956.INIT = 16'h7F80;
LUT6 ix1360z45015 (.O (nx1360z12), .I0 (\path_cost(7) [3]), .I1 (
\path_cost(7) [2]), .I2 (\path_cost(7) [1]), .I3 (\path_cost(7) [0]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix1360z45015.INIT = 64'h6A6A6AAA6AAAAAAA;
LUT4 ix363z33955 (.O (nx363z2), .I0 (\path_cost(6) [3]), .I1 (
\path_cost(6) [2]), .I2 (nx2357z4), .I3 (\path_cost(6) [4])) ;
defparam ix363z33955.INIT = 16'h7F80;
LUT5 ix1360z12073 (.O (nx1360z6), .I0 (\validity(1) ), .I1 (nx1360z7), .I2 (
nx1360z9), .I3 (nx2357z3), .I4 (nx1360z10)) ;
defparam ix1360z12073.INIT = 32'hAA222A02;
(* HLUTNM = "LUT62_3_7" *)
LUT5 ix1360z34084 (.O (nx1360z3), .I0 (\path_cost(7) [4]), .I1 (
\path_cost(7) [3]), .I2 (\path_cost(7) [2]), .I3 (nx1360z4), .I4 (
\path_cost(7) [5])) ;
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defparam ix1360z34084.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_6" *)
LUT5 ix1360z34083 (.O (nx1360z2), .I0 (\path_cost(6) [4]), .I1 (
\path_cost(6) [3]), .I2 (\path_cost(6) [2]), .I3 (nx2357z4), .I4 (
\path_cost(6) [5])) ;
defparam ix1360z34083.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_89" *)
LUT2 ix2357z1318 (.O (nx2357z1), .I0 (nx2357z2), .I1 (enable_int)) ;
defparam ix2357z1318.INIT = 4'h4;
LUT6 ix2357z5410 (.O (nx2357z2), .I0 (nx2357z3), .I1 (nx1360z7), .I2 (
nx1360z10), .I3 (\validity(2) ), .I4 (nx1360z9), .I5 (\validity(1) )
) ;
defparam ix2357z5410.INIT = 64'h0D00BFFF0FFF0FFF;
LUT6 ix1360z45012 (.O (nx1360z9), .I0 (\path_cost(7) [6]), .I1 (
\path_cost(7) [5]), .I2 (\path_cost(7) [4]), .I3 (\path_cost(7) [3]), .I4 (
\path_cost(7) [2]), .I5 (nx1360z4)) ;
defparam ix1360z45012.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT6 ix1360z45013 (.O (nx1360z10), .I0 (\path_cost(6) [6]), .I1 (
\path_cost(6) [5]), .I2 (\path_cost(6) [4]), .I3 (\path_cost(6) [3]), .I4 (
\path_cost(6) [2]), .I5 (nx2357z4)) ;
defparam ix1360z45013.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT5 ix3354z1314 (.O (nx3354z1), .I0 (\validity(2) ), .I1 (nx2357z3), .I2 (
nx1360z7), .I3 (\validity(1) ), .I4 (nx17186z2)) ;
defparam ix3354z1314.INIT = 32'hD0880000;
(* HLUTNM = "LUT62_3_54" *)
LUT4 ix1360z33960 (.O (nx1360z7), .I0 (\path_cost(7) [6]), .I1 (nx1360z4), .I2 (
nx1360z8), .I3 (\path_cost(7) [7])) ;
defparam ix1360z33960.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_7" *)
LUT4 ix1360z34089 (.O (nx1360z8), .I0 (\path_cost(7) [5]), .I1 (
\path_cost(7) [4]), .I2 (\path_cost(7) [3]), .I3 (\path_cost(7) [2])) ;
defparam ix1360z34089.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_22" *)
LUT4 ix1360z60709 (.O (nx1360z4), .I0 (\path_cost(7) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(7) [1])) ;
defparam ix1360z60709.INIT = 16'hE800;
LUT4 ix2357z33956 (.O (nx2357z3), .I0 (\path_cost(6) [6]), .I1 (nx2357z4), .I2 (
nx18183z3), .I3 (\path_cost(6) [7])) ;
defparam ix2357z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_6" *)
LUT4 ix18183z34084 (.O (nx18183z3), .I0 (\path_cost(6) [5]), .I1 (
\path_cost(6) [4]), .I2 (\path_cost(6) [3]), .I3 (\path_cost(6) [2])) ;
defparam ix18183z34084.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_32" *)
LUT4 ix2357z12325 (.O (nx2357z4), .I0 (\path_cost(6) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(6) [1])) ;
defparam ix2357z12325.INIT = 16'h2B00;
LUT6 ix1510z1314 (.O (nx1510z1), .I0 (\path_cost(5) [0]), .I1 (
\path_cost(4) [0]), .I2 (\validity(4) ), .I3 (nx1510z2), .I4 (
nx1510z10), .I5 (enable_int)) ;
defparam ix1510z1314.INIT = 64'hAAC0553000000000;
LUT6 ix2507z1314 (.O (nx2507z1), .I0 (nx2507z2), .I1 (\path_cost(4) [1]), .I2 (
nx2507z3), .I3 (\validity(4) ), .I4 (nx1510z2), .I5 (enable_int)) ;
defparam ix2507z1314.INIT = 64'hAAAA3C0000000000;
(* HLUTNM = "LUT62_3_84" *)
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LUT3 ix2507z1458 (.O (nx2507z3), .I0 (\path_cost(4) [0]), .I1 (d_in_int[0])
, .I2 (d_in_int[1])) ;
defparam ix2507z1458.INIT = 8'h8E;
(* HLUTNM = "LUT62_3_35" *)
LUT4 ix2507z30385 (.O (nx2507z2), .I0 (\path_cost(5) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(5) [1])) ;
defparam ix2507z30385.INIT = 16'h718E;
LUT5 ix3504z1314 (.O (nx3504z1), .I0 (nx3504z2), .I1 (\path_cost(5) [2]), .I2 (
nx7492z4), .I3 (nx1510z2), .I4 (nx42042z1)) ;
defparam ix3504z1314.INIT = 32'h3CAA0000;
(* HLUTNM = "LUT62_3_86" *)
LUT3 ix42042z1538 (.O (nx42042z1), .I0 (\validity(4) ), .I1 (\validity(1)
), .I2 (enable_int)) ;
defparam ix42042z1538.INIT = 8'hE0;
LUT5 ix4501z1314 (.O (nx4501z1), .I0 (nx4501z2), .I1 (\validity(4) ), .I2 (
nx4501z3), .I3 (nx1510z2), .I4 (enable_int)) ;
defparam ix4501z1314.INIT = 32'hF0880000;
LUT6 ix4501z28622 (.O (nx4501z3), .I0 (\path_cost(5) [3]), .I1 (
\path_cost(5) [2]), .I2 (\path_cost(5) [1]), .I3 (\path_cost(5) [0]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix4501z28622.INIT = 64'h6AAAAAAA6A6A6AAA;
LUT5 ix5498z1314 (.O (nx5498z1), .I0 (nx5498z2), .I1 (\validity(4) ), .I2 (
nx5498z3), .I3 (nx1510z2), .I4 (enable_int)) ;
defparam ix5498z1314.INIT = 32'hF0880000;
LUT5 ix6495z1314 (.O (nx6495z1), .I0 (nx6495z2), .I1 (\validity(4) ), .I2 (
nx1510z4), .I3 (nx1510z2), .I4 (enable_int)) ;
defparam ix6495z1314.INIT = 32'hF0880000;
LUT5 ix1510z1315 (.O (nx1510z2), .I0 (nx1510z3), .I1 (nx1510z7), .I2 (
nx1510z9), .I3 (\validity(4) ), .I4 (\validity(1) )) ;
defparam ix1510z1315.INIT = 32'h8FFF0000;
LUT5 ix1510z22068 (.O (nx1510z3), .I0 (nx1510z4), .I1 (nx5498z3), .I2 (
nx5498z2), .I3 (nx1510z5), .I4 (nx6495z2)) ;
defparam ix1510z22068.INIT = 32'hF7755110;
LUT6 ix1510z54819 (.O (nx1510z5), .I0 (nx1510z6), .I1 (nx3504z2), .I2 (
nx4501z2), .I3 (\path_cost(5) [3]), .I4 (\path_cost(5) [2]), .I5 (
nx7492z4)) ;
defparam ix1510z54819.INIT = 64'hFDD040F440F4D0FD;
LUT6 ix4501z1315 (.O (nx4501z2), .I0 (\path_cost(4) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(4) [2]), .I4 (\path_cost(4) [1]),
.I5 (\path_cost(4) [3])) ;
defparam ix4501z1315.INIT = 64'h4DFFFFFFB2000000;
(* HLUTNM = "LUT62_3_2" *)
LUT5 ix3504z46883 (.O (nx3504z2), .I0 (\path_cost(4) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(4) [1]), .I4 (\path_cost(4) [2])
) ;
defparam ix3504z46883.INIT = 32'h4DFFB200;
LUT6 ix1510z63374 (.O (nx1510z6), .I0 (\path_cost(5) [1]), .I1 (
\path_cost(5) [0]), .I2 (\path_cost(4) [0]), .I3 (\path_cost(4) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix1510z63374.INIT = 64'hF267EF8A45DFF267;
(* HLUTNM = "LUT62_3_23" *)
LUT4 ix5498z33955 (.O (nx5498z2), .I0 (\path_cost(4) [3]), .I1 (
\path_cost(4) [2]), .I2 (nx8489z3), .I3 (\path_cost(4) [4])) ;
defparam ix5498z33955.INIT = 16'h7F80;
LUT4 ix5498z33956 (.O (nx5498z3), .I0 (\path_cost(5) [3]), .I1 (
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\path_cost(5) [2]), .I2 (nx7492z4), .I3 (\path_cost(5) [4])) ;
defparam ix5498z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_9" *)
LUT5 ix1510z34085 (.O (nx1510z4), .I0 (\path_cost(5) [4]), .I1 (
\path_cost(5) [3]), .I2 (\path_cost(5) [2]), .I3 (nx7492z4), .I4 (
\path_cost(5) [5])) ;
defparam ix1510z34085.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_23" *)
LUT5 ix6495z34083 (.O (nx6495z2), .I0 (\path_cost(4) [4]), .I1 (
\path_cost(4) [3]), .I2 (\path_cost(4) [2]), .I3 (nx8489z3), .I4 (
\path_cost(4) [5])) ;
defparam ix6495z34083.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_91" *)
LUT2 ix7492z1318 (.O (nx7492z1), .I0 (nx7492z2), .I1 (enable_int)) ;
defparam ix7492z1318.INIT = 4'h4;
LUT6 ix7492z5410 (.O (nx7492z2), .I0 (nx7492z3), .I1 (nx8489z2), .I2 (
nx1510z8), .I3 (\validity(1) ), .I4 (nx7492z5), .I5 (\validity(4) )
) ;
defparam ix7492z5410.INIT = 64'h0D00BFFF0FFF0FFF;
LUT6 ix7492z45008 (.O (nx7492z5), .I0 (\path_cost(4) [6]), .I1 (
\path_cost(4) [5]), .I2 (\path_cost(4) [4]), .I3 (\path_cost(4) [3]), .I4 (
\path_cost(4) [2]), .I5 (nx8489z3)) ;
defparam ix7492z45008.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT6 ix1510z45011 (.O (nx1510z8), .I0 (\path_cost(5) [6]), .I1 (
\path_cost(5) [5]), .I2 (\path_cost(5) [4]), .I3 (\path_cost(5) [3]), .I4 (
\path_cost(5) [2]), .I5 (nx7492z4)) ;
defparam ix1510z45011.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT5 ix8489z1314 (.O (nx8489z1), .I0 (\validity(1) ), .I1 (nx7492z3), .I2 (
nx8489z2), .I3 (\validity(4) ), .I4 (nx17186z2)) ;
defparam ix8489z1314.INIT = 32'hD0880000;
(* HLUTNM = "LUT62_3_55" *)
LUT4 ix8489z33955 (.O (nx8489z2), .I0 (\path_cost(4) [6]), .I1 (nx8489z3), .I2 (
nx8489z4), .I3 (\path_cost(4) [7])) ;
defparam ix8489z33955.INIT = 16'h7F80;
LUT4 ix8489z34085 (.O (nx8489z4), .I0 (\path_cost(4) [5]), .I1 (
\path_cost(4) [4]), .I2 (\path_cost(4) [3]), .I3 (\path_cost(4) [2])) ;
defparam ix8489z34085.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_25" *)
LUT4 ix8489z46884 (.O (nx8489z3), .I0 (\path_cost(4) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(4) [1])) ;
defparam ix8489z46884.INIT = 16'hB200;
LUT4 ix7492z33956 (.O (nx7492z3), .I0 (\path_cost(5) [6]), .I1 (nx7492z4), .I2 (
nx18033z4), .I3 (\path_cost(5) [7])) ;
defparam ix7492z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_9" *)
LUT4 ix18033z34085 (.O (nx18033z4), .I0 (\path_cost(5) [5]), .I1 (
\path_cost(5) [4]), .I2 (\path_cost(5) [3]), .I3 (\path_cost(5) [2])) ;
defparam ix18033z34085.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_35" *)
LUT4 ix7492z46885 (.O (nx7492z4), .I0 (\path_cost(5) [0]), .I1 (d_in_int[0])
, .I2 (d_in_int[1]), .I3 (\path_cost(5) [1])) ;
defparam ix7492z46885.INIT = 16'hB200;
LUT6 ix6645z1314 (.O (nx6645z1), .I0 (\path_cost(2) [0]), .I1 (
\path_cost(3) [0]), .I2 (\validity(1) ), .I3 (nx10633z6), .I4 (
nx1510z10), .I5 (enable_int)) ;
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defparam ix6645z1314.INIT = 64'hAAC0553000000000;
LUT6 ix7642z1314 (.O (nx7642z1), .I0 (nx7642z2), .I1 (nx10633z6), .I2 (
nx7642z3), .I3 (\path_cost(3) [1]), .I4 (nx7642z4), .I5 (enable_int)) ;
defparam ix7642z1314.INIT = 64'h8FF8888800000000;
(* HLUTNM = "LUT62_3_106" *)
LUT3 ix7642z1458 (.O (nx7642z3), .I0 (\path_cost(3) [0]), .I1 (d_in_int[0])
, .I2 (d_in_int[1])) ;
defparam ix7642z1458.INIT = 8'h8E;
(* HLUTNM = "LUT62_3_83" *)
LUT4 ix7642z30385 (.O (nx7642z2), .I0 (\path_cost(2) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(2) [1])) ;
defparam ix7642z30385.INIT = 16'h718E;
LUT5 ix8639z1314 (.O (nx8639z1), .I0 (nx8639z2), .I1 (\path_cost(2) [2]), .I2 (
nx10633z3), .I3 (nx10633z6), .I4 (nx41045z1)) ;
defparam ix8639z1314.INIT = 32'h3CAA0000;
LUT6 ix9636z1314 (.O (nx9636z1), .I0 (nx9636z2), .I1 (\path_cost(2) [2]), .I2 (
nx10633z3), .I3 (\path_cost(2) [3]), .I4 (nx10633z6), .I5 (nx41045z1)
) ;
defparam ix9636z1314.INIT = 64'h3FC0AAAA00000000;
LUT6 ix10633z1314 (.O (nx10633z1), .I0 (\validity(2) ), .I1 (\validity(1)
), .I2 (enable_int), .I3 (nx10633z2), .I4 (nx10633z4), .I5 (nx10633z6)
) ;
defparam ix10633z1314.INIT = 64'hE000E000E0E00000;
LUT6 ix11630z1314 (.O (nx11630z1), .I0 (\validity(2) ), .I1 (\validity(1)
), .I2 (enable_int), .I3 (nx11630z2), .I4 (nx10633z12), .I5 (nx10633z6)
) ;
defparam ix11630z1314.INIT = 64'hE000E000E0E00000;
(* HLUTNM = "LUT62_3_86" *)
LUT3 ix41045z1538 (.O (nx41045z1), .I0 (\validity(1) ), .I1 (\validity(2)
), .I2 (enable_int)) ;
defparam ix41045z1538.INIT = 8'hE0;
(* HLUTNM = "LUT62_3_72" *)
LUT5 ix12627z1314 (.O (nx12627z1), .I0 (nx12627z2), .I1 (nx7642z4), .I2 (
nx12627z3), .I3 (nx10633z6), .I4 (enable_int)) ;
defparam ix12627z1314.INIT = 32'hF8880000;
LUT6 ix10633z1319 (.O (nx10633z6), .I0 (\validity(1) ), .I1 (nx10633z7), .I2 (
nx10633z8), .I3 (nx10633z9), .I4 (nx10633z14), .I5 (\validity(2) )) ;
defparam ix10633z1319.INIT = 64'h555DDDDF00000000;
LUT6 ix7642z45007 (.O (nx7642z4), .I0 (\validity(1) ), .I1 (nx10633z7), .I2 (
nx10633z8), .I3 (nx10633z9), .I4 (nx10633z14), .I5 (\validity(2) )) ;
defparam ix7642z45007.INIT = 64'hAAA22220AAAAAAAA;
LUT6 ix10633z46105 (.O (nx10633z9), .I0 (nx11630z2), .I1 (nx10633z2), .I2 (
nx10633z4), .I3 (nx10633z10), .I4 (nx10633z12), .I5 (nx10633z13)) ;
defparam ix10633z46105.INIT = 64'h00000000088AAEEF;
LUT6 ix10633z3510 (.O (nx10633z13), .I0 (nx10633z5), .I1 (nx53635z6), .I2 (
nx10633z3), .I3 (nx52638z4), .I4 (\path_cost(3) [6]), .I5 (
\path_cost(2) [6])) ;
defparam ix10633z3510.INIT = 64'h7000800007770888;
(* HLUTNM = "LUT62_3_4" *)
LUT5 ix10633z34093 (.O (nx10633z12), .I0 (\path_cost(3) [4]), .I1 (
\path_cost(3) [3]), .I2 (\path_cost(3) [2]), .I3 (nx10633z5), .I4 (
\path_cost(3) [5])) ;
defparam ix10633z34093.INIT = 32'h7FFF8000;
LUT6 ix10633z54824 (.O (nx10633z10), .I0 (nx10633z11), .I1 (nx8639z2), .I2 (
nx9636z2), .I3 (\path_cost(2) [3]), .I4 (\path_cost(2) [2]), .I5 (
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nx10633z3)) ;
defparam ix10633z54824.INIT = 64'hFDD040F440F4D0FD;
LUT6 ix9636z1315 (.O (nx9636z2), .I0 (\path_cost(3) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(3) [2]), .I4 (\path_cost(3) [1]),
.I5 (\path_cost(3) [3])) ;
defparam ix9636z1315.INIT = 64'h4DFFFFFFB2000000;
(* HLUTNM = "LUT62_3_24" *)
LUT5 ix8639z46883 (.O (nx8639z2), .I0 (\path_cost(3) [0]), .I1 (d_in_int[1])
, .I2 (d_in_int[0]), .I3 (\path_cost(3) [1]), .I4 (\path_cost(3) [2])
) ;
defparam ix8639z46883.INIT = 32'h4DFFB200;
LUT6 ix10633z38149 (.O (nx10633z11), .I0 (\path_cost(3) [1]), .I1 (
\path_cost(3) [0]), .I2 (\path_cost(2) [0]), .I3 (\path_cost(2) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix10633z38149.INIT = 64'h8FD9FBA251F78FD9;
(* HLUTNM = "LUT62_3_4" *)
LUT4 ix10633z33957 (.O (nx10633z4), .I0 (\path_cost(3) [3]), .I1 (
\path_cost(3) [2]), .I2 (nx10633z5), .I3 (\path_cost(3) [4])) ;
defparam ix10633z33957.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_5" *)
LUT4 ix10633z33955 (.O (nx10633z2), .I0 (\path_cost(2) [3]), .I1 (
\path_cost(2) [2]), .I2 (nx10633z3), .I3 (\path_cost(2) [4])) ;
defparam ix10633z33955.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_5" *)
LUT5 ix11630z34083 (.O (nx11630z2), .I0 (\path_cost(2) [4]), .I1 (
\path_cost(2) [3]), .I2 (\path_cost(2) [2]), .I3 (nx10633z3), .I4 (
\path_cost(2) [5])) ;
defparam ix11630z34083.INIT = 32'h7FFF8000;
LUT6 ix10633z6509 (.O (nx10633z8), .I0 (\path_cost(3) [6]), .I1 (
\path_cost(2) [6]), .I2 (nx10633z3), .I3 (nx52638z4), .I4 (nx10633z5),
.I5 (nx53635z6)) ;
defparam ix10633z6509.INIT = 64'h2888144414441444;
LUT6 ix12627z45006 (.O (nx12627z3), .I0 (\path_cost(2) [6]), .I1 (
\path_cost(2) [5]), .I2 (\path_cost(2) [4]), .I3 (\path_cost(2) [3]), .I4 (
\path_cost(2) [2]), .I5 (nx10633z3)) ;
defparam ix12627z45006.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT6 ix12627z45005 (.O (nx12627z2), .I0 (\path_cost(3) [6]), .I1 (
\path_cost(3) [5]), .I2 (\path_cost(3) [4]), .I3 (\path_cost(3) [3]), .I4 (
\path_cost(3) [2]), .I5 (nx10633z5)) ;
defparam ix12627z45005.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT5 ix13624z1314 (.O (nx13624z1), .I0 (\validity(2) ), .I1 (nx10633z14), .I2 (
nx10633z7), .I3 (\validity(1) ), .I4 (nx17186z2)) ;
defparam ix13624z1314.INIT = 32'hD0880000;
(* HLUTNM = "LUT62_3_52" *)
LUT4 ix10633z33960 (.O (nx10633z7), .I0 (\path_cost(3) [6]), .I1 (nx10633z5)
, .I2 (nx53635z6), .I3 (\path_cost(3) [7])) ;
defparam ix10633z33960.INIT = 16'h7F80;
LUT4 ix53635z34087 (.O (nx53635z6), .I0 (\path_cost(3) [5]), .I1 (
\path_cost(3) [4]), .I2 (\path_cost(3) [3]), .I3 (\path_cost(3) [2])) ;
defparam ix53635z34087.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_24" *)
LUT4 ix10633z46886 (.O (nx10633z5), .I0 (\path_cost(3) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(3) [1])) ;
defparam ix10633z46886.INIT = 16'hB200;
LUT4 ix10633z33967 (.O (nx10633z14), .I0 (\path_cost(2) [6]), .I1 (nx10633z3
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), .I2 (nx52638z4), .I3 (\path_cost(2) [7])) ;
defparam ix10633z33967.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_19" *)
LUT4 ix52638z34085 (.O (nx52638z4), .I0 (\path_cost(2) [5]), .I1 (
\path_cost(2) [4]), .I2 (\path_cost(2) [3]), .I3 (\path_cost(2) [2])) ;
defparam ix52638z34085.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_18" *)
LUT4 ix10633z46884 (.O (nx10633z3), .I0 (\path_cost(2) [0]), .I1 (
d_in_int[0]), .I2 (d_in_int[1]), .I3 (\path_cost(2) [1])) ;
defparam ix10633z46884.INIT = 16'hB200;
LUT6 ix11780z13794 (.O (nx11780z1), .I0 (\path_cost(1) [0]), .I1 (
\path_cost(0) [0]), .I2 (enable_int), .I3 (d_in_int[1]), .I4 (
d_in_int[0]), .I5 (nx11780z2)) ;
defparam ix11780z13794.INIT = 64'hA05050A0C03030C0;
(* HLUTNM = "LUT62_3_107" *)
LUT2 ix1510z1329 (.O (nx1510z10), .I0 (d_in_int[1]), .I1 (d_in_int[0])) ;
defparam ix1510z1329.INIT = 4'h6;
LUT6 ix12777z1314 (.O (nx12777z1), .I0 (\path_cost(1) [1]), .I1 (nx12777z2)
, .I2 (\path_cost(0) [1]), .I3 (nx12777z3), .I4 (nx11780z2), .I5 (
enable_int)) ;
defparam ix12777z1314.INIT = 64'h99990FF000000000;
(* HLUTNM = "LUT62_3_82" *)
LUT3 ix12777z1548 (.O (nx12777z3), .I0 (\path_cost(0) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix12777z1548.INIT = 8'hE8;
(* HLUTNM = "LUT62_3_79" *)
LUT3 ix12777z1527 (.O (nx12777z2), .I0 (\path_cost(1) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0])) ;
defparam ix12777z1527.INIT = 8'hD4;
(* HLUTNM = "LUT62_3_74" *)
LUT5 ix13774z1314 (.O (nx13774z1), .I0 (nx13774z2), .I1 (\path_cost(1) [2])
, .I2 (nx17762z2), .I3 (nx11780z2), .I4 (enable_int)) ;
defparam ix13774z1314.INIT = 32'h3CAA0000;
LUT6 ix14771z1314 (.O (nx14771z1), .I0 (nx14771z2), .I1 (\path_cost(1) [2])
, .I2 (nx17762z2), .I3 (\path_cost(1) [3]), .I4 (nx11780z2), .I5 (
enable_int)) ;
defparam ix14771z1314.INIT = 64'h3FC0AAAA00000000;
(* HLUTNM = "LUT62_3_89" *)
LUT4 ix15768z45346 (.O (nx15768z1), .I0 (nx15768z2), .I1 (nx15768z3), .I2 (
nx11780z2), .I3 (enable_int)) ;
defparam ix15768z45346.INIT = 16'hAC00;
(* HLUTNM = "LUT62_3_90" *)
LUT4 ix16765z45346 (.O (nx16765z1), .I0 (nx16765z2), .I1 (nx16765z3), .I2 (
nx11780z2), .I3 (enable_int)) ;
defparam ix16765z45346.INIT = 16'hAC00;
LUT6 ix17762z1314 (.O (nx17762z1), .I0 (\path_cost(1) [6]), .I1 (enable_int)
, .I2 (nx17762z2), .I3 (nx17762z3), .I4 (nx17762z4), .I5 (nx11780z2)) ;
defparam ix17762z1314.INIT = 64'h48884888CCCC0000;
LUT6 ix11780z1315 (.O (nx11780z2), .I0 (nx11780z3), .I1 (nx11780z4), .I2 (
nx17762z4), .I3 (nx18759z2), .I4 (nx11780z5), .I5 (\validity(1) )) ;
defparam ix11780z1315.INIT = 64'h7510F75100000000;
LUT5 ix11780z6695 (.O (nx11780z5), .I0 (nx16765z3), .I1 (nx15768z3), .I2 (
nx11780z6), .I3 (nx15768z2), .I4 (nx16765z2)) ;
defparam ix11780z6695.INIT = 32'h7F571501;
(* HLUTNM = "LUT62_3_11" *)
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LUT5 ix16765z34083 (.O (nx16765z2), .I0 (\path_cost(1) [4]), .I1 (
\path_cost(1) [3]), .I2 (\path_cost(1) [2]), .I3 (nx17762z2), .I4 (
\path_cost(1) [5])) ;
defparam ix16765z34083.INIT = 32'h7FFF8000;
(* HLUTNM = "LUT62_3_11" *)
LUT4 ix15768z33955 (.O (nx15768z2), .I0 (\path_cost(1) [3]), .I1 (
\path_cost(1) [2]), .I2 (nx17762z2), .I3 (\path_cost(1) [4])) ;
defparam ix15768z33955.INIT = 16'h7F80;
LUT6 ix11780z54820 (.O (nx11780z6), .I0 (nx11780z7), .I1 (nx13774z2), .I2 (
nx14771z2), .I3 (\path_cost(1) [3]), .I4 (\path_cost(1) [2]), .I5 (
nx17762z2)) ;
defparam ix11780z54820.INIT = 64'hFDD040F440F4D0FD;
LUT6 ix14771z1315 (.O (nx14771z2), .I0 (\path_cost(0) [0]), .I1 (d_in_int[1]
), .I2 (d_in_int[0]), .I3 (\path_cost(0) [2]), .I4 (\path_cost(0) [1])
, .I5 (\path_cost(0) [3])) ;
defparam ix14771z1315.INIT = 64'h17FFFFFE8000000;
(* HLUTNM = "LUT62_3_1" *)
LUT5 ix13774z60707 (.O (nx13774z2), .I0 (\path_cost(0) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(0) [1]), .I4 (
\path_cost(0) [2])) ;
defparam ix13774z60707.INIT = 32'h17FFE800;
LUT6 ix11780z19207 (.O (nx11780z7), .I0 (\path_cost(1) [1]), .I1 (
\path_cost(1) [0]), .I2 (\path_cost(0) [0]), .I3 (\path_cost(0) [1]), .I4 (
d_in_int[1]), .I5 (d_in_int[0])) ;
defparam ix11780z19207.INIT = 64'hEF8AF267F26745DF;
(* HLUTNM = "LUT62_3_12" *)
LUT4 ix15768z33956 (.O (nx15768z3), .I0 (\path_cost(0) [3]), .I1 (
\path_cost(0) [2]), .I2 (nx16765z4), .I3 (\path_cost(0) [4])) ;
defparam ix15768z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_12" *)
LUT5 ix16765z34084 (.O (nx16765z3), .I0 (\path_cost(0) [4]), .I1 (
\path_cost(0) [3]), .I2 (\path_cost(0) [2]), .I3 (nx16765z4), .I4 (
\path_cost(0) [5])) ;
defparam ix16765z34084.INIT = 32'h7FFF8000;
LUT6 ix11780z45007 (.O (nx11780z4), .I0 (\path_cost(1) [6]), .I1 (
\path_cost(1) [5]), .I2 (\path_cost(1) [4]), .I3 (\path_cost(1) [3]), .I4 (
\path_cost(1) [2]), .I5 (nx17762z2)) ;
defparam ix11780z45007.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT6 ix17762z45007 (.O (nx17762z4), .I0 (\path_cost(0) [6]), .I1 (
\path_cost(0) [5]), .I2 (\path_cost(0) [4]), .I3 (\path_cost(0) [3]), .I4 (
\path_cost(0) [2]), .I5 (nx16765z4)) ;
defparam ix17762z45007.INIT = 64'h6AAAAAAAAAAAAAAA;
LUT4 ix18759z46370 (.O (nx18759z1), .I0 (nx11780z3), .I1 (\validity(1) ), .I2 (
nx18759z2), .I3 (nx17186z2)) ;
defparam ix18759z46370.INIT = 16'hB000;
LUT6 ix17186z1315 (.O (nx17186z2), .I0 (\path_cost(2) [7]), .I1 (
\path_cost(1) [7]), .I2 (\path_cost(0) [7]), .I3 (\path_cost(3) [7]), .I4 (
nx17186z3), .I5 (enable_int)) ;
defparam ix17186z1315.INIT = 64'h7FFFFFF00000000;
LUT4 ix17186z34084 (.O (nx17186z3), .I0 (\path_cost(7) [7]), .I1 (
\path_cost(6) [7]), .I2 (\path_cost(5) [7]), .I3 (\path_cost(4) [7])) ;
defparam ix17186z34084.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_56" *)
LUT4 ix18759z33955 (.O (nx18759z2), .I0 (\path_cost(0) [6]), .I1 (nx16765z4)
, .I2 (nx63755z4), .I3 (\path_cost(0) [7])) ;
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defparam ix18759z33955.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_20" *)
LUT4 ix63755z34085 (.O (nx63755z4), .I0 (\path_cost(0) [5]), .I1 (
\path_cost(0) [4]), .I2 (\path_cost(0) [3]), .I3 (\path_cost(0) [2])) ;
defparam ix63755z34085.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_27" *)
LUT4 ix16765z60709 (.O (nx16765z4), .I0 (\path_cost(0) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(0) [1])) ;
defparam ix16765z60709.INIT = 16'hE800;
(* HLUTNM = "LUT62_3_53" *)
LUT4 ix11780z33956 (.O (nx11780z3), .I0 (\path_cost(1) [6]), .I1 (nx17762z2)
, .I2 (nx17762z3), .I3 (\path_cost(1) [7])) ;
defparam ix11780z33956.INIT = 16'h7F80;
(* HLUTNM = "LUT62_3_21" *)
LUT4 ix17762z34084 (.O (nx17762z3), .I0 (\path_cost(1) [5]), .I1 (
\path_cost(1) [4]), .I2 (\path_cost(1) [3]), .I3 (\path_cost(1) [2])) ;
defparam ix17762z34084.INIT = 16'h8000;
(* HLUTNM = "LUT62_3_28" *)
LUT4 ix17762z12323 (.O (nx17762z2), .I0 (\path_cost(1) [0]), .I1 (
d_in_int[1]), .I2 (d_in_int[0]), .I3 (\path_cost(1) [1])) ;
defparam ix17762z12323.INIT = 16'h2B00;
(* HLUTNM = "LUT62_3_85" *)
LUT2 ix11312z1315 (.O (nx11312z1), .I0 (mem_bank[1]), .I1 (mem_bank[0])) ;
defparam ix11312z1315.INIT = 4'h1;
(* HLUTNM = "LUT62_3_105" *)
LUT2 ix11313z1318 (.O (nx11313z1), .I0 (mem_bank[1]), .I1 (mem_bank[0])) ;
defparam ix11313z1318.INIT = 4'h4;
(* HLUTNM = "LUT62_3_107" *)
LUT2 ix11315z1322 (.O (nx11315z1), .I0 (mem_bank[1]), .I1 (mem_bank[0])) ;
defparam ix11315z1322.INIT = 4'h8;
(* HLUTNM = "LUT62_3_106" *)
LUT2 ix11314z1318 (.O (nx11314z1), .I0 (mem_bank[0]), .I1 (mem_bank[1])) ;
defparam ix11314z1318.INIT = 4'h4;
(* HLUTNM = "LUT62_3_102" *)
LUT3 ix64075z1443 (.O (nx64075z5), .I0 (nx1530z2), .I1 (nx1530z3), .I2 (
nx2527z2)) ;
defparam ix64075z1443.INIT = 8'h80;
(* HLUTNM = "LUT62_3_3" *)
LUT5 ix2527z1315 (.O (nx2527z2), .I0 (\rd_mem_counter_disp(0) ), .I1 (
enable_int), .I2 (nx3524z2), .I3 (nx3524z3), .I4 (nx3524z4)) ;
defparam ix2527z1315.INIT = 32'h80000000;
LUT2 ix60837z1318 (.O (nx60837z2), .I0 (wr_mem_counter_disp[1]), .I1 (
rst_int)) ;
defparam ix60837z1318.INIT = 4'h4;
(* HLUTNM = "LUT62_3_90" *)
LUT2 ix51679z1322 (.O (nx51679z2), .I0 (wr_mem_counter[9]), .I1 (enable_int)
) ;
defparam ix51679z1322.INIT = 4'h8;
LUT2 ix8481z1322 (.O (nx8481z2), .I0 (wr_mem_counter[8]), .I1 (enable_int)
) ;
defparam ix8481z1322.INIT = 4'h8;
LUT2 ix8480z1322 (.O (nx8480z2), .I0 (wr_mem_counter[7]), .I1 (enable_int)
) ;
defparam ix8480z1322.INIT = 4'h8;
LUT2 ix8479z1322 (.O (nx8479z2), .I0 (wr_mem_counter[6]), .I1 (enable_int)
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) ;
defparam ix8479z1322.INIT = 4'h8;
LUT2 ix8478z1322 (.O (nx8478z2), .I0 (wr_mem_counter[5]), .I1 (enable_int)
) ;
defparam ix8478z1322.INIT = 4'h8;
LUT2 ix8477z1322 (.O (nx8477z2), .I0 (wr_mem_counter[4]), .I1 (enable_int)
) ;
defparam ix8477z1322.INIT = 4'h8;
LUT2 ix8476z1322 (.O (nx8476z2), .I0 (wr_mem_counter[3]), .I1 (enable_int)
) ;
defparam ix8476z1322.INIT = 4'h8;
LUT2 ix8475z1322 (.O (nx8475z2), .I0 (wr_mem_counter[2]), .I1 (enable_int)
) ;
defparam ix8475z1322.INIT = 4'h8;
LUT2 ix8474z1322 (.O (nx8474z2), .I0 (wr_mem_counter[1]), .I1 (enable_int)
) ;
defparam ix8474z1322.INIT = 4'h8;
LUT2 ix8473z1322 (.O (nx8473z1), .I0 (wr_mem_counter[0]), .I1 (enable_int)
) ;
defparam ix8473z1322.INIT = 4'h8;
tbu_0 tbu_0 (.p_nbus_d_in_1_tbu_0 ({d_in_1_tbu_0[3],d_in_1_tbu_0[7],
d_in_1_tbu_0[2],d_in_1_tbu_0[6],d_in_1_tbu_0[1],d_in_1_tbu_0[5],
d_in_1_tbu_0[0],d_in_1_tbu_0[4]}), .p_nbus_d_in_0_tbu_0 ({
d_in_0_tbu_0[3],d_in_0_tbu_0[7],d_in_0_tbu_0[2],d_in_0_tbu_0[6],
d_in_0_tbu_0[1],d_in_0_tbu_0[5],d_in_0_tbu_0[0],d_in_0_tbu_0[4]}), .p_enable_tbu_0 (
enable_tbu_0), .p_rst_int (rst_int), .p_not_rst (not_rst), .p_mem_bank_Q3 (
mem_bank_Q3), .p_wr_disp_mem_0 (wr_disp_mem_0), .p_clk_int (clk_int),
.p_d_in_disp_mem_0 (d_in_disp_mem_0)) ;
tbu_1 tbu_1 (.p_nbus_d_in_1_tbu_1 ({d_in_1_tbu_1[3],d_in_1_tbu_1[7],
d_in_1_tbu_1[2],d_in_1_tbu_1[6],d_in_1_tbu_1[1],d_in_1_tbu_1[5],
d_in_1_tbu_1[0],d_in_1_tbu_1[4]}), .p_nbus_d_in_0_tbu_1 ({
d_in_0_tbu_1[3],d_in_0_tbu_1[7],d_in_0_tbu_1[2],d_in_0_tbu_1[6],
d_in_0_tbu_1[1],d_in_0_tbu_1[5],d_in_0_tbu_1[0],d_in_0_tbu_1[4]}), .p_enable_tbu_1 (
enable_tbu_1), .p_not_rst (not_rst), .p_selection_tbu_1 (
selection_tbu_1), .p_wr_disp_mem_1 (wr_disp_mem_1), .p_clk_int (
clk_int), .p_d_in_disp_mem_1 (d_in_disp_mem_1)) ;
RAMB18E1 disp_mem_0_mem_1 (.DOADO ({\$dummy [0],\$dummy [1],\$dummy [2],
\$dummy [3],\$dummy [4],\$dummy [5],\$dummy [6],\$dummy [7],
\$dummy [8],\$dummy [9],\$dummy [10],\$dummy [11],\$dummy [12],
\$dummy [13],\$dummy [14],d_o_disp_mem_0}), .ADDRARDADDR ({nx63z1,
nx63z1,nx63z1,nx63z1,addr_disp_mem_0[9],addr_disp_mem_0[8],
addr_disp_mem_0[7],addr_disp_mem_0[6],addr_disp_mem_0[5],
addr_disp_mem_0[4],addr_disp_mem_0[3],addr_disp_mem_0[2],
addr_disp_mem_0[1],addr_disp_mem_0[0]}), .ADDRBWRADDR ({nx56042z2,
nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,
nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,
nx56042z2}), .CLKARDCLK (clk_int), .CLKBWRCLK (nx63z1), .DIADI ({
nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,
nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,d_in_disp_mem_0}), .DIBDI (
{nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,
nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1}), .DIPADIP ({
nx63z1,nx63z1}), .DIPBDIP ({nx63z1,nx63z1}), .ENARDEN (nx56042z2),
.ENBWREN (nx63z1), .REGCEAREGCE (nx63z1), .REGCEB (nx63z1), .RSTRAMARSTRAM (
nx63z1), .RSTRAMB (nx63z1), .RSTREGARSTREG (nx63z1), .RSTREGB (
nx63z1), .WEA ({wr_disp_mem_0,wr_disp_mem_0}), .WEBWE ({nx63z1,
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nx63z1,nx63z1,nx63z1})) ;
defparam disp_mem_0_mem_1.DOA_REG = 0;
defparam disp_mem_0_mem_1.DOB_REG = 0;
defparam disp_mem_0_mem_1.INIT_A = 20'h00000;
defparam disp_mem_0_mem_1.INIT_B = 20'h00000;
defparam disp_mem_0_mem_1.RAM_MODE = "TDP";
defparam disp_mem_0_mem_1.READ_WIDTH_A = 1;
defparam disp_mem_0_mem_1.READ_WIDTH_B = 0;
defparam disp_mem_0_mem_1.RSTREG_PRIORITY_A = "RSTREG";
defparam disp_mem_0_mem_1.RSTREG_PRIORITY_B = "RSTREG";
defparam disp_mem_0_mem_1.SIM_COLLISION_CHECK = "ALL";
defparam disp_mem_0_mem_1.SIM_DEVICE = "7SERIES";
defparam disp_mem_0_mem_1.SRVAL_A = 36'h000000000;
defparam disp_mem_0_mem_1.SRVAL_B = 36'h000000000;
defparam disp_mem_0_mem_1.WRITE_MODE_A = "READ_FIRST";
defparam disp_mem_0_mem_1.WRITE_MODE_B = "NO_CHANGE";
defparam disp_mem_0_mem_1.WRITE_WIDTH_A = 1;
defparam disp_mem_0_mem_1.WRITE_WIDTH_B = 0;
RAMB18E1 disp_mem_1_mem_1 (.DOADO ({\$dummy [15],\$dummy [16],\$dummy [17],
\$dummy [18],\$dummy [19],\$dummy [20],\$dummy [21],\$dummy [22],
\$dummy [23],\$dummy [24],\$dummy [25],\$dummy [26],\$dummy [27],
\$dummy [28],\$dummy [29],d_o_disp_mem_1}), .ADDRARDADDR ({nx63z1,
nx63z1,nx63z1,nx63z1,addr_disp_mem_1[9],addr_disp_mem_1[8],
addr_disp_mem_1[7],addr_disp_mem_1[6],addr_disp_mem_1[5],
addr_disp_mem_1[4],addr_disp_mem_1[3],addr_disp_mem_1[2],
addr_disp_mem_1[1],addr_disp_mem_1[0]}), .ADDRBWRADDR ({nx56042z2,
nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,
nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,nx56042z2,
nx56042z2}), .CLKARDCLK (clk_int), .CLKBWRCLK (nx63z1), .DIADI ({
nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,
nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,d_in_disp_mem_1}), .DIBDI (
{nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,
nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1,nx63z1}), .DIPADIP ({
nx63z1,nx63z1}), .DIPBDIP ({nx63z1,nx63z1}), .ENARDEN (nx56042z2),
.ENBWREN (nx63z1), .REGCEAREGCE (nx63z1), .REGCEB (nx63z1), .RSTRAMARSTRAM (
nx63z1), .RSTRAMB (nx63z1), .RSTREGARSTREG (nx63z1), .RSTREGB (
nx63z1), .WEA ({wr_disp_mem_1,wr_disp_mem_1}), .WEBWE ({nx63z1,
nx63z1,nx63z1,nx63z1})) ;
defparam disp_mem_1_mem_1.DOA_REG = 0;
defparam disp_mem_1_mem_1.DOB_REG = 0;
defparam disp_mem_1_mem_1.INIT_A = 20'h00000;
defparam disp_mem_1_mem_1.INIT_B = 20'h00000;
defparam disp_mem_1_mem_1.RAM_MODE = "TDP";
defparam disp_mem_1_mem_1.READ_WIDTH_A = 1;
defparam disp_mem_1_mem_1.READ_WIDTH_B = 0;
defparam disp_mem_1_mem_1.RSTREG_PRIORITY_A = "RSTREG";
defparam disp_mem_1_mem_1.RSTREG_PRIORITY_B = "RSTREG";
defparam disp_mem_1_mem_1.SIM_COLLISION_CHECK = "ALL";
defparam disp_mem_1_mem_1.SIM_DEVICE = "7SERIES";
defparam disp_mem_1_mem_1.SRVAL_A = 36'h000000000;
defparam disp_mem_1_mem_1.SRVAL_B = 36'h000000000;
defparam disp_mem_1_mem_1.WRITE_MODE_A = "READ_FIRST";
defparam disp_mem_1_mem_1.WRITE_MODE_B = "NO_CHANGE";
defparam disp_mem_1_mem_1.WRITE_WIDTH_A = 1;
defparam disp_mem_1_mem_1.WRITE_WIDTH_B = 0;
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BUFGP clk_ibuf (.O (clk_int), .I (clk)) ;
LUT6 ix1510z54521 (.O (nx1510z9), .I0 (nx1510z8), .I1 (nx7492z3), .I2 (
\path_cost(4) [7]), .I3 (nx8489z4), .I4 (nx8489z3), .I5 (
\path_cost(4) [6])) ;
defparam ix1510z54521.INIT = 64'hFC8E8E8E8ECFCFCF;
LUT6 ix1510z22137 (.O (nx1510z7), .I0 (nx1510z8), .I1 (nx7492z3), .I2 (
\path_cost(4) [7]), .I3 (nx8489z4), .I4 (nx8489z3), .I5 (
\path_cost(4) [6])) ;
defparam ix1510z22137.INIT = 64'h15F3F3F3F3515151;
LUT6 ix1360z22146 (.O (nx1360z16), .I0 (nx1360z10), .I1 (nx2357z3), .I2 (
\path_cost(7) [7]), .I3 (nx1360z8), .I4 (nx1360z4), .I5 (
\path_cost(7) [6])) ;
defparam ix1360z22146.INIT = 64'h15F3F3F3F3515151;
FDSE \reg_rd_mem_counter_disp(0) (.Q (\rd_mem_counter_disp(0) ), .C (
clk_int), .CE (nx56042z2), .D (nx7512z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(1) (.Q (nx3524z2), .C (clk_int), .CE (
nx56042z2), .D (nx6515z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(2) (.Q (nx3524z3), .C (clk_int), .CE (
nx56042z2), .D (nx5518z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(3) (.Q (nx3524z4), .C (clk_int), .CE (
nx56042z2), .D (nx4521z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(4) (.Q (nx1530z2), .C (clk_int), .CE (
nx56042z2), .D (nx3524z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(5) (.Q (nx1530z3), .C (clk_int), .CE (
nx56042z2), .D (nx2527z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(6) (.Q (nx64075z2), .C (clk_int), .CE (
nx56042z2), .D (nx1530z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(7) (.Q (nx64075z3), .C (clk_int), .CE (
nx56042z2), .D (nx533z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(8) (.Q (nx64075z4), .C (clk_int), .CE (
nx56042z2), .D (nx65072z1), .S (not_rst)) ;
FDSE \reg_rd_mem_counter_disp(9) (.Q (nx62798z2), .C (clk_int), .CE (
nx56042z2), .D (nx64075z1), .S (not_rst)) ;
FDRE \reg_d_in_mem_C(7) (.Q (d_in_mem_C[7]), .C (clk_int), .CE (nx56042z2)
, .D (selection[7]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(6) (.Q (d_in_mem_C[6]), .C (clk_int), .CE (nx56042z2)
, .D (selection[6]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(5) (.Q (d_in_mem_C[5]), .C (clk_int), .CE (nx56042z2)
, .D (selection[5]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(4) (.Q (d_in_mem_C[4]), .C (clk_int), .CE (nx56042z2)
, .D (selection[4]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(3) (.Q (d_in_mem_C[3]), .C (clk_int), .CE (nx56042z2)
, .D (selection[3]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(2) (.Q (d_in_mem_C[2]), .C (clk_int), .CE (nx56042z2)
, .D (selection[2]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(1) (.Q (d_in_mem_C[1]), .C (clk_int), .CE (nx56042z2)
, .D (selection[1]), .R (nx63z1)) ;
FDRE \reg_d_in_mem_C(0) (.Q (d_in_mem_C[0]), .C (clk_int), .CE (nx56042z2)
, .D (selection[0]), .R (nx63z1)) ;
FDRE reg_wr_mem_C (.Q (wr_mem_C), .C (clk_int), .CE (nx56042z2), .D (
nx11314z1), .R (nx63z1)) ;
FDRE reg_wr_mem_D (.Q (wr_mem_D), .C (clk_int), .CE (nx56042z2), .D (
nx11315z1), .R (nx63z1)) ;
FDRE reg_wr_mem_B (.Q (wr_mem_B), .C (clk_int), .CE (nx56042z2), .D (
nx11313z1), .R (nx63z1)) ;
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FDRE \reg_addr_mem_D(9) (.Q (addr_mem_D[9]), .C (clk_int), .CE (nx56042z2)
, .D (nx29097z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(8) (.Q (addr_mem_D[8]), .C (clk_int), .CE (nx56042z2)
, .D (nx30094z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(7) (.Q (addr_mem_D[7]), .C (clk_int), .CE (nx56042z2)
, .D (nx31091z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(6) (.Q (addr_mem_D[6]), .C (clk_int), .CE (nx56042z2)
, .D (nx32088z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(5) (.Q (addr_mem_D[5]), .C (clk_int), .CE (nx56042z2)
, .D (nx33085z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(4) (.Q (addr_mem_D[4]), .C (clk_int), .CE (nx56042z2)
, .D (nx34082z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(3) (.Q (addr_mem_D[3]), .C (clk_int), .CE (nx56042z2)
, .D (nx35079z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(2) (.Q (addr_mem_D[2]), .C (clk_int), .CE (nx56042z2)
, .D (nx36076z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(1) (.Q (addr_mem_D[1]), .C (clk_int), .CE (nx56042z2)
, .D (nx37073z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_D(0) (.Q (addr_mem_D[0]), .C (clk_int), .CE (nx56042z2)
, .D (nx38070z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(9) (.Q (addr_mem_B[9]), .C (clk_int), .CE (nx56042z2)
, .D (nx53277z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(8) (.Q (addr_mem_B[8]), .C (clk_int), .CE (nx56042z2)
, .D (nx52280z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(7) (.Q (addr_mem_B[7]), .C (clk_int), .CE (nx56042z2)
, .D (nx51283z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(6) (.Q (addr_mem_B[6]), .C (clk_int), .CE (nx56042z2)
, .D (nx50286z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(5) (.Q (addr_mem_B[5]), .C (clk_int), .CE (nx56042z2)
, .D (nx49289z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(4) (.Q (addr_mem_B[4]), .C (clk_int), .CE (nx56042z2)
, .D (nx48292z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(3) (.Q (addr_mem_B[3]), .C (clk_int), .CE (nx56042z2)
, .D (nx47295z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(2) (.Q (addr_mem_B[2]), .C (clk_int), .CE (nx56042z2)
, .D (nx46298z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(1) (.Q (addr_mem_B[1]), .C (clk_int), .CE (nx56042z2)
, .D (nx45301z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_B(0) (.Q (addr_mem_B[0]), .C (clk_int), .CE (nx56042z2)
, .D (nx44304z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(9) (.Q (addr_mem_A[9]), .C (clk_int), .CE (nx56042z2)
, .D (nx61696z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(8) (.Q (addr_mem_A[8]), .C (clk_int), .CE (nx56042z2)
, .D (nx60699z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(7) (.Q (addr_mem_A[7]), .C (clk_int), .CE (nx56042z2)
, .D (nx59702z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(6) (.Q (addr_mem_A[6]), .C (clk_int), .CE (nx56042z2)
, .D (nx58705z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(5) (.Q (addr_mem_A[5]), .C (clk_int), .CE (nx56042z2)
, .D (nx57708z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(4) (.Q (addr_mem_A[4]), .C (clk_int), .CE (nx56042z2)
, .D (nx56711z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(3) (.Q (addr_mem_A[3]), .C (clk_int), .CE (nx56042z2)
, .D (nx55714z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(2) (.Q (addr_mem_A[2]), .C (clk_int), .CE (nx56042z2)
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, .D (nx54717z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(1) (.Q (addr_mem_A[1]), .C (clk_int), .CE (nx56042z2)
, .D (nx53720z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_A(0) (.Q (addr_mem_A[0]), .C (clk_int), .CE (nx56042z2)
, .D (nx52723z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(9) (.Q (addr_mem_C[9]), .C (clk_int), .CE (nx56042z2)
, .D (nx20678z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(8) (.Q (addr_mem_C[8]), .C (clk_int), .CE (nx56042z2)
, .D (nx21675z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(7) (.Q (addr_mem_C[7]), .C (clk_int), .CE (nx56042z2)
, .D (nx22672z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(6) (.Q (addr_mem_C[6]), .C (clk_int), .CE (nx56042z2)
, .D (nx23669z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(5) (.Q (addr_mem_C[5]), .C (clk_int), .CE (nx56042z2)
, .D (nx24666z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(4) (.Q (addr_mem_C[4]), .C (clk_int), .CE (nx56042z2)
, .D (nx25663z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(3) (.Q (addr_mem_C[3]), .C (clk_int), .CE (nx56042z2)
, .D (nx26660z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(2) (.Q (addr_mem_C[2]), .C (clk_int), .CE (nx56042z2)
, .D (nx27657z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(1) (.Q (addr_mem_C[1]), .C (clk_int), .CE (nx56042z2)
, .D (nx28654z1), .R (nx63z1)) ;
FDRE \reg_addr_mem_C(0) (.Q (addr_mem_C[0]), .C (clk_int), .CE (nx56042z2)
, .D (nx29651z1), .R (nx63z1)) ;
FDRE reg_wr_mem_A (.Q (wr_mem_A), .C (clk_int), .CE (nx56042z2), .D (
nx11312z1), .R (nx63z1)) ;
FDRE \reg_mem_bank_Q1(1) (.Q (mem_bank_Q1[1]), .C (clk_int), .CE (nx56042z2
), .D (mem_bank[1]), .R (nx63z1)) ;
FDRE \reg_mem_bank_Q1(0) (.Q (mem_bank_Q1[0]), .C (clk_int), .CE (nx56042z2
), .D (mem_bank[0]), .R (nx63z1)) ;
FDRE \reg_mem_bank_Q2(1) (.Q (mem_bank_Q2[1]), .C (clk_int), .CE (nx56042z2
), .D (mem_bank_Q1[1]), .R (nx63z1)) ;
FDRE \reg_mem_bank_Q2(0) (.Q (mem_bank_Q2[0]), .C (clk_int), .CE (nx56042z2
), .D (mem_bank_Q1[0]), .R (nx63z1)) ;
FDRE reg_selection_tbu_1 (.Q (selection_tbu_1), .C (clk_int), .CE (nx56042z2
), .D (not_mem_bank_Q2_0), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(7) (.Q (d_in_1_tbu_1[7]), .C (clk_int), .CE (
nx56042z2), .D (nx52733z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(6) (.Q (d_in_1_tbu_1[6]), .C (clk_int), .CE (
nx56042z2), .D (nx51736z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(5) (.Q (d_in_1_tbu_1[5]), .C (clk_int), .CE (
nx56042z2), .D (nx50739z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(4) (.Q (d_in_1_tbu_1[4]), .C (clk_int), .CE (
nx56042z2), .D (nx49742z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(3) (.Q (d_in_1_tbu_1[3]), .C (clk_int), .CE (
nx56042z2), .D (nx48745z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(2) (.Q (d_in_1_tbu_1[2]), .C (clk_int), .CE (
nx56042z2), .D (nx47748z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(1) (.Q (d_in_1_tbu_1[1]), .C (clk_int), .CE (
nx56042z2), .D (nx46751z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_1(0) (.Q (d_in_1_tbu_1[0]), .C (clk_int), .CE (
nx56042z2), .D (nx45754z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(7) (.Q (d_in_0_tbu_1[7]), .C (clk_int), .CE (
nx56042z2), .D (nx22088z1), .R (nx63z1)) ;
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FDRE \reg_d_in_0_tbu_1(6) (.Q (d_in_0_tbu_1[6]), .C (clk_int), .CE (
nx56042z2), .D (nx23085z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(5) (.Q (d_in_0_tbu_1[5]), .C (clk_int), .CE (
nx56042z2), .D (nx24082z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(4) (.Q (d_in_0_tbu_1[4]), .C (clk_int), .CE (
nx56042z2), .D (nx25079z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(3) (.Q (d_in_0_tbu_1[3]), .C (clk_int), .CE (
nx56042z2), .D (nx26076z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(2) (.Q (d_in_0_tbu_1[2]), .C (clk_int), .CE (
nx56042z2), .D (nx27073z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(1) (.Q (d_in_0_tbu_1[1]), .C (clk_int), .CE (
nx56042z2), .D (nx28070z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_1(0) (.Q (d_in_0_tbu_1[0]), .C (clk_int), .CE (
nx56042z2), .D (nx29067z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(7) (.Q (d_in_1_tbu_0[7]), .C (clk_int), .CE (
nx56042z2), .D (nx61152z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(6) (.Q (d_in_1_tbu_0[6]), .C (clk_int), .CE (
nx56042z2), .D (nx60155z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(5) (.Q (d_in_1_tbu_0[5]), .C (clk_int), .CE (
nx56042z2), .D (nx59158z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(4) (.Q (d_in_1_tbu_0[4]), .C (clk_int), .CE (
nx56042z2), .D (nx58161z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(3) (.Q (d_in_1_tbu_0[3]), .C (clk_int), .CE (
nx56042z2), .D (nx57164z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(2) (.Q (d_in_1_tbu_0[2]), .C (clk_int), .CE (
nx56042z2), .D (nx56167z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(1) (.Q (d_in_1_tbu_0[1]), .C (clk_int), .CE (
nx56042z2), .D (nx55170z1), .R (nx63z1)) ;
FDRE \reg_d_in_1_tbu_0(0) (.Q (d_in_1_tbu_0[0]), .C (clk_int), .CE (
nx56042z2), .D (nx54173z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(7) (.Q (d_in_0_tbu_0[7]), .C (clk_int), .CE (
nx56042z2), .D (nx13669z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(6) (.Q (d_in_0_tbu_0[6]), .C (clk_int), .CE (
nx56042z2), .D (nx14666z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(5) (.Q (d_in_0_tbu_0[5]), .C (clk_int), .CE (
nx56042z2), .D (nx15663z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(4) (.Q (d_in_0_tbu_0[4]), .C (clk_int), .CE (
nx56042z2), .D (nx16660z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(3) (.Q (d_in_0_tbu_0[3]), .C (clk_int), .CE (
nx56042z2), .D (nx17657z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(2) (.Q (d_in_0_tbu_0[2]), .C (clk_int), .CE (
nx56042z2), .D (nx18654z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(1) (.Q (d_in_0_tbu_0[1]), .C (clk_int), .CE (
nx56042z2), .D (nx19651z1), .R (nx63z1)) ;
FDRE \reg_d_in_0_tbu_0(0) (.Q (d_in_0_tbu_0[0]), .C (clk_int), .CE (
nx56042z2), .D (nx20648z1), .R (nx63z1)) ;
FDRE reg_mem_bank_Q3 (.Q (mem_bank_Q3), .C (clk_int), .CE (nx56042z2), .D (
mem_bank_Q2[0]), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(9) (.Q (addr_disp_mem_1[9]), .C (clk_int), .CE (
nx56042z2), .D (nx5681z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(8) (.Q (addr_disp_mem_1[8]), .C (clk_int), .CE (
nx56042z2), .D (nx6678z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(7) (.Q (addr_disp_mem_1[7]), .C (clk_int), .CE (
nx56042z2), .D (nx7675z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(6) (.Q (addr_disp_mem_1[6]), .C (clk_int), .CE (
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nx56042z2), .D (nx8672z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(5) (.Q (addr_disp_mem_1[5]), .C (clk_int), .CE (
nx56042z2), .D (nx9669z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(4) (.Q (addr_disp_mem_1[4]), .C (clk_int), .CE (
nx56042z2), .D (nx10666z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(3) (.Q (addr_disp_mem_1[3]), .C (clk_int), .CE (
nx56042z2), .D (nx11663z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(2) (.Q (addr_disp_mem_1[2]), .C (clk_int), .CE (
nx56042z2), .D (nx12660z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(1) (.Q (addr_disp_mem_1[1]), .C (clk_int), .CE (
nx56042z2), .D (nx13657z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_1(0) (.Q (addr_disp_mem_1[0]), .C (clk_int), .CE (
nx56042z2), .D (nx14654z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(9) (.Q (addr_disp_mem_0[9]), .C (clk_int), .CE (
nx56042z2), .D (nx62798z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(8) (.Q (addr_disp_mem_0[8]), .C (clk_int), .CE (
nx56042z2), .D (nx63795z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(7) (.Q (addr_disp_mem_0[7]), .C (clk_int), .CE (
nx56042z2), .D (nx64792z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(6) (.Q (addr_disp_mem_0[6]), .C (clk_int), .CE (
nx56042z2), .D (nx253z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(5) (.Q (addr_disp_mem_0[5]), .C (clk_int), .CE (
nx56042z2), .D (nx1250z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(4) (.Q (addr_disp_mem_0[4]), .C (clk_int), .CE (
nx56042z2), .D (nx2247z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(3) (.Q (addr_disp_mem_0[3]), .C (clk_int), .CE (
nx56042z2), .D (nx3244z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(2) (.Q (addr_disp_mem_0[2]), .C (clk_int), .CE (
nx56042z2), .D (nx4241z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(1) (.Q (addr_disp_mem_0[1]), .C (clk_int), .CE (
nx56042z2), .D (nx5238z1), .R (nx63z1)) ;
FDRE \reg_addr_disp_mem_0(0) (.Q (addr_disp_mem_0[0]), .C (clk_int), .CE (
nx56042z2), .D (nx6235z1), .R (nx63z1)) ;
FDRE reg_mem_bank_Q5 (.Q (mem_bank_Q5), .C (clk_int), .CE (nx56042z2), .D (
wr_disp_mem_0), .R (nx63z1)) ;
FDRE reg_decoder_o_reg (.Q (d_out_1_0), .C (clk_int), .CE (nx56042z2), .D (
nx65470z1), .R (nx63z1)) ;
FDCE \reg_path_cost(0)(7) (.Q (\path_cost(0) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx18759z1)) ;
FDCE \reg_path_cost(0)(6) (.Q (\path_cost(0) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx17762z1)) ;
FDCE \reg_path_cost(0)(5) (.Q (\path_cost(0) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx16765z1)) ;
FDCE \reg_path_cost(0)(4) (.Q (\path_cost(0) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx15768z1)) ;
FDCE \reg_path_cost(0)(3) (.Q (\path_cost(0) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx14771z1)) ;
FDCE \reg_path_cost(0)(2) (.Q (\path_cost(0) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx13774z1)) ;
FDCE \reg_path_cost(0)(1) (.Q (\path_cost(0) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx12777z1)) ;
FDCE \reg_path_cost(0)(0) (.Q (\path_cost(0) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx11780z1)) ;
FDCE \reg_path_cost(1)(7) (.Q (\path_cost(1) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx13624z1)) ;
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FDCE \reg_path_cost(1)(6) (.Q (\path_cost(1) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx12627z1)) ;
FDCE \reg_path_cost(1)(5) (.Q (\path_cost(1) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx11630z1)) ;
FDCE \reg_path_cost(1)(4) (.Q (\path_cost(1) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx10633z1)) ;
FDCE \reg_path_cost(1)(3) (.Q (\path_cost(1) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx9636z1)) ;
FDCE \reg_path_cost(1)(2) (.Q (\path_cost(1) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx8639z1)) ;
FDCE \reg_path_cost(1)(1) (.Q (\path_cost(1) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx7642z1)) ;
FDCE \reg_path_cost(1)(0) (.Q (\path_cost(1) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx6645z1)) ;
FDCE \reg_path_cost(2)(7) (.Q (\path_cost(2) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx8489z1)) ;
FDCE \reg_path_cost(2)(6) (.Q (\path_cost(2) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx7492z1)) ;
FDCE \reg_path_cost(2)(5) (.Q (\path_cost(2) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx6495z1)) ;
FDCE \reg_path_cost(2)(4) (.Q (\path_cost(2) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx5498z1)) ;
FDCE \reg_path_cost(2)(3) (.Q (\path_cost(2) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx4501z1)) ;
FDCE \reg_path_cost(2)(2) (.Q (\path_cost(2) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx3504z1)) ;
FDCE \reg_path_cost(2)(1) (.Q (\path_cost(2) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx2507z1)) ;
FDCE \reg_path_cost(2)(0) (.Q (\path_cost(2) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx1510z1)) ;
FDCE \reg_path_cost(3)(7) (.Q (\path_cost(3) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx3354z1)) ;
FDCE \reg_path_cost(3)(6) (.Q (\path_cost(3) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx2357z1)) ;
FDCE \reg_path_cost(3)(5) (.Q (\path_cost(3) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx1360z1)) ;
FDCE \reg_path_cost(3)(4) (.Q (\path_cost(3) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx363z1)) ;
FDCE \reg_path_cost(3)(3) (.Q (\path_cost(3) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx64902z1)) ;
FDCE \reg_path_cost(3)(2) (.Q (\path_cost(3) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx63905z1)) ;
FDCE \reg_path_cost(3)(1) (.Q (\path_cost(3) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx62908z1)) ;
FDCE \reg_path_cost(3)(0) (.Q (\path_cost(3) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx61911z1)) ;
FDCE \reg_path_cost(4)(7) (.Q (\path_cost(4) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx63755z1)) ;
FDCE \reg_path_cost(4)(6) (.Q (\path_cost(4) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx62758z1)) ;
FDCE \reg_path_cost(4)(5) (.Q (\path_cost(4) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx61761z1)) ;
FDCE \reg_path_cost(4)(4) (.Q (\path_cost(4) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx60764z1)) ;
FDCE \reg_path_cost(4)(3) (.Q (\path_cost(4) [3]), .C (clk_int), .CE (
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nx56042z2), .CLR (not_rst), .D (nx59767z1)) ;
FDCE \reg_path_cost(4)(2) (.Q (\path_cost(4) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx58770z1)) ;
FDCE \reg_path_cost(4)(1) (.Q (\path_cost(4) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx57773z1)) ;
FDCE \reg_path_cost(4)(0) (.Q (\path_cost(4) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx56776z1)) ;
FDCE \reg_path_cost(5)(7) (.Q (\path_cost(5) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx58620z1)) ;
FDCE \reg_path_cost(5)(6) (.Q (\path_cost(5) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx57623z1)) ;
FDCE \reg_path_cost(5)(5) (.Q (\path_cost(5) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx56626z1)) ;
FDCE \reg_path_cost(5)(4) (.Q (\path_cost(5) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx55629z1)) ;
FDCE \reg_path_cost(5)(3) (.Q (\path_cost(5) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx54632z1)) ;
FDCE \reg_path_cost(5)(2) (.Q (\path_cost(5) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx53635z1)) ;
FDCE \reg_path_cost(5)(1) (.Q (\path_cost(5) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx52638z1)) ;
FDCE \reg_path_cost(5)(0) (.Q (\path_cost(5) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx51641z1)) ;
FDCE \reg_path_cost(6)(7) (.Q (\path_cost(6) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx12051z1)) ;
FDCE \reg_path_cost(6)(6) (.Q (\path_cost(6) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx13048z1)) ;
FDCE \reg_path_cost(6)(5) (.Q (\path_cost(6) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx14045z1)) ;
FDCE \reg_path_cost(6)(4) (.Q (\path_cost(6) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx15042z1)) ;
FDCE \reg_path_cost(6)(3) (.Q (\path_cost(6) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx16039z1)) ;
FDCE \reg_path_cost(6)(2) (.Q (\path_cost(6) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx17036z1)) ;
FDCE \reg_path_cost(6)(1) (.Q (\path_cost(6) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx18033z1)) ;
FDCE \reg_path_cost(6)(0) (.Q (\path_cost(6) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx19030z1)) ;
FDCE \reg_path_cost(7)(7) (.Q (\path_cost(7) [7]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx17186z1)) ;
FDCE \reg_path_cost(7)(6) (.Q (\path_cost(7) [6]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx18183z1)) ;
FDCE \reg_path_cost(7)(5) (.Q (\path_cost(7) [5]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx19180z1)) ;
FDCE \reg_path_cost(7)(4) (.Q (\path_cost(7) [4]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx20177z1)) ;
FDCE \reg_path_cost(7)(3) (.Q (\path_cost(7) [3]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx21174z1)) ;
FDCE \reg_path_cost(7)(2) (.Q (\path_cost(7) [2]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx22171z1)) ;
FDCE \reg_path_cost(7)(1) (.Q (\path_cost(7) [1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx23168z1)) ;
FDCE \reg_path_cost(7)(0) (.Q (\path_cost(7) [0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx24165z1)) ;
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FDCE \reg_selection(7) (.Q (selection[7]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx51057z1)) ;
FDCE \reg_selection(6) (.Q (selection[6]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx52054z1)) ;
FDCE \reg_selection(5) (.Q (selection[5]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx53051z1)) ;
FDCE \reg_selection(4) (.Q (selection[4]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx54048z1)) ;
FDCE \reg_selection(3) (.Q (selection[3]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx55045z1)) ;
FDCE \reg_selection(2) (.Q (selection[2]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx56042z1)) ;
FDCE \reg_selection(1) (.Q (selection[1]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx57039z1)) ;
FDCE \reg_selection(0) (.Q (selection[0]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx58036z1)) ;
FDCE \reg_validity(6) (.Q (\validity(2) ), .C (clk_int), .CE (nx56042z2),
.CLR (not_rst), .D (nx42042z1)) ;
FDCE \reg_validity(5) (.Q (\validity(1) ), .C (clk_int), .CE (nx56042z2),
.CLR (not_rst), .D (nx41045z1)) ;
FDCE \reg_validity(4) (.Q (\validity(4) ), .C (clk_int), .CE (nx56042z2),
.CLR (not_rst), .D (enable_int)) ;
FDCE reg_enable_tbu_0 (.Q (enable_tbu_0), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx48297z1)) ;
FDCE reg_enable_tbu_1 (.Q (enable_tbu_1), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (nx48298z1)) ;
FDCE \reg_q(9) (.Q (wr_mem_counter[9]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_9_dup_134)) ;
FDCE \reg_q(8) (.Q (wr_mem_counter[8]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_8_dup_133)) ;
FDCE \reg_q(7) (.Q (wr_mem_counter[7]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_7_dup_132)) ;
FDCE \reg_q(6) (.Q (wr_mem_counter[6]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_6_dup_131)) ;
FDCE \reg_q(5) (.Q (wr_mem_counter[5]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_5_dup_130)) ;
FDCE \reg_q(4) (.Q (wr_mem_counter[4]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_4_dup_129)) ;
FDCE \reg_q(3) (.Q (wr_mem_counter[3]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_3_dup_128)) ;
FDCE \reg_q(2) (.Q (wr_mem_counter[2]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_2_dup_127)) ;
FDCE \reg_q(1) (.Q (wr_mem_counter[1]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_1_dup_126)) ;
FDCE \reg_q(0) (.Q (wr_mem_counter[0]), .C (clk_int), .CE (nx56042z2), .CLR (
not_rst), .D (sclear_and_0_dup_125)) ;
FDPE \reg_q(9)_1_0 (.Q (rd_mem_counter[9]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(9) ), .PRE (not_rst)) ;
FDPE \reg_q(8)_1_1 (.Q (rd_mem_counter[8]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(8) ), .PRE (not_rst)) ;
FDPE \reg_q(7)_1_2 (.Q (rd_mem_counter[7]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(7) ), .PRE (not_rst)) ;
FDPE \reg_q(6)_1_3 (.Q (rd_mem_counter[6]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(6) ), .PRE (not_rst)) ;
FDPE \reg_q(5)_1_4 (.Q (rd_mem_counter[5]), .C (clk_int), .CE (nx56042z2),
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.D (\dec_d(5) ), .PRE (not_rst)) ;
FDPE \reg_q(4)_1_5 (.Q (rd_mem_counter[4]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(4) ), .PRE (not_rst)) ;
FDPE \reg_q(3)_1_6 (.Q (rd_mem_counter[3]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(3) ), .PRE (not_rst)) ;
FDPE \reg_q(2)_1_7 (.Q (rd_mem_counter[2]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(2) ), .PRE (not_rst)) ;
FDPE \reg_q(1)_1_8 (.Q (rd_mem_counter[1]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(1) ), .PRE (not_rst)) ;
FDPE \reg_q(0)_1_9 (.Q (rd_mem_counter[0]), .C (clk_int), .CE (nx56042z2),
.D (\dec_d(0) ), .PRE (not_rst)) ;
FDRE \reg_q(9)_1_29 (.Q (wr_mem_counter_disp[9]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_9_dup_128), .R (nx63z1)) ;
FDRE \reg_q(8)_1_30 (.Q (wr_mem_counter_disp[8]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_8_dup_127), .R (nx63z1)) ;
FDRE \reg_q(7)_1_31 (.Q (wr_mem_counter_disp[7]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_7_dup_126), .R (nx63z1)) ;
FDRE \reg_q(6)_1_32 (.Q (wr_mem_counter_disp[6]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_6_dup_125), .R (nx63z1)) ;
FDRE \reg_q(5)_1_33 (.Q (wr_mem_counter_disp[5]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_5_dup_124), .R (nx63z1)) ;
FDRE \reg_q(4)_1_34 (.Q (wr_mem_counter_disp[4]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_4_dup_123), .R (nx63z1)) ;
FDRE \reg_q(3)_1_35 (.Q (wr_mem_counter_disp[3]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_3_dup_122), .R (nx63z1)) ;
FDRE \reg_q(2)_1_36 (.Q (wr_mem_counter_disp[2]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_2_dup_121), .R (nx63z1)) ;
FDRE \reg_q(1)_1_37 (.Q (wr_mem_counter_disp[1]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_1_dup_120), .R (nx63z1)) ;
FDRE \reg_q(0)_1_38 (.Q (wr_mem_counter_disp[0]), .C (clk_int), .CE (
nx56042z2), .D (sload_mux_0_dup_119), .R (nx63z1)) ;
(* no_enable_dff = "TRUE" *)
FDCE \modgen_counter_mem_bank_reg_q(1) (.Q (mem_bank[1]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx37341z1)) ;
(* no_enable_dff = "TRUE" *)
FDCE \modgen_counter_mem_bank_reg_q(0) (.Q (mem_bank[0]), .C (clk_int), .CE (
nx56042z2), .CLR (not_rst), .D (nx36344z1)) ;
MUXCY muxcy_0 (.O (nx8474z1), .CI (enable_int), .DI (nx63z1), .S (nx8473z1)
) ;
MUXCY muxcy_1 (.O (nx8475z1), .CI (nx8474z1), .DI (nx63z1), .S (nx8474z2)) ;
MUXCY muxcy_2 (.O (nx8476z1), .CI (nx8475z1), .DI (nx63z1), .S (nx8475z2)) ;
MUXCY muxcy_3 (.O (nx8477z1), .CI (nx8476z1), .DI (nx63z1), .S (nx8476z2)) ;
MUXCY muxcy_4 (.O (nx8478z1), .CI (nx8477z1), .DI (nx63z1), .S (nx8477z2)) ;
MUXCY muxcy_5 (.O (nx8479z1), .CI (nx8478z1), .DI (nx63z1), .S (nx8478z2)) ;
MUXCY muxcy_6 (.O (nx8480z1), .CI (nx8479z1), .DI (nx63z1), .S (nx8479z2)) ;
MUXCY muxcy_7 (.O (nx8481z1), .CI (nx8480z1), .DI (nx63z1), .S (nx8480z2)) ;
MUXCY muxcy_8 (.O (nx51679z1), .CI (nx8481z1), .DI (nx63z1), .S (nx8481z2)
) ;
MUXCY muxcy_0_1_11 (.O (nx63827z1), .CI (nx63z1), .DI (nx56042z2), .S (
nx56042z3)) ;
MUXCY muxcy_1_1_13 (.O (nx6076z1), .CI (nx63827z1), .DI (nx56042z2), .S (
nx63827z2)) ;
MUXCY muxcy_2_1_15 (.O (nx51675z1), .CI (nx6076z1), .DI (nx56042z2), .S (
nx6076z2)) ;
MUXCY muxcy_3_1_17 (.O (nx43890z1), .CI (nx51675z1), .DI (nx56042z2), .S (
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nx51675z2)) ;
MUXCY muxcy_4_1_19 (.O (nx37092z1), .CI (nx43890z1), .DI (nx56042z2), .S (
nx43890z2)) ;
MUXCY muxcy_5_1_21 (.O (nx29307z1), .CI (nx37092z1), .DI (nx56042z2), .S (
nx37092z2)) ;
MUXCY muxcy_6_1_23 (.O (nx21522z1), .CI (nx29307z1), .DI (nx56042z2), .S (
nx29307z2)) ;
MUXCY muxcy_7_1_25 (.O (nx13737z1), .CI (nx21522z1), .DI (nx56042z2), .S (
nx21522z2)) ;
MUXCY muxcy_8_1_27 (.O (nx37206z1), .CI (nx13737z1), .DI (nx56042z2), .S (
nx13737z2)) ;
MUXCY muxcy_0_1_40 (.O (nx60837z1), .CI (nx53052z1), .DI (nx56042z2), .S (
nx53052z2)) ;
MUXCY muxcy_1_1_42 (.O (nx3086z1), .CI (nx60837z1), .DI (nx56042z2), .S (
nx60837z2)) ;
MUXCY muxcy_2_1_44 (.O (nx54665z1), .CI (nx3086z1), .DI (nx56042z2), .S (
nx3086z2)) ;
MUXCY muxcy_3_1_46 (.O (nx46880z1), .CI (nx54665z1), .DI (nx56042z2), .S (
nx54665z2)) ;
MUXCY muxcy_4_1_48 (.O (nx40082z1), .CI (nx46880z1), .DI (nx56042z2), .S (
nx46880z2)) ;
MUXCY muxcy_5_1_50 (.O (nx32297z1), .CI (nx40082z1), .DI (nx56042z2), .S (
nx40082z2)) ;
MUXCY muxcy_6_1_52 (.O (nx24512z1), .CI (nx32297z1), .DI (nx56042z2), .S (
nx32297z2)) ;
MUXCY muxcy_7_1_54 (.O (nx16727z1), .CI (nx24512z1), .DI (nx56042z2), .S (
nx24512z2)) ;
MUXCY muxcy_8_1_56 (.O (nx34216z1), .CI (nx16727z1), .DI (nx56042z2), .S (
nx16727z2)) ;
endmodule


module tbu_1 ( p_nbus_d_in_1_tbu_1, p_nbus_d_in_0_tbu_1, p_enable_tbu_1,
p_not_rst, p_selection_tbu_1, p_wr_disp_mem_1, p_clk_int,
p_d_in_disp_mem_1 ) ;

input [7:0]p_nbus_d_in_1_tbu_1 ;
input [7:0]p_nbus_d_in_0_tbu_1 ;
input p_enable_tbu_1 ;
input p_not_rst ;
input p_selection_tbu_1 ;
output p_wr_disp_mem_1 ;
input p_clk_int ;
output p_d_in_disp_mem_1 ;

wire nx2867z4, nx2867z3, nx33535z3, nx33535z2, nx34532z3, nx34532z2,
nx37523z3, nx37523z2, nx38520z2, nx2867z5, nx2867z2, nx33535z4,
nx39517z2, nx39517z1, nx38520z1, nx37523z1, nx36526z1, nx35529z1,
nx34532z1, nx33535z1, nx32538z1, not_selection, nx2867z1, nx45, nx49;
wire [7:0]tbu_1_pstate;



INV ix2867z1320 (.O (not_selection), .I (p_selection_tbu_1)) ;
LUT6 ix39517z1313 (.O (nx39517z1), .I0 (p_nbus_d_in_0_tbu_1[0]), .I1 (
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p_nbus_d_in_1_tbu_1[0]), .I2 (tbu_1_pstate[3]), .I3 (p_selection_tbu_1)
, .I4 (nx39517z2), .I5 (nx33535z4)) ;
defparam ix39517z1313.INIT = 64'hFFFFC0A0FFFFFFFF;
LUT6 ix2867z1316 (.O (nx2867z1), .I0 (nx2867z2), .I1 (tbu_1_pstate[0]), .I2 (
tbu_1_pstate[1]), .I3 (tbu_1_pstate[2]), .I4 (tbu_1_pstate[3]), .I5 (
nx2867z5)) ;
defparam ix2867z1316.INIT = 64'hFFFFFFFE00000002;
LUT4 ix39517z14883 (.O (nx39517z2), .I0 (p_nbus_d_in_0_tbu_1[1]), .I1 (
p_nbus_d_in_1_tbu_1[1]), .I2 (p_selection_tbu_1), .I3 (tbu_1_pstate[7])
) ;
defparam ix39517z14883.INIT = 16'h3500;
LUT6 ix38520z1314 (.O (nx38520z1), .I0 (p_nbus_d_in_0_tbu_1[1]), .I1 (
p_nbus_d_in_1_tbu_1[1]), .I2 (nx38520z2), .I3 (p_selection_tbu_1), .I4 (
tbu_1_pstate[7]), .I5 (nx33535z4)) ;
defparam ix38520z1314.INIT = 64'hFCFAF0F000000000;
LUT4 ix38520z14883 (.O (nx38520z2), .I0 (p_nbus_d_in_0_tbu_1[0]), .I1 (
p_nbus_d_in_1_tbu_1[0]), .I2 (p_selection_tbu_1), .I3 (tbu_1_pstate[3])
) ;
defparam ix38520z14883.INIT = 16'h3500;
(* HLUTNM = "LUT62_2_3" *)
LUT5 ix37523z1314 (.O (nx37523z1), .I0 (nx37523z2), .I1 (tbu_1_pstate[6]), .I2 (
tbu_1_pstate[2]), .I3 (nx37523z3), .I4 (nx33535z4)) ;
defparam ix37523z1314.INIT = 32'h88F80000;
(* HLUTNM = "LUT62_2_3" *)
LUT5 ix36526z1314 (.O (nx36526z1), .I0 (tbu_1_pstate[6]), .I1 (nx37523z2), .I2 (
nx37523z3), .I3 (tbu_1_pstate[2]), .I4 (nx33535z4)) ;
defparam ix36526z1314.INIT = 32'hF2220000;
LUT3 ix37523z1518 (.O (nx37523z3), .I0 (p_nbus_d_in_0_tbu_1[2]), .I1 (
p_nbus_d_in_1_tbu_1[2]), .I2 (p_selection_tbu_1)) ;
defparam ix37523z1518.INIT = 8'hCA;
(* HLUTNM = "LUT62_2_6" *)
LUT3 ix37523z1517 (.O (nx37523z2), .I0 (p_nbus_d_in_0_tbu_1[3]), .I1 (
p_nbus_d_in_1_tbu_1[3]), .I2 (p_selection_tbu_1)) ;
defparam ix37523z1517.INIT = 8'hCA;
(* HLUTNM = "LUT62_2_2" *)
LUT5 ix35529z1314 (.O (nx35529z1), .I0 (tbu_1_pstate[5]), .I1 (nx34532z2), .I2 (
nx34532z3), .I3 (tbu_1_pstate[1]), .I4 (nx33535z4)) ;
defparam ix35529z1314.INIT = 32'hF2220000;
(* HLUTNM = "LUT62_2_2" *)
LUT5 ix34532z1314 (.O (nx34532z1), .I0 (nx34532z2), .I1 (tbu_1_pstate[5]), .I2 (
tbu_1_pstate[1]), .I3 (nx34532z3), .I4 (nx33535z4)) ;
defparam ix34532z1314.INIT = 32'h88F80000;
(* HLUTNM = "LUT62_2_6" *)
LUT3 ix34532z1518 (.O (nx34532z3), .I0 (p_nbus_d_in_0_tbu_1[4]), .I1 (
p_nbus_d_in_1_tbu_1[4]), .I2 (p_selection_tbu_1)) ;
defparam ix34532z1518.INIT = 8'hCA;
(* HLUTNM = "LUT62_2_5" *)
LUT3 ix34532z1517 (.O (nx34532z2), .I0 (p_nbus_d_in_0_tbu_1[5]), .I1 (
p_nbus_d_in_1_tbu_1[5]), .I2 (p_selection_tbu_1)) ;
defparam ix34532z1517.INIT = 8'hCA;
(* HLUTNM = "LUT62_2_1" *)
LUT5 ix33535z1314 (.O (nx33535z1), .I0 (nx33535z2), .I1 (tbu_1_pstate[4]), .I2 (
tbu_1_pstate[0]), .I3 (nx33535z3), .I4 (nx33535z4)) ;
defparam ix33535z1314.INIT = 32'h88F80000;
(* HLUTNM = "LUT62_2_1" *)
```

```
LUT5 ix32538z1314 (.O (nx32538z1), .I0 (tbu_1_pstate[4]), .I1 (nx33535z2), .I2 (
nx33535z3), .I3 (tbu_1_pstate[0]), .I4 (nx33535z4)) ;
defparam ix32538z1314.INIT = 32'hF2220000;
(* HLUTNM = "LUT62_2_4" *)
LUT3 ix33535z1493 (.O (nx33535z4), .I0 (p_selection_tbu_1), .I1 (
p_wr_disp_mem_1), .I2 (p_enable_tbu_1)) ;
defparam ix33535z1493.INIT = 8'hB0;
(* HLUTNM = "LUT62_2_5" *)
LUT3 ix33535z1518 (.O (nx33535z3), .I0 (p_nbus_d_in_0_tbu_1[6]), .I1 (
p_nbus_d_in_1_tbu_1[6]), .I2 (p_selection_tbu_1)) ;
defparam ix33535z1518.INIT = 8'hCA;
(* HLUTNM = "LUT62_2_4" *)
LUT3 ix33535z1517 (.O (nx33535z2), .I0 (p_nbus_d_in_0_tbu_1[7]), .I1 (
p_nbus_d_in_1_tbu_1[7]), .I2 (p_selection_tbu_1)) ;
defparam ix33535z1517.INIT = 8'hCA;
LUT6 ix2867z1062 (.O (nx2867z5), .I0 (p_nbus_d_in_1_tbu_1[0]), .I1 (
p_nbus_d_in_1_tbu_1[2]), .I2 (p_nbus_d_in_1_tbu_1[4]), .I3 (
p_nbus_d_in_1_tbu_1[6]), .I4 (nx2867z3), .I5 (nx2867z4)) ;
defparam ix2867z1062.INIT = 64'hAAAACCCCF0F0FF00;
LUT6 ix2867z1059 (.O (nx2867z2), .I0 (p_nbus_d_in_1_tbu_1[1]), .I1 (
p_nbus_d_in_1_tbu_1[3]), .I2 (p_nbus_d_in_1_tbu_1[5]), .I3 (
p_nbus_d_in_1_tbu_1[7]), .I4 (nx2867z3), .I5 (nx2867z4)) ;
defparam ix2867z1059.INIT = 64'hAAAACCCCF0F0FF00;
LUT4 ix2867z1318 (.O (nx2867z4), .I0 (tbu_1_pstate[0]), .I1 (tbu_1_pstate[1]
), .I2 (tbu_1_pstate[4]), .I3 (tbu_1_pstate[5])) ;
defparam ix2867z1318.INIT = 16'h0001;
LUT4 ix2867z1317 (.O (nx2867z3), .I0 (tbu_1_pstate[0]), .I1 (tbu_1_pstate[2]
), .I2 (tbu_1_pstate[4]), .I3 (tbu_1_pstate[6])) ;
defparam ix2867z1317.INIT = 16'h0001;
VCC ps_vcc (.P (nx45)) ;
FDRE reg_d_o (.Q (p_d_in_disp_mem_1), .C (p_clk_int), .CE (nx45), .D (
nx2867z1), .R (not_selection)) ;
GND ps_gnd (.G (nx49)) ;
FDRE reg_selection_buf (.Q (p_wr_disp_mem_1), .C (p_clk_int), .CE (nx45), .D (
p_selection_tbu_1), .R (nx49)) ;
FDCE \reg_pstate(7) (.Q (tbu_1_pstate[0]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx32538z1)) ;
FDCE \reg_pstate(6) (.Q (tbu_1_pstate[1]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx33535z1)) ;
FDCE \reg_pstate(5) (.Q (tbu_1_pstate[2]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx34532z1)) ;
FDCE \reg_pstate(4) (.Q (tbu_1_pstate[3]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx35529z1)) ;
FDCE \reg_pstate(3) (.Q (tbu_1_pstate[4]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx36526z1)) ;
FDCE \reg_pstate(2) (.Q (tbu_1_pstate[5]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx37523z1)) ;
FDCE \reg_pstate(1) (.Q (tbu_1_pstate[6]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx38520z1)) ;
FDPE \reg_pstate(0) (.Q (tbu_1_pstate[7]), .C (p_clk_int), .CE (nx45), .D (
nx39517z1), .PRE (p_not_rst)) ;
endmodule


module tbu_0 ( p_nbus_d_in_1_tbu_0, p_nbus_d_in_0_tbu_0, p_enable_tbu_0,
```

```
p_rst_int, p_not_rst, p_mem_bank_Q3, p_wr_disp_mem_0, p_clk_int,
p_d_in_disp_mem_0 ) ;

input [7:0]p_nbus_d_in_1_tbu_0 ;
input [7:0]p_nbus_d_in_0_tbu_0 ;
input p_enable_tbu_0 ;
input p_rst_int ;
output p_not_rst ;
input p_mem_bank_Q3 ;
output p_wr_disp_mem_0 ;
input p_clk_int ;
output p_d_in_disp_mem_0 ;

wire nx2867z4, nx2867z3, nx33535z3, nx33535z2, nx34532z3, nx34532z2,
nx37523z3, nx37523z2, nx38520z2, nx2867z5, nx2867z2, nx33535z4,
nx39517z2, nx39517z1, nx38520z1, nx37523z1, nx36526z1, nx35529z1,
nx34532z1, nx33535z1, nx32538z1, not_selection, nx2867z1, nx45, nx49;
wire [7:0]tbu_0_pstate;



INV ix3304z1315 (.O (p_not_rst), .I (p_rst_int)) ;
INV ix2867z1320 (.O (not_selection), .I (p_mem_bank_Q3)) ;
LUT6 ix39517z1313 (.O (nx39517z1), .I0 (p_nbus_d_in_0_tbu_0[0]), .I1 (
p_nbus_d_in_1_tbu_0[0]), .I2 (tbu_0_pstate[3]), .I3 (p_mem_bank_Q3), .I4 (
nx39517z2), .I5 (nx33535z4)) ;
defparam ix39517z1313.INIT = 64'hFFFFC0A0FFFFFFFF;
LUT6 ix2867z1316 (.O (nx2867z1), .I0 (nx2867z2), .I1 (tbu_0_pstate[0]), .I2 (
tbu_0_pstate[1]), .I3 (tbu_0_pstate[2]), .I4 (tbu_0_pstate[3]), .I5 (
nx2867z5)) ;
defparam ix2867z1316.INIT = 64'hFFFFFFFE00000002;
LUT4 ix39517z14883 (.O (nx39517z2), .I0 (p_nbus_d_in_0_tbu_0[1]), .I1 (
p_nbus_d_in_1_tbu_0[1]), .I2 (p_mem_bank_Q3), .I3 (tbu_0_pstate[7])) ;
defparam ix39517z14883.INIT = 16'h3500;
LUT6 ix38520z1314 (.O (nx38520z1), .I0 (p_nbus_d_in_0_tbu_0[1]), .I1 (
p_nbus_d_in_1_tbu_0[1]), .I2 (nx38520z2), .I3 (p_mem_bank_Q3), .I4 (
tbu_0_pstate[7]), .I5 (nx33535z4)) ;
defparam ix38520z1314.INIT = 64'hFCFAF0F000000000;
LUT4 ix38520z14883 (.O (nx38520z2), .I0 (p_nbus_d_in_0_tbu_0[0]), .I1 (
p_nbus_d_in_1_tbu_0[0]), .I2 (p_mem_bank_Q3), .I3 (tbu_0_pstate[3])) ;
defparam ix38520z14883.INIT = 16'h3500;
(* HLUTNM = "LUT62_1_3" *)
LUT5 ix37523z1314 (.O (nx37523z1), .I0 (nx37523z2), .I1 (tbu_0_pstate[6]), .I2 (
tbu_0_pstate[2]), .I3 (nx37523z3), .I4 (nx33535z4)) ;
defparam ix37523z1314.INIT = 32'h88F80000;
(* HLUTNM = "LUT62_1_3" *)
LUT5 ix36526z1314 (.O (nx36526z1), .I0 (tbu_0_pstate[6]), .I1 (nx37523z2), .I2 (
nx37523z3), .I3 (tbu_0_pstate[2]), .I4 (nx33535z4)) ;
defparam ix36526z1314.INIT = 32'hF2220000;
LUT3 ix37523z1518 (.O (nx37523z3), .I0 (p_nbus_d_in_0_tbu_0[2]), .I1 (
p_nbus_d_in_1_tbu_0[2]), .I2 (p_mem_bank_Q3)) ;
defparam ix37523z1518.INIT = 8'hCA;
(* HLUTNM = "LUT62_1_6" *)
LUT3 ix37523z1517 (.O (nx37523z2), .I0 (p_nbus_d_in_0_tbu_0[3]), .I1 (
p_nbus_d_in_1_tbu_0[3]), .I2 (p_mem_bank_Q3)) ;
```

```
defparam ix37523z1517.INIT = 8'hCA;
(* HLUTNM = "LUT62_1_2" *)
LUT5 ix35529z1314 (.O (nx35529z1), .I0 (tbu_0_pstate[5]), .I1 (nx34532z2), .I2 (
nx34532z3), .I3 (tbu_0_pstate[1]), .I4 (nx33535z4)) ;
defparam ix35529z1314.INIT = 32'hF2220000;
(* HLUTNM = "LUT62_1_2" *)
LUT5 ix34532z1314 (.O (nx34532z1), .I0 (nx34532z2), .I1 (tbu_0_pstate[5]), .I2 (
tbu_0_pstate[1]), .I3 (nx34532z3), .I4 (nx33535z4)) ;
defparam ix34532z1314.INIT = 32'h88F80000;
(* HLUTNM = "LUT62_1_6" *)
LUT3 ix34532z1518 (.O (nx34532z3), .I0 (p_nbus_d_in_0_tbu_0[4]), .I1 (
p_nbus_d_in_1_tbu_0[4]), .I2 (p_mem_bank_Q3)) ;
defparam ix34532z1518.INIT = 8'hCA;
(* HLUTNM = "LUT62_1_5" *)
LUT3 ix34532z1517 (.O (nx34532z2), .I0 (p_nbus_d_in_0_tbu_0[5]), .I1 (
p_nbus_d_in_1_tbu_0[5]), .I2 (p_mem_bank_Q3)) ;
defparam ix34532z1517.INIT = 8'hCA;
(* HLUTNM = "LUT62_1_1" *)
LUT5 ix33535z1314 (.O (nx33535z1), .I0 (nx33535z2), .I1 (tbu_0_pstate[4]), .I2 (
tbu_0_pstate[0]), .I3 (nx33535z3), .I4 (nx33535z4)) ;
defparam ix33535z1314.INIT = 32'h88F80000;
(* HLUTNM = "LUT62_1_1" *)
LUT5 ix32538z1314 (.O (nx32538z1), .I0 (tbu_0_pstate[4]), .I1 (nx33535z2), .I2 (
nx33535z3), .I3 (tbu_0_pstate[0]), .I4 (nx33535z4)) ;
defparam ix32538z1314.INIT = 32'hF2220000;
(* HLUTNM = "LUT62_1_4" *)
LUT3 ix33535z1493 (.O (nx33535z4), .I0 (p_mem_bank_Q3), .I1 (p_wr_disp_mem_0
), .I2 (p_enable_tbu_0)) ;
defparam ix33535z1493.INIT = 8'hB0;
(* HLUTNM = "LUT62_1_5" *)
LUT3 ix33535z1518 (.O (nx33535z3), .I0 (p_nbus_d_in_0_tbu_0[6]), .I1 (
p_nbus_d_in_1_tbu_0[6]), .I2 (p_mem_bank_Q3)) ;
defparam ix33535z1518.INIT = 8'hCA;
(* HLUTNM = "LUT62_1_4" *)
LUT3 ix33535z1517 (.O (nx33535z2), .I0 (p_nbus_d_in_0_tbu_0[7]), .I1 (
p_nbus_d_in_1_tbu_0[7]), .I2 (p_mem_bank_Q3)) ;
defparam ix33535z1517.INIT = 8'hCA;
LUT6 ix2867z1062 (.O (nx2867z5), .I0 (p_nbus_d_in_1_tbu_0[0]), .I1 (
p_nbus_d_in_1_tbu_0[2]), .I2 (p_nbus_d_in_1_tbu_0[4]), .I3 (
p_nbus_d_in_1_tbu_0[6]), .I4 (nx2867z3), .I5 (nx2867z4)) ;
defparam ix2867z1062.INIT = 64'hAAAACCCCF0F0FF00;
LUT6 ix2867z1059 (.O (nx2867z2), .I0 (p_nbus_d_in_1_tbu_0[1]), .I1 (
p_nbus_d_in_1_tbu_0[3]), .I2 (p_nbus_d_in_1_tbu_0[5]), .I3 (
p_nbus_d_in_1_tbu_0[7]), .I4 (nx2867z3), .I5 (nx2867z4)) ;
defparam ix2867z1059.INIT = 64'hAAAACCCCF0F0FF00;
LUT4 ix2867z1318 (.O (nx2867z4), .I0 (tbu_0_pstate[0]), .I1 (tbu_0_pstate[1]
), .I2 (tbu_0_pstate[4]), .I3 (tbu_0_pstate[5])) ;
defparam ix2867z1318.INIT = 16'h0001;
LUT4 ix2867z1317 (.O (nx2867z3), .I0 (tbu_0_pstate[0]), .I1 (tbu_0_pstate[2]
), .I2 (tbu_0_pstate[4]), .I3 (tbu_0_pstate[6])) ;
defparam ix2867z1317.INIT = 16'h0001;
VCC ps_vcc (.P (nx45)) ;
FDRE reg_d_o (.Q (p_d_in_disp_mem_0), .C (p_clk_int), .CE (nx45), .D (
nx2867z1), .R (not_selection)) ;
GND ps_gnd (.G (nx49)) ;
```

```
FDRE reg_selection_buf (.Q (p_wr_disp_mem_0), .C (p_clk_int), .CE (nx45), .D (
p_mem_bank_Q3), .R (nx49)) ;
FDCE \reg_pstate(7) (.Q (tbu_0_pstate[0]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx32538z1)) ;
FDCE \reg_pstate(6) (.Q (tbu_0_pstate[1]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx33535z1)) ;
FDCE \reg_pstate(5) (.Q (tbu_0_pstate[2]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx34532z1)) ;
FDCE \reg_pstate(4) (.Q (tbu_0_pstate[3]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx35529z1)) ;
FDCE \reg_pstate(3) (.Q (tbu_0_pstate[4]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx36526z1)) ;
FDCE \reg_pstate(2) (.Q (tbu_0_pstate[5]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx37523z1)) ;
FDCE \reg_pstate(1) (.Q (tbu_0_pstate[6]), .C (p_clk_int), .CE (nx45), .CLR (
p_not_rst), .D (nx38520z1)) ;
FDPE \reg_pstate(0) (.Q (tbu_0_pstate[7]), .C (p_clk_int), .CE (nx45), .D (
nx39517z1), .PRE (p_not_rst)) ;
endmodule


module ram_dq_8_3 ( wr_data1, rd_data1, addr1, wr_clk1, rd_clk1, wr_ena1,
rd_ena1, ena1, rst1, regce1, regrst1, rd_access1 ) ;

input [7:0]wr_data1 ;
output [7:0]rd_data1 ;
input [9:0]addr1 ;
input wr_clk1 ;
input rd_clk1 ;
input wr_ena1 ;
input rd_ena1 ;
input ena1 ;
input rst1 ;
input regce1 ;
input regrst1 ;
output rd_access1 ;

wire rst1_rename23, ena1_rename24;
wire [27:0] \$dummy ;




GND ps_gnd (.G (rst1_rename23)) ;
VCC ps_vcc (.P (ena1_rename24)) ;
RAMB18E1 trelis_mem_D_mem_1 (.DOADO ({\$dummy [0],\$dummy [1],\$dummy [2],
\$dummy [3],\$dummy [4],\$dummy [5],\$dummy [6],\$dummy [7],
rd_data1[7],rd_data1[6],rd_data1[5],rd_data1[4],rd_data1[3],
rd_data1[2],rd_data1[1],rd_data1[0]}), .ADDRARDADDR ({rst1_rename23
,addr1[9],addr1[8],addr1[7],addr1[6],addr1[5],addr1[4],addr1[3],
addr1[2],addr1[1],addr1[0],ena1_rename24,ena1_rename24,
ena1_rename24}), .ADDRBWRADDR ({ena1_rename24,ena1_rename24,
ena1_rename24,ena1_rename24,ena1_rename24,ena1_rename24,
ena1_rename24,ena1_rename24,ena1_rename24,ena1_rename24,
ena1_rename24,ena1_rename24,ena1_rename24,ena1_rename24}), .CLKARDCLK (
```

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wr_clk1), .CLKBWRCLK (rst1_rename23), .DIADI ({rst1_rename23,
rst1_rename23,rst1_rename23,rst1_rename23,rst1_rename23,
rst1_rename23,rst1_rename23,rst1_rename23,wr_data1[7],wr_data1[6],
wr_data1[5],wr_data1[4],wr_data1[3],wr_data1[2],wr_data1[1],
wr_data1[0]}), .DIBDI ({rst1_rename23,rst1_rename23,rst1_rename23,
rst1_rename23,rst1_rename23,rst1_rename23,rst1_rename23,
rst1_rename23,rst1_rename23,rst1_rename23,rst1_rename23,
rst1_rename23,rst1_rename23,rst1_rename23,rst1_rename23,
rst1_rename23}), .DIPADIP ({rst1_rename23,rst1_rename23}), .DIPBDIP (
{rst1_rename23,rst1_rename23}), .ENARDEN (ena1_rename24), .ENBWREN (
rst1_rename23), .REGCEAREGCE (rst1_rename23), .REGCEB (
rst1_rename23), .RSTRAMARSTRAM (rst1_rename23), .RSTRAMB (
rst1_rename23), .RSTREGARSTREG (rst1_rename23), .RSTREGB (
rst1_rename23), .WEA ({wr_ena1,wr_ena1}), .WEBWE ({rst1_rename23,
rst1_rename23,rst1_rename23,rst1_rename23})) ;
defparam trelis_mem_D_mem_1.DOA_REG = 0;
defparam trelis_mem_D_mem_1.DOB_REG = 0;
defparam trelis_mem_D_mem_1.INIT_A = 20'h00000;
defparam trelis_mem_D_mem_1.INIT_B = 20'h00000;
defparam trelis_mem_D_mem_1.RAM_MODE = "TDP";
defparam trelis_mem_D_mem_1.READ_WIDTH_A = 9;
defparam trelis_mem_D_mem_1.READ_WIDTH_B = 0;
defparam trelis_mem_D_mem_1.RSTREG_PRIORITY_A = "RSTREG";
defparam trelis_mem_D_mem_1.RSTREG_PRIORITY_B = "RSTREG";
defparam trelis_mem_D_mem_1.SIM_COLLISION_CHECK = "ALL";
defparam trelis_mem_D_mem_1.SIM_DEVICE = "7SERIES";
defparam trelis_mem_D_mem_1.SRVAL_A = 36'h000000000;
defparam trelis_mem_D_mem_1.SRVAL_B = 36'h000000000;
defparam trelis_mem_D_mem_1.WRITE_MODE_A = "READ_FIRST";
defparam trelis_mem_D_mem_1.WRITE_MODE_B = "NO_CHANGE";
defparam trelis_mem_D_mem_1.WRITE_WIDTH_A = 9;
defparam trelis_mem_D_mem_1.WRITE_WIDTH_B = 0;
endmodule


module ram_dq_8_2 ( wr_data1, rd_data1, addr1, wr_clk1, rd_clk1, wr_ena1,
rd_ena1, ena1, rst1, regce1, regrst1, rd_access1 ) ;

input [7:0]wr_data1 ;
output [7:0]rd_data1 ;
input [9:0]addr1 ;
input wr_clk1 ;
input rd_clk1 ;
input wr_ena1 ;
input rd_ena1 ;
input ena1 ;
input rst1 ;
input regce1 ;
input regrst1 ;
output rd_access1 ;

wire rst1_rename16, ena1_rename17;
wire [27:0] \$dummy ;
```

```
GND ps_gnd (.G (rst1_rename16)) ;
VCC ps_vcc (.P (ena1_rename17)) ;
RAMB18E1 trelis_mem_C_mem_1 (.DOADO ({\$dummy [0],\$dummy [1],\$dummy [2],
\$dummy [3],\$dummy [4],\$dummy [5],\$dummy [6],\$dummy [7],
rd_data1[7],rd_data1[6],rd_data1[5],rd_data1[4],rd_data1[3],
rd_data1[2],rd_data1[1],rd_data1[0]}), .ADDRARDADDR ({rst1_rename16
,addr1[9],addr1[8],addr1[7],addr1[6],addr1[5],addr1[4],addr1[3],
addr1[2],addr1[1],addr1[0],ena1_rename17,ena1_rename17,
ena1_rename17}), .ADDRBWRADDR ({ena1_rename17,ena1_rename17,
ena1_rename17,ena1_rename17,ena1_rename17,ena1_rename17,
ena1_rename17,ena1_rename17,ena1_rename17,ena1_rename17,
ena1_rename17,ena1_rename17,ena1_rename17,ena1_rename17}), .CLKARDCLK (
wr_clk1), .CLKBWRCLK (rst1_rename16), .DIADI ({rst1_rename16,
rst1_rename16,rst1_rename16,rst1_rename16,rst1_rename16,
rst1_rename16,rst1_rename16,rst1_rename16,wr_data1[7],wr_data1[6],
wr_data1[5],wr_data1[4],wr_data1[3],wr_data1[2],wr_data1[1],
wr_data1[0]}), .DIBDI ({rst1_rename16,rst1_rename16,rst1_rename16,
rst1_rename16,rst1_rename16,rst1_rename16,rst1_rename16,
rst1_rename16,rst1_rename16,rst1_rename16,rst1_rename16,
rst1_rename16,rst1_rename16,rst1_rename16,rst1_rename16,
rst1_rename16}), .DIPADIP ({rst1_rename16,rst1_rename16}), .DIPBDIP (
{rst1_rename16,rst1_rename16}), .ENARDEN (ena1_rename17), .ENBWREN (
rst1_rename16), .REGCEAREGCE (rst1_rename16), .REGCEB (
rst1_rename16), .RSTRAMARSTRAM (rst1_rename16), .RSTRAMB (
rst1_rename16), .RSTREGARSTREG (rst1_rename16), .RSTREGB (
rst1_rename16), .WEA ({wr_ena1,wr_ena1}), .WEBWE ({rst1_rename16,
rst1_rename16,rst1_rename16,rst1_rename16})) ;
defparam trelis_mem_C_mem_1.DOA_REG = 0;
defparam trelis_mem_C_mem_1.DOB_REG = 0;
defparam trelis_mem_C_mem_1.INIT_A = 20'h00000;
defparam trelis_mem_C_mem_1.INIT_B = 20'h00000;
defparam trelis_mem_C_mem_1.RAM_MODE = "TDP";
defparam trelis_mem_C_mem_1.READ_WIDTH_A = 9;
defparam trelis_mem_C_mem_1.READ_WIDTH_B = 0;
defparam trelis_mem_C_mem_1.RSTREG_PRIORITY_A = "RSTREG";
defparam trelis_mem_C_mem_1.RSTREG_PRIORITY_B = "RSTREG";
defparam trelis_mem_C_mem_1.SIM_COLLISION_CHECK = "ALL";
defparam trelis_mem_C_mem_1.SIM_DEVICE = "7SERIES";
defparam trelis_mem_C_mem_1.SRVAL_A = 36'h000000000;
defparam trelis_mem_C_mem_1.SRVAL_B = 36'h000000000;
defparam trelis_mem_C_mem_1.WRITE_MODE_A = "READ_FIRST";
defparam trelis_mem_C_mem_1.WRITE_MODE_B = "NO_CHANGE";
defparam trelis_mem_C_mem_1.WRITE_WIDTH_A = 9;
defparam trelis_mem_C_mem_1.WRITE_WIDTH_B = 0;
endmodule


module ram_dq_8_1 ( wr_data1, rd_data1, addr1, wr_clk1, rd_clk1, wr_ena1,
rd_ena1, ena1, rst1, regce1, regrst1, rd_access1 ) ;

input [7:0]wr_data1 ;
output [7:0]rd_data1 ;
input [9:0]addr1 ;
```

```
input wr_clk1 ;
input rd_clk1 ;
input wr_ena1 ;
input rd_ena1 ;
input ena1 ;
input rst1 ;
input regce1 ;
input regrst1 ;
output rd_access1 ;

wire rst1_rename9, ena1_rename10;
wire [27:0] \$dummy ;



GND ps_gnd (.G (rst1_rename9)) ;
VCC ps_vcc (.P (ena1_rename10)) ;
RAMB18E1 trelis_mem_B_mem_1 (.DOADO ({\$dummy [0],\$dummy [1],\$dummy [2],
\$dummy [3],\$dummy [4],\$dummy [5],\$dummy [6],\$dummy [7],
rd_data1[7],rd_data1[6],rd_data1[5],rd_data1[4],rd_data1[3],
rd_data1[2],rd_data1[1],rd_data1[0]}), .ADDRARDADDR ({rst1_rename9,
addr1[9],addr1[8],addr1[7],addr1[6],addr1[5],addr1[4],addr1[3],
addr1[2],addr1[1],addr1[0],ena1_rename10,ena1_rename10,
ena1_rename10}), .ADDRBWRADDR ({ena1_rename10,ena1_rename10,
ena1_rename10,ena1_rename10,ena1_rename10,ena1_rename10,
ena1_rename10,ena1_rename10,ena1_rename10,ena1_rename10,
ena1_rename10,ena1_rename10,ena1_rename10,ena1_rename10}), .CLKARDCLK (
wr_clk1), .CLKBWRCLK (rst1_rename9), .DIADI ({rst1_rename9,
rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,
rst1_rename9,rst1_rename9,wr_data1[7],wr_data1[6],wr_data1[5],
wr_data1[4],wr_data1[3],wr_data1[2],wr_data1[1],wr_data1[0]}), .DIBDI (
{rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,
rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,
rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,
rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9,
rst1_rename9}), .DIPADIP ({rst1_rename9,rst1_rename9}), .DIPBDIP ({
rst1_rename9,rst1_rename9}), .ENARDEN (ena1_rename10), .ENBWREN (
rst1_rename9), .REGCEAREGCE (rst1_rename9), .REGCEB (rst1_rename9)
, .RSTRAMARSTRAM (rst1_rename9), .RSTRAMB (rst1_rename9), .RSTREGARSTREG (
rst1_rename9), .RSTREGB (rst1_rename9), .WEA ({wr_ena1,wr_ena1}), .WEBWE (
{rst1_rename9,rst1_rename9,rst1_rename9,rst1_rename9})) ;
defparam trelis_mem_B_mem_1.DOA_REG = 0;
defparam trelis_mem_B_mem_1.DOB_REG = 0;
defparam trelis_mem_B_mem_1.INIT_A = 20'h00000;
defparam trelis_mem_B_mem_1.INIT_B = 20'h00000;
defparam trelis_mem_B_mem_1.RAM_MODE = "TDP";
defparam trelis_mem_B_mem_1.READ_WIDTH_A = 9;
defparam trelis_mem_B_mem_1.READ_WIDTH_B = 0;
defparam trelis_mem_B_mem_1.RSTREG_PRIORITY_A = "RSTREG";
defparam trelis_mem_B_mem_1.RSTREG_PRIORITY_B = "RSTREG";
defparam trelis_mem_B_mem_1.SIM_COLLISION_CHECK = "ALL";
defparam trelis_mem_B_mem_1.SIM_DEVICE = "7SERIES";
defparam trelis_mem_B_mem_1.SRVAL_A = 36'h000000000;
defparam trelis_mem_B_mem_1.SRVAL_B = 36'h000000000;
defparam trelis_mem_B_mem_1.WRITE_MODE_A = "READ_FIRST";
```

```
defparam trelis_mem_B_mem_1.WRITE_MODE_B = "NO_CHANGE";
defparam trelis_mem_B_mem_1.WRITE_WIDTH_A = 9;
defparam trelis_mem_B_mem_1.WRITE_WIDTH_B = 0;
endmodule


module ram_dq_8_0 ( wr_data1, rd_data1, addr1, wr_clk1, rd_clk1, wr_ena1,
rd_ena1, ena1, rst1, regce1, regrst1, rd_access1 ) ;

input [7:0]wr_data1 ;
output [7:0]rd_data1 ;
input [9:0]addr1 ;
input wr_clk1 ;
input rd_clk1 ;
input wr_ena1 ;
input rd_ena1 ;
input ena1 ;
input rst1 ;
input regce1 ;
input regrst1 ;
output rd_access1 ;

wire rst1_rename2, ena1_rename3;
wire [27:0] \$dummy ;



GND ps_gnd (.G (rst1_rename2)) ;
VCC ps_vcc (.P (ena1_rename3)) ;
RAMB18E1 trelis_mem_A_mem_1 (.DOADO ({\$dummy [0],\$dummy [1],\$dummy [2],
\$dummy [3],\$dummy [4],\$dummy [5],\$dummy [6],\$dummy [7],
rd_data1[7],rd_data1[6],rd_data1[5],rd_data1[4],rd_data1[3],
rd_data1[2],rd_data1[1],rd_data1[0]}), .ADDRARDADDR ({rst1_rename2,
addr1[9],addr1[8],addr1[7],addr1[6],addr1[5],addr1[4],addr1[3],
addr1[2],addr1[1],addr1[0],ena1_rename3,ena1_rename3,ena1_rename3})
, .ADDRBWRADDR ({ena1_rename3,ena1_rename3,ena1_rename3,
ena1_rename3,ena1_rename3,ena1_rename3,ena1_rename3,ena1_rename3,
ena1_rename3,ena1_rename3,ena1_rename3,ena1_rename3,ena1_rename3,
ena1_rename3}), .CLKARDCLK (wr_clk1), .CLKBWRCLK (rst1_rename2), .DIADI (
{rst1_rename2,rst1_rename2,rst1_rename2,rst1_rename2,rst1_rename2,
rst1_rename2,rst1_rename2,rst1_rename2,wr_data1[7],wr_data1[6],
wr_data1[5],wr_data1[4],wr_data1[3],wr_data1[2],wr_data1[1],
wr_data1[0]}), .DIBDI ({rst1_rename2,rst1_rename2,rst1_rename2,
rst1_rename2,rst1_rename2,rst1_rename2,rst1_rename2,rst1_rename2,
rst1_rename2,rst1_rename2,rst1_rename2,rst1_rename2,rst1_rename2,
rst1_rename2,rst1_rename2,rst1_rename2}), .DIPADIP ({rst1_rename2,
rst1_rename2}), .DIPBDIP ({rst1_rename2,rst1_rename2}), .ENARDEN (
ena1_rename3), .ENBWREN (rst1_rename2), .REGCEAREGCE (rst1_rename2)
, .REGCEB (rst1_rename2), .RSTRAMARSTRAM (rst1_rename2), .RSTRAMB (
rst1_rename2), .RSTREGARSTREG (rst1_rename2), .RSTREGB (
rst1_rename2), .WEA ({wr_ena1,wr_ena1}), .WEBWE ({rst1_rename2,
rst1_rename2,rst1_rename2,rst1_rename2})) ;
defparam trelis_mem_A_mem_1.DOA_REG = 0;
defparam trelis_mem_A_mem_1.DOB_REG = 0;
```

```
defparam trelis_mem_A_mem_1.INIT_A = 20'h00000;
defparam trelis_mem_A_mem_1.INIT_B = 20'h00000;
defparam trelis_mem_A_mem_1.RAM_MODE = "TDP";
defparam trelis_mem_A_mem_1.READ_WIDTH_A = 9;
defparam trelis_mem_A_mem_1.READ_WIDTH_B = 0;
defparam trelis_mem_A_mem_1.RSTREG_PRIORITY_A = "RSTREG";
defparam trelis_mem_A_mem_1.RSTREG_PRIORITY_B = "RSTREG";
defparam trelis_mem_A_mem_1.SIM_COLLISION_CHECK = "ALL";
defparam trelis_mem_A_mem_1.SIM_DEVICE = "7SERIES";
defparam trelis_mem_A_mem_1.SRVAL_A = 36'h000000000;
defparam trelis_mem_A_mem_1.SRVAL_B = 36'h000000000;
defparam trelis_mem_A_mem_1.WRITE_MODE_A = "READ_FIRST";
defparam trelis_mem_A_mem_1.WRITE_MODE_B = "NO_CHANGE";
defparam trelis_mem_A_mem_1.WRITE_WIDTH_A = 9;
defparam trelis_mem_A_mem_1.WRITE_WIDTH_B = 0;
endmodule
```

2) This convolutional encoder differs from the encoder in homework 7 in that its output is calculated where the MSB of the output is XOR of input and the 2nd and 1st bits of the present state and the LSB is the input. In contrast, the bits of the output for homework 7 was calculated by the XOR reduction and bitwise and of the corresponding mask and shift register. The shift register is updated by shifting right by 1 bit and filling in the MSB with the input while the next state is updated XORing the input and 1st and 0th bits of the present state. However, both are non-recursive and non-systematic.

3) The decoder works by calculating the branch metrics along the paths using the bmc module. Then the ACS modules are used to find the path costs by accumulating the bmc metrics and making selections for the best path at each stage. Then the traceback module traces the optimal path through the ACS's to select the path or sequence of bits with the least amount of cost or error.

The s

Problem:	# of Errors	Pattern of Errors	# of Errors Uncorrected
2a	16	00000000000000010000000000000001…	0
2a1	32	0000000100000001…	0
2a2	32	0000000100000001…	0
2a3	16	00000000000000000000000000000011…	0
2a4	32	00000000000000110000000000000011…	0

2a5	32	00000000000000110000000000000011…	0
2a6	32	00000000000000000000000000001111…	0
2a7	32	00000000000000000000000000000000000000000000000000000000010101010	0
2a8	16	00000000000000000000000000000011	0
2b, 2b3	18	00010000000000000000000000000000000000000000000010000000000000001	0
2b1, b2	30	Random bad bits with a random spacing of 0 to 8 bits	0
2b4 &b5	16	Random bad bits with two consecutive injections	0
2b6&7	32	Random bad bits with 4 consecutive injections	12
2b8	8	Random bad bits with 2 consecutive injections with a spacing of 32	0
2c	80	00000000000000000000001010101	42
2d	80	00000000000000000000010101010	26
2e	24	000000000000000000000000000000000000000000000000000000000111111	18

2A) Transcript:

```
# yaa! in = 0, out = 0, w_ct =          0, err = 00
# yaa! in = 0, out = 0, w_ct =          1, err = 00
# yaa! in = 0, out = 0, w_ct =          2, err = 00
# yaa! in = 0, out = 0, w_ct =          3, err = 00
# yaa! in = 0, out = 0, w_ct =          4, err = 00
# yaa! in = 0, out = 0, w_ct =          5, err = 00
# yaa! in = 0, out = 0, w_ct =          6, err = 00
# yaa! in = 0, out = 0, w_ct =          7, err = 00
# yaa! in = 0, out = 0, w_ct =          8, err = 00
# yaa! in = 0, out = 0, w_ct =          9, err = 00
# yaa! in = 0, out = 0, w_ct =         10, err = 00
# yaa! in = 1, out = 1, w_ct =         11, err = 00
# yaa! in = 0, out = 0, w_ct =         12, err = 00
# yaa! in = 0, out = 0, w_ct =         13, err = 00
# yaa! in = 1, out = 1, w_ct =         14, err = 00
# yaa! in = 1, out = 1, w_ct =         15, err = 00
# yaa! in = 0, out = 0, w_ct =         16, err = 00
# yaa! in = 0, out = 0, w_ct =         17, err = 00
# yaa! in = 0, out = 0, w_ct =         18, err = 00
# yaa! in = 1, out = 1, w_ct =         19, err = 00
# yaa! in = 1, out = 1, w_ct =         20, err = 00
# yaa! in = 1, out = 1, w_ct =         21, err = 00
# yaa! in = 0, out = 0, w_ct =         22, err = 00
# yaa! in = 0, out = 0, w_ct =         23, err = 00
# yaa! in = 0, out = 0, w_ct =         24, err = 00
# yaa! in = 0, out = 0, w_ct =         25, err = 00
# yaa! in = 1, out = 1, w_ct =         26, err = 00
# yaa! in = 1, out = 1, w_ct =         27, err = 00
# yaa! in = 1, out = 1, w_ct =         28, err = 00
# yaa! in = 1, out = 1, w_ct =         29, err = 00
# yaa! in = 0, out = 0, w_ct =         30, err = 00
# yaa! in = 0, out = 0, w_ct =         31, err = 00
# yaa! in = 0, out = 0, w_ct =         32, err = 00
# yaa! in = 0, out = 0, w_ct =         33, err = 00
# yaa! in = 0, out = 0, w_ct =         34, err = 00
# yaa! in = 1, out = 1, w_ct =         35, err = 00
# yaa! in = 1, out = 1, w_ct =         36, err = 00
# yaa! in = 1, out = 1, w_ct =         37, err = 00
# yaa! in = 1, out = 1, w_ct =         38, err = 00
# yaa! in = 1, out = 1, w_ct =         39, err = 00
# yaa! in = 0, out = 0, w_ct =         40, err = 00
# yaa! in = 1, out = 1, w_ct =         41, err = 00
# yaa! in = 0, out = 0, w_ct =         42, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000000 00          0         12
# error_counter,err_inj = 00000000 00          0         13
# error_counter,err_inj = 00000000 00          0         14
# error_counter,err_inj = 00000000 00          0         15
# error_counter,err_inj = 00000001 01          0         16
# error_counter,err_inj = 00000001 00          0         17
# error_counter,err_inj = 00000001 00          0         18
# error_counter,err_inj = 00000001 00          0         19
# error_counter,err_inj = 00000001 00          0         20
# error_counter,err_inj = 00000001 00          0         21
# error_counter,err_inj = 00000001 00          0         22
# error_counter,err_inj = 00000001 00          0         23
# error_counter,err_inj = 00000001 00          0         24
# error_counter,err_inj = 00000001 00          0         25
# error_counter,err_inj = 00000001 00          0         26
# error_counter,err_inj = 00000001 00          0         27
# error_counter,err_inj = 00000001 00          0         28
# error_counter,err_inj = 00000001 00          0         29
# error_counter,err_inj = 00000001 00          0         30
# error_counter,err_inj = 00000001 00          0         31
# error_counter,err_inj = 00000002 01          0         32
# error_counter,err_inj = 00000002 00          0         33
# error_counter,err_inj = 00000002 00          0         34
# error_counter,err_inj = 00000002 00          0         35
# error_counter,err_inj = 00000002 00          0         36
# error_counter,err_inj = 00000002 00          0         37
# error_counter,err_inj = 00000002 00          0         38
# error_counter,err_inj = 00000002 00          0         39
# error_counter,err_inj = 00000002 00          0         40
# error_counter,err_inj = 00000002 00          0         41
# error_counter,err_inj = 00000002 00          0         42
```

```
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
```

```
# error_counter,err_inj = 00000002 00          0          43
# error_counter,err_inj = 00000002 00          0          44
# error_counter,err_inj = 00000002 00          0          45
# error_counter,err_inj = 00000002 00          0          46
# error_counter,err_inj = 00000002 00          0          47
# error_counter,err_inj = 00000003 01          0          48
# error_counter,err_inj = 00000003 00          0          49
# error_counter,err_inj = 00000003 00          0          50
# error_counter,err_inj = 00000003 00          0          51
# error_counter,err_inj = 00000003 00          0          52
# error_counter,err_inj = 00000003 00          0          53
# error_counter,err_inj = 00000003 00          0          54
# error_counter,err_inj = 00000003 00          0          55
# error_counter,err_inj = 00000003 00          0          56
# error_counter,err_inj = 00000003 00          0          57
# error_counter,err_inj = 00000003 00          0          58
# error_counter,err_inj = 00000003 00          0          59
# error_counter,err_inj = 00000003 00          0          60
# error_counter,err_inj = 00000003 00          0          61
# error_counter,err_inj = 00000003 00          0          62
# error_counter,err_inj = 00000003 00          0          63
# error_counter,err_inj = 00000004 01          0          64
# error_counter,err_inj = 00000004 00          0          65
# error_counter,err_inj = 00000004 00          0          66
# error_counter,err_inj = 00000004 00          0          67
# error_counter,err_inj = 00000004 00          0          68
# error_counter,err_inj = 00000004 00          0          69
# error_counter,err_inj = 00000004 00          0          70
# error_counter,err_inj = 00000004 00          0          71
# error_counter,err_inj = 00000004 00          0          72
# error_counter,err_inj = 00000004 00          0          73
# error_counter,err_inj = 00000004 00          0          74
# error_counter,err_inj = 00000004 00          0          75
# error_counter,err_inj = 00000004 00          0          76
# error_counter,err_inj = 00000004 00          0          77
# error_counter,err_inj = 00000004 00          0          78
# error_counter,err_inj = 00000004 00          0          79
# error_counter,err_inj = 00000005 01          0          80
# error_counter,err_inj = 00000005 00          0          81
# error_counter,err_inj = 00000005 00          0          82
# error_counter,err_inj = 00000005 00          0          83
# error_counter,err_inj = 00000005 00          0          84
# error_counter,err_inj = 00000005 00          0          85
# error_counter,err_inj = 00000005 00          0          86
# error_counter,err_inj = 00000005 00          0          87
# error_counter,err_inj = 00000005 00          0          88
# error_counter,err_inj = 00000005 00          0          89
# error_counter,err_inj = 00000005 00          0          90
# error_counter,err_inj = 00000005 00          0          91
# error_counter,err_inj = 00000005 00          0          92
# error_counter,err_inj = 00000005 00          0          93
# error_counter,err_inj = 00000005 00          0          94
# error_counter,err_inj = 00000005 00          0          95
# error_counter,err_inj = 00000006 01          0          96
# error_counter,err_inj = 00000006 00          0          97
# error_counter,err_inj = 00000006 00          0          98
# error_counter,err_inj = 00000006 00          0          99
# error_counter,err_inj = 00000006 00          0         100
# error_counter,err_inj = 00000006 00          0         101
# error_counter,err_inj = 00000006 00          0         102
# error_counter,err_inj = 00000006 00          0         103
# error_counter,err_inj = 00000006 00          0         104
# error_counter,err_inj = 00000006 00          0         105
# error_counter,err_inj = 00000006 00          0         106
# error_counter,err_inj = 00000006 00          0         107
# error_counter,err_inj = 00000006 00          0         108
# error_counter,err_inj = 00000006 00          0         109
# error_counter,err_inj = 00000006 00          0         110
# error_counter,err_inj = 00000006 00          0         111
# error_counter,err_inj = 00000007 01          0         112
# error_counter,err_inj = 00000007 00          0         113
# error_counter,err_inj = 00000007 00          0         114
# error_counter,err_inj = 00000007 00          0         115
# error_counter,err_inj = 00000007 00          0         116
# error_counter,err_inj = 00000007 00          0         117
# error_counter,err_inj = 00000007 00          0         118
# error_counter,err_inj = 00000007 00          0         119
# error_counter,err_inj = 00000007 00          0         120
# error_counter,err_inj = 00000007 00          0         121
# error_counter,err_inj = 00000007 00          0         122
# error_counter,err_inj = 00000007 00          0         123
# error_counter,err_inj = 00000007 00          0         124
# error_counter,err_inj = 00000007 00          0         125
# error_counter,err_inj = 00000007 00          0         126
# error_counter,err_inj = 00000007 00          0         127
# error_counter,err_inj = 00000008 01          0         128
# error_counter,err_inj = 00000008 00          0         129
# error_counter,err_inj = 00000008 00          0         130
# error_counter,err_inj = 00000008 00          0         131
# error_counter,err_inj = 00000008 00          0         132
# error_counter,err_inj = 00000008 00          0         133
# error_counter,err_inj = 00000008 00          0         134
# error_counter,err_inj = 00000008 00          0         135
```

```
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
```

```
# error_counter,err_inj = 00000008 00          0         136
# error_counter,err_inj = 00000008 00          0         137
# error_counter,err_inj = 00000008 00          0         138
# error_counter,err_inj = 00000008 00          0         139
# error_counter,err_inj = 00000008 00          0         140
# error_counter,err_inj = 00000008 00          0         141
# error_counter,err_inj = 00000008 00          0         142
# error_counter,err_inj = 00000008 00          0         143
# error_counter,err_inj = 00000009 01          0         144
# error_counter,err_inj = 00000009 00          0         145
# error_counter,err_inj = 00000009 00          0         146
# error_counter,err_inj = 00000009 00          0         147
# error_counter,err_inj = 00000009 00          0         148
# error_counter,err_inj = 00000009 00          0         149
# error_counter,err_inj = 00000009 00          0         150
# error_counter,err_inj = 00000009 00          0         151
# error_counter,err_inj = 00000009 00          0         152
# error_counter,err_inj = 00000009 00          0         153
# error_counter,err_inj = 00000009 00          0         154
# error_counter,err_inj = 00000009 00          0         155
# error_counter,err_inj = 00000009 00          0         156
# error_counter,err_inj = 00000009 00          0         157
# error_counter,err_inj = 00000009 00          0         158
# error_counter,err_inj = 00000009 00          0         159
# error_counter,err_inj = 0000000a 01          0         160
# error_counter,err_inj = 0000000a 00          0         161
# error_counter,err_inj = 0000000a 00          0         162
# error_counter,err_inj = 0000000a 00          0         163
# error_counter,err_inj = 0000000a 00          0         164
# error_counter,err_inj = 0000000a 00          0         165
# error_counter,err_inj = 0000000a 00          0         166
# error_counter,err_inj = 0000000a 00          0         167
# error_counter,err_inj = 0000000a 00          0         168
# error_counter,err_inj = 0000000a 00          0         169
# error_counter,err_inj = 0000000a 00          0         170
# error_counter,err_inj = 0000000a 00          0         171
# error_counter,err_inj = 0000000a 00          0         172
# error_counter,err_inj = 0000000a 00          0         173
# error_counter,err_inj = 0000000a 00          0         174
# error_counter,err_inj = 0000000a 00          0         175
# error_counter,err_inj = 0000000b 01          0         176
# error_counter,err_inj = 0000000b 00          0         177
# error_counter,err_inj = 0000000b 00          0         178
# error_counter,err_inj = 0000000b 00          0         179
# error_counter,err_inj = 0000000b 00          0         180
# error_counter,err_inj = 0000000b 00          0         181
# error_counter,err_inj = 0000000b 00          0         182
# error_counter,err_inj = 0000000b 00          0         183
# error_counter,err_inj = 0000000b 00          0         184
# error_counter,err_inj = 0000000b 00          0         185
# error_counter,err_inj = 0000000b 00          0         186
# error_counter,err_inj = 0000000b 00          0         187
# error_counter,err_inj = 0000000b 00          0         188
# error_counter,err_inj = 0000000b 00          0         189
# error_counter,err_inj = 0000000b 00          0         190
# error_counter,err_inj = 0000000b 00          0         191
# error_counter,err_inj = 0000000c 01          0         192
# error_counter,err_inj = 0000000c 00          0         193
# error_counter,err_inj = 0000000c 00          0         194
# error_counter,err_inj = 0000000c 00          0         195
# error_counter,err_inj = 0000000c 00          0         196
# error_counter,err_inj = 0000000c 00          0         197
# error_counter,err_inj = 0000000c 00          0         198
# error_counter,err_inj = 0000000c 00          0         199
# error_counter,err_inj = 0000000c 00          0         200
# error_counter,err_inj = 0000000c 00          0         201
# error_counter,err_inj = 0000000c 00          0         202
# error_counter,err_inj = 0000000c 00          0         203
# error_counter,err_inj = 0000000c 00          0         204
# error_counter,err_inj = 0000000c 00          0         205
# error_counter,err_inj = 0000000c 00          0         206
# error_counter,err_inj = 0000000c 00          0         207
# error_counter,err_inj = 0000000d 01          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000d 00          0         213
# error_counter,err_inj = 0000000d 00          0         214
# error_counter,err_inj = 0000000d 00          0         215
# error_counter,err_inj = 0000000d 00          0         216
# error_counter,err_inj = 0000000d 00          0         217
# error_counter,err_inj = 0000000d 00          0         218
# error_counter,err_inj = 0000000d 00          0         219
# error_counter,err_inj = 0000000d 00          0         220
# error_counter,err_inj = 0000000d 00          0         221
# error_counter,err_inj = 0000000d 00          0         222
# error_counter,err_inj = 0000000d 00          0         223
# error_counter,err_inj = 0000000e 01          0         224
# error_counter,err_inj = 0000000e 00          0         225
# error_counter,err_inj = 0000000e 00          0         226
# error_counter,err_inj = 0000000e 00          0         227
# error_counter,err_inj = 0000000e 00          0         228
```

```
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000e 00           0         229
# error_counter,err_inj = 0000000e 00           0         230
# error_counter,err_inj = 0000000e 00           0         231
# error_counter,err_inj = 0000000e 00           0         232
# error_counter,err_inj = 0000000e 00           0         233
# error_counter,err_inj = 0000000e 00           0         234
# error_counter,err_inj = 0000000e 00           0         235
# error_counter,err_inj = 0000000e 00           0         236
# error_counter,err_inj = 0000000e 00           0         237
# error_counter,err_inj = 0000000e 00           0         238
# error_counter,err_inj = 0000000e 00           0         239
# error_counter,err_inj = 0000000f 01           0         240
# error_counter,err_inj = 0000000f 00           0         241
# error_counter,err_inj = 0000000f 00           0         242
# error_counter,err_inj = 0000000f 00           0         243
# error_counter,err_inj = 0000000f 00           0         244
# error_counter,err_inj = 0000000f 00           0         245
# error_counter,err_inj = 0000000f 00           0         246
# error_counter,err_inj = 0000000f 00           0         247
# error_counter,err_inj = 0000000f 00           0         248
# error_counter,err_inj = 0000000f 00           0         249
# error_counter,err_inj = 0000000f 00           0         250
# error_counter,err_inj = 0000000f 00           0         251
# error_counter,err_inj = 0000000f 00           0         252
# error_counter,err_inj = 0000000f 00           0         253
# error_counter,err_inj = 0000000f 00           0         254
# error_counter,err_inj = 0000000f 00           0         255
# word_count =       10440
```

2a1) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000001 01           0           8
# error_counter,err_inj = 00000001 00           0           9
# error_counter,err_inj = 00000001 00           0          10
# error_counter,err_inj = 00000001 00           0          11
# error_counter,err_inj = 00000001 00           0          12
# error_counter,err_inj = 00000001 00           0          13
# error_counter,err_inj = 00000001 00           0          14
# error_counter,err_inj = 00000001 00           0          15
# error_counter,err_inj = 00000002 01           0          16
# error_counter,err_inj = 00000002 00           0          17
# error_counter,err_inj = 00000002 00           0          18
# error_counter,err_inj = 00000002 00           0          19
# error_counter,err_inj = 00000002 00           0          20
# error_counter,err_inj = 00000002 00           0          21
# error_counter,err_inj = 00000002 00           0          22
# error_counter,err_inj = 00000002 00           0          23
# error_counter,err_inj = 00000003 01           0          24
# error_counter,err_inj = 00000003 00           0          25
# error_counter,err_inj = 00000003 00           0          26
# error_counter,err_inj = 00000003 00           0          27
# error_counter,err_inj = 00000003 00           0          28
# error_counter,err_inj = 00000003 00           0          29
# error_counter,err_inj = 00000003 00           0          30
# error_counter,err_inj = 00000003 00           0          31
# error_counter,err_inj = 00000004 01           0          32
# error_counter,err_inj = 00000004 00           0          33
# error_counter,err_inj = 00000004 00           0          34
# error_counter,err_inj = 00000004 00           0          35
# error_counter,err_inj = 00000004 00           0          36
# error_counter,err_inj = 00000004 00           0          37
# error_counter,err_inj = 00000004 00           0          38
# error_counter,err_inj = 00000004 00           0          39
# error_counter,err_inj = 00000005 01           0          40
# error_counter,err_inj = 00000005 00           0          41
# error_counter,err_inj = 00000005 00           0          42
# error_counter,err_inj = 00000005 00           0          43
# error_counter,err_inj = 00000005 00           0          44
# error_counter,err_inj = 00000005 00           0          45
# error_counter,err_inj = 00000005 00           0          46
# error_counter,err_inj = 00000005 00           0          47
# error_counter,err_inj = 00000006 01           0          48
# error_counter,err_inj = 00000006 00           0          49
# error_counter,err_inj = 00000006 00           0          50
# error_counter,err_inj = 00000006 00           0          51
```

```
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
```

```
# error_counter,err_inj = 00000006 00          0          52
# error_counter,err_inj = 00000006 00          0          53
# error_counter,err_inj = 00000006 00          0          54
# error_counter,err_inj = 00000006 00          0          55
# error_counter,err_inj = 00000007 01          0          56
# error_counter,err_inj = 00000007 00          0          57
# error_counter,err_inj = 00000007 00          0          58
# error_counter,err_inj = 00000007 00          0          59
# error_counter,err_inj = 00000007 00          0          60
# error_counter,err_inj = 00000007 00          0          61
# error_counter,err_inj = 00000007 00          0          62
# error_counter,err_inj = 00000007 00          0          63
# error_counter,err_inj = 00000008 01          0          64
# error_counter,err_inj = 00000008 00          0          65
# error_counter,err_inj = 00000008 00          0          66
# error_counter,err_inj = 00000008 00          0          67
# error_counter,err_inj = 00000008 00          0          68
# error_counter,err_inj = 00000008 00          0          69
# error_counter,err_inj = 00000008 00          0          70
# error_counter,err_inj = 00000008 00          0          71
# error_counter,err_inj = 00000009 01          0          72
# error_counter,err_inj = 00000009 00          0          73
# error_counter,err_inj = 00000009 00          0          74
# error_counter,err_inj = 00000009 00          0          75
# error_counter,err_inj = 00000009 00          0          76
# error_counter,err_inj = 00000009 00          0          77
# error_counter,err_inj = 00000009 00          0          78
# error_counter,err_inj = 00000009 00          0          79
# error_counter,err_inj = 0000000a 01          0          80
# error_counter,err_inj = 0000000a 00          0          81
# error_counter,err_inj = 0000000a 00          0          82
# error_counter,err_inj = 0000000a 00          0          83
# error_counter,err_inj = 0000000a 00          0          84
# error_counter,err_inj = 0000000a 00          0          85
# error_counter,err_inj = 0000000a 00          0          86
# error_counter,err_inj = 0000000a 00          0          87
# error_counter,err_inj = 0000000b 01          0          88
# error_counter,err_inj = 0000000b 00          0          89
# error_counter,err_inj = 0000000b 00          0          90
# error_counter,err_inj = 0000000b 00          0          91
# error_counter,err_inj = 0000000b 00          0          92
# error_counter,err_inj = 0000000b 00          0          93
# error_counter,err_inj = 0000000b 00          0          94
# error_counter,err_inj = 0000000b 00          0          95
# error_counter,err_inj = 0000000c 01          0          96
# error_counter,err_inj = 0000000c 00          0          97
# error_counter,err_inj = 0000000c 00          0          98
# error_counter,err_inj = 0000000c 00          0          99
# error_counter,err_inj = 0000000c 00          0         100
# error_counter,err_inj = 0000000c 00          0         101
# error_counter,err_inj = 0000000c 00          0         102
# error_counter,err_inj = 0000000c 00          0         103
# error_counter,err_inj = 0000000d 01          0         104
# error_counter,err_inj = 0000000d 00          0         105
# error_counter,err_inj = 0000000d 00          0         106
# error_counter,err_inj = 0000000d 00          0         107
# error_counter,err_inj = 0000000d 00          0         108
# error_counter,err_inj = 0000000d 00          0         109
# error_counter,err_inj = 0000000d 00          0         110
# error_counter,err_inj = 0000000d 00          0         111
# error_counter,err_inj = 0000000e 01          0         112
# error_counter,err_inj = 0000000e 00          0         113
# error_counter,err_inj = 0000000e 00          0         114
# error_counter,err_inj = 0000000e 00          0         115
# error_counter,err_inj = 0000000e 00          0         116
# error_counter,err_inj = 0000000e 00          0         117
# error_counter,err_inj = 0000000e 00          0         118
# error_counter,err_inj = 0000000e 00          0         119
# error_counter,err_inj = 0000000f 01          0         120
# error_counter,err_inj = 0000000f 00          0         121
# error_counter,err_inj = 0000000f 00          0         122
# error_counter,err_inj = 0000000f 00          0         123
# error_counter,err_inj = 0000000f 00          0         124
# error_counter,err_inj = 0000000f 00          0         125
# error_counter,err_inj = 0000000f 00          0         126
# error_counter,err_inj = 0000000f 00          0         127
# error_counter,err_inj = 00000010 01          0         128
# error_counter,err_inj = 00000010 00          0         129
# error_counter,err_inj = 00000010 00          0         130
# error_counter,err_inj = 00000010 00          0         131
# error_counter,err_inj = 00000010 00          0         132
# error_counter,err_inj = 00000010 00          0         133
# error_counter,err_inj = 00000010 00          0         134
# error_counter,err_inj = 00000010 00          0         135
# error_counter,err_inj = 00000011 01          0         136
# error_counter,err_inj = 00000011 00          0         137
# error_counter,err_inj = 00000011 00          0         138
# error_counter,err_inj = 00000011 00          0         139
# error_counter,err_inj = 00000011 00          0         140
# error_counter,err_inj = 00000011 00          0         141
# error_counter,err_inj = 00000011 00          0         142
# error_counter,err_inj = 00000011 00          0         143
# error_counter,err_inj = 00000012 01          0         144
```

```
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
```

```
# error_counter,err_inj = 00000012 00          0         145
# error_counter,err_inj = 00000012 00          0         146
# error_counter,err_inj = 00000012 00          0         147
# error_counter,err_inj = 00000012 00          0         148
# error_counter,err_inj = 00000012 00          0         149
# error_counter,err_inj = 00000012 00          0         150
# error_counter,err_inj = 00000012 00          0         151
# error_counter,err_inj = 00000013 01          0         152
# error_counter,err_inj = 00000013 00          0         153
# error_counter,err_inj = 00000013 00          0         154
# error_counter,err_inj = 00000013 00          0         155
# error_counter,err_inj = 00000013 00          0         156
# error_counter,err_inj = 00000013 00          0         157
# error_counter,err_inj = 00000013 00          0         158
# error_counter,err_inj = 00000013 00          0         159
# error_counter,err_inj = 00000014 01          0         160
# error_counter,err_inj = 00000014 00          0         161
# error_counter,err_inj = 00000014 00          0         162
# error_counter,err_inj = 00000014 00          0         163
# error_counter,err_inj = 00000014 00          0         164
# error_counter,err_inj = 00000014 00          0         165
# error_counter,err_inj = 00000014 00          0         166
# error_counter,err_inj = 00000014 00          0         167
# error_counter,err_inj = 00000015 01          0         168
# error_counter,err_inj = 00000015 00          0         169
# error_counter,err_inj = 00000015 00          0         170
# error_counter,err_inj = 00000015 00          0         171
# error_counter,err_inj = 00000015 00          0         172
# error_counter,err_inj = 00000015 00          0         173
# error_counter,err_inj = 00000015 00          0         174
# error_counter,err_inj = 00000015 00          0         175
# error_counter,err_inj = 00000016 01          0         176
# error_counter,err_inj = 00000016 00          0         177
# error_counter,err_inj = 00000016 00          0         178
# error_counter,err_inj = 00000016 00          0         179
# error_counter,err_inj = 00000016 00          0         180
# error_counter,err_inj = 00000016 00          0         181
# error_counter,err_inj = 00000016 00          0         182
# error_counter,err_inj = 00000016 00          0         183
# error_counter,err_inj = 00000017 01          0         184
# error_counter,err_inj = 00000017 00          0         185
# error_counter,err_inj = 00000017 00          0         186
# error_counter,err_inj = 00000017 00          0         187
# error_counter,err_inj = 00000017 00          0         188
# error_counter,err_inj = 00000017 00          0         189
# error_counter,err_inj = 00000017 00          0         190
# error_counter,err_inj = 00000017 00          0         191
# error_counter,err_inj = 00000018 01          0         192
# error_counter,err_inj = 00000018 00          0         193
# error_counter,err_inj = 00000018 00          0         194
# error_counter,err_inj = 00000018 00          0         195
# error_counter,err_inj = 00000018 00          0         196
# error_counter,err_inj = 00000018 00          0         197
# error_counter,err_inj = 00000018 00          0         198
# error_counter,err_inj = 00000018 00          0         199
# error_counter,err_inj = 00000019 01          0         200
# error_counter,err_inj = 00000019 00          0         201
# error_counter,err_inj = 00000019 00          0         202
# error_counter,err_inj = 00000019 00          0         203
# error_counter,err_inj = 00000019 00          0         204
# error_counter,err_inj = 00000019 00          0         205
# error_counter,err_inj = 00000019 00          0         206
# error_counter,err_inj = 00000019 00          0         207
# error_counter,err_inj = 0000001a 01          0         208
# error_counter,err_inj = 0000001a 00          0         209
# error_counter,err_inj = 0000001a 00          0         210
# error_counter,err_inj = 0000001a 00          0         211
# error_counter,err_inj = 0000001a 00          0         212
# error_counter,err_inj = 0000001a 00          0         213
# error_counter,err_inj = 0000001a 00          0         214
# error_counter,err_inj = 0000001a 00          0         215
# error_counter,err_inj = 0000001b 01          0         216
# error_counter,err_inj = 0000001b 00          0         217
# error_counter,err_inj = 0000001b 00          0         218
# error_counter,err_inj = 0000001b 00          0         219
# error_counter,err_inj = 0000001b 00          0         220
# error_counter,err_inj = 0000001b 00          0         221
# error_counter,err_inj = 0000001b 00          0         222
# error_counter,err_inj = 0000001b 00          0         223
# error_counter,err_inj = 0000001c 01          0         224
# error_counter,err_inj = 0000001c 00          0         225
# error_counter,err_inj = 0000001c 00          0         226
# error_counter,err_inj = 0000001c 00          0         227
# error_counter,err_inj = 0000001c 00          0         228
# error_counter,err_inj = 0000001c 00          0         229
# error_counter,err_inj = 0000001c 00          0         230
# error_counter,err_inj = 0000001c 00          0         231
# error_counter,err_inj = 0000001d 01          0         232
# error_counter,err_inj = 0000001d 00          0         233
# error_counter,err_inj = 0000001d 00          0         234
# error_counter,err_inj = 0000001d 00          0         235
# error_counter,err_inj = 0000001d 00          0         236
# error_counter,err_inj = 0000001d 00          0         237
```

```
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001d 00          0         238
# error_counter,err_inj = 0000001d 00          0         239
# error_counter,err_inj = 0000001e 01          0         240
# error_counter,err_inj = 0000001e 00          0         241
# error_counter,err_inj = 0000001e 00          0         242
# error_counter,err_inj = 0000001e 00          0         243
# error_counter,err_inj = 0000001e 00          0         244
# error_counter,err_inj = 0000001e 00          0         245
# error_counter,err_inj = 0000001e 00          0         246
# error_counter,err_inj = 0000001e 00          0         247
# error_counter,err_inj = 0000001f 01          0         248
# error_counter,err_inj = 0000001f 00          0         249
# error_counter,err_inj = 0000001f 00          0         250
# error_counter,err_inj = 0000001f 00          0         251
# error_counter,err_inj = 0000001f 00          0         252
# error_counter,err_inj = 0000001f 00          0         253
# error_counter,err_inj = 0000001f 00          0         254
# error_counter,err_inj = 0000001f 00          0         255
# word_count =       10440
```

2a2) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000001 10          0           8
# error_counter,err_inj = 00000001 00          0           9
# error_counter,err_inj = 00000001 00          0          10
# error_counter,err_inj = 00000001 00          0          11
# error_counter,err_inj = 00000001 00          0          12
# error_counter,err_inj = 00000001 00          0          13
# error_counter,err_inj = 00000001 00          0          14
# error_counter,err_inj = 00000001 00          0          15
# error_counter,err_inj = 00000002 10          0          16
# error_counter,err_inj = 00000002 00          0          17
# error_counter,err_inj = 00000002 00          0          18
# error_counter,err_inj = 00000002 00          0          19
# error_counter,err_inj = 00000002 00          0          20
# error_counter,err_inj = 00000002 00          0          21
# error_counter,err_inj = 00000002 00          0          22
# error_counter,err_inj = 00000002 00          0          23
# error_counter,err_inj = 00000003 10          0          24
# error_counter,err_inj = 00000003 00          0          25
# error_counter,err_inj = 00000003 00          0          26
# error_counter,err_inj = 00000003 00          0          27
# error_counter,err_inj = 00000003 00          0          28
# error_counter,err_inj = 00000003 00          0          29
# error_counter,err_inj = 00000003 00          0          30
# error_counter,err_inj = 00000003 00          0          31
# error_counter,err_inj = 00000004 10          0          32
# error_counter,err_inj = 00000004 00          0          33
# error_counter,err_inj = 00000004 00          0          34
# error_counter,err_inj = 00000004 00          0          35
# error_counter,err_inj = 00000004 00          0          36
# error_counter,err_inj = 00000004 00          0          37
# error_counter,err_inj = 00000004 00          0          38
# error_counter,err_inj = 00000004 00          0          39
# error_counter,err_inj = 00000005 10          0          40
# error_counter,err_inj = 00000005 00          0          41
# error_counter,err_inj = 00000005 00          0          42
# error_counter,err_inj = 00000005 00          0          43
# error_counter,err_inj = 00000005 00          0          44
# error_counter,err_inj = 00000005 00          0          45
# error_counter,err_inj = 00000005 00          0          46
# error_counter,err_inj = 00000005 00          0          47
# error_counter,err_inj = 00000006 10          0          48
# error_counter,err_inj = 00000006 00          0          49
# error_counter,err_inj = 00000006 00          0          50
# error_counter,err_inj = 00000006 00          0          51
# error_counter,err_inj = 00000006 00          0          52
# error_counter,err_inj = 00000006 00          0          53
# error_counter,err_inj = 00000006 00          0          54
# error_counter,err_inj = 00000006 00          0          55
# error_counter,err_inj = 00000007 10          0          56
# error_counter,err_inj = 00000007 00          0          57
# error_counter,err_inj = 00000007 00          0          58
# error_counter,err_inj = 00000007 00          0          59
# error_counter,err_inj = 00000007 00          0          60
# error_counter,err_inj = 00000007 00          0          61
# error_counter,err_inj = 00000007 00          0          62
# error_counter,err_inj = 00000007 00          0          63
# error_counter,err_inj = 00000008 10          0          64
# error_counter,err_inj = 00000008 00          0          65
# error_counter,err_inj = 00000008 00          0          66
# error_counter,err_inj = 00000008 00          0          67
```

```
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
```

```
# error_counter,err_inj = 00000008 00          0          68
# error_counter,err_inj = 00000008 00          0          69
# error_counter,err_inj = 00000008 00          0          70
# error_counter,err_inj = 00000008 00          0          71
# error_counter,err_inj = 00000009 10          0          72
# error_counter,err_inj = 00000009 00          0          73
# error_counter,err_inj = 00000009 00          0          74
# error_counter,err_inj = 00000009 00          0          75
# error_counter,err_inj = 00000009 00          0          76
# error_counter,err_inj = 00000009 00          0          77
# error_counter,err_inj = 00000009 00          0          78
# error_counter,err_inj = 00000009 00          0          79
# error_counter,err_inj = 0000000a 10          0          80
# error_counter,err_inj = 0000000a 00          0          81
# error_counter,err_inj = 0000000a 00          0          82
# error_counter,err_inj = 0000000a 00          0          83
# error_counter,err_inj = 0000000a 00          0          84
# error_counter,err_inj = 0000000a 00          0          85
# error_counter,err_inj = 0000000a 00          0          86
# error_counter,err_inj = 0000000a 00          0          87
# error_counter,err_inj = 0000000b 10          0          88
# error_counter,err_inj = 0000000b 00          0          89
# error_counter,err_inj = 0000000b 00          0          90
# error_counter,err_inj = 0000000b 00          0          91
# error_counter,err_inj = 0000000b 00          0          92
# error_counter,err_inj = 0000000b 00          0          93
# error_counter,err_inj = 0000000b 00          0          94
# error_counter,err_inj = 0000000b 00          0          95
# error_counter,err_inj = 0000000c 10          0          96
# error_counter,err_inj = 0000000c 00          0          97
# error_counter,err_inj = 0000000c 00          0          98
# error_counter,err_inj = 0000000c 00          0          99
# error_counter,err_inj = 0000000c 00          0         100
# error_counter,err_inj = 0000000c 00          0         101
# error_counter,err_inj = 0000000c 00          0         102
# error_counter,err_inj = 0000000c 00          0         103
# error_counter,err_inj = 0000000d 10          0         104
# error_counter,err_inj = 0000000d 00          0         105
# error_counter,err_inj = 0000000d 00          0         106
# error_counter,err_inj = 0000000d 00          0         107
# error_counter,err_inj = 0000000d 00          0         108
# error_counter,err_inj = 0000000d 00          0         109
# error_counter,err_inj = 0000000d 00          0         110
# error_counter,err_inj = 0000000d 00          0         111
# error_counter,err_inj = 0000000e 10          0         112
# error_counter,err_inj = 0000000e 00          0         113
# error_counter,err_inj = 0000000e 00          0         114
# error_counter,err_inj = 0000000e 00          0         115
# error_counter,err_inj = 0000000e 00          0         116
# error_counter,err_inj = 0000000e 00          0         117
# error_counter,err_inj = 0000000e 00          0         118
# error_counter,err_inj = 0000000e 00          0         119
# error_counter,err_inj = 0000000f 10          0         120
# error_counter,err_inj = 0000000f 00          0         121
# error_counter,err_inj = 0000000f 00          0         122
# error_counter,err_inj = 0000000f 00          0         123
# error_counter,err_inj = 0000000f 00          0         124
# error_counter,err_inj = 0000000f 00          0         125
# error_counter,err_inj = 0000000f 00          0         126
# error_counter,err_inj = 0000000f 00          0         127
# error_counter,err_inj = 00000010 10          0         128
# error_counter,err_inj = 00000010 00          0         129
# error_counter,err_inj = 00000010 00          0         130
# error_counter,err_inj = 00000010 00          0         131
# error_counter,err_inj = 00000010 00          0         132
# error_counter,err_inj = 00000010 00          0         133
# error_counter,err_inj = 00000010 00          0         134
# error_counter,err_inj = 00000010 00          0         135
# error_counter,err_inj = 00000011 10          0         136
# error_counter,err_inj = 00000011 00          0         137
# error_counter,err_inj = 00000011 00          0         138
# error_counter,err_inj = 00000011 00          0         139
# error_counter,err_inj = 00000011 00          0         140
# error_counter,err_inj = 00000011 00          0         141
# error_counter,err_inj = 00000011 00          0         142
# error_counter,err_inj = 00000011 00          0         143
# error_counter,err_inj = 00000012 10          0         144
# error_counter,err_inj = 00000012 00          0         145
# error_counter,err_inj = 00000012 00          0         146
# error_counter,err_inj = 00000012 00          0         147
# error_counter,err_inj = 00000012 00          0         148
# error_counter,err_inj = 00000012 00          0         149
# error_counter,err_inj = 00000012 00          0         150
# error_counter,err_inj = 00000012 00          0         151
# error_counter,err_inj = 00000013 10          0         152
# error_counter,err_inj = 00000013 00          0         153
# error_counter,err_inj = 00000013 00          0         154
# error_counter,err_inj = 00000013 00          0         155
# error_counter,err_inj = 00000013 00          0         156
# error_counter,err_inj = 00000013 00          0         157
# error_counter,err_inj = 00000013 00          0         158
# error_counter,err_inj = 00000013 00          0         159
# error_counter,err_inj = 00000014 10          0         160
```

```
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
```

```
# error_counter,err_inj = 00000014 00          0         161
# error_counter,err_inj = 00000014 00          0         162
# error_counter,err_inj = 00000014 00          0         163
# error_counter,err_inj = 00000014 00          0         164
# error_counter,err_inj = 00000014 00          0         165
# error_counter,err_inj = 00000014 00          0         166
# error_counter,err_inj = 00000014 00          0         167
# error_counter,err_inj = 00000015 10          0         168
# error_counter,err_inj = 00000015 00          0         169
# error_counter,err_inj = 00000015 00          0         170
# error_counter,err_inj = 00000015 00          0         171
# error_counter,err_inj = 00000015 00          0         172
# error_counter,err_inj = 00000015 00          0         173
# error_counter,err_inj = 00000015 00          0         174
# error_counter,err_inj = 00000015 00          0         175
# error_counter,err_inj = 00000016 10          0         176
# error_counter,err_inj = 00000016 00          0         177
# error_counter,err_inj = 00000016 00          0         178
# error_counter,err_inj = 00000016 00          0         179
# error_counter,err_inj = 00000016 00          0         180
# error_counter,err_inj = 00000016 00          0         181
# error_counter,err_inj = 00000016 00          0         182
# error_counter,err_inj = 00000016 00          0         183
# error_counter,err_inj = 00000017 10          0         184
# error_counter,err_inj = 00000017 00          0         185
# error_counter,err_inj = 00000017 00          0         186
# error_counter,err_inj = 00000017 00          0         187
# error_counter,err_inj = 00000017 00          0         188
# error_counter,err_inj = 00000017 00          0         189
# error_counter,err_inj = 00000017 00          0         190
# error_counter,err_inj = 00000017 00          0         191
# error_counter,err_inj = 00000018 10          0         192
# error_counter,err_inj = 00000018 00          0         193
# error_counter,err_inj = 00000018 00          0         194
# error_counter,err_inj = 00000018 00          0         195
# error_counter,err_inj = 00000018 00          0         196
# error_counter,err_inj = 00000018 00          0         197
# error_counter,err_inj = 00000018 00          0         198
# error_counter,err_inj = 00000018 00          0         199
# error_counter,err_inj = 00000019 10          0         200
# error_counter,err_inj = 00000019 00          0         201
# error_counter,err_inj = 00000019 00          0         202
# error_counter,err_inj = 00000019 00          0         203
# error_counter,err_inj = 00000019 00          0         204
# error_counter,err_inj = 00000019 00          0         205
# error_counter,err_inj = 00000019 00          0         206
# error_counter,err_inj = 00000019 00          0         207
# error_counter,err_inj = 0000001a 10          0         208
# error_counter,err_inj = 0000001a 00          0         209
# error_counter,err_inj = 0000001a 00          0         210
# error_counter,err_inj = 0000001a 00          0         211
# error_counter,err_inj = 0000001a 00          0         212
# error_counter,err_inj = 0000001a 00          0         213
# error_counter,err_inj = 0000001a 00          0         214
# error_counter,err_inj = 0000001a 00          0         215
# error_counter,err_inj = 0000001b 10          0         216
# error_counter,err_inj = 0000001b 00          0         217
# error_counter,err_inj = 0000001b 00          0         218
# error_counter,err_inj = 0000001b 00          0         219
# error_counter,err_inj = 0000001b 00          0         220
# error_counter,err_inj = 0000001b 00          0         221
# error_counter,err_inj = 0000001b 00          0         222
# error_counter,err_inj = 0000001b 00          0         223
# error_counter,err_inj = 0000001c 10          0         224
# error_counter,err_inj = 0000001c 00          0         225
# error_counter,err_inj = 0000001c 00          0         226
# error_counter,err_inj = 0000001c 00          0         227
# error_counter,err_inj = 0000001c 00          0         228
# error_counter,err_inj = 0000001c 00          0         229
# error_counter,err_inj = 0000001c 00          0         230
# error_counter,err_inj = 0000001c 00          0         231
# error_counter,err_inj = 0000001d 10          0         232
# error_counter,err_inj = 0000001d 00          0         233
# error_counter,err_inj = 0000001d 00          0         234
# error_counter,err_inj = 0000001d 00          0         235
# error_counter,err_inj = 0000001d 00          0         236
# error_counter,err_inj = 0000001d 00          0         237
# error_counter,err_inj = 0000001d 00          0         238
# error_counter,err_inj = 0000001d 00          0         239
# error_counter,err_inj = 0000001e 10          0         240
# error_counter,err_inj = 0000001e 00          0         241
# error_counter,err_inj = 0000001e 00          0         242
# error_counter,err_inj = 0000001e 00          0         243
# error_counter,err_inj = 0000001e 00          0         244
# error_counter,err_inj = 0000001e 00          0         245
# error_counter,err_inj = 0000001e 00          0         246
# error_counter,err_inj = 0000001e 00          0         247
# error_counter,err_inj = 0000001f 10          0         248
# error_counter,err_inj = 0000001f 00          0         249
# error_counter,err_inj = 0000001f 00          0         250
# error_counter,err_inj = 0000001f 00          0         251
# error_counter,err_inj = 0000001f 00          0         252
# error_counter,err_inj = 0000001f 00          0         253
```

```
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001f 00           0         254
# error_counter,err_inj = 0000001f 00           0         255
```

2a3) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000000 00           0           8
# error_counter,err_inj = 00000000 00           0           9
# error_counter,err_inj = 00000000 00           0          10
# error_counter,err_inj = 00000000 00           0          11
# error_counter,err_inj = 00000000 00           0          12
# error_counter,err_inj = 00000000 00           0          13
# error_counter,err_inj = 00000000 00           0          14
# error_counter,err_inj = 00000000 00           0          15
# error_counter,err_inj = 00000001 11           0          16
# error_counter,err_inj = 00000001 00           0          17
# error_counter,err_inj = 00000001 00           0          18
# error_counter,err_inj = 00000001 00           0          19
# error_counter,err_inj = 00000001 00           0          20
# error_counter,err_inj = 00000001 00           0          21
# error_counter,err_inj = 00000001 00           0          22
# error_counter,err_inj = 00000001 00           0          23
# error_counter,err_inj = 00000001 00           0          24
# error_counter,err_inj = 00000001 00           0          25
# error_counter,err_inj = 00000001 00           0          26
# error_counter,err_inj = 00000001 00           0          27
# error_counter,err_inj = 00000001 00           0          28
# error_counter,err_inj = 00000001 00           0          29
# error_counter,err_inj = 00000001 00           0          30
# error_counter,err_inj = 00000001 00           0          31
# error_counter,err_inj = 00000002 11           0          32
# error_counter,err_inj = 00000002 00           0          33
# error_counter,err_inj = 00000002 00           0          34
# error_counter,err_inj = 00000002 00           0          35
# error_counter,err_inj = 00000002 00           0          36
# error_counter,err_inj = 00000002 00           0          37
# error_counter,err_inj = 00000002 00           0          38
# error_counter,err_inj = 00000002 00           0          39
# error_counter,err_inj = 00000002 00           0          40
# error_counter,err_inj = 00000002 00           0          41
# error_counter,err_inj = 00000002 00           0          42
# error_counter,err_inj = 00000002 00           0          43
# error_counter,err_inj = 00000002 00           0          44
# error_counter,err_inj = 00000002 00           0          45
# error_counter,err_inj = 00000002 00           0          46
# error_counter,err_inj = 00000002 00           0          47
# error_counter,err_inj = 00000003 11           0          48
# error_counter,err_inj = 00000003 00           0          49
# error_counter,err_inj = 00000003 00           0          50
# error_counter,err_inj = 00000003 00           0          51
# error_counter,err_inj = 00000003 00           0          52
# error_counter,err_inj = 00000003 00           0          53
# error_counter,err_inj = 00000003 00           0          54
# error_counter,err_inj = 00000003 00           0          55
# error_counter,err_inj = 00000003 00           0          56
# error_counter,err_inj = 00000003 00           0          57
# error_counter,err_inj = 00000003 00           0          58
# error_counter,err_inj = 00000003 00           0          59
# error_counter,err_inj = 00000003 00           0          60
# error_counter,err_inj = 00000003 00           0          61
# error_counter,err_inj = 00000003 00           0          62
# error_counter,err_inj = 00000003 00           0          63
# error_counter,err_inj = 00000004 11           0          64
# error_counter,err_inj = 00000004 00           0          65
# error_counter,err_inj = 00000004 00           0          66
# error_counter,err_inj = 00000004 00           0          67
# error_counter,err_inj = 00000004 00           0          68
# error_counter,err_inj = 00000004 00           0          69
# error_counter,err_inj = 00000004 00           0          70
# error_counter,err_inj = 00000004 00           0          71
# error_counter,err_inj = 00000004 00           0          72
# error_counter,err_inj = 00000004 00           0          73
# error_counter,err_inj = 00000004 00           0          74
# error_counter,err_inj = 00000004 00           0          75
# error_counter,err_inj = 00000004 00           0          76
# error_counter,err_inj = 00000004 00           0          77
# error_counter,err_inj = 00000004 00           0          78
# error_counter,err_inj = 00000004 00           0          79
# error_counter,err_inj = 00000005 11           0          80
# error_counter,err_inj = 00000005 00           0          81
# error_counter,err_inj = 00000005 00           0          82
# error_counter,err_inj = 00000005 00           0          83
# error_counter,err_inj = 00000005 00           0          84
# error_counter,err_inj = 00000005 00           0          85
```

```
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
```

```
# error_counter,err_inj = 00000005 00          0          86
# error_counter,err_inj = 00000005 00          0          87
# error_counter,err_inj = 00000005 00          0          88
# error_counter,err_inj = 00000005 00          0          89
# error_counter,err_inj = 00000005 00          0          90
# error_counter,err_inj = 00000005 00          0          91
# error_counter,err_inj = 00000005 00          0          92
# error_counter,err_inj = 00000005 00          0          93
# error_counter,err_inj = 00000005 00          0          94
# error_counter,err_inj = 00000005 00          0          95
# error_counter,err_inj = 00000006 11          0          96
# error_counter,err_inj = 00000006 00          0          97
# error_counter,err_inj = 00000006 00          0          98
# error_counter,err_inj = 00000006 00          0          99
# error_counter,err_inj = 00000006 00          0         100
# error_counter,err_inj = 00000006 00          0         101
# error_counter,err_inj = 00000006 00          0         102
# error_counter,err_inj = 00000006 00          0         103
# error_counter,err_inj = 00000006 00          0         104
# error_counter,err_inj = 00000006 00          0         105
# error_counter,err_inj = 00000006 00          0         106
# error_counter,err_inj = 00000006 00          0         107
# error_counter,err_inj = 00000006 00          0         108
# error_counter,err_inj = 00000006 00          0         109
# error_counter,err_inj = 00000006 00          0         110
# error_counter,err_inj = 00000006 00          0         111
# error_counter,err_inj = 00000007 11          0         112
# error_counter,err_inj = 00000007 00          0         113
# error_counter,err_inj = 00000007 00          0         114
# error_counter,err_inj = 00000007 00          0         115
# error_counter,err_inj = 00000007 00          0         116
# error_counter,err_inj = 00000007 00          0         117
# error_counter,err_inj = 00000007 00          0         118
# error_counter,err_inj = 00000007 00          0         119
# error_counter,err_inj = 00000007 00          0         120
# error_counter,err_inj = 00000007 00          0         121
# error_counter,err_inj = 00000007 00          0         122
# error_counter,err_inj = 00000007 00          0         123
# error_counter,err_inj = 00000007 00          0         124
# error_counter,err_inj = 00000007 00          0         125
# error_counter,err_inj = 00000007 00          0         126
# error_counter,err_inj = 00000007 00          0         127
# error_counter,err_inj = 00000008 11          0         128
# error_counter,err_inj = 00000008 00          0         129
# error_counter,err_inj = 00000008 00          0         130
# error_counter,err_inj = 00000008 00          0         131
# error_counter,err_inj = 00000008 00          0         132
# error_counter,err_inj = 00000008 00          0         133
# error_counter,err_inj = 00000008 00          0         134
# error_counter,err_inj = 00000008 00          0         135
# error_counter,err_inj = 00000008 00          0         136
# error_counter,err_inj = 00000008 00          0         137
# error_counter,err_inj = 00000008 00          0         138
# error_counter,err_inj = 00000008 00          0         139
# error_counter,err_inj = 00000008 00          0         140
# error_counter,err_inj = 00000008 00          0         141
# error_counter,err_inj = 00000008 00          0         142
# error_counter,err_inj = 00000008 00          0         143
# error_counter,err_inj = 00000009 11          0         144
# error_counter,err_inj = 00000009 00          0         145
# error_counter,err_inj = 00000009 00          0         146
# error_counter,err_inj = 00000009 00          0         147
# error_counter,err_inj = 00000009 00          0         148
# error_counter,err_inj = 00000009 00          0         149
# error_counter,err_inj = 00000009 00          0         150
# error_counter,err_inj = 00000009 00          0         151
# error_counter,err_inj = 00000009 00          0         152
# error_counter,err_inj = 00000009 00          0         153
# error_counter,err_inj = 00000009 00          0         154
# error_counter,err_inj = 00000009 00          0         155
# error_counter,err_inj = 00000009 00          0         156
# error_counter,err_inj = 00000009 00          0         157
# error_counter,err_inj = 00000009 00          0         158
# error_counter,err_inj = 00000009 00          0         159
# error_counter,err_inj = 0000000a 11          0         160
# error_counter,err_inj = 0000000a 00          0         161
# error_counter,err_inj = 0000000a 00          0         162
# error_counter,err_inj = 0000000a 00          0         163
# error_counter,err_inj = 0000000a 00          0         164
# error_counter,err_inj = 0000000a 00          0         165
# error_counter,err_inj = 0000000a 00          0         166
# error_counter,err_inj = 0000000a 00          0         167
# error_counter,err_inj = 0000000a 00          0         168
# error_counter,err_inj = 0000000a 00          0         169
# error_counter,err_inj = 0000000a 00          0         170
# error_counter,err_inj = 0000000a 00          0         171
# error_counter,err_inj = 0000000a 00          0         172
# error_counter,err_inj = 0000000a 00          0         173
# error_counter,err_inj = 0000000a 00          0         174
# error_counter,err_inj = 0000000a 00          0         175
# error_counter,err_inj = 0000000b 11          0         176
# error_counter,err_inj = 0000000b 00          0         177
# error_counter,err_inj = 0000000b 00          0         178
```

```
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =         16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000b 00          0         179
# error_counter,err_inj = 0000000b 00          0         180
# error_counter,err_inj = 0000000b 00          0         181
# error_counter,err_inj = 0000000b 00          0         182
# error_counter,err_inj = 0000000b 00          0         183
# error_counter,err_inj = 0000000b 00          0         184
# error_counter,err_inj = 0000000b 00          0         185
# error_counter,err_inj = 0000000b 00          0         186
# error_counter,err_inj = 0000000b 00          0         187
# error_counter,err_inj = 0000000b 00          0         188
# error_counter,err_inj = 0000000b 00          0         189
# error_counter,err_inj = 0000000b 00          0         190
# error_counter,err_inj = 0000000b 00          0         191
# error_counter,err_inj = 0000000c 11          0         192
# error_counter,err_inj = 0000000c 00          0         193
# error_counter,err_inj = 0000000c 00          0         194
# error_counter,err_inj = 0000000c 00          0         195
# error_counter,err_inj = 0000000c 00          0         196
# error_counter,err_inj = 0000000c 00          0         197
# error_counter,err_inj = 0000000c 00          0         198
# error_counter,err_inj = 0000000c 00          0         199
# error_counter,err_inj = 0000000c 00          0         200
# error_counter,err_inj = 0000000c 00          0         201
# error_counter,err_inj = 0000000c 00          0         202
# error_counter,err_inj = 0000000c 00          0         203
# error_counter,err_inj = 0000000c 00          0         204
# error_counter,err_inj = 0000000c 00          0         205
# error_counter,err_inj = 0000000c 00          0         206
# error_counter,err_inj = 0000000c 00          0         207
# error_counter,err_inj = 0000000d 11          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000d 00          0         213
# error_counter,err_inj = 0000000d 00          0         214
# error_counter,err_inj = 0000000d 00          0         215
# error_counter,err_inj = 0000000d 00          0         216
# error_counter,err_inj = 0000000d 00          0         217
# error_counter,err_inj = 0000000d 00          0         218
# error_counter,err_inj = 0000000d 00          0         219
# error_counter,err_inj = 0000000d 00          0         220
# error_counter,err_inj = 0000000d 00          0         221
# error_counter,err_inj = 0000000d 00          0         222
# error_counter,err_inj = 0000000d 00          0         223
# error_counter,err_inj = 0000000e 11          0         224
# error_counter,err_inj = 0000000e 00          0         225
# error_counter,err_inj = 0000000e 00          0         226
# error_counter,err_inj = 0000000e 00          0         227
# error_counter,err_inj = 0000000e 00          0         228
# error_counter,err_inj = 0000000e 00          0         229
# error_counter,err_inj = 0000000e 00          0         230
# error_counter,err_inj = 0000000e 00          0         231
# error_counter,err_inj = 0000000e 00          0         232
# error_counter,err_inj = 0000000e 00          0         233
# error_counter,err_inj = 0000000e 00          0         234
# error_counter,err_inj = 0000000e 00          0         235
# error_counter,err_inj = 0000000e 00          0         236
# error_counter,err_inj = 0000000e 00          0         237
# error_counter,err_inj = 0000000e 00          0         238
# error_counter,err_inj = 0000000e 00          0         239
# error_counter,err_inj = 0000000f 11          0         240
# error_counter,err_inj = 0000000f 00          0         241
# error_counter,err_inj = 0000000f 00          0         242
# error_counter,err_inj = 0000000f 00          0         243
# error_counter,err_inj = 0000000f 00          0         244
# error_counter,err_inj = 0000000f 00          0         245
# error_counter,err_inj = 0000000f 00          0         246
# error_counter,err_inj = 0000000f 00          0         247
# error_counter,err_inj = 0000000f 00          0         248
# error_counter,err_inj = 0000000f 00          0         249
# error_counter,err_inj = 0000000f 00          0         250
# error_counter,err_inj = 0000000f 00          0         251
# error_counter,err_inj = 0000000f 00          0         252
# error_counter,err_inj = 0000000f 00          0         253
# error_counter,err_inj = 0000000f 00          0         254
# error_counter,err_inj = 0000000f 00          0         255
```

2a4) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
```

```
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
```

```
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000001 01          0          15
# error_counter,err_inj = 00000002 01          0          16
# error_counter,err_inj = 00000002 00          1          17
# error_counter,err_inj = 00000002 00          1          18
# error_counter,err_inj = 00000002 00          1          19
# error_counter,err_inj = 00000002 00          1          20
# error_counter,err_inj = 00000002 00          1          21
# error_counter,err_inj = 00000002 00          1          22
# error_counter,err_inj = 00000002 00          1          23
# error_counter,err_inj = 00000002 00          1          24
# error_counter,err_inj = 00000002 00          1          25
# error_counter,err_inj = 00000002 00          1          26
# error_counter,err_inj = 00000002 00          1          27
# error_counter,err_inj = 00000002 00          1          28
# error_counter,err_inj = 00000002 00          1          29
# error_counter,err_inj = 00000002 00          1          30
# error_counter,err_inj = 00000003 01          1          31
# error_counter,err_inj = 00000004 01          1          32
# error_counter,err_inj = 00000004 00          2          33
# error_counter,err_inj = 00000004 00          2          34
# error_counter,err_inj = 00000004 00          2          35
# error_counter,err_inj = 00000004 00          2          36
# error_counter,err_inj = 00000004 00          2          37
# error_counter,err_inj = 00000004 00          2          38
# error_counter,err_inj = 00000004 00          2          39
# error_counter,err_inj = 00000004 00          2          40
# error_counter,err_inj = 00000004 00          2          41
# error_counter,err_inj = 00000004 00          2          42
# error_counter,err_inj = 00000004 00          2          43
# error_counter,err_inj = 00000004 00          2          44
# error_counter,err_inj = 00000004 00          2          45
# error_counter,err_inj = 00000004 00          2          46
# error_counter,err_inj = 00000005 01          2          47
# error_counter,err_inj = 00000006 01          2          48
# error_counter,err_inj = 00000006 00          3          49
# error_counter,err_inj = 00000006 00          3          50
# error_counter,err_inj = 00000006 00          3          51
# error_counter,err_inj = 00000006 00          3          52
# error_counter,err_inj = 00000006 00          3          53
# error_counter,err_inj = 00000006 00          3          54
# error_counter,err_inj = 00000006 00          3          55
# error_counter,err_inj = 00000006 00          3          56
# error_counter,err_inj = 00000006 00          3          57
# error_counter,err_inj = 00000006 00          3          58
# error_counter,err_inj = 00000006 00          3          59
# error_counter,err_inj = 00000006 00          3          60
# error_counter,err_inj = 00000006 00          3          61
# error_counter,err_inj = 00000006 00          3          62
# error_counter,err_inj = 00000007 01          3          63
# error_counter,err_inj = 00000008 01          3          64
# error_counter,err_inj = 00000008 00          4          65
# error_counter,err_inj = 00000008 00          4          66
# error_counter,err_inj = 00000008 00          4          67
# error_counter,err_inj = 00000008 00          4          68
# error_counter,err_inj = 00000008 00          4          69
# error_counter,err_inj = 00000008 00          4          70
# error_counter,err_inj = 00000008 00          4          71
# error_counter,err_inj = 00000008 00          4          72
# error_counter,err_inj = 00000008 00          4          73
# error_counter,err_inj = 00000008 00          4          74
# error_counter,err_inj = 00000008 00          4          75
# error_counter,err_inj = 00000008 00          4          76
# error_counter,err_inj = 00000008 00          4          77
# error_counter,err_inj = 00000008 00          4          78
# error_counter,err_inj = 00000009 01          4          79
# error_counter,err_inj = 0000000a 01          4          80
# error_counter,err_inj = 0000000a 00          5          81
# error_counter,err_inj = 0000000a 00          5          82
# error_counter,err_inj = 0000000a 00          5          83
# error_counter,err_inj = 0000000a 00          5          84
# error_counter,err_inj = 0000000a 00          5          85
# error_counter,err_inj = 0000000a 00          5          86
# error_counter,err_inj = 0000000a 00          5          87
# error_counter,err_inj = 0000000a 00          5          88
# error_counter,err_inj = 0000000a 00          5          89
# error_counter,err_inj = 0000000a 00          5          90
# error_counter,err_inj = 0000000a 00          5          91
# error_counter,err_inj = 0000000a 00          5          92
# error_counter,err_inj = 0000000a 00          5          93
# error_counter,err_inj = 0000000a 00          5          94
# error_counter,err_inj = 0000000b 01          5          95
# error_counter,err_inj = 0000000c 01          5          96
# error_counter,err_inj = 0000000c 00          6          97
# error_counter,err_inj = 0000000c 00          6          98
# error_counter,err_inj = 0000000c 00          6          99
# error_counter,err_inj = 0000000c 00          6         100
# error_counter,err_inj = 0000000c 00          6         101
# error_counter,err_inj = 0000000c 00          6         102
```

```
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
```

```
# error_counter,err_inj = 0000000c 00          6         103
# error_counter,err_inj = 0000000c 00          6         104
# error_counter,err_inj = 0000000c 00          6         105
# error_counter,err_inj = 0000000c 00          6         106
# error_counter,err_inj = 0000000c 00          6         107
# error_counter,err_inj = 0000000c 00          6         108
# error_counter,err_inj = 0000000c 00          6         109
# error_counter,err_inj = 0000000c 00          6         110
# error_counter,err_inj = 0000000d 01          6         111
# error_counter,err_inj = 0000000e 01          6         112
# error_counter,err_inj = 0000000e 00          7         113
# error_counter,err_inj = 0000000e 00          7         114
# error_counter,err_inj = 0000000e 00          7         115
# error_counter,err_inj = 0000000e 00          7         116
# error_counter,err_inj = 0000000e 00          7         117
# error_counter,err_inj = 0000000e 00          7         118
# error_counter,err_inj = 0000000e 00          7         119
# error_counter,err_inj = 0000000e 00          7         120
# error_counter,err_inj = 0000000e 00          7         121
# error_counter,err_inj = 0000000e 00          7         122
# error_counter,err_inj = 0000000e 00          7         123
# error_counter,err_inj = 0000000e 00          7         124
# error_counter,err_inj = 0000000e 00          7         125
# error_counter,err_inj = 0000000e 00          7         126
# error_counter,err_inj = 0000000f 01          7         127
# error_counter,err_inj = 00000010 01          7         128
# error_counter,err_inj = 00000010 00          8         129
# error_counter,err_inj = 00000010 00          8         130
# error_counter,err_inj = 00000010 00          8         131
# error_counter,err_inj = 00000010 00          8         132
# error_counter,err_inj = 00000010 00          8         133
# error_counter,err_inj = 00000010 00          8         134
# error_counter,err_inj = 00000010 00          8         135
# error_counter,err_inj = 00000010 00          8         136
# error_counter,err_inj = 00000010 00          8         137
# error_counter,err_inj = 00000010 00          8         138
# error_counter,err_inj = 00000010 00          8         139
# error_counter,err_inj = 00000010 00          8         140
# error_counter,err_inj = 00000010 00          8         141
# error_counter,err_inj = 00000010 00          8         142
# error_counter,err_inj = 00000011 01          8         143
# error_counter,err_inj = 00000012 01          8         144
# error_counter,err_inj = 00000012 00          9         145
# error_counter,err_inj = 00000012 00          9         146
# error_counter,err_inj = 00000012 00          9         147
# error_counter,err_inj = 00000012 00          9         148
# error_counter,err_inj = 00000012 00          9         149
# error_counter,err_inj = 00000012 00          9         150
# error_counter,err_inj = 00000012 00          9         151
# error_counter,err_inj = 00000012 00          9         152
# error_counter,err_inj = 00000012 00          9         153
# error_counter,err_inj = 00000012 00          9         154
# error_counter,err_inj = 00000012 00          9         155
# error_counter,err_inj = 00000012 00          9         156
# error_counter,err_inj = 00000012 00          9         157
# error_counter,err_inj = 00000012 00          9         158
# error_counter,err_inj = 00000013 01          9         159
# error_counter,err_inj = 00000014 01          9         160
# error_counter,err_inj = 00000014 00         10         161
# error_counter,err_inj = 00000014 00         10         162
# error_counter,err_inj = 00000014 00         10         163
# error_counter,err_inj = 00000014 00         10         164
# error_counter,err_inj = 00000014 00         10         165
# error_counter,err_inj = 00000014 00         10         166
# error_counter,err_inj = 00000014 00         10         167
# error_counter,err_inj = 00000014 00         10         168
# error_counter,err_inj = 00000014 00         10         169
# error_counter,err_inj = 00000014 00         10         170
# error_counter,err_inj = 00000014 00         10         171
# error_counter,err_inj = 00000014 00         10         172
# error_counter,err_inj = 00000014 00         10         173
# error_counter,err_inj = 00000014 00         10         174
# error_counter,err_inj = 00000015 01         10         175
# error_counter,err_inj = 00000016 01         10         176
# error_counter,err_inj = 00000016 00         11         177
# error_counter,err_inj = 00000016 00         11         178
# error_counter,err_inj = 00000016 00         11         179
# error_counter,err_inj = 00000016 00         11         180
# error_counter,err_inj = 00000016 00         11         181
# error_counter,err_inj = 00000016 00         11         182
# error_counter,err_inj = 00000016 00         11         183
# error_counter,err_inj = 00000016 00         11         184
# error_counter,err_inj = 00000016 00         11         185
# error_counter,err_inj = 00000016 00         11         186
# error_counter,err_inj = 00000016 00         11         187
# error_counter,err_inj = 00000016 00         11         188
# error_counter,err_inj = 00000016 00         11         189
# error_counter,err_inj = 00000016 00         11         190
# error_counter,err_inj = 00000017 01         11         191
# error_counter,err_inj = 00000018 01         11         192
# error_counter,err_inj = 00000018 00         12         193
# error_counter,err_inj = 00000018 00         12         194
# error_counter,err_inj = 00000018 00         12         195
```

```
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000018 00          12         196
# error_counter,err_inj = 00000018 00          12         197
# error_counter,err_inj = 00000018 00          12         198
# error_counter,err_inj = 00000018 00          12         199
# error_counter,err_inj = 00000018 00          12         200
# error_counter,err_inj = 00000018 00          12         201
# error_counter,err_inj = 00000018 00          12         202
# error_counter,err_inj = 00000018 00          12         203
# error_counter,err_inj = 00000018 00          12         204
# error_counter,err_inj = 00000018 00          12         205
# error_counter,err_inj = 00000018 00          12         206
# error_counter,err_inj = 00000019 01          12         207
# error_counter,err_inj = 0000001a 01          12         208
# error_counter,err_inj = 0000001a 00          13         209
# error_counter,err_inj = 0000001a 00          13         210
# error_counter,err_inj = 0000001a 00          13         211
# error_counter,err_inj = 0000001a 00          13         212
# error_counter,err_inj = 0000001a 00          13         213
# error_counter,err_inj = 0000001a 00          13         214
# error_counter,err_inj = 0000001a 00          13         215
# error_counter,err_inj = 0000001a 00          13         216
# error_counter,err_inj = 0000001a 00          13         217
# error_counter,err_inj = 0000001a 00          13         218
# error_counter,err_inj = 0000001a 00          13         219
# error_counter,err_inj = 0000001a 00          13         220
# error_counter,err_inj = 0000001a 00          13         221
# error_counter,err_inj = 0000001a 00          13         222
# error_counter,err_inj = 0000001b 01          13         223
# error_counter,err_inj = 0000001c 01          13         224
# error_counter,err_inj = 0000001c 00          14         225
# error_counter,err_inj = 0000001c 00          14         226
# error_counter,err_inj = 0000001c 00          14         227
# error_counter,err_inj = 0000001c 00          14         228
# error_counter,err_inj = 0000001c 00          14         229
# error_counter,err_inj = 0000001c 00          14         230
# error_counter,err_inj = 0000001c 00          14         231
# error_counter,err_inj = 0000001c 00          14         232
# error_counter,err_inj = 0000001c 00          14         233
# error_counter,err_inj = 0000001c 00          14         234
# error_counter,err_inj = 0000001c 00          14         235
# error_counter,err_inj = 0000001c 00          14         236
# error_counter,err_inj = 0000001c 00          14         237
# error_counter,err_inj = 0000001c 00          14         238
# error_counter,err_inj = 0000001d 01          14         239
# error_counter,err_inj = 0000001e 01          14         240
# error_counter,err_inj = 0000001e 00          15         241
# error_counter,err_inj = 0000001e 00          15         242
# error_counter,err_inj = 0000001e 00          15         243
# error_counter,err_inj = 0000001e 00          15         244
# error_counter,err_inj = 0000001e 00          15         245
# error_counter,err_inj = 0000001e 00          15         246
# error_counter,err_inj = 0000001e 00          15         247
# error_counter,err_inj = 0000001e 00          15         248
# error_counter,err_inj = 0000001e 00          15         249
# error_counter,err_inj = 0000001e 00          15         250
# error_counter,err_inj = 0000001e 00          15         251
# error_counter,err_inj = 0000001e 00          15         252
# error_counter,err_inj = 0000001e 00          15         253
# error_counter,err_inj = 0000001e 00          15         254
# error_counter,err_inj = 0000001f 01          15         255
# word_count =       10440
```

2a5) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000000 00           0           8
# error_counter,err_inj = 00000000 00           0           9
# error_counter,err_inj = 00000000 00           0          10
# error_counter,err_inj = 00000000 00           0          11
# error_counter,err_inj = 00000000 00           0          12
# error_counter,err_inj = 00000000 00           0          13
# error_counter,err_inj = 00000000 00           0          14
# error_counter,err_inj = 00000001 10           0          15
# error_counter,err_inj = 00000002 10           0          16
# error_counter,err_inj = 00000002 00           1          17
# error_counter,err_inj = 00000002 00           1          18
# error_counter,err_inj = 00000002 00           1          19
# error_counter,err_inj = 00000002 00           1          20
# error_counter,err_inj = 00000002 00           1          21
# error_counter,err_inj = 00000002 00           1          22
# error_counter,err_inj = 00000002 00           1          23
# error_counter,err_inj = 00000002 00           1          24
# error_counter,err_inj = 00000002 00           1          25
# error_counter,err_inj = 00000002 00           1          26
```

```
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
```

```
# error_counter,err_inj = 00000002 00          1          27
# error_counter,err_inj = 00000002 00          1          28
# error_counter,err_inj = 00000002 00          1          29
# error_counter,err_inj = 00000002 00          1          30
# error_counter,err_inj = 00000003 10          1          31
# error_counter,err_inj = 00000004 10          1          32
# error_counter,err_inj = 00000004 00          2          33
# error_counter,err_inj = 00000004 00          2          34
# error_counter,err_inj = 00000004 00          2          35
# error_counter,err_inj = 00000004 00          2          36
# error_counter,err_inj = 00000004 00          2          37
# error_counter,err_inj = 00000004 00          2          38
# error_counter,err_inj = 00000004 00          2          39
# error_counter,err_inj = 00000004 00          2          40
# error_counter,err_inj = 00000004 00          2          41
# error_counter,err_inj = 00000004 00          2          42
# error_counter,err_inj = 00000004 00          2          43
# error_counter,err_inj = 00000004 00          2          44
# error_counter,err_inj = 00000004 00          2          45
# error_counter,err_inj = 00000004 00          2          46
# error_counter,err_inj = 00000005 10          2          47
# error_counter,err_inj = 00000006 10          2          48
# error_counter,err_inj = 00000006 00          3          49
# error_counter,err_inj = 00000006 00          3          50
# error_counter,err_inj = 00000006 00          3          51
# error_counter,err_inj = 00000006 00          3          52
# error_counter,err_inj = 00000006 00          3          53
# error_counter,err_inj = 00000006 00          3          54
# error_counter,err_inj = 00000006 00          3          55
# error_counter,err_inj = 00000006 00          3          56
# error_counter,err_inj = 00000006 00          3          57
# error_counter,err_inj = 00000006 00          3          58
# error_counter,err_inj = 00000006 00          3          59
# error_counter,err_inj = 00000006 00          3          60
# error_counter,err_inj = 00000006 00          3          61
# error_counter,err_inj = 00000006 00          3          62
# error_counter,err_inj = 00000007 10          3          63
# error_counter,err_inj = 00000008 10          3          64
# error_counter,err_inj = 00000008 00          4          65
# error_counter,err_inj = 00000008 00          4          66
# error_counter,err_inj = 00000008 00          4          67
# error_counter,err_inj = 00000008 00          4          68
# error_counter,err_inj = 00000008 00          4          69
# error_counter,err_inj = 00000008 00          4          70
# error_counter,err_inj = 00000008 00          4          71
# error_counter,err_inj = 00000008 00          4          72
# error_counter,err_inj = 00000008 00          4          73
# error_counter,err_inj = 00000008 00          4          74
# error_counter,err_inj = 00000008 00          4          75
# error_counter,err_inj = 00000008 00          4          76
# error_counter,err_inj = 00000008 00          4          77
# error_counter,err_inj = 00000008 00          4          78
# error_counter,err_inj = 00000009 10          4          79
# error_counter,err_inj = 0000000a 10          4          80
# error_counter,err_inj = 0000000a 00          5          81
# error_counter,err_inj = 0000000a 00          5          82
# error_counter,err_inj = 0000000a 00          5          83
# error_counter,err_inj = 0000000a 00          5          84
# error_counter,err_inj = 0000000a 00          5          85
# error_counter,err_inj = 0000000a 00          5          86
# error_counter,err_inj = 0000000a 00          5          87
# error_counter,err_inj = 0000000a 00          5          88
# error_counter,err_inj = 0000000a 00          5          89
# error_counter,err_inj = 0000000a 00          5          90
# error_counter,err_inj = 0000000a 00          5          91
# error_counter,err_inj = 0000000a 00          5          92
# error_counter,err_inj = 0000000a 00          5          93
# error_counter,err_inj = 0000000a 00          5          94
# error_counter,err_inj = 0000000b 10          5          95
# error_counter,err_inj = 0000000c 10          5          96
# error_counter,err_inj = 0000000c 00          6          97
# error_counter,err_inj = 0000000c 00          6          98
# error_counter,err_inj = 0000000c 00          6          99
# error_counter,err_inj = 0000000c 00          6         100
# error_counter,err_inj = 0000000c 00          6         101
# error_counter,err_inj = 0000000c 00          6         102
# error_counter,err_inj = 0000000c 00          6         103
# error_counter,err_inj = 0000000c 00          6         104
# error_counter,err_inj = 0000000c 00          6         105
# error_counter,err_inj = 0000000c 00          6         106
# error_counter,err_inj = 0000000c 00          6         107
# error_counter,err_inj = 0000000c 00          6         108
# error_counter,err_inj = 0000000c 00          6         109
# error_counter,err_inj = 0000000c 00          6         110
# error_counter,err_inj = 0000000d 10          6         111
# error_counter,err_inj = 0000000e 10          6         112
# error_counter,err_inj = 0000000e 00          7         113
# error_counter,err_inj = 0000000e 00          7         114
# error_counter,err_inj = 0000000e 00          7         115
# error_counter,err_inj = 0000000e 00          7         116
# error_counter,err_inj = 0000000e 00          7         117
# error_counter,err_inj = 0000000e 00          7         118
# error_counter,err_inj = 0000000e 00          7         119
```

```
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
```

```
# error_counter,err_inj = 0000000e 00          7         120
# error_counter,err_inj = 0000000e 00          7         121
# error_counter,err_inj = 0000000e 00          7         122
# error_counter,err_inj = 0000000e 00          7         123
# error_counter,err_inj = 0000000e 00          7         124
# error_counter,err_inj = 0000000e 00          7         125
# error_counter,err_inj = 0000000e 00          7         126
# error_counter,err_inj = 0000000f 10          7         127
# error_counter,err_inj = 00000010 10          7         128
# error_counter,err_inj = 00000010 00          8         129
# error_counter,err_inj = 00000010 00          8         130
# error_counter,err_inj = 00000010 00          8         131
# error_counter,err_inj = 00000010 00          8         132
# error_counter,err_inj = 00000010 00          8         133
# error_counter,err_inj = 00000010 00          8         134
# error_counter,err_inj = 00000010 00          8         135
# error_counter,err_inj = 00000010 00          8         136
# error_counter,err_inj = 00000010 00          8         137
# error_counter,err_inj = 00000010 00          8         138
# error_counter,err_inj = 00000010 00          8         139
# error_counter,err_inj = 00000010 00          8         140
# error_counter,err_inj = 00000010 00          8         141
# error_counter,err_inj = 00000010 00          8         142
# error_counter,err_inj = 00000011 10          8         143
# error_counter,err_inj = 00000012 10          8         144
# error_counter,err_inj = 00000012 00          9         145
# error_counter,err_inj = 00000012 00          9         146
# error_counter,err_inj = 00000012 00          9         147
# error_counter,err_inj = 00000012 00          9         148
# error_counter,err_inj = 00000012 00          9         149
# error_counter,err_inj = 00000012 00          9         150
# error_counter,err_inj = 00000012 00          9         151
# error_counter,err_inj = 00000012 00          9         152
# error_counter,err_inj = 00000012 00          9         153
# error_counter,err_inj = 00000012 00          9         154
# error_counter,err_inj = 00000012 00          9         155
# error_counter,err_inj = 00000012 00          9         156
# error_counter,err_inj = 00000012 00          9         157
# error_counter,err_inj = 00000012 00          9         158
# error_counter,err_inj = 00000013 10          9         159
# error_counter,err_inj = 00000014 10          9         160
# error_counter,err_inj = 00000014 00         10         161
# error_counter,err_inj = 00000014 00         10         162
# error_counter,err_inj = 00000014 00         10         163
# error_counter,err_inj = 00000014 00         10         164
# error_counter,err_inj = 00000014 00         10         165
# error_counter,err_inj = 00000014 00         10         166
# error_counter,err_inj = 00000014 00         10         167
# error_counter,err_inj = 00000014 00         10         168
# error_counter,err_inj = 00000014 00         10         169
# error_counter,err_inj = 00000014 00         10         170
# error_counter,err_inj = 00000014 00         10         171
# error_counter,err_inj = 00000014 00         10         172
# error_counter,err_inj = 00000014 00         10         173
# error_counter,err_inj = 00000014 00         10         174
# error_counter,err_inj = 00000015 10         10         175
# error_counter,err_inj = 00000016 10         10         176
# error_counter,err_inj = 00000016 00         11         177
# error_counter,err_inj = 00000016 00         11         178
# error_counter,err_inj = 00000016 00         11         179
# error_counter,err_inj = 00000016 00         11         180
# error_counter,err_inj = 00000016 00         11         181
# error_counter,err_inj = 00000016 00         11         182
# error_counter,err_inj = 00000016 00         11         183
# error_counter,err_inj = 00000016 00         11         184
# error_counter,err_inj = 00000016 00         11         185
# error_counter,err_inj = 00000016 00         11         186
# error_counter,err_inj = 00000016 00         11         187
# error_counter,err_inj = 00000016 00         11         188
# error_counter,err_inj = 00000016 00         11         189
# error_counter,err_inj = 00000016 00         11         190
# error_counter,err_inj = 00000017 10         11         191
# error_counter,err_inj = 00000018 10         11         192
# error_counter,err_inj = 00000018 00         12         193
# error_counter,err_inj = 00000018 00         12         194
# error_counter,err_inj = 00000018 00         12         195
# error_counter,err_inj = 00000018 00         12         196
# error_counter,err_inj = 00000018 00         12         197
# error_counter,err_inj = 00000018 00         12         198
# error_counter,err_inj = 00000018 00         12         199
# error_counter,err_inj = 00000018 00         12         200
# error_counter,err_inj = 00000018 00         12         201
# error_counter,err_inj = 00000018 00         12         202
# error_counter,err_inj = 00000018 00         12         203
# error_counter,err_inj = 00000018 00         12         204
# error_counter,err_inj = 00000018 00         12         205
# error_counter,err_inj = 00000018 00         12         206
# error_counter,err_inj = 00000019 10         12         207
# error_counter,err_inj = 0000001a 10         12         208
# error_counter,err_inj = 0000001a 00         13         209
# error_counter,err_inj = 0000001a 00         13         210
# error_counter,err_inj = 0000001a 00         13         211
# error_counter,err_inj = 0000001a 00         13         212
```

```
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001a 00          13          213
# error_counter,err_inj = 0000001a 00          13          214
# error_counter,err_inj = 0000001a 00          13          215
# error_counter,err_inj = 0000001a 00          13          216
# error_counter,err_inj = 0000001a 00          13          217
# error_counter,err_inj = 0000001a 00          13          218
# error_counter,err_inj = 0000001a 00          13          219
# error_counter,err_inj = 0000001a 00          13          220
# error_counter,err_inj = 0000001a 00          13          221
# error_counter,err_inj = 0000001a 00          13          222
# error_counter,err_inj = 0000001b 10          13          223
# error_counter,err_inj = 0000001c 10          13          224
# error_counter,err_inj = 0000001c 00          14          225
# error_counter,err_inj = 0000001c 00          14          226
# error_counter,err_inj = 0000001c 00          14          227
# error_counter,err_inj = 0000001c 00          14          228
# error_counter,err_inj = 0000001c 00          14          229
# error_counter,err_inj = 0000001c 00          14          230
# error_counter,err_inj = 0000001c 00          14          231
# error_counter,err_inj = 0000001c 00          14          232
# error_counter,err_inj = 0000001c 00          14          233
# error_counter,err_inj = 0000001c 00          14          234
# error_counter,err_inj = 0000001c 00          14          235
# error_counter,err_inj = 0000001c 00          14          236
# error_counter,err_inj = 0000001c 00          14          237
# error_counter,err_inj = 0000001c 00          14          238
# error_counter,err_inj = 0000001d 10          14          239
# error_counter,err_inj = 0000001e 10          14          240
# error_counter,err_inj = 0000001e 00          15          241
# error_counter,err_inj = 0000001e 00          15          242
# error_counter,err_inj = 0000001e 00          15          243
# error_counter,err_inj = 0000001e 00          15          244
# error_counter,err_inj = 0000001e 00          15          245
# error_counter,err_inj = 0000001e 00          15          246
# error_counter,err_inj = 0000001e 00          15          247
# error_counter,err_inj = 0000001e 00          15          248
# error_counter,err_inj = 0000001e 00          15          249
# error_counter,err_inj = 0000001e 00          15          250
# error_counter,err_inj = 0000001e 00          15          251
# error_counter,err_inj = 0000001e 00          15          252
# error_counter,err_inj = 0000001e 00          15          253
# error_counter,err_inj = 0000001e 00          15          254
# error_counter,err_inj = 0000001f 10          15          255
# word_count =       10440
```

2a6) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000001 01          0          29
# error_counter,err_inj = 00000002 01          0          30
# error_counter,err_inj = 00000003 01          1          31
# error_counter,err_inj = 00000004 01          2          32
# error_counter,err_inj = 00000004 00          3          33
# error_counter,err_inj = 00000004 00          3          34
# error_counter,err_inj = 00000004 00          3          35
# error_counter,err_inj = 00000004 00          3          36
# error_counter,err_inj = 00000004 00          3          37
# error_counter,err_inj = 00000004 00          3          38
# error_counter,err_inj = 00000004 00          3          39
# error_counter,err_inj = 00000004 00          3          40
# error_counter,err_inj = 00000004 00          3          41
# error_counter,err_inj = 00000004 00          3          42
# error_counter,err_inj = 00000004 00          3          43
```

```
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
```

```
# error_counter,err_inj = 00000004 00          3          44
# error_counter,err_inj = 00000004 00          3          45
# error_counter,err_inj = 00000004 00          3          46
# error_counter,err_inj = 00000004 00          3          47
# error_counter,err_inj = 00000004 00          3          48
# error_counter,err_inj = 00000004 00          3          49
# error_counter,err_inj = 00000004 00          3          50
# error_counter,err_inj = 00000004 00          3          51
# error_counter,err_inj = 00000004 00          3          52
# error_counter,err_inj = 00000004 00          3          53
# error_counter,err_inj = 00000004 00          3          54
# error_counter,err_inj = 00000004 00          3          55
# error_counter,err_inj = 00000004 00          3          56
# error_counter,err_inj = 00000004 00          3          57
# error_counter,err_inj = 00000004 00          3          58
# error_counter,err_inj = 00000004 00          3          59
# error_counter,err_inj = 00000004 00          3          60
# error_counter,err_inj = 00000005 01          3          61
# error_counter,err_inj = 00000006 01          3          62
# error_counter,err_inj = 00000007 01          4          63
# error_counter,err_inj = 00000008 01          5          64
# error_counter,err_inj = 00000008 00          6          65
# error_counter,err_inj = 00000008 00          6          66
# error_counter,err_inj = 00000008 00          6          67
# error_counter,err_inj = 00000008 00          6          68
# error_counter,err_inj = 00000008 00          6          69
# error_counter,err_inj = 00000008 00          6          70
# error_counter,err_inj = 00000008 00          6          71
# error_counter,err_inj = 00000008 00          6          72
# error_counter,err_inj = 00000008 00          6          73
# error_counter,err_inj = 00000008 00          6          74
# error_counter,err_inj = 00000008 00          6          75
# error_counter,err_inj = 00000008 00          6          76
# error_counter,err_inj = 00000008 00          6          77
# error_counter,err_inj = 00000008 00          6          78
# error_counter,err_inj = 00000008 00          6          79
# error_counter,err_inj = 00000008 00          6          80
# error_counter,err_inj = 00000008 00          6          81
# error_counter,err_inj = 00000008 00          6          82
# error_counter,err_inj = 00000008 00          6          83
# error_counter,err_inj = 00000008 00          6          84
# error_counter,err_inj = 00000008 00          6          85
# error_counter,err_inj = 00000008 00          6          86
# error_counter,err_inj = 00000008 00          6          87
# error_counter,err_inj = 00000008 00          6          88
# error_counter,err_inj = 00000008 00          6          89
# error_counter,err_inj = 00000008 00          6          90
# error_counter,err_inj = 00000008 00          6          91
# error_counter,err_inj = 00000008 00          6          92
# error_counter,err_inj = 00000009 01          6          93
# error_counter,err_inj = 0000000a 01          6          94
# error_counter,err_inj = 0000000b 01          7          95
# error_counter,err_inj = 0000000c 01          8          96
# error_counter,err_inj = 0000000c 00          9          97
# error_counter,err_inj = 0000000c 00          9          98
# error_counter,err_inj = 0000000c 00          9          99
# error_counter,err_inj = 0000000c 00          9         100
# error_counter,err_inj = 0000000c 00          9         101
# error_counter,err_inj = 0000000c 00          9         102
# error_counter,err_inj = 0000000c 00          9         103
# error_counter,err_inj = 0000000c 00          9         104
# error_counter,err_inj = 0000000c 00          9         105
# error_counter,err_inj = 0000000c 00          9         106
# error_counter,err_inj = 0000000c 00          9         107
# error_counter,err_inj = 0000000c 00          9         108
# error_counter,err_inj = 0000000c 00          9         109
# error_counter,err_inj = 0000000c 00          9         110
# error_counter,err_inj = 0000000c 00          9         111
# error_counter,err_inj = 0000000c 00          9         112
# error_counter,err_inj = 0000000c 00          9         113
# error_counter,err_inj = 0000000c 00          9         114
# error_counter,err_inj = 0000000c 00          9         115
# error_counter,err_inj = 0000000c 00          9         116
# error_counter,err_inj = 0000000c 00          9         117
# error_counter,err_inj = 0000000c 00          9         118
# error_counter,err_inj = 0000000c 00          9         119
# error_counter,err_inj = 0000000c 00          9         120
# error_counter,err_inj = 0000000c 00          9         121
# error_counter,err_inj = 0000000c 00          9         122
# error_counter,err_inj = 0000000c 00          9         123
# error_counter,err_inj = 0000000c 00          9         124
# error_counter,err_inj = 0000000d 01          9         125
# error_counter,err_inj = 0000000e 01          9         126
# error_counter,err_inj = 0000000f 01         10         127
# error_counter,err_inj = 00000010 01         11         128
# error_counter,err_inj = 00000010 00         12         129
# error_counter,err_inj = 00000010 00         12         130
# error_counter,err_inj = 00000010 00         12         131
# error_counter,err_inj = 00000010 00         12         132
# error_counter,err_inj = 00000010 00         12         133
# error_counter,err_inj = 00000010 00         12         134
# error_counter,err_inj = 00000010 00         12         135
# error_counter,err_inj = 00000010 00         12         136
```

```
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
```

```
# error_counter,err_inj = 00000010 00          12          137
# error_counter,err_inj = 00000010 00          12          138
# error_counter,err_inj = 00000010 00          12          139
# error_counter,err_inj = 00000010 00          12          140
# error_counter,err_inj = 00000010 00          12          141
# error_counter,err_inj = 00000010 00          12          142
# error_counter,err_inj = 00000010 00          12          143
# error_counter,err_inj = 00000010 00          12          144
# error_counter,err_inj = 00000010 00          12          145
# error_counter,err_inj = 00000010 00          12          146
# error_counter,err_inj = 00000010 00          12          147
# error_counter,err_inj = 00000010 00          12          148
# error_counter,err_inj = 00000010 00          12          149
# error_counter,err_inj = 00000010 00          12          150
# error_counter,err_inj = 00000010 00          12          151
# error_counter,err_inj = 00000010 00          12          152
# error_counter,err_inj = 00000010 00          12          153
# error_counter,err_inj = 00000010 00          12          154
# error_counter,err_inj = 00000010 00          12          155
# error_counter,err_inj = 00000010 00          12          156
# error_counter,err_inj = 00000011 01          12          157
# error_counter,err_inj = 00000012 01          12          158
# error_counter,err_inj = 00000013 01          13          159
# error_counter,err_inj = 00000014 01          14          160
# error_counter,err_inj = 00000014 00          15          161
# error_counter,err_inj = 00000014 00          15          162
# error_counter,err_inj = 00000014 00          15          163
# error_counter,err_inj = 00000014 00          15          164
# error_counter,err_inj = 00000014 00          15          165
# error_counter,err_inj = 00000014 00          15          166
# error_counter,err_inj = 00000014 00          15          167
# error_counter,err_inj = 00000014 00          15          168
# error_counter,err_inj = 00000014 00          15          169
# error_counter,err_inj = 00000014 00          15          170
# error_counter,err_inj = 00000014 00          15          171
# error_counter,err_inj = 00000014 00          15          172
# error_counter,err_inj = 00000014 00          15          173
# error_counter,err_inj = 00000014 00          15          174
# error_counter,err_inj = 00000014 00          15          175
# error_counter,err_inj = 00000014 00          15          176
# error_counter,err_inj = 00000014 00          15          177
# error_counter,err_inj = 00000014 00          15          178
# error_counter,err_inj = 00000014 00          15          179
# error_counter,err_inj = 00000014 00          15          180
# error_counter,err_inj = 00000014 00          15          181
# error_counter,err_inj = 00000014 00          15          182
# error_counter,err_inj = 00000014 00          15          183
# error_counter,err_inj = 00000014 00          15          184
# error_counter,err_inj = 00000014 00          15          185
# error_counter,err_inj = 00000014 00          15          186
# error_counter,err_inj = 00000014 00          15          187
# error_counter,err_inj = 00000014 00          15          188
# error_counter,err_inj = 00000015 01          15          189
# error_counter,err_inj = 00000016 01          15          190
# error_counter,err_inj = 00000017 01          16          191
# error_counter,err_inj = 00000018 01          17          192
# error_counter,err_inj = 00000018 00          18          193
# error_counter,err_inj = 00000018 00          18          194
# error_counter,err_inj = 00000018 00          18          195
# error_counter,err_inj = 00000018 00          18          196
# error_counter,err_inj = 00000018 00          18          197
# error_counter,err_inj = 00000018 00          18          198
# error_counter,err_inj = 00000018 00          18          199
# error_counter,err_inj = 00000018 00          18          200
# error_counter,err_inj = 00000018 00          18          201
# error_counter,err_inj = 00000018 00          18          202
# error_counter,err_inj = 00000018 00          18          203
# error_counter,err_inj = 00000018 00          18          204
# error_counter,err_inj = 00000018 00          18          205
# error_counter,err_inj = 00000018 00          18          206
# error_counter,err_inj = 00000018 00          18          207
# error_counter,err_inj = 00000018 00          18          208
# error_counter,err_inj = 00000018 00          18          209
# error_counter,err_inj = 00000018 00          18          210
# error_counter,err_inj = 00000018 00          18          211
# error_counter,err_inj = 00000018 00          18          212
# error_counter,err_inj = 00000018 00          18          213
# error_counter,err_inj = 00000018 00          18          214
# error_counter,err_inj = 00000018 00          18          215
# error_counter,err_inj = 00000018 00          18          216
# error_counter,err_inj = 00000018 00          18          217
# error_counter,err_inj = 00000018 00          18          218
# error_counter,err_inj = 00000018 00          18          219
# error_counter,err_inj = 00000018 00          18          220
# error_counter,err_inj = 00000019 01          18          221
# error_counter,err_inj = 0000001a 01          18          222
# error_counter,err_inj = 0000001b 01          19          223
# error_counter,err_inj = 0000001c 01          20          224
# error_counter,err_inj = 0000001c 00          21          225
# error_counter,err_inj = 0000001c 00          21          226
# error_counter,err_inj = 0000001c 00          21          227
# error_counter,err_inj = 0000001c 00          21          228
# error_counter,err_inj = 0000001c 00          21          229
```

```
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001c 00          21         230
# error_counter,err_inj = 0000001c 00          21         231
# error_counter,err_inj = 0000001c 00          21         232
# error_counter,err_inj = 0000001c 00          21         233
# error_counter,err_inj = 0000001c 00          21         234
# error_counter,err_inj = 0000001c 00          21         235
# error_counter,err_inj = 0000001c 00          21         236
# error_counter,err_inj = 0000001c 00          21         237
# error_counter,err_inj = 0000001c 00          21         238
# error_counter,err_inj = 0000001c 00          21         239
# error_counter,err_inj = 0000001c 00          21         240
# error_counter,err_inj = 0000001c 00          21         241
# error_counter,err_inj = 0000001c 00          21         242
# error_counter,err_inj = 0000001c 00          21         243
# error_counter,err_inj = 0000001c 00          21         244
# error_counter,err_inj = 0000001c 00          21         245
# error_counter,err_inj = 0000001c 00          21         246
# error_counter,err_inj = 0000001c 00          21         247
# error_counter,err_inj = 0000001c 00          21         248
# error_counter,err_inj = 0000001c 00          21         249
# error_counter,err_inj = 0000001c 00          21         250
# error_counter,err_inj = 0000001c 00          21         251
# error_counter,err_inj = 0000001c 00          21         252
# error_counter,err_inj = 0000001d 01          21         253
# error_counter,err_inj = 0000001e 01          21         254
# error_counter,err_inj = 0000001f 01          22         255
```

2a7) Transcript:

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000000 00           0           8
# error_counter,err_inj = 00000000 00           0           9
# error_counter,err_inj = 00000000 00           0          10
# error_counter,err_inj = 00000000 00           0          11
# error_counter,err_inj = 00000000 00           0          12
# error_counter,err_inj = 00000000 00           0          13
# error_counter,err_inj = 00000000 00           0          14
# error_counter,err_inj = 00000000 00           0          15
# error_counter,err_inj = 00000000 00           0          16
# error_counter,err_inj = 00000000 00           0          17
# error_counter,err_inj = 00000000 00           0          18
# error_counter,err_inj = 00000000 00           0          19
# error_counter,err_inj = 00000000 00           0          20
# error_counter,err_inj = 00000000 00           0          21
# error_counter,err_inj = 00000000 00           0          22
# error_counter,err_inj = 00000000 00           0          23
# error_counter,err_inj = 00000000 00           0          24
# error_counter,err_inj = 00000000 00           0          25
# error_counter,err_inj = 00000000 00           0          26
# error_counter,err_inj = 00000000 00           0          27
# error_counter,err_inj = 00000000 00           0          28
# error_counter,err_inj = 00000001 10           0          29
# error_counter,err_inj = 00000002 10           0          30
# error_counter,err_inj = 00000003 10           1          31
# error_counter,err_inj = 00000004 10           2          32
# error_counter,err_inj = 00000004 00           3          33
# error_counter,err_inj = 00000004 00           3          34
# error_counter,err_inj = 00000004 00           3          35
# error_counter,err_inj = 00000004 00           3          36
# error_counter,err_inj = 00000004 00           3          37
# error_counter,err_inj = 00000004 00           3          38
# error_counter,err_inj = 00000004 00           3          39
# error_counter,err_inj = 00000004 00           3          40
# error_counter,err_inj = 00000004 00           3          41
# error_counter,err_inj = 00000004 00           3          42
# error_counter,err_inj = 00000004 00           3          43
# error_counter,err_inj = 00000004 00           3          44
# error_counter,err_inj = 00000004 00           3          45
# error_counter,err_inj = 00000004 00           3          46
# error_counter,err_inj = 00000004 00           3          47
# error_counter,err_inj = 00000004 00           3          48
# error_counter,err_inj = 00000004 00           3          49
# error_counter,err_inj = 00000004 00           3          50
# error_counter,err_inj = 00000004 00           3          51
# error_counter,err_inj = 00000004 00           3          52
# error_counter,err_inj = 00000004 00           3          53
# error_counter,err_inj = 00000004 00           3          54
# error_counter,err_inj = 00000004 00           3          55
# error_counter,err_inj = 00000004 00           3          56
# error_counter,err_inj = 00000004 00           3          57
# error_counter,err_inj = 00000004 00           3          58
# error_counter,err_inj = 00000004 00           3          59
# error_counter,err_inj = 00000004 00           3          60
```

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
```

```
# error_counter,err_inj = 00000005 10           3          61   # yaa! in = 0, out = 0, w_ct =          61, err = 00
# error_counter,err_inj = 00000006 10           3          62   # yaa! in = 0, out = 0, w_ct =          62, err = 00
# error_counter,err_inj = 00000007 10           4          63   # yaa! in = 0, out = 0, w_ct =          63, err = 00
# error_counter,err_inj = 00000008 10           5          64   # yaa! in = 1, out = 1, w_ct =          64, err = 00
# error_counter,err_inj = 00000008 00           6          65   # yaa! in = 1, out = 1, w_ct =          65, err = 00
# error_counter,err_inj = 00000008 00           6          66   # yaa! in = 1, out = 1, w_ct =          66, err = 00
# error_counter,err_inj = 00000008 00           6          67   # yaa! in = 1, out = 1, w_ct =          67, err = 00
# error_counter,err_inj = 00000008 00           6          68   # yaa! in = 0, out = 0, w_ct =          68, err = 00
# error_counter,err_inj = 00000008 00           6          69   # yaa! in = 0, out = 0, w_ct =          69, err = 00
# error_counter,err_inj = 00000008 00           6          70   # yaa! in = 0, out = 0, w_ct =          70, err = 00
# error_counter,err_inj = 00000008 00           6          71   # yaa! in = 0, out = 0, w_ct =          71, err = 00
# error_counter,err_inj = 00000008 00           6          72   # yaa! in = 0, out = 0, w_ct =          72, err = 00
# error_counter,err_inj = 00000008 00           6          73   # yaa! in = 1, out = 1, w_ct =          73, err = 00
# error_counter,err_inj = 00000008 00           6          74   # yaa! in = 1, out = 1, w_ct =          74, err = 00
# error_counter,err_inj = 00000008 00           6          75   # yaa! in = 1, out = 1, w_ct =          75, err = 00
# error_counter,err_inj = 00000008 00           6          76   # yaa! in = 1, out = 1, w_ct =          76, err = 00
# error_counter,err_inj = 00000008 00           6          77   # yaa! in = 1, out = 1, w_ct =          77, err = 00
# error_counter,err_inj = 00000008 00           6          78   # yaa! in = 0, out = 0, w_ct =          78, err = 00
# error_counter,err_inj = 00000008 00           6          79   # yaa! in = 1, out = 1, w_ct =          79, err = 00
# error_counter,err_inj = 00000008 00           6          80   # yaa! in = 0, out = 0, w_ct =          80, err = 00
# error_counter,err_inj = 00000008 00           6          81   # yaa! in = 1, out = 1, w_ct =          81, err = 00
# error_counter,err_inj = 00000008 00           6          82   # yaa! in = 0, out = 0, w_ct =          82, err = 00
# error_counter,err_inj = 00000008 00           6          83   # yaa! in = 1, out = 1, w_ct =          83, err = 00
# error_counter,err_inj = 00000008 00           6          84   # yaa! in = 0, out = 0, w_ct =          84, err = 00
# error_counter,err_inj = 00000008 00           6          85   # yaa! in = 1, out = 1, w_ct =          85, err = 00
# error_counter,err_inj = 00000008 00           6          86   # yaa! in = 0, out = 0, w_ct =          86, err = 00
# error_counter,err_inj = 00000008 00           6          87   # yaa! in = 0, out = 0, w_ct =          87, err = 00
# error_counter,err_inj = 00000008 00           6          88   # yaa! in = 0, out = 0, w_ct =          88, err = 00
# error_counter,err_inj = 00000008 00           6          89   # yaa! in = 0, out = 0, w_ct =          89, err = 00
# error_counter,err_inj = 00000008 00           6          90   # yaa! in = 0, out = 0, w_ct =          90, err = 00
# error_counter,err_inj = 00000008 00           6          91   # yaa! in = 0, out = 0, w_ct =          91, err = 00
# error_counter,err_inj = 00000008 00           6          92   # yaa! in = 0, out = 0, w_ct =          92, err = 00
# error_counter,err_inj = 00000009 10           6          93   # yaa! in = 0, out = 0, w_ct =          93, err = 00
# error_counter,err_inj = 0000000a 10           6          94   # yaa! in = 0, out = 0, w_ct =          94, err = 00
# error_counter,err_inj = 0000000b 10           7          95   # yaa! in = 0, out = 0, w_ct =          95, err = 00
# error_counter,err_inj = 0000000c 10           8          96   # yaa! in = 1, out = 1, w_ct =          96, err = 00
# error_counter,err_inj = 0000000c 00           9          97   # yaa! in = 1, out = 1, w_ct =          97, err = 00
# error_counter,err_inj = 0000000c 00           9          98   # yaa! in = 1, out = 1, w_ct =          98, err = 00
# error_counter,err_inj = 0000000c 00           9          99   # yaa! in = 1, out = 1, w_ct =          99, err = 00
# error_counter,err_inj = 0000000c 00           9         100   # yaa! in = 1, out = 1, w_ct =         100, err = 00
# error_counter,err_inj = 0000000c 00           9         101   # yaa! in = 1, out = 1, w_ct =         101, err = 00
# error_counter,err_inj = 0000000c 00           9         102   # yaa! in = 1, out = 1, w_ct =         102, err = 00
# error_counter,err_inj = 0000000c 00           9         103   # yaa! in = 1, out = 1, w_ct =         103, err = 00
# error_counter,err_inj = 0000000c 00           9         104   # yaa! in = 1, out = 1, w_ct =         104, err = 00
# error_counter,err_inj = 0000000c 00           9         105   # yaa! in = 1, out = 1, w_ct =         105, err = 00
# error_counter,err_inj = 0000000c 00           9         106   # yaa! in = 0, out = 0, w_ct =         106, err = 00
# error_counter,err_inj = 0000000c 00           9         107   # yaa! in = 0, out = 0, w_ct =         107, err = 00
# error_counter,err_inj = 0000000c 00           9         108   # yaa! in = 1, out = 1, w_ct =         108, err = 00
# error_counter,err_inj = 0000000c 00           9         109   # yaa! in = 1, out = 1, w_ct =         109, err = 00
# error_counter,err_inj = 0000000c 00           9         110   # yaa! in = 1, out = 1, w_ct =         110, err = 00
# error_counter,err_inj = 0000000c 00           9         111   # yaa! in = 0, out = 0, w_ct =         111, err = 00
# error_counter,err_inj = 0000000c 00           9         112   # yaa! in = 1, out = 1, w_ct =         112, err = 00
# error_counter,err_inj = 0000000c 00           9         113   # yaa! in = 0, out = 0, w_ct =         113, err = 00
# error_counter,err_inj = 0000000c 00           9         114   # yaa! in = 0, out = 0, w_ct =         114, err = 00
# error_counter,err_inj = 0000000c 00           9         115   # yaa! in = 0, out = 0, w_ct =         115, err = 00
# error_counter,err_inj = 0000000c 00           9         116   # yaa! in = 0, out = 0, w_ct =         116, err = 00
# error_counter,err_inj = 0000000c 00           9         117   # yaa! in = 1, out = 1, w_ct =         117, err = 00
# error_counter,err_inj = 0000000c 00           9         118   # yaa! in = 1, out = 1, w_ct =         118, err = 00
# error_counter,err_inj = 0000000c 00           9         119   # yaa! in = 1, out = 1, w_ct =         119, err = 00
# error_counter,err_inj = 0000000c 00           9         120   # yaa! in = 0, out = 0, w_ct =         120, err = 00
# error_counter,err_inj = 0000000c 00           9         121   # yaa! in = 1, out = 1, w_ct =         121, err = 00
# error_counter,err_inj = 0000000c 00           9         122   # yaa! in = 0, out = 0, w_ct =         122, err = 00
# error_counter,err_inj = 0000000c 00           9         123   # yaa! in = 1, out = 1, w_ct =         123, err = 00
# error_counter,err_inj = 0000000c 00           9         124   # yaa! in = 1, out = 1, w_ct =         124, err = 00
# error_counter,err_inj = 0000000d 10           9         125   # yaa! in = 1, out = 1, w_ct =         125, err = 00
# error_counter,err_inj = 0000000e 10           9         126   # yaa! in = 0, out = 0, w_ct =         126, err = 00
# error_counter,err_inj = 0000000f 10          10         127   # yaa! in = 0, out = 0, w_ct =         127, err = 00
# error_counter,err_inj = 00000010 10          11         128   # yaa! in = 0, out = 0, w_ct =         128, err = 00
# error_counter,err_inj = 00000010 00          12         129   # yaa! in = 0, out = 0, w_ct =         129, err = 00
# error_counter,err_inj = 00000010 00          12         130   # yaa! in = 0, out = 0, w_ct =         130, err = 00
# error_counter,err_inj = 00000010 00          12         131   # yaa! in = 0, out = 0, w_ct =         131, err = 00
# error_counter,err_inj = 00000010 00          12         132   # yaa! in = 0, out = 0, w_ct =         132, err = 00
# error_counter,err_inj = 00000010 00          12         133   # yaa! in = 0, out = 0, w_ct =         133, err = 00
# error_counter,err_inj = 00000010 00          12         134   # yaa! in = 0, out = 0, w_ct =         134, err = 00
# error_counter,err_inj = 00000010 00          12         135   # yaa! in = 0, out = 0, w_ct =         135, err = 00
# error_counter,err_inj = 00000010 00          12         136   # yaa! in = 0, out = 0, w_ct =         136, err = 00
# error_counter,err_inj = 00000010 00          12         137   # yaa! in = 1, out = 1, w_ct =         137, err = 00
# error_counter,err_inj = 00000010 00          12         138   # yaa! in = 1, out = 1, w_ct =         138, err = 00
# error_counter,err_inj = 00000010 00          12         139   # yaa! in = 1, out = 1, w_ct =         139, err = 00
# error_counter,err_inj = 00000010 00          12         140   # yaa! in = 1, out = 1, w_ct =         140, err = 00
# error_counter,err_inj = 00000010 00          12         141   # yaa! in = 1, out = 1, w_ct =         141, err = 00
# error_counter,err_inj = 00000010 00          12         142   # yaa! in = 1, out = 1, w_ct =         142, err = 00
# error_counter,err_inj = 00000010 00          12         143   # yaa! in = 1, out = 1, w_ct =         143, err = 00
# error_counter,err_inj = 00000010 00          12         144   # yaa! in = 1, out = 1, w_ct =         144, err = 00
# error_counter,err_inj = 00000010 00          12         145   # yaa! in = 1, out = 1, w_ct =         145, err = 00
# error_counter,err_inj = 00000010 00          12         146   # yaa! in = 1, out = 1, w_ct =         146, err = 00
# error_counter,err_inj = 00000010 00          12         147   # yaa! in = 0, out = 0, w_ct =         147, err = 00
# error_counter,err_inj = 00000010 00          12         148   # yaa! in = 1, out = 1, w_ct =         148, err = 00
# error_counter,err_inj = 00000010 00          12         149   # yaa! in = 1, out = 1, w_ct =         149, err = 00
# error_counter,err_inj = 00000010 00          12         150   # yaa! in = 1, out = 1, w_ct =         150, err = 00
# error_counter,err_inj = 00000010 00          12         151   # yaa! in = 1, out = 1, w_ct =         151, err = 00
# error_counter,err_inj = 00000010 00          12         152   # yaa! in = 1, out = 1, w_ct =         152, err = 00
# error_counter,err_inj = 00000010 00          12         153   # yaa! in = 1, out = 1, w_ct =         153, err = 00
```

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# error_counter,err_inj = 00000010 00          12          154    # yaa! in = 1, out = 1, w_ct =          154, err = 00
# error_counter,err_inj = 00000010 00          12          155    # yaa! in = 1, out = 1, w_ct =          155, err = 00
# error_counter,err_inj = 00000010 00          12          156    # yaa! in = 1, out = 1, w_ct =          156, err = 00
# error_counter,err_inj = 00000011 10          12          157    # yaa! in = 1, out = 1, w_ct =          157, err = 00
# error_counter,err_inj = 00000012 10          12          158    # yaa! in = 1, out = 1, w_ct =          158, err = 00
# error_counter,err_inj = 00000013 10          13          159    # yaa! in = 1, out = 1, w_ct =          159, err = 00
# error_counter,err_inj = 00000014 10          14          160    # yaa! in = 1, out = 1, w_ct =          160, err = 00
# error_counter,err_inj = 00000014 00          15          161    # yaa! in = 1, out = 1, w_ct =          161, err = 00
# error_counter,err_inj = 00000014 00          15          162    # yaa! in = 1, out = 1, w_ct =          162, err = 00
# error_counter,err_inj = 00000014 00          15          163    # yaa! in = 1, out = 1, w_ct =          163, err = 00
# error_counter,err_inj = 00000014 00          15          164    # yaa! in = 1, out = 1, w_ct =          164, err = 00
# error_counter,err_inj = 00000014 00          15          165    # yaa! in = 1, out = 1, w_ct =          165, err = 00
# error_counter,err_inj = 00000014 00          15          166    # yaa! in = 1, out = 1, w_ct =          166, err = 00
# error_counter,err_inj = 00000014 00          15          167    # yaa! in = 1, out = 1, w_ct =          167, err = 00
# error_counter,err_inj = 00000014 00          15          168    # yaa! in = 1, out = 1, w_ct =          168, err = 00
# error_counter,err_inj = 00000014 00          15          169    # yaa! in = 1, out = 1, w_ct =          169, err = 00
# error_counter,err_inj = 00000014 00          15          170    # yaa! in = 1, out = 1, w_ct =          170, err = 00
# error_counter,err_inj = 00000014 00          15          171    # yaa! in = 1, out = 1, w_ct =          171, err = 00
# error_counter,err_inj = 00000014 00          15          172    # yaa! in = 1, out = 1, w_ct =          172, err = 00
# error_counter,err_inj = 00000014 00          15          173    # yaa! in = 1, out = 1, w_ct =          173, err = 00
# error_counter,err_inj = 00000014 00          15          174    # yaa! in = 1, out = 1, w_ct =          174, err = 00
# error_counter,err_inj = 00000014 00          15          175    # yaa! in = 1, out = 1, w_ct =          175, err = 00
# error_counter,err_inj = 00000014 00          15          176    # yaa! in = 1, out = 1, w_ct =          176, err = 00
# error_counter,err_inj = 00000014 00          15          177    # yaa! in = 1, out = 1, w_ct =          177, err = 00
# error_counter,err_inj = 00000014 00          15          178    # yaa! in = 1, out = 1, w_ct =          178, err = 00
# error_counter,err_inj = 00000014 00          15          179    # yaa! in = 1, out = 1, w_ct =          179, err = 00
# error_counter,err_inj = 00000014 00          15          180    # yaa! in = 1, out = 1, w_ct =          180, err = 00
# error_counter,err_inj = 00000014 00          15          181    # yaa! in = 1, out = 1, w_ct =          181, err = 00
# error_counter,err_inj = 00000014 00          15          182    # yaa! in = 1, out = 1, w_ct =          182, err = 00
# error_counter,err_inj = 00000014 00          15          183    # yaa! in = 1, out = 1, w_ct =          183, err = 00
# error_counter,err_inj = 00000014 00          15          184    # yaa! in = 1, out = 1, w_ct =          184, err = 00
# error_counter,err_inj = 00000014 00          15          185    # yaa! in = 1, out = 1, w_ct =          185, err = 00
# error_counter,err_inj = 00000014 00          15          186    # yaa! in = 1, out = 1, w_ct =          186, err = 00
# error_counter,err_inj = 00000014 00          15          187    # yaa! in = 1, out = 1, w_ct =          187, err = 00
# error_counter,err_inj = 00000014 00          15          188    # yaa! in = 1, out = 1, w_ct =          188, err = 00
# error_counter,err_inj = 00000015 10          15          189    # yaa! in = 1, out = 1, w_ct =          189, err = 00
# error_counter,err_inj = 00000016 10          15          190    # yaa! in = 1, out = 1, w_ct =          190, err = 00
# error_counter,err_inj = 00000017 10          16          191    # yaa! in = 1, out = 1, w_ct =          191, err = 00
# error_counter,err_inj = 00000018 10          17          192    # yaa! in = 1, out = 1, w_ct =          192, err = 00
# error_counter,err_inj = 00000018 00          18          193    # yaa! in = 1, out = 1, w_ct =          193, err = 00
# error_counter,err_inj = 00000018 00          18          194    # yaa! in = 1, out = 1, w_ct =          194, err = 00
# error_counter,err_inj = 00000018 00          18          195    # yaa! in = 1, out = 1, w_ct =          195, err = 00
# error_counter,err_inj = 00000018 00          18          196    # yaa! in = 1, out = 1, w_ct =          196, err = 00
# error_counter,err_inj = 00000018 00          18          197    # yaa! in = 1, out = 1, w_ct =          197, err = 00
# error_counter,err_inj = 00000018 00          18          198    # yaa! in = 1, out = 1, w_ct =          198, err = 00
# error_counter,err_inj = 00000018 00          18          199    # yaa! in = 1, out = 1, w_ct =          199, err = 00
# error_counter,err_inj = 00000018 00          18          200    # yaa! in = 1, out = 1, w_ct =          200, err = 00
# error_counter,err_inj = 00000018 00          18          201    # yaa! in = 1, out = 1, w_ct =          201, err = 00
# error_counter,err_inj = 00000018 00          18          202    # yaa! in = 1, out = 1, w_ct =          202, err = 00
# error_counter,err_inj = 00000018 00          18          203    # yaa! in = 1, out = 1, w_ct =          203, err = 00
# error_counter,err_inj = 00000018 00          18          204    # yaa! in = 1, out = 1, w_ct =          204, err = 00
# error_counter,err_inj = 00000018 00          18          205    # yaa! in = 1, out = 1, w_ct =          205, err = 00
# error_counter,err_inj = 00000018 00          18          206    # yaa! in = 1, out = 1, w_ct =          206, err = 00
# error_counter,err_inj = 00000018 00          18          207    # yaa! in = 1, out = 1, w_ct =          207, err = 00
# error_counter,err_inj = 00000018 00          18          208    # yaa! in = 1, out = 1, w_ct =          208, err = 00
# error_counter,err_inj = 00000018 00          18          209    # yaa! in = 1, out = 1, w_ct =          209, err = 00
# error_counter,err_inj = 00000018 00          18          210    # yaa! in = 1, out = 1, w_ct =          210, err = 00
# error_counter,err_inj = 00000018 00          18          211    # yaa! in = 1, out = 1, w_ct =          211, err = 00
# error_counter,err_inj = 00000018 00          18          212    # yaa! in = 1, out = 1, w_ct =          212, err = 00
# error_counter,err_inj = 00000018 00          18          213    # yaa! in = 1, out = 1, w_ct =          213, err = 00
# error_counter,err_inj = 00000018 00          18          214    # yaa! in = 1, out = 1, w_ct =          214, err = 00
# error_counter,err_inj = 00000018 00          18          215    # yaa! in = 1, out = 1, w_ct =          215, err = 00
# error_counter,err_inj = 00000018 00          18          216    # yaa! in = 1, out = 1, w_ct =          216, err = 00
# error_counter,err_inj = 00000018 00          18          217    # yaa! in = 1, out = 1, w_ct =          217, err = 00
# error_counter,err_inj = 00000018 00          18          218    # yaa! in = 1, out = 1, w_ct =          218, err = 00
# error_counter,err_inj = 00000018 00          18          219    # yaa! in = 1, out = 1, w_ct =          219, err = 00
# error_counter,err_inj = 00000018 00          18          220    # yaa! in = 1, out = 1, w_ct =          220, err = 00
# error_counter,err_inj = 00000019 10          18          221    # yaa! in = 1, out = 1, w_ct =          221, err = 00
# error_counter,err_inj = 0000001a 10          18          222    # yaa! in = 1, out = 1, w_ct =          222, err = 00
# error_counter,err_inj = 0000001b 10          19          223    # yaa! in = 1, out = 1, w_ct =          223, err = 00
# error_counter,err_inj = 0000001c 10          20          224    # yaa! in = 1, out = 1, w_ct =          224, err = 00
# error_counter,err_inj = 0000001c 00          21          225    # yaa! in = 1, out = 1, w_ct =          225, err = 00
# error_counter,err_inj = 0000001c 00          21          226    # yaa! in = 1, out = 1, w_ct =          226, err = 00
# error_counter,err_inj = 0000001c 00          21          227    # yaa! in = 1, out = 1, w_ct =          227, err = 00
# error_counter,err_inj = 0000001c 00          21          228    # yaa! in = 1, out = 1, w_ct =          228, err = 00
# error_counter,err_inj = 0000001c 00          21          229    # yaa! in = 1, out = 1, w_ct =          229, err = 00
# error_counter,err_inj = 0000001c 00          21          230    # yaa! in = 1, out = 1, w_ct =          230, err = 00
# error_counter,err_inj = 0000001c 00          21          231    # yaa! in = 1, out = 1, w_ct =          231, err = 00
# error_counter,err_inj = 0000001c 00          21          232    # yaa! in = 1, out = 1, w_ct =          232, err = 00
# error_counter,err_inj = 0000001c 00          21          233    # yaa! in = 1, out = 1, w_ct =          233, err = 00
# error_counter,err_inj = 0000001c 00          21          234    # yaa! in = 1, out = 1, w_ct =          234, err = 00
# error_counter,err_inj = 0000001c 00          21          235    # yaa! in = 1, out = 1, w_ct =          235, err = 00
# error_counter,err_inj = 0000001c 00          21          236    # yaa! in = 1, out = 1, w_ct =          236, err = 00
# error_counter,err_inj = 0000001c 00          21          237    # yaa! in = 1, out = 1, w_ct =          237, err = 00
# error_counter,err_inj = 0000001c 00          21          238    # yaa! in = 1, out = 1, w_ct =          238, err = 00
# error_counter,err_inj = 0000001c 00          21          239    # yaa! in = 1, out = 1, w_ct =          239, err = 00
# error_counter,err_inj = 0000001c 00          21          240    # yaa! in = 1, out = 1, w_ct =          240, err = 00
# error_counter,err_inj = 0000001c 00          21          241    # yaa! in = 1, out = 1, w_ct =          241, err = 00
# error_counter,err_inj = 0000001c 00          21          242    # yaa! in = 1, out = 1, w_ct =          242, err = 00
# error_counter,err_inj = 0000001c 00          21          243    # yaa! in = 1, out = 1, w_ct =          243, err = 00
# error_counter,err_inj = 0000001c 00          21          244    # yaa! in = 1, out = 1, w_ct =          244, err = 00
# error_counter,err_inj = 0000001c 00          21          245    # yaa! in = 1, out = 1, w_ct =          245, err = 00
# error_counter,err_inj = 0000001c 00          21          246    # yaa! in = 1, out = 1, w_ct =          246, err = 00
```

```
# error_counter,err_inj = 0000001c 00          21          247
# error_counter,err_inj = 0000001c 00          21          248
# error_counter,err_inj = 0000001c 00          21          249
# error_counter,err_inj = 0000001c 00          21          250
# error_counter,err_inj = 0000001c 00          21          251
# error_counter,err_inj = 0000001c 00          21          252
# error_counter,err_inj = 0000001d 10          21          253
# error_counter,err_inj = 0000001e 10          21          254
# error_counter,err_inj = 0000001f 10          22          255
```

```
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

2a8) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000000 00          0          29
# error_counter,err_inj = 00000000 00          0          30
# error_counter,err_inj = 00000001 11          0          31
# error_counter,err_inj = 00000002 11          0          32
# error_counter,err_inj = 00000002 00          2          33
# error_counter,err_inj = 00000002 00          2          34
# error_counter,err_inj = 00000002 00          2          35
# error_counter,err_inj = 00000002 00          2          36
# error_counter,err_inj = 00000002 00          2          37
# error_counter,err_inj = 00000002 00          2          38
# error_counter,err_inj = 00000002 00          2          39
# error_counter,err_inj = 00000002 00          2          40
# error_counter,err_inj = 00000002 00          2          41
# error_counter,err_inj = 00000002 00          2          42
# error_counter,err_inj = 00000002 00          2          43
# error_counter,err_inj = 00000002 00          2          44
# error_counter,err_inj = 00000002 00          2          45
# error_counter,err_inj = 00000002 00          2          46
# error_counter,err_inj = 00000002 00          2          47
# error_counter,err_inj = 00000002 00          2          48
# error_counter,err_inj = 00000002 00          2          49
# error_counter,err_inj = 00000002 00          2          50
# error_counter,err_inj = 00000002 00          2          51
# error_counter,err_inj = 00000002 00          2          52
# error_counter,err_inj = 00000002 00          2          53
# error_counter,err_inj = 00000002 00          2          54
# error_counter,err_inj = 00000002 00          2          55
# error_counter,err_inj = 00000002 00          2          56
# error_counter,err_inj = 00000002 00          2          57
# error_counter,err_inj = 00000002 00          2          58
# error_counter,err_inj = 00000002 00          2          59
# error_counter,err_inj = 00000002 00          2          60
# error_counter,err_inj = 00000002 00          2          61
# error_counter,err_inj = 00000002 00          2          62
# error_counter,err_inj = 00000003 11          2          63
# error_counter,err_inj = 00000004 11          2          64
# error_counter,err_inj = 00000004 00          4          65
# error_counter,err_inj = 00000004 00          4          66
# error_counter,err_inj = 00000004 00          4          67
# error_counter,err_inj = 00000004 00          4          68
# error_counter,err_inj = 00000004 00          4          69
# error_counter,err_inj = 00000004 00          4          70
# error_counter,err_inj = 00000004 00          4          71
# error_counter,err_inj = 00000004 00          4          72
# error_counter,err_inj = 00000004 00          4          73
# error_counter,err_inj = 00000004 00          4          74
# error_counter,err_inj = 00000004 00          4          75
# error_counter,err_inj = 00000004 00          4          76
# error_counter,err_inj = 00000004 00          4          77
```

```
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
```

```
# error_counter,err_inj = 00000004 00          4          78
# error_counter,err_inj = 00000004 00          4          79
# error_counter,err_inj = 00000004 00          4          80
# error_counter,err_inj = 00000004 00          4          81
# error_counter,err_inj = 00000004 00          4          82
# error_counter,err_inj = 00000004 00          4          83
# error_counter,err_inj = 00000004 00          4          84
# error_counter,err_inj = 00000004 00          4          85
# error_counter,err_inj = 00000004 00          4          86
# error_counter,err_inj = 00000004 00          4          87
# error_counter,err_inj = 00000004 00          4          88
# error_counter,err_inj = 00000004 00          4          89
# error_counter,err_inj = 00000004 00          4          90
# error_counter,err_inj = 00000004 00          4          91
# error_counter,err_inj = 00000004 00          4          92
# error_counter,err_inj = 00000004 00          4          93
# error_counter,err_inj = 00000004 00          4          94
# error_counter,err_inj = 00000005 11          4          95
# error_counter,err_inj = 00000006 11          4          96
# error_counter,err_inj = 00000006 00          6          97
# error_counter,err_inj = 00000006 00          6          98
# error_counter,err_inj = 00000006 00          6          99
# error_counter,err_inj = 00000006 00          6         100
# error_counter,err_inj = 00000006 00          6         101
# error_counter,err_inj = 00000006 00          6         102
# error_counter,err_inj = 00000006 00          6         103
# error_counter,err_inj = 00000006 00          6         104
# error_counter,err_inj = 00000006 00          6         105
# error_counter,err_inj = 00000006 00          6         106
# error_counter,err_inj = 00000006 00          6         107
# error_counter,err_inj = 00000006 00          6         108
# error_counter,err_inj = 00000006 00          6         109
# error_counter,err_inj = 00000006 00          6         110
# error_counter,err_inj = 00000006 00          6         111
# error_counter,err_inj = 00000006 00          6         112
# error_counter,err_inj = 00000006 00          6         113
# error_counter,err_inj = 00000006 00          6         114
# error_counter,err_inj = 00000006 00          6         115
# error_counter,err_inj = 00000006 00          6         116
# error_counter,err_inj = 00000006 00          6         117
# error_counter,err_inj = 00000006 00          6         118
# error_counter,err_inj = 00000006 00          6         119
# error_counter,err_inj = 00000006 00          6         120
# error_counter,err_inj = 00000006 00          6         121
# error_counter,err_inj = 00000006 00          6         122
# error_counter,err_inj = 00000006 00          6         123
# error_counter,err_inj = 00000006 00          6         124
# error_counter,err_inj = 00000006 00          6         125
# error_counter,err_inj = 00000006 00          6         126
# error_counter,err_inj = 00000007 11          6         127
# error_counter,err_inj = 00000008 11          6         128
# error_counter,err_inj = 00000008 00          8         129
# error_counter,err_inj = 00000008 00          8         130
# error_counter,err_inj = 00000008 00          8         131
# error_counter,err_inj = 00000008 00          8         132
# error_counter,err_inj = 00000008 00          8         133
# error_counter,err_inj = 00000008 00          8         134
# error_counter,err_inj = 00000008 00          8         135
# error_counter,err_inj = 00000008 00          8         136
# error_counter,err_inj = 00000008 00          8         137
# error_counter,err_inj = 00000008 00          8         138
# error_counter,err_inj = 00000008 00          8         139
# error_counter,err_inj = 00000008 00          8         140
# error_counter,err_inj = 00000008 00          8         141
# error_counter,err_inj = 00000008 00          8         142
# error_counter,err_inj = 00000008 00          8         143
# error_counter,err_inj = 00000008 00          8         144
# error_counter,err_inj = 00000008 00          8         145
# error_counter,err_inj = 00000008 00          8         146
# error_counter,err_inj = 00000008 00          8         147
# error_counter,err_inj = 00000008 00          8         148
# error_counter,err_inj = 00000008 00          8         149
# error_counter,err_inj = 00000008 00          8         150
# error_counter,err_inj = 00000008 00          8         151
# error_counter,err_inj = 00000008 00          8         152
# error_counter,err_inj = 00000008 00          8         153
# error_counter,err_inj = 00000008 00          8         154
# error_counter,err_inj = 00000008 00          8         155
# error_counter,err_inj = 00000008 00          8         156
# error_counter,err_inj = 00000008 00          8         157
# error_counter,err_inj = 00000008 00          8         158
# error_counter,err_inj = 00000009 11          8         159
# error_counter,err_inj = 0000000a 11          8         160
# error_counter,err_inj = 0000000a 00         10         161
# error_counter,err_inj = 0000000a 00         10         162
# error_counter,err_inj = 0000000a 00         10         163
# error_counter,err_inj = 0000000a 00         10         164
# error_counter,err_inj = 0000000a 00         10         165
# error_counter,err_inj = 0000000a 00         10         166
# error_counter,err_inj = 0000000a 00         10         167
# error_counter,err_inj = 0000000a 00         10         168
# error_counter,err_inj = 0000000a 00         10         169
# error_counter,err_inj = 0000000a 00         10         170
```

```
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000a 00          10          171
# error_counter,err_inj = 0000000a 00          10          172
# error_counter,err_inj = 0000000a 00          10          173
# error_counter,err_inj = 0000000a 00          10          174
# error_counter,err_inj = 0000000a 00          10          175
# error_counter,err_inj = 0000000a 00          10          176
# error_counter,err_inj = 0000000a 00          10          177
# error_counter,err_inj = 0000000a 00          10          178
# error_counter,err_inj = 0000000a 00          10          179
# error_counter,err_inj = 0000000a 00          10          180
# error_counter,err_inj = 0000000a 00          10          181
# error_counter,err_inj = 0000000a 00          10          182
# error_counter,err_inj = 0000000a 00          10          183
# error_counter,err_inj = 0000000a 00          10          184
# error_counter,err_inj = 0000000a 00          10          185
# error_counter,err_inj = 0000000a 00          10          186
# error_counter,err_inj = 0000000a 00          10          187
# error_counter,err_inj = 0000000a 00          10          188
# error_counter,err_inj = 0000000a 00          10          189
# error_counter,err_inj = 0000000a 00          10          190
# error_counter,err_inj = 0000000b 11          10          191
# error_counter,err_inj = 0000000c 11          10          192
# error_counter,err_inj = 0000000c 00          12          193
# error_counter,err_inj = 0000000c 00          12          194
# error_counter,err_inj = 0000000c 00          12          195
# error_counter,err_inj = 0000000c 00          12          196
# error_counter,err_inj = 0000000c 00          12          197
# error_counter,err_inj = 0000000c 00          12          198
# error_counter,err_inj = 0000000c 00          12          199
# error_counter,err_inj = 0000000c 00          12          200
# error_counter,err_inj = 0000000c 00          12          201
# error_counter,err_inj = 0000000c 00          12          202
# error_counter,err_inj = 0000000c 00          12          203
# error_counter,err_inj = 0000000c 00          12          204
# error_counter,err_inj = 0000000c 00          12          205
# error_counter,err_inj = 0000000c 00          12          206
# error_counter,err_inj = 0000000c 00          12          207
# error_counter,err_inj = 0000000c 00          12          208
# error_counter,err_inj = 0000000c 00          12          209
# error_counter,err_inj = 0000000c 00          12          210
# error_counter,err_inj = 0000000c 00          12          211
# error_counter,err_inj = 0000000c 00          12          212
# error_counter,err_inj = 0000000c 00          12          213
# error_counter,err_inj = 0000000c 00          12          214
# error_counter,err_inj = 0000000c 00          12          215
# error_counter,err_inj = 0000000c 00          12          216
# error_counter,err_inj = 0000000c 00          12          217
# error_counter,err_inj = 0000000c 00          12          218
# error_counter,err_inj = 0000000c 00          12          219
# error_counter,err_inj = 0000000c 00          12          220
# error_counter,err_inj = 0000000c 00          12          221
# error_counter,err_inj = 0000000c 00          12          222
# error_counter,err_inj = 0000000d 11          12          223
# error_counter,err_inj = 0000000e 11          12          224
# error_counter,err_inj = 0000000e 00          14          225
# error_counter,err_inj = 0000000e 00          14          226
# error_counter,err_inj = 0000000e 00          14          227
# error_counter,err_inj = 0000000e 00          14          228
# error_counter,err_inj = 0000000e 00          14          229
# error_counter,err_inj = 0000000e 00          14          230
# error_counter,err_inj = 0000000e 00          14          231
# error_counter,err_inj = 0000000e 00          14          232
# error_counter,err_inj = 0000000e 00          14          233
# error_counter,err_inj = 0000000e 00          14          234
# error_counter,err_inj = 0000000e 00          14          235
# error_counter,err_inj = 0000000e 00          14          236
# error_counter,err_inj = 0000000e 00          14          237
# error_counter,err_inj = 0000000e 00          14          238
# error_counter,err_inj = 0000000e 00          14          239
# error_counter,err_inj = 0000000e 00          14          240
# error_counter,err_inj = 0000000e 00          14          241
# error_counter,err_inj = 0000000e 00          14          242
# error_counter,err_inj = 0000000e 00          14          243
# error_counter,err_inj = 0000000e 00          14          244
# error_counter,err_inj = 0000000e 00          14          245
# error_counter,err_inj = 0000000e 00          14          246
# error_counter,err_inj = 0000000e 00          14          247
# error_counter,err_inj = 0000000e 00          14          248
# error_counter,err_inj = 0000000e 00          14          249
# error_counter,err_inj = 0000000e 00          14          250
# error_counter,err_inj = 0000000e 00          14          251
# error_counter,err_inj = 0000000e 00          14          252
# error_counter,err_inj = 0000000e 00          14          253
# error_counter,err_inj = 0000000e 00          14          254
# error_counter,err_inj = 0000000f 11          14          255
# word_count =       10440
```

2b) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
```

```
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
```

```
# error_counter,err_inj = 00000001 01          0           2
# error_counter,err_inj = 00000001 00          0           3
# error_counter,err_inj = 00000002 00          0           4
# error_counter,err_inj = 00000002 00          0           5
# error_counter,err_inj = 00000002 00          0           6
# error_counter,err_inj = 00000002 00          0           7
# error_counter,err_inj = 00000002 00          0           8
# error_counter,err_inj = 00000002 00          0           9
# error_counter,err_inj = 00000002 00          0          10
# error_counter,err_inj = 00000002 00          0          11
# error_counter,err_inj = 00000002 00          0          12
# error_counter,err_inj = 00000002 00          0          13
# error_counter,err_inj = 00000002 00          0          14
# error_counter,err_inj = 00000002 00          0          15
# error_counter,err_inj = 00000002 00          0          16
# error_counter,err_inj = 00000002 00          0          17
# error_counter,err_inj = 00000002 00          0          18
# error_counter,err_inj = 00000002 00          0          19
# error_counter,err_inj = 00000002 00          0          20
# error_counter,err_inj = 00000002 00          0          21
# error_counter,err_inj = 00000002 00          0          22
# error_counter,err_inj = 00000002 00          0          23
# error_counter,err_inj = 00000002 00          0          24
# error_counter,err_inj = 00000003 01          0          25
# error_counter,err_inj = 00000003 00          0          26
# error_counter,err_inj = 00000003 00          0          27
# error_counter,err_inj = 00000003 00          0          28
# error_counter,err_inj = 00000003 00          0          29
# error_counter,err_inj = 00000003 00          0          30
# error_counter,err_inj = 00000003 00          0          31
# error_counter,err_inj = 00000003 00          0          32
# error_counter,err_inj = 00000004 01          0          33
# error_counter,err_inj = 00000004 00          0          34
# error_counter,err_inj = 00000004 00          0          35
# error_counter,err_inj = 00000004 00          0          36
# error_counter,err_inj = 00000004 00          0          37
# error_counter,err_inj = 00000004 00          0          38
# error_counter,err_inj = 00000004 00          0          39
# error_counter,err_inj = 00000004 00          0          40
# error_counter,err_inj = 00000004 00          0          41
# error_counter,err_inj = 00000004 00          0          42
# error_counter,err_inj = 00000004 00          0          43
# error_counter,err_inj = 00000004 00          0          44
# error_counter,err_inj = 00000004 00          0          45
# error_counter,err_inj = 00000004 00          0          46
# error_counter,err_inj = 00000004 00          0          47
# error_counter,err_inj = 00000004 00          0          48
# error_counter,err_inj = 00000004 00          0          49
# error_counter,err_inj = 00000004 00          0          50
# error_counter,err_inj = 00000005 11          0          51
# error_counter,err_inj = 00000005 00          0          52
# error_counter,err_inj = 00000005 00          0          53
# error_counter,err_inj = 00000005 00          0          54
# error_counter,err_inj = 00000005 00          0          55
# error_counter,err_inj = 00000005 00          0          56
# error_counter,err_inj = 00000005 00          0          57
# error_counter,err_inj = 00000005 00          0          58
# error_counter,err_inj = 00000005 00          0          59
# error_counter,err_inj = 00000005 00          0          60
# error_counter,err_inj = 00000006 01          0          61
# error_counter,err_inj = 00000006 00          0          62
# error_counter,err_inj = 00000006 00          0          63
# error_counter,err_inj = 00000006 00          0          64
# error_counter,err_inj = 00000006 00          0          65
# error_counter,err_inj = 00000006 00          0          66
# error_counter,err_inj = 00000006 00          0          67
# error_counter,err_inj = 00000006 00          0          68
# error_counter,err_inj = 00000006 00          0          69
# error_counter,err_inj = 00000007 10          0          70
# error_counter,err_inj = 00000007 00          0          71
# error_counter,err_inj = 00000007 00          0          72
# error_counter,err_inj = 00000007 00          0          73
# error_counter,err_inj = 00000007 00          0          74
# error_counter,err_inj = 00000007 00          0          75
# error_counter,err_inj = 00000007 00          0          76
# error_counter,err_inj = 00000007 00          0          77
# error_counter,err_inj = 00000007 00          0          78
# error_counter,err_inj = 00000007 00          0          79
# error_counter,err_inj = 00000007 00          0          80
# error_counter,err_inj = 00000007 00          0          81
# error_counter,err_inj = 00000007 00          0          82
# error_counter,err_inj = 00000007 00          0          83
# error_counter,err_inj = 00000007 00          0          84
# error_counter,err_inj = 00000007 00          0          85
# error_counter,err_inj = 00000007 00          0          86
# error_counter,err_inj = 00000008 11          0          87
# error_counter,err_inj = 00000008 00          0          88
# error_counter,err_inj = 00000008 00          0          89
# error_counter,err_inj = 00000008 00          0          90
# error_counter,err_inj = 00000008 00          0          91
# error_counter,err_inj = 00000008 00          0          92
# error_counter,err_inj = 00000008 00          0          93
# error_counter,err_inj = 00000008 00          0          94
```

```
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
```

```
# error_counter,err_inj = 00000008 00          0          95
# error_counter,err_inj = 00000008 00          0          96
# error_counter,err_inj = 00000008 00          0          97
# error_counter,err_inj = 00000008 00          0          98
# error_counter,err_inj = 00000008 00          0          99
# error_counter,err_inj = 00000008 00          0         100
# error_counter,err_inj = 00000008 00          0         101
# error_counter,err_inj = 00000008 00          0         102
# error_counter,err_inj = 00000008 00          0         103
# error_counter,err_inj = 00000008 00          0         104
# error_counter,err_inj = 00000008 00          0         105
# error_counter,err_inj = 00000008 00          0         106
# error_counter,err_inj = 00000008 00          0         107
# error_counter,err_inj = 00000008 00          0         108
# error_counter,err_inj = 00000008 00          0         109
# error_counter,err_inj = 00000008 00          0         110
# error_counter,err_inj = 00000008 00          0         111
# error_counter,err_inj = 00000008 00          0         112
# error_counter,err_inj = 00000009 01          0         113
# error_counter,err_inj = 00000009 00          0         114
# error_counter,err_inj = 00000009 00          0         115
# error_counter,err_inj = 00000009 00          0         116
# error_counter,err_inj = 00000009 00          0         117
# error_counter,err_inj = 00000009 00          0         118
# error_counter,err_inj = 00000009 00          0         119
# error_counter,err_inj = 00000009 00          0         120
# error_counter,err_inj = 00000009 00          0         121
# error_counter,err_inj = 00000009 00          0         122
# error_counter,err_inj = 00000009 00          0         123
# error_counter,err_inj = 00000009 00          0         124
# error_counter,err_inj = 00000009 00          0         125
# error_counter,err_inj = 00000009 00          0         126
# error_counter,err_inj = 00000009 00          0         127
# error_counter,err_inj = 00000009 00          0         128
# error_counter,err_inj = 00000009 00          0         129
# error_counter,err_inj = 00000009 00          0         130
# error_counter,err_inj = 00000009 00          0         131
# error_counter,err_inj = 00000009 00          0         132
# error_counter,err_inj = 00000009 00          0         133
# error_counter,err_inj = 00000009 00          0         134
# error_counter,err_inj = 0000000a 11          0         135
# error_counter,err_inj = 0000000a 00          0         136
# error_counter,err_inj = 0000000a 00          0         137
# error_counter,err_inj = 0000000a 00          0         138
# error_counter,err_inj = 0000000a 00          0         139
# error_counter,err_inj = 0000000a 00          0         140
# error_counter,err_inj = 0000000a 00          0         141
# error_counter,err_inj = 0000000a 00          0         142
# error_counter,err_inj = 0000000a 00          0         143
# error_counter,err_inj = 0000000a 00          0         144
# error_counter,err_inj = 0000000a 00          0         145
# error_counter,err_inj = 0000000a 00          0         146
# error_counter,err_inj = 0000000a 00          0         147
# error_counter,err_inj = 0000000a 00          0         148
# error_counter,err_inj = 0000000a 00          0         149
# error_counter,err_inj = 0000000a 00          0         150
# error_counter,err_inj = 0000000a 00          0         151
# error_counter,err_inj = 0000000a 00          0         152
# error_counter,err_inj = 0000000a 00          0         153
# error_counter,err_inj = 0000000a 00          0         154
# error_counter,err_inj = 0000000a 00          0         155
# error_counter,err_inj = 0000000a 00          0         156
# error_counter,err_inj = 0000000a 00          0         157
# error_counter,err_inj = 0000000a 00          0         158
# error_counter,err_inj = 0000000b 10          0         159
# error_counter,err_inj = 0000000b 00          0         160
# error_counter,err_inj = 0000000b 00          0         161
# error_counter,err_inj = 0000000b 00          0         162
# error_counter,err_inj = 0000000b 00          0         163
# error_counter,err_inj = 0000000b 00          0         164
# error_counter,err_inj = 0000000b 00          0         165
# error_counter,err_inj = 0000000b 00          0         166
# error_counter,err_inj = 0000000c 10          0         167
# error_counter,err_inj = 0000000c 00          0         168
# error_counter,err_inj = 0000000c 00          0         169
# error_counter,err_inj = 0000000c 00          0         170
# error_counter,err_inj = 0000000c 00          0         171
# error_counter,err_inj = 0000000c 00          0         172
# error_counter,err_inj = 0000000c 00          0         173
# error_counter,err_inj = 0000000c 00          0         174
# error_counter,err_inj = 0000000c 00          0         175
# error_counter,err_inj = 0000000c 00          0         176
# error_counter,err_inj = 0000000c 00          0         177
# error_counter,err_inj = 0000000d 00          0         178
# error_counter,err_inj = 0000000d 00          0         179
# error_counter,err_inj = 0000000d 00          0         180
# error_counter,err_inj = 0000000d 00          0         181
# error_counter,err_inj = 0000000d 00          0         182
# error_counter,err_inj = 0000000d 00          0         183
# error_counter,err_inj = 0000000d 00          0         184
# error_counter,err_inj = 0000000d 00          0         185
# error_counter,err_inj = 0000000d 00          0         186
# error_counter,err_inj = 0000000d 00          0         187
```

```
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          18, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000d 00          0         188
# error_counter,err_inj = 0000000d 00          0         189
# error_counter,err_inj = 0000000d 00          0         190
# error_counter,err_inj = 0000000d 00          0         191
# error_counter,err_inj = 0000000d 00          0         192
# error_counter,err_inj = 0000000d 00          0         193
# error_counter,err_inj = 0000000d 00          0         194
# error_counter,err_inj = 0000000d 00          0         195
# error_counter,err_inj = 0000000d 00          0         196
# error_counter,err_inj = 0000000d 00          0         197
# error_counter,err_inj = 0000000d 00          0         198
# error_counter,err_inj = 0000000d 00          0         199
# error_counter,err_inj = 0000000d 00          0         200
# error_counter,err_inj = 0000000d 00          0         201
# error_counter,err_inj = 0000000d 00          0         202
# error_counter,err_inj = 0000000d 00          0         203
# error_counter,err_inj = 0000000d 00          0         204
# error_counter,err_inj = 0000000d 00          0         205
# error_counter,err_inj = 0000000d 00          0         206
# error_counter,err_inj = 0000000d 00          0         207
# error_counter,err_inj = 0000000d 00          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000e 11          0         213
# error_counter,err_inj = 0000000e 00          0         214
# error_counter,err_inj = 0000000e 00          0         215
# error_counter,err_inj = 0000000e 00          0         216
# error_counter,err_inj = 0000000e 00          0         217
# error_counter,err_inj = 0000000e 00          0         218
# error_counter,err_inj = 0000000f 11          0         219
# error_counter,err_inj = 0000000f 00          0         220
# error_counter,err_inj = 0000000f 00          0         221
# error_counter,err_inj = 00000010 01          0         222
# error_counter,err_inj = 00000010 00          0         223
# error_counter,err_inj = 00000010 00          0         224
# error_counter,err_inj = 00000010 00          0         225
# error_counter,err_inj = 00000010 00          0         226
# error_counter,err_inj = 00000010 00          0         227
# error_counter,err_inj = 00000010 00          0         228
# error_counter,err_inj = 00000010 00          0         229
# error_counter,err_inj = 00000010 00          0         230
# error_counter,err_inj = 00000010 00          0         231
# error_counter,err_inj = 00000010 00          0         232
# error_counter,err_inj = 00000010 00          0         233
# error_counter,err_inj = 00000010 00          0         234
# error_counter,err_inj = 00000010 00          0         235
# error_counter,err_inj = 00000011 10          0         236
# error_counter,err_inj = 00000011 00          0         237
# error_counter,err_inj = 00000011 00          0         238
# error_counter,err_inj = 00000011 00          0         239
# error_counter,err_inj = 00000011 00          0         240
# error_counter,err_inj = 00000011 00          0         241
# error_counter,err_inj = 00000011 00          0         242
# error_counter,err_inj = 00000011 00          0         243
# error_counter,err_inj = 00000011 00          0         244
# error_counter,err_inj = 00000011 00          0         245
# error_counter,err_inj = 00000011 00          0         246
# error_counter,err_inj = 00000011 00          0         247
# error_counter,err_inj = 00000011 00          0         248
# error_counter,err_inj = 00000011 00          0         249
# error_counter,err_inj = 00000011 00          0         250
# error_counter,err_inj = 00000011 00          0         251
# error_counter,err_inj = 00000012 00          0         252
# error_counter,err_inj = 00000012 00          0         253
# error_counter,err_inj = 00000012 00          0         254
# error_counter,err_inj = 00000012 00          0         255
```

2b1, 2b2) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000001 01          0           2
# error_counter,err_inj = 00000001 00          0           3
# error_counter,err_inj = 00000002 00          0           4
# error_counter,err_inj = 00000002 00          0           5
# error_counter,err_inj = 00000003 11          0           6
# error_counter,err_inj = 00000003 00          0           7
# error_counter,err_inj = 00000003 00          0           8
# error_counter,err_inj = 00000003 00          0           9
# error_counter,err_inj = 00000003 00          0          10
# error_counter,err_inj = 00000003 00          0          11
# error_counter,err_inj = 00000003 00          0          12
# error_counter,err_inj = 00000003 00          0          13
# error_counter,err_inj = 00000003 00          0          14
# error_counter,err_inj = 00000003 00          0          15
# error_counter,err_inj = 00000003 00          0          16
# error_counter,err_inj = 00000003 00          0          17
# error_counter,err_inj = 00000003 00          0          18
```

```
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
```

```
# error_counter,err_inj = 00000003 00          0          19
# error_counter,err_inj = 00000003 00          0          20
# error_counter,err_inj = 00000003 00          0          21
# error_counter,err_inj = 00000003 00          0          22
# error_counter,err_inj = 00000003 00          0          23
# error_counter,err_inj = 00000003 00          0          24
# error_counter,err_inj = 00000004 01          0          25
# error_counter,err_inj = 00000004 00          0          26
# error_counter,err_inj = 00000004 00          0          27
# error_counter,err_inj = 00000004 00          0          28
# error_counter,err_inj = 00000004 00          0          29
# error_counter,err_inj = 00000005 00          0          30
# error_counter,err_inj = 00000005 00          0          31
# error_counter,err_inj = 00000005 00          0          32
# error_counter,err_inj = 00000006 01          0          33
# error_counter,err_inj = 00000006 00          0          34
# error_counter,err_inj = 00000007 11          0          35
# error_counter,err_inj = 00000007 00          0          36
# error_counter,err_inj = 00000007 00          0          37
# error_counter,err_inj = 00000007 00          0          38
# error_counter,err_inj = 00000007 00          0          39
# error_counter,err_inj = 00000007 00          0          40
# error_counter,err_inj = 00000007 00          0          41
# error_counter,err_inj = 00000007 00          0          42
# error_counter,err_inj = 00000008 00          0          43
# error_counter,err_inj = 00000008 00          0          44
# error_counter,err_inj = 00000008 00          0          45
# error_counter,err_inj = 00000008 00          0          46
# error_counter,err_inj = 00000008 00          0          47
# error_counter,err_inj = 00000008 00          0          48
# error_counter,err_inj = 00000008 00          0          49
# error_counter,err_inj = 00000008 00          0          50
# error_counter,err_inj = 00000009 11          0          51
# error_counter,err_inj = 00000009 00          0          52
# error_counter,err_inj = 00000009 00          0          53
# error_counter,err_inj = 00000009 00          0          54
# error_counter,err_inj = 00000009 00          0          55
# error_counter,err_inj = 00000009 00          0          56
# error_counter,err_inj = 00000009 00          0          57
# error_counter,err_inj = 00000009 00          0          58
# error_counter,err_inj = 00000009 00          0          59
# error_counter,err_inj = 00000009 00          0          60
# error_counter,err_inj = 0000000a 01          0          61
# error_counter,err_inj = 0000000a 00          0          62
# error_counter,err_inj = 0000000a 00          0          63
# error_counter,err_inj = 0000000a 00          0          64
# error_counter,err_inj = 0000000a 00          0          65
# error_counter,err_inj = 0000000a 00          0          66
# error_counter,err_inj = 0000000a 00          0          67
# error_counter,err_inj = 0000000a 00          0          68
# error_counter,err_inj = 0000000a 00          0          69
# error_counter,err_inj = 0000000b 10          0          70
# error_counter,err_inj = 0000000b 00          0          71
# error_counter,err_inj = 0000000b 00          0          72
# error_counter,err_inj = 0000000b 00          0          73
# error_counter,err_inj = 0000000b 00          0          74
# error_counter,err_inj = 0000000b 00          0          75
# error_counter,err_inj = 0000000b 00          0          76
# error_counter,err_inj = 0000000b 00          0          77
# error_counter,err_inj = 0000000b 00          0          78
# error_counter,err_inj = 0000000b 00          0          79
# error_counter,err_inj = 0000000b 00          0          80
# error_counter,err_inj = 0000000b 00          0          81
# error_counter,err_inj = 0000000c 11          0          82
# error_counter,err_inj = 0000000c 00          0          83
# error_counter,err_inj = 0000000c 00          0          84
# error_counter,err_inj = 0000000c 00          0          85
# error_counter,err_inj = 0000000c 00          0          86
# error_counter,err_inj = 0000000d 11          0          87
# error_counter,err_inj = 0000000d 00          0          88
# error_counter,err_inj = 0000000d 00          0          89
# error_counter,err_inj = 0000000d 00          0          90
# error_counter,err_inj = 0000000d 00          0          91
# error_counter,err_inj = 0000000d 00          0          92
# error_counter,err_inj = 0000000d 00          0          93
# error_counter,err_inj = 0000000d 00          0          94
# error_counter,err_inj = 0000000d 00          0          95
# error_counter,err_inj = 0000000d 00          0          96
# error_counter,err_inj = 0000000d 00          0          97
# error_counter,err_inj = 0000000d 00          0          98
# error_counter,err_inj = 0000000d 00          0          99
# error_counter,err_inj = 0000000d 00          0         100
# error_counter,err_inj = 0000000d 00          0         101
# error_counter,err_inj = 0000000d 00          0         102
# error_counter,err_inj = 0000000d 00          0         103
# error_counter,err_inj = 0000000d 00          0         104
# error_counter,err_inj = 0000000d 00          0         105
# error_counter,err_inj = 0000000d 00          0         106
# error_counter,err_inj = 0000000d 00          0         107
# error_counter,err_inj = 0000000d 00          0         108
# error_counter,err_inj = 0000000d 00          0         109
# error_counter,err_inj = 0000000d 00          0         110
# error_counter,err_inj = 0000000d 00          0         111
```

```
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
```

```
# error_counter,err_inj = 0000000d 00          0         112
# error_counter,err_inj = 0000000e 01          0         113
# error_counter,err_inj = 0000000e 00          0         114
# error_counter,err_inj = 0000000e 00          0         115
# error_counter,err_inj = 0000000e 00          0         116
# error_counter,err_inj = 0000000e 00          0         117
# error_counter,err_inj = 0000000e 00          0         118
# error_counter,err_inj = 0000000e 00          0         119
# error_counter,err_inj = 0000000e 00          0         120
# error_counter,err_inj = 0000000e 00          0         121
# error_counter,err_inj = 0000000e 00          0         122
# error_counter,err_inj = 0000000e 00          0         123
# error_counter,err_inj = 0000000e 00          0         124
# error_counter,err_inj = 0000000f 11          0         125
# error_counter,err_inj = 0000000f 00          0         126
# error_counter,err_inj = 0000000f 00          0         127
# error_counter,err_inj = 0000000f 00          0         128
# error_counter,err_inj = 0000000f 00          0         129
# error_counter,err_inj = 0000000f 00          0         130
# error_counter,err_inj = 0000000f 00          0         131
# error_counter,err_inj = 0000000f 00          0         132
# error_counter,err_inj = 0000000f 00          0         133
# error_counter,err_inj = 0000000f 00          0         134
# error_counter,err_inj = 00000010 11          0         135
# error_counter,err_inj = 00000010 00          0         136
# error_counter,err_inj = 00000010 00          0         137
# error_counter,err_inj = 00000010 00          0         138
# error_counter,err_inj = 00000010 00          0         139
# error_counter,err_inj = 00000010 00          0         140
# error_counter,err_inj = 00000011 01          0         141
# error_counter,err_inj = 00000011 00          0         142
# error_counter,err_inj = 00000011 00          0         143
# error_counter,err_inj = 00000011 00          0         144
# error_counter,err_inj = 00000011 00          0         145
# error_counter,err_inj = 00000011 00          0         146
# error_counter,err_inj = 00000011 00          0         147
# error_counter,err_inj = 00000011 00          0         148
# error_counter,err_inj = 00000011 00          0         149
# error_counter,err_inj = 00000011 00          0         150
# error_counter,err_inj = 00000011 00          0         151
# error_counter,err_inj = 00000011 00          0         152
# error_counter,err_inj = 00000011 00          0         153
# error_counter,err_inj = 00000011 00          0         154
# error_counter,err_inj = 00000011 00          0         155
# error_counter,err_inj = 00000011 00          0         156
# error_counter,err_inj = 00000011 00          0         157
# error_counter,err_inj = 00000011 00          0         158
# error_counter,err_inj = 00000012 10          0         159
# error_counter,err_inj = 00000012 00          0         160
# error_counter,err_inj = 00000012 00          0         161
# error_counter,err_inj = 00000012 00          0         162
# error_counter,err_inj = 00000012 00          0         163
# error_counter,err_inj = 00000012 00          0         164
# error_counter,err_inj = 00000012 00          0         165
# error_counter,err_inj = 00000012 00          0         166
# error_counter,err_inj = 00000013 10          0         167
# error_counter,err_inj = 00000013 00          0         168
# error_counter,err_inj = 00000013 00          0         169
# error_counter,err_inj = 00000013 00          0         170
# error_counter,err_inj = 00000013 00          0         171
# error_counter,err_inj = 00000013 00          0         172
# error_counter,err_inj = 00000013 00          0         173
# error_counter,err_inj = 00000013 00          0         174
# error_counter,err_inj = 00000013 00          0         175
# error_counter,err_inj = 00000013 00          0         176
# error_counter,err_inj = 00000013 00          0         177
# error_counter,err_inj = 00000014 00          0         178
# error_counter,err_inj = 00000014 00          0         179
# error_counter,err_inj = 00000014 00          0         180
# error_counter,err_inj = 00000014 00          0         181
# error_counter,err_inj = 00000014 00          0         182
# error_counter,err_inj = 00000014 00          0         183
# error_counter,err_inj = 00000014 00          0         184
# error_counter,err_inj = 00000014 00          0         185
# error_counter,err_inj = 00000015 11          0         186
# error_counter,err_inj = 00000015 00          0         187
# error_counter,err_inj = 00000015 00          0         188
# error_counter,err_inj = 00000015 00          0         189
# error_counter,err_inj = 00000015 00          0         190
# error_counter,err_inj = 00000015 00          0         191
# error_counter,err_inj = 00000015 00          0         192
# error_counter,err_inj = 00000015 00          0         193
# error_counter,err_inj = 00000015 00          0         194
# error_counter,err_inj = 00000015 00          0         195
# error_counter,err_inj = 00000015 00          0         196
# error_counter,err_inj = 00000015 00          0         197
# error_counter,err_inj = 00000015 00          0         198
# error_counter,err_inj = 00000015 00          0         199
# error_counter,err_inj = 00000015 00          0         200
# error_counter,err_inj = 00000015 00          0         201
# error_counter,err_inj = 00000015 00          0         202
# error_counter,err_inj = 00000015 00          0         203
# error_counter,err_inj = 00000015 00          0         204
```

```
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          30, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000015 00          0         205
# error_counter,err_inj = 00000015 00          0         206
# error_counter,err_inj = 00000015 00          0         207
# error_counter,err_inj = 00000015 00          0         208
# error_counter,err_inj = 00000015 00          0         209
# error_counter,err_inj = 00000015 00          0         210
# error_counter,err_inj = 00000015 00          0         211
# error_counter,err_inj = 00000015 00          0         212
# error_counter,err_inj = 00000016 11          0         213
# error_counter,err_inj = 00000016 00          0         214
# error_counter,err_inj = 00000016 00          0         215
# error_counter,err_inj = 00000016 00          0         216
# error_counter,err_inj = 00000016 00          0         217
# error_counter,err_inj = 00000016 00          0         218
# error_counter,err_inj = 00000017 11          0         219
# error_counter,err_inj = 00000017 00          0         220
# error_counter,err_inj = 00000017 00          0         221
# error_counter,err_inj = 00000018 01          0         222
# error_counter,err_inj = 00000018 00          0         223
# error_counter,err_inj = 00000018 00          0         224
# error_counter,err_inj = 00000018 00          0         225
# error_counter,err_inj = 00000019 11          0         226
# error_counter,err_inj = 00000019 00          0         227
# error_counter,err_inj = 00000019 00          0         228
# error_counter,err_inj = 00000019 00          0         229
# error_counter,err_inj = 00000019 00          0         230
# error_counter,err_inj = 00000019 00          0         231
# error_counter,err_inj = 00000019 00          0         232
# error_counter,err_inj = 00000019 00          0         233
# error_counter,err_inj = 00000019 00          0         234
# error_counter,err_inj = 00000019 00          0         235
# error_counter,err_inj = 0000001a 10          0         236
# error_counter,err_inj = 0000001a 00          0         237
# error_counter,err_inj = 0000001a 00          0         238
# error_counter,err_inj = 0000001a 00          0         239
# error_counter,err_inj = 0000001a 00          0         240
# error_counter,err_inj = 0000001b 10          0         241
# error_counter,err_inj = 0000001b 00          0         242
# error_counter,err_inj = 0000001b 00          0         243
# error_counter,err_inj = 0000001b 00          0         244
# error_counter,err_inj = 0000001b 00          0         245
# error_counter,err_inj = 0000001c 10          0         246
# error_counter,err_inj = 0000001c 00          0         247
# error_counter,err_inj = 0000001c 00          0         248
# error_counter,err_inj = 0000001c 00          0         249
# error_counter,err_inj = 0000001c 00          0         250
# error_counter,err_inj = 0000001c 00          0         251
# error_counter,err_inj = 0000001d 00          0         252
# error_counter,err_inj = 0000001d 00          0         253
# error_counter,err_inj = 0000001d 00          0         254
# error_counter,err_inj = 0000001e 00          0         255
```

2b4 &5) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000001 01          0          16
# error_counter,err_inj = 00000001 00          0          17
# error_counter,err_inj = 00000001 00          0          18
# error_counter,err_inj = 00000001 00          0          19
# error_counter,err_inj = 00000001 00          0          20
# error_counter,err_inj = 00000001 00          0          21
# error_counter,err_inj = 00000001 00          0          22
# error_counter,err_inj = 00000001 00          0          23
# error_counter,err_inj = 00000001 00          0          24
# error_counter,err_inj = 00000001 00          0          25
# error_counter,err_inj = 00000001 00          0          26
# error_counter,err_inj = 00000001 00          0          27
# error_counter,err_inj = 00000001 00          0          28
# error_counter,err_inj = 00000001 00          0          29
# error_counter,err_inj = 00000001 00          0          30
# error_counter,err_inj = 00000001 00          0          31
# error_counter,err_inj = 00000002 10          0          32
# error_counter,err_inj = 00000002 00          0          33
# error_counter,err_inj = 00000002 00          0          34
# error_counter,err_inj = 00000002 00          0          35
```

```
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
```

```
# error_counter,err_inj = 00000002 00          0          36
# error_counter,err_inj = 00000002 00          0          37
# error_counter,err_inj = 00000002 00          0          38
# error_counter,err_inj = 00000002 00          0          39
# error_counter,err_inj = 00000002 00          0          40
# error_counter,err_inj = 00000002 00          0          41
# error_counter,err_inj = 00000002 00          0          42
# error_counter,err_inj = 00000002 00          0          43
# error_counter,err_inj = 00000002 00          0          44
# error_counter,err_inj = 00000002 00          0          45
# error_counter,err_inj = 00000002 00          0          46
# error_counter,err_inj = 00000002 00          0          47
# error_counter,err_inj = 00000003 10          0          48
# error_counter,err_inj = 00000003 00          0          49
# error_counter,err_inj = 00000003 00          0          50
# error_counter,err_inj = 00000003 00          0          51
# error_counter,err_inj = 00000003 00          0          52
# error_counter,err_inj = 00000003 00          0          53
# error_counter,err_inj = 00000003 00          0          54
# error_counter,err_inj = 00000003 00          0          55
# error_counter,err_inj = 00000003 00          0          56
# error_counter,err_inj = 00000003 00          0          57
# error_counter,err_inj = 00000003 00          0          58
# error_counter,err_inj = 00000003 00          0          59
# error_counter,err_inj = 00000003 00          0          60
# error_counter,err_inj = 00000003 00          0          61
# error_counter,err_inj = 00000003 00          0          62
# error_counter,err_inj = 00000003 00          0          63
# error_counter,err_inj = 00000004 00          0          64
# error_counter,err_inj = 00000004 00          0          65
# error_counter,err_inj = 00000004 00          0          66
# error_counter,err_inj = 00000004 00          0          67
# error_counter,err_inj = 00000004 00          0          68
# error_counter,err_inj = 00000004 00          0          69
# error_counter,err_inj = 00000004 00          0          70
# error_counter,err_inj = 00000004 00          0          71
# error_counter,err_inj = 00000004 00          0          72
# error_counter,err_inj = 00000004 00          0          73
# error_counter,err_inj = 00000004 00          0          74
# error_counter,err_inj = 00000004 00          0          75
# error_counter,err_inj = 00000004 00          0          76
# error_counter,err_inj = 00000004 00          0          77
# error_counter,err_inj = 00000004 00          0          78
# error_counter,err_inj = 00000004 00          0          79
# error_counter,err_inj = 00000005 00          0          80
# error_counter,err_inj = 00000005 00          0          81
# error_counter,err_inj = 00000005 00          0          82
# error_counter,err_inj = 00000005 00          0          83
# error_counter,err_inj = 00000005 00          0          84
# error_counter,err_inj = 00000005 00          0          85
# error_counter,err_inj = 00000005 00          0          86
# error_counter,err_inj = 00000005 00          0          87
# error_counter,err_inj = 00000005 00          0          88
# error_counter,err_inj = 00000005 00          0          89
# error_counter,err_inj = 00000005 00          0          90
# error_counter,err_inj = 00000005 00          0          91
# error_counter,err_inj = 00000005 00          0          92
# error_counter,err_inj = 00000005 00          0          93
# error_counter,err_inj = 00000005 00          0          94
# error_counter,err_inj = 00000005 00          0          95
# error_counter,err_inj = 00000006 10          0          96
# error_counter,err_inj = 00000006 00          0          97
# error_counter,err_inj = 00000006 00          0          98
# error_counter,err_inj = 00000006 00          0          99
# error_counter,err_inj = 00000006 00          0         100
# error_counter,err_inj = 00000006 00          0         101
# error_counter,err_inj = 00000006 00          0         102
# error_counter,err_inj = 00000006 00          0         103
# error_counter,err_inj = 00000006 00          0         104
# error_counter,err_inj = 00000006 00          0         105
# error_counter,err_inj = 00000006 00          0         106
# error_counter,err_inj = 00000006 00          0         107
# error_counter,err_inj = 00000006 00          0         108
# error_counter,err_inj = 00000006 00          0         109
# error_counter,err_inj = 00000006 00          0         110
# error_counter,err_inj = 00000006 00          0         111
# error_counter,err_inj = 00000007 10          0         112
# error_counter,err_inj = 00000007 00          0         113
# error_counter,err_inj = 00000007 00          0         114
# error_counter,err_inj = 00000007 00          0         115
# error_counter,err_inj = 00000007 00          0         116
# error_counter,err_inj = 00000007 00          0         117
# error_counter,err_inj = 00000007 00          0         118
# error_counter,err_inj = 00000007 00          0         119
# error_counter,err_inj = 00000007 00          0         120
# error_counter,err_inj = 00000007 00          0         121
# error_counter,err_inj = 00000007 00          0         122
# error_counter,err_inj = 00000007 00          0         123
# error_counter,err_inj = 00000007 00          0         124
# error_counter,err_inj = 00000007 00          0         125
# error_counter,err_inj = 00000007 00          0         126
# error_counter,err_inj = 00000007 00          0         127
# error_counter,err_inj = 00000008 00          0         128
```

```
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
```

```
# error_counter,err_inj = 00000008 00          0         129
# error_counter,err_inj = 00000008 00          0         130
# error_counter,err_inj = 00000008 00          0         131
# error_counter,err_inj = 00000008 00          0         132
# error_counter,err_inj = 00000008 00          0         133
# error_counter,err_inj = 00000008 00          0         134
# error_counter,err_inj = 00000008 00          0         135
# error_counter,err_inj = 00000008 00          0         136
# error_counter,err_inj = 00000008 00          0         137
# error_counter,err_inj = 00000008 00          0         138
# error_counter,err_inj = 00000008 00          0         139
# error_counter,err_inj = 00000008 00          0         140
# error_counter,err_inj = 00000008 00          0         141
# error_counter,err_inj = 00000008 00          0         142
# error_counter,err_inj = 00000008 00          0         143
# error_counter,err_inj = 00000009 00          0         144
# error_counter,err_inj = 00000009 00          0         145
# error_counter,err_inj = 00000009 00          0         146
# error_counter,err_inj = 00000009 00          0         147
# error_counter,err_inj = 00000009 00          0         148
# error_counter,err_inj = 00000009 00          0         149
# error_counter,err_inj = 00000009 00          0         150
# error_counter,err_inj = 00000009 00          0         151
# error_counter,err_inj = 00000009 00          0         152
# error_counter,err_inj = 00000009 00          0         153
# error_counter,err_inj = 00000009 00          0         154
# error_counter,err_inj = 00000009 00          0         155
# error_counter,err_inj = 00000009 00          0         156
# error_counter,err_inj = 00000009 00          0         157
# error_counter,err_inj = 00000009 00          0         158
# error_counter,err_inj = 00000009 00          0         159
# error_counter,err_inj = 0000000a 11          0         160
# error_counter,err_inj = 0000000a 00          0         161
# error_counter,err_inj = 0000000a 00          0         162
# error_counter,err_inj = 0000000a 00          0         163
# error_counter,err_inj = 0000000a 00          0         164
# error_counter,err_inj = 0000000a 00          0         165
# error_counter,err_inj = 0000000a 00          0         166
# error_counter,err_inj = 0000000a 00          0         167
# error_counter,err_inj = 0000000a 00          0         168
# error_counter,err_inj = 0000000a 00          0         169
# error_counter,err_inj = 0000000a 00          0         170
# error_counter,err_inj = 0000000a 00          0         171
# error_counter,err_inj = 0000000a 00          0         172
# error_counter,err_inj = 0000000a 00          0         173
# error_counter,err_inj = 0000000a 00          0         174
# error_counter,err_inj = 0000000a 00          0         175
# error_counter,err_inj = 0000000b 10          0         176
# error_counter,err_inj = 0000000b 00          0         177
# error_counter,err_inj = 0000000b 00          0         178
# error_counter,err_inj = 0000000b 00          0         179
# error_counter,err_inj = 0000000b 00          0         180
# error_counter,err_inj = 0000000b 00          0         181
# error_counter,err_inj = 0000000b 00          0         182
# error_counter,err_inj = 0000000b 00          0         183
# error_counter,err_inj = 0000000b 00          0         184
# error_counter,err_inj = 0000000b 00          0         185
# error_counter,err_inj = 0000000b 00          0         186
# error_counter,err_inj = 0000000b 00          0         187
# error_counter,err_inj = 0000000b 00          0         188
# error_counter,err_inj = 0000000b 00          0         189
# error_counter,err_inj = 0000000b 00          0         190
# error_counter,err_inj = 0000000b 00          0         191
# error_counter,err_inj = 0000000c 01          0         192
# error_counter,err_inj = 0000000c 00          0         193
# error_counter,err_inj = 0000000c 00          0         194
# error_counter,err_inj = 0000000c 00          0         195
# error_counter,err_inj = 0000000c 00          0         196
# error_counter,err_inj = 0000000c 00          0         197
# error_counter,err_inj = 0000000c 00          0         198
# error_counter,err_inj = 0000000c 00          0         199
# error_counter,err_inj = 0000000c 00          0         200
# error_counter,err_inj = 0000000c 00          0         201
# error_counter,err_inj = 0000000c 00          0         202
# error_counter,err_inj = 0000000c 00          0         203
# error_counter,err_inj = 0000000c 00          0         204
# error_counter,err_inj = 0000000c 00          0         205
# error_counter,err_inj = 0000000c 00          0         206
# error_counter,err_inj = 0000000c 00          0         207
# error_counter,err_inj = 0000000d 10          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000d 00          0         213
# error_counter,err_inj = 0000000d 00          0         214
# error_counter,err_inj = 0000000d 00          0         215
# error_counter,err_inj = 0000000d 00          0         216
# error_counter,err_inj = 0000000d 00          0         217
# error_counter,err_inj = 0000000d 00          0         218
# error_counter,err_inj = 0000000d 00          0         219
# error_counter,err_inj = 0000000d 00          0         220
# error_counter,err_inj = 0000000d 00          0         221
```

```
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000d 00          0         222
# error_counter,err_inj = 0000000d 00          0         223
# error_counter,err_inj = 0000000e 01          0         224
# error_counter,err_inj = 0000000e 00          0         225
# error_counter,err_inj = 0000000e 00          0         226
# error_counter,err_inj = 0000000e 00          0         227
# error_counter,err_inj = 0000000e 00          0         228
# error_counter,err_inj = 0000000e 00          0         229
# error_counter,err_inj = 0000000e 00          0         230
# error_counter,err_inj = 0000000e 00          0         231
# error_counter,err_inj = 0000000e 00          0         232
# error_counter,err_inj = 0000000e 00          0         233
# error_counter,err_inj = 0000000e 00          0         234
# error_counter,err_inj = 0000000e 00          0         235
# error_counter,err_inj = 0000000e 00          0         236
# error_counter,err_inj = 0000000e 00          0         237
# error_counter,err_inj = 0000000e 00          0         238
# error_counter,err_inj = 0000000e 00          0         239
# error_counter,err_inj = 0000000f 11          0         240
# error_counter,err_inj = 0000000f 00          0         241
# error_counter,err_inj = 0000000f 00          0         242
# error_counter,err_inj = 0000000f 00          0         243
# error_counter,err_inj = 0000000f 00          0         244
# error_counter,err_inj = 0000000f 00          0         245
# error_counter,err_inj = 0000000f 00          0         246
# error_counter,err_inj = 0000000f 00          0         247
# error_counter,err_inj = 0000000f 00          0         248
# error_counter,err_inj = 0000000f 00          0         249
# error_counter,err_inj = 0000000f 00          0         250
# error_counter,err_inj = 0000000f 00          0         251
# error_counter,err_inj = 0000000f 00          0         252
# error_counter,err_inj = 0000000f 00          0         253
# error_counter,err_inj = 0000000f 00          0         254
# error_counter,err_inj = 0000000f 00          0         255
# word_count =       10440
```

2b6 & 7) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000001 00          0          29
# error_counter,err_inj = 00000002 11          0          30
# error_counter,err_inj = 00000003 01          0          31
# error_counter,err_inj = 00000004 11          2          32
# error_counter,err_inj = 00000004 00          3          33
# error_counter,err_inj = 00000004 00          3          34
# error_counter,err_inj = 00000004 00          3          35
# error_counter,err_inj = 00000004 00          3          36
# error_counter,err_inj = 00000004 00          3          37
# error_counter,err_inj = 00000004 00          3          38
# error_counter,err_inj = 00000004 00          3          39
# error_counter,err_inj = 00000004 00          3          40
# error_counter,err_inj = 00000004 00          3          41
# error_counter,err_inj = 00000004 00          3          42
# error_counter,err_inj = 00000004 00          3          43
# error_counter,err_inj = 00000004 00          3          44
# error_counter,err_inj = 00000004 00          3          45
# error_counter,err_inj = 00000004 00          3          46
# error_counter,err_inj = 00000004 00          3          47
# error_counter,err_inj = 00000004 00          3          48
# error_counter,err_inj = 00000004 00          3          49
# error_counter,err_inj = 00000004 00          3          50
# error_counter,err_inj = 00000004 00          3          51
# error_counter,err_inj = 00000004 00          3          52
```

```
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# boo! in = 0, out = 1, w_ct =          70, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# boo! in = 0, out = 1, w_ct =          72, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =          73, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# boo! in = 0, out = 1, w_ct =         134, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# boo! in = 1, out = 0, w_ct =         141, err = 00,
```

```
# error_counter,err_inj = 00000004 00          3          53
# error_counter,err_inj = 00000004 00          3          54
# error_counter,err_inj = 00000004 00          3          55
# error_counter,err_inj = 00000004 00          3          56
# error_counter,err_inj = 00000004 00          3          57
# error_counter,err_inj = 00000004 00          3          58
# error_counter,err_inj = 00000004 00          3          59
# error_counter,err_inj = 00000004 00          3          60
# error_counter,err_inj = 00000005 01          3          61
# error_counter,err_inj = 00000006 10          3          62
# error_counter,err_inj = 00000007 01          4          63
# error_counter,err_inj = 00000008 11          5          64
# error_counter,err_inj = 00000008 00          6          65
# error_counter,err_inj = 00000008 00          6          66
# error_counter,err_inj = 00000008 00          6          67
# error_counter,err_inj = 00000008 00          6          68
# error_counter,err_inj = 00000008 00          6          69
# error_counter,err_inj = 00000008 00          6          70
# error_counter,err_inj = 00000008 00          6          71
# error_counter,err_inj = 00000008 00          6          72
# error_counter,err_inj = 00000008 00          6          73
# error_counter,err_inj = 00000008 00          6          74
# error_counter,err_inj = 00000008 00          6          75
# error_counter,err_inj = 00000008 00          6          76
# error_counter,err_inj = 00000008 00          6          77
# error_counter,err_inj = 00000008 00          6          78
# error_counter,err_inj = 00000008 00          6          79
# error_counter,err_inj = 00000008 00          6          80
# error_counter,err_inj = 00000008 00          6          81
# error_counter,err_inj = 00000008 00          6          82
# error_counter,err_inj = 00000008 00          6          83
# error_counter,err_inj = 00000008 00          6          84
# error_counter,err_inj = 00000008 00          6          85
# error_counter,err_inj = 00000008 00          6          86
# error_counter,err_inj = 00000008 00          6          87
# error_counter,err_inj = 00000008 00          6          88
# error_counter,err_inj = 00000008 00          6          89
# error_counter,err_inj = 00000008 00          6          90
# error_counter,err_inj = 00000008 00          6          91
# error_counter,err_inj = 00000008 00          6          92
# error_counter,err_inj = 00000009 01          6          93
# error_counter,err_inj = 0000000a 00          6          94
# error_counter,err_inj = 0000000b 00          7          95
# error_counter,err_inj = 0000000c 11          7          96
# error_counter,err_inj = 0000000c 00          7          97
# error_counter,err_inj = 0000000c 00          7          98
# error_counter,err_inj = 0000000c 00          7          99
# error_counter,err_inj = 0000000c 00          7         100
# error_counter,err_inj = 0000000c 00          7         101
# error_counter,err_inj = 0000000c 00          7         102
# error_counter,err_inj = 0000000c 00          7         103
# error_counter,err_inj = 0000000c 00          7         104
# error_counter,err_inj = 0000000c 00          7         105
# error_counter,err_inj = 0000000c 00          7         106
# error_counter,err_inj = 0000000c 00          7         107
# error_counter,err_inj = 0000000c 00          7         108
# error_counter,err_inj = 0000000c 00          7         109
# error_counter,err_inj = 0000000c 00          7         110
# error_counter,err_inj = 0000000c 00          7         111
# error_counter,err_inj = 0000000c 00          7         112
# error_counter,err_inj = 0000000c 00          7         113
# error_counter,err_inj = 0000000c 00          7         114
# error_counter,err_inj = 0000000c 00          7         115
# error_counter,err_inj = 0000000c 00          7         116
# error_counter,err_inj = 0000000c 00          7         117
# error_counter,err_inj = 0000000c 00          7         118
# error_counter,err_inj = 0000000c 00          7         119
# error_counter,err_inj = 0000000c 00          7         120
# error_counter,err_inj = 0000000c 00          7         121
# error_counter,err_inj = 0000000c 00          7         122
# error_counter,err_inj = 0000000c 00          7         123
# error_counter,err_inj = 0000000c 00          7         124
# error_counter,err_inj = 0000000d 11          7         125
# error_counter,err_inj = 0000000e 10          7         126
# error_counter,err_inj = 0000000f 10          9         127
# error_counter,err_inj = 00000010 01         10         128
# error_counter,err_inj = 00000010 00         11         129
# error_counter,err_inj = 00000010 00         11         130
# error_counter,err_inj = 00000010 00         11         131
# error_counter,err_inj = 00000010 00         11         132
# error_counter,err_inj = 00000010 00         11         133
# error_counter,err_inj = 00000010 00         11         134
# error_counter,err_inj = 00000010 00         11         135
# error_counter,err_inj = 00000010 00         11         136
# error_counter,err_inj = 00000010 00         11         137
# error_counter,err_inj = 00000010 00         11         138
# error_counter,err_inj = 00000010 00         11         139
# error_counter,err_inj = 00000010 00         11         140
# error_counter,err_inj = 00000010 00         11         141
# error_counter,err_inj = 00000010 00         11         142
# error_counter,err_inj = 00000010 00         11         143
# error_counter,err_inj = 00000010 00         11         144
# error_counter,err_inj = 00000010 00         11         145
```

```
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# boo! in = 1, out = 0, w_ct =         198, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         199, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         200, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# boo! in = 1, out = 0, w_ct =         202, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
```

```
# error_counter,err_inj = 00000010 00          11          146
# error_counter,err_inj = 00000010 00          11          147
# error_counter,err_inj = 00000010 00          11          148
# error_counter,err_inj = 00000010 00          11          149
# error_counter,err_inj = 00000010 00          11          150
# error_counter,err_inj = 00000010 00          11          151
# error_counter,err_inj = 00000010 00          11          152
# error_counter,err_inj = 00000010 00          11          153
# error_counter,err_inj = 00000010 00          11          154
# error_counter,err_inj = 00000010 00          11          155
# error_counter,err_inj = 00000010 00          11          156
# error_counter,err_inj = 00000011 10          11          157
# error_counter,err_inj = 00000012 00          11          158
# error_counter,err_inj = 00000013 01          12          159
# error_counter,err_inj = 00000014 00          12          160
# error_counter,err_inj = 00000014 00          13          161
# error_counter,err_inj = 00000014 00          13          162
# error_counter,err_inj = 00000014 00          13          163
# error_counter,err_inj = 00000014 00          13          164
# error_counter,err_inj = 00000014 00          13          165
# error_counter,err_inj = 00000014 00          13          166
# error_counter,err_inj = 00000014 00          13          167
# error_counter,err_inj = 00000014 00          13          168
# error_counter,err_inj = 00000014 00          13          169
# error_counter,err_inj = 00000014 00          13          170
# error_counter,err_inj = 00000014 00          13          171
# error_counter,err_inj = 00000014 00          13          172
# error_counter,err_inj = 00000014 00          13          173
# error_counter,err_inj = 00000014 00          13          174
# error_counter,err_inj = 00000014 00          13          175
# error_counter,err_inj = 00000014 00          13          176
# error_counter,err_inj = 00000014 00          13          177
# error_counter,err_inj = 00000014 00          13          178
# error_counter,err_inj = 00000014 00          13          179
# error_counter,err_inj = 00000014 00          13          180
# error_counter,err_inj = 00000014 00          13          181
# error_counter,err_inj = 00000014 00          13          182
# error_counter,err_inj = 00000014 00          13          183
# error_counter,err_inj = 00000014 00          13          184
# error_counter,err_inj = 00000014 00          13          185
# error_counter,err_inj = 00000014 00          13          186
# error_counter,err_inj = 00000014 00          13          187
# error_counter,err_inj = 00000014 00          13          188
# error_counter,err_inj = 00000015 01          13          189
# error_counter,err_inj = 00000016 01          13          190
# error_counter,err_inj = 00000017 11          14          191
# error_counter,err_inj = 00000018 11          15          192
# error_counter,err_inj = 00000018 00          17          193
# error_counter,err_inj = 00000018 00          17          194
# error_counter,err_inj = 00000018 00          17          195
# error_counter,err_inj = 00000018 00          17          196
# error_counter,err_inj = 00000018 00          17          197
# error_counter,err_inj = 00000018 00          17          198
# error_counter,err_inj = 00000018 00          17          199
# error_counter,err_inj = 00000018 00          17          200
# error_counter,err_inj = 00000018 00          17          201
# error_counter,err_inj = 00000018 00          17          202
# error_counter,err_inj = 00000018 00          17          203
# error_counter,err_inj = 00000018 00          17          204
# error_counter,err_inj = 00000018 00          17          205
# error_counter,err_inj = 00000018 00          17          206
# error_counter,err_inj = 00000018 00          17          207
# error_counter,err_inj = 00000018 00          17          208
# error_counter,err_inj = 00000018 00          17          209
# error_counter,err_inj = 00000018 00          17          210
# error_counter,err_inj = 00000018 00          17          211
# error_counter,err_inj = 00000018 00          17          212
# error_counter,err_inj = 00000018 00          17          213
# error_counter,err_inj = 00000018 00          17          214
# error_counter,err_inj = 00000018 00          17          215
# error_counter,err_inj = 00000018 00          17          216
# error_counter,err_inj = 00000018 00          17          217
# error_counter,err_inj = 00000018 00          17          218
# error_counter,err_inj = 00000018 00          17          219
# error_counter,err_inj = 00000018 00          17          220
# error_counter,err_inj = 00000019 00          17          221
# error_counter,err_inj = 0000001a 11          17          222
# error_counter,err_inj = 0000001b 10          17          223
# error_counter,err_inj = 0000001c 10          19          224
# error_counter,err_inj = 0000001c 00          20          225
# error_counter,err_inj = 0000001c 00          20          226
# error_counter,err_inj = 0000001c 00          20          227
# error_counter,err_inj = 0000001c 00          20          228
# error_counter,err_inj = 0000001c 00          20          229
# error_counter,err_inj = 0000001c 00          20          230
# error_counter,err_inj = 0000001c 00          20          231
# error_counter,err_inj = 0000001c 00          20          232
# error_counter,err_inj = 0000001c 00          20          233
# error_counter,err_inj = 0000001c 00          20          234
# error_counter,err_inj = 0000001c 00          20          235
# error_counter,err_inj = 0000001c 00          20          236
# error_counter,err_inj = 0000001c 00          20          237
# error_counter,err_inj = 0000001c 00          20          238
```

```
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# boo! in = 1, out = 0, w_ct =         231, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# boo! in = 1, out = 0, w_ct =         233, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         234, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         244, bad =
12
```

```
# error_counter,err_inj = 0000001c 00          20          239
# error_counter,err_inj = 0000001c 00          20          240
# error_counter,err_inj = 0000001c 00          20          241
# error_counter,err_inj = 0000001c 00          20          242
# error_counter,err_inj = 0000001c 00          20          243
# error_counter,err_inj = 0000001c 00          20          244
# error_counter,err_inj = 0000001c 00          20          245
# error_counter,err_inj = 0000001c 00          20          246
# error_counter,err_inj = 0000001c 00          20          247
# error_counter,err_inj = 0000001c 00          20          248
# error_counter,err_inj = 0000001c 00          20          249
# error_counter,err_inj = 0000001c 00          20          250
# error_counter,err_inj = 0000001c 00          20          251
# error_counter,err_inj = 0000001c 00          20          252
# error_counter,err_inj = 0000001d 00          20          253
# error_counter,err_inj = 0000001e 00          20          254
# error_counter,err_inj = 0000001f 01          20          255
# word_count =       10440
```

2b8) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000000 00          0          29
# error_counter,err_inj = 00000000 00          0          30
# error_counter,err_inj = 00000000 00          0          31
# error_counter,err_inj = 00000001 10          0          32
# error_counter,err_inj = 00000001 00          0          33
# error_counter,err_inj = 00000001 00          0          34
# error_counter,err_inj = 00000001 00          0          35
# error_counter,err_inj = 00000001 00          0          36
# error_counter,err_inj = 00000001 00          0          37
# error_counter,err_inj = 00000001 00          0          38
# error_counter,err_inj = 00000001 00          0          39
# error_counter,err_inj = 00000001 00          0          40
# error_counter,err_inj = 00000001 00          0          41
# error_counter,err_inj = 00000001 00          0          42
# error_counter,err_inj = 00000001 00          0          43
# error_counter,err_inj = 00000001 00          0          44
# error_counter,err_inj = 00000001 00          0          45
# error_counter,err_inj = 00000001 00          0          46
# error_counter,err_inj = 00000001 00          0          47
# error_counter,err_inj = 00000001 00          0          48
# error_counter,err_inj = 00000001 00          0          49
# error_counter,err_inj = 00000001 00          0          50
# error_counter,err_inj = 00000001 00          0          51
# error_counter,err_inj = 00000001 00          0          52
# error_counter,err_inj = 00000001 00          0          53
# error_counter,err_inj = 00000001 00          0          54
# error_counter,err_inj = 00000001 00          0          55
# error_counter,err_inj = 00000001 00          0          56
# error_counter,err_inj = 00000001 00          0          57
```

```
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
```

```
# error_counter,err_inj = 00000001 00          0          58
# error_counter,err_inj = 00000001 00          0          59
# error_counter,err_inj = 00000001 00          0          60
# error_counter,err_inj = 00000001 00          0          61
# error_counter,err_inj = 00000001 00          0          62
# error_counter,err_inj = 00000001 00          0          63
# error_counter,err_inj = 00000002 00          0          64
# error_counter,err_inj = 00000002 00          0          65
# error_counter,err_inj = 00000002 00          0          66
# error_counter,err_inj = 00000002 00          0          67
# error_counter,err_inj = 00000002 00          0          68
# error_counter,err_inj = 00000002 00          0          69
# error_counter,err_inj = 00000002 00          0          70
# error_counter,err_inj = 00000002 00          0          71
# error_counter,err_inj = 00000002 00          0          72
# error_counter,err_inj = 00000002 00          0          73
# error_counter,err_inj = 00000002 00          0          74
# error_counter,err_inj = 00000002 00          0          75
# error_counter,err_inj = 00000002 00          0          76
# error_counter,err_inj = 00000002 00          0          77
# error_counter,err_inj = 00000002 00          0          78
# error_counter,err_inj = 00000002 00          0          79
# error_counter,err_inj = 00000002 00          0          80
# error_counter,err_inj = 00000002 00          0          81
# error_counter,err_inj = 00000002 00          0          82
# error_counter,err_inj = 00000002 00          0          83
# error_counter,err_inj = 00000002 00          0          84
# error_counter,err_inj = 00000002 00          0          85
# error_counter,err_inj = 00000002 00          0          86
# error_counter,err_inj = 00000002 00          0          87
# error_counter,err_inj = 00000002 00          0          88
# error_counter,err_inj = 00000002 00          0          89
# error_counter,err_inj = 00000002 00          0          90
# error_counter,err_inj = 00000002 00          0          91
# error_counter,err_inj = 00000002 00          0          92
# error_counter,err_inj = 00000002 00          0          93
# error_counter,err_inj = 00000002 00          0          94
# error_counter,err_inj = 00000002 00          0          95
# error_counter,err_inj = 00000003 10          0          96
# error_counter,err_inj = 00000003 00          0          97
# error_counter,err_inj = 00000003 00          0          98
# error_counter,err_inj = 00000003 00          0          99
# error_counter,err_inj = 00000003 00          0         100
# error_counter,err_inj = 00000003 00          0         101
# error_counter,err_inj = 00000003 00          0         102
# error_counter,err_inj = 00000003 00          0         103
# error_counter,err_inj = 00000003 00          0         104
# error_counter,err_inj = 00000003 00          0         105
# error_counter,err_inj = 00000003 00          0         106
# error_counter,err_inj = 00000003 00          0         107
# error_counter,err_inj = 00000003 00          0         108
# error_counter,err_inj = 00000003 00          0         109
# error_counter,err_inj = 00000003 00          0         110
# error_counter,err_inj = 00000003 00          0         111
# error_counter,err_inj = 00000003 00          0         112
# error_counter,err_inj = 00000003 00          0         113
# error_counter,err_inj = 00000003 00          0         114
# error_counter,err_inj = 00000003 00          0         115
# error_counter,err_inj = 00000003 00          0         116
# error_counter,err_inj = 00000003 00          0         117
# error_counter,err_inj = 00000003 00          0         118
# error_counter,err_inj = 00000003 00          0         119
# error_counter,err_inj = 00000003 00          0         120
# error_counter,err_inj = 00000003 00          0         121
# error_counter,err_inj = 00000003 00          0         122
# error_counter,err_inj = 00000003 00          0         123
# error_counter,err_inj = 00000003 00          0         124
# error_counter,err_inj = 00000003 00          0         125
# error_counter,err_inj = 00000003 00          0         126
# error_counter,err_inj = 00000003 00          0         127
# error_counter,err_inj = 00000004 00          0         128
# error_counter,err_inj = 00000004 00          0         129
# error_counter,err_inj = 00000004 00          0         130
# error_counter,err_inj = 00000004 00          0         131
# error_counter,err_inj = 00000004 00          0         132
# error_counter,err_inj = 00000004 00          0         133
# error_counter,err_inj = 00000004 00          0         134
# error_counter,err_inj = 00000004 00          0         135
# error_counter,err_inj = 00000004 00          0         136
# error_counter,err_inj = 00000004 00          0         137
# error_counter,err_inj = 00000004 00          0         138
# error_counter,err_inj = 00000004 00          0         139
# error_counter,err_inj = 00000004 00          0         140
# error_counter,err_inj = 00000004 00          0         141
# error_counter,err_inj = 00000004 00          0         142
# error_counter,err_inj = 00000004 00          0         143
# error_counter,err_inj = 00000004 00          0         144
# error_counter,err_inj = 00000004 00          0         145
# error_counter,err_inj = 00000004 00          0         146
# error_counter,err_inj = 00000004 00          0         147
# error_counter,err_inj = 00000004 00          0         148
# error_counter,err_inj = 00000004 00          0         149
# error_counter,err_inj = 00000004 00          0         150
```

```
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
```

```
# error_counter,err_inj = 00000004 00           0         151
# error_counter,err_inj = 00000004 00           0         152
# error_counter,err_inj = 00000004 00           0         153
# error_counter,err_inj = 00000004 00           0         154
# error_counter,err_inj = 00000004 00           0         155
# error_counter,err_inj = 00000004 00           0         156
# error_counter,err_inj = 00000004 00           0         157
# error_counter,err_inj = 00000004 00           0         158
# error_counter,err_inj = 00000004 00           0         159
# error_counter,err_inj = 00000005 11           0         160
# error_counter,err_inj = 00000005 00           0         161
# error_counter,err_inj = 00000005 00           0         162
# error_counter,err_inj = 00000005 00           0         163
# error_counter,err_inj = 00000005 00           0         164
# error_counter,err_inj = 00000005 00           0         165
# error_counter,err_inj = 00000005 00           0         166
# error_counter,err_inj = 00000005 00           0         167
# error_counter,err_inj = 00000005 00           0         168
# error_counter,err_inj = 00000005 00           0         169
# error_counter,err_inj = 00000005 00           0         170
# error_counter,err_inj = 00000005 00           0         171
# error_counter,err_inj = 00000005 00           0         172
# error_counter,err_inj = 00000005 00           0         173
# error_counter,err_inj = 00000005 00           0         174
# error_counter,err_inj = 00000005 00           0         175
# error_counter,err_inj = 00000005 00           0         176
# error_counter,err_inj = 00000005 00           0         177
# error_counter,err_inj = 00000005 00           0         178
# error_counter,err_inj = 00000005 00           0         179
# error_counter,err_inj = 00000005 00           0         180
# error_counter,err_inj = 00000005 00           0         181
# error_counter,err_inj = 00000005 00           0         182
# error_counter,err_inj = 00000005 00           0         183
# error_counter,err_inj = 00000005 00           0         184
# error_counter,err_inj = 00000005 00           0         185
# error_counter,err_inj = 00000005 00           0         186
# error_counter,err_inj = 00000005 00           0         187
# error_counter,err_inj = 00000005 00           0         188
# error_counter,err_inj = 00000005 00           0         189
# error_counter,err_inj = 00000005 00           0         190
# error_counter,err_inj = 00000005 00           0         191
# error_counter,err_inj = 00000006 01           0         192
# error_counter,err_inj = 00000006 00           0         193
# error_counter,err_inj = 00000006 00           0         194
# error_counter,err_inj = 00000006 00           0         195
# error_counter,err_inj = 00000006 00           0         196
# error_counter,err_inj = 00000006 00           0         197
# error_counter,err_inj = 00000006 00           0         198
# error_counter,err_inj = 00000006 00           0         199
# error_counter,err_inj = 00000006 00           0         200
# error_counter,err_inj = 00000006 00           0         201
# error_counter,err_inj = 00000006 00           0         202
# error_counter,err_inj = 00000006 00           0         203
# error_counter,err_inj = 00000006 00           0         204
# error_counter,err_inj = 00000006 00           0         205
# error_counter,err_inj = 00000006 00           0         206
# error_counter,err_inj = 00000006 00           0         207
# error_counter,err_inj = 00000006 00           0         208
# error_counter,err_inj = 00000006 00           0         209
# error_counter,err_inj = 00000006 00           0         210
# error_counter,err_inj = 00000006 00           0         211
# error_counter,err_inj = 00000006 00           0         212
# error_counter,err_inj = 00000006 00           0         213
# error_counter,err_inj = 00000006 00           0         214
# error_counter,err_inj = 00000006 00           0         215
# error_counter,err_inj = 00000006 00           0         216
# error_counter,err_inj = 00000006 00           0         217
# error_counter,err_inj = 00000006 00           0         218
# error_counter,err_inj = 00000006 00           0         219
# error_counter,err_inj = 00000006 00           0         220
# error_counter,err_inj = 00000006 00           0         221
# error_counter,err_inj = 00000006 00           0         222
# error_counter,err_inj = 00000006 00           0         223
# error_counter,err_inj = 00000007 01           0         224
# error_counter,err_inj = 00000007 00           0         225
# error_counter,err_inj = 00000007 00           0         226
# error_counter,err_inj = 00000007 00           0         227
# error_counter,err_inj = 00000007 00           0         228
# error_counter,err_inj = 00000007 00           0         229
# error_counter,err_inj = 00000007 00           0         230
# error_counter,err_inj = 00000007 00           0         231
# error_counter,err_inj = 00000007 00           0         232
# error_counter,err_inj = 00000007 00           0         233
# error_counter,err_inj = 00000007 00           0         234
# error_counter,err_inj = 00000007 00           0         235
# error_counter,err_inj = 00000007 00           0         236
# error_counter,err_inj = 00000007 00           0         237
# error_counter,err_inj = 00000007 00           0         238
# error_counter,err_inj = 00000007 00           0         239
# error_counter,err_inj = 00000007 00           0         240
# error_counter,err_inj = 00000007 00           0         241
# error_counter,err_inj = 00000007 00           0         242
# error_counter,err_inj = 00000007 00           0         243
```

```
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =           8, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000007 00           0         244
# error_counter,err_inj = 00000007 00           0         245
# error_counter,err_inj = 00000007 00           0         246
# error_counter,err_inj = 00000007 00           0         247
# error_counter,err_inj = 00000007 00           0         248
# error_counter,err_inj = 00000007 00           0         249
# error_counter,err_inj = 00000007 00           0         250
# error_counter,err_inj = 00000007 00           0         251
# error_counter,err_inj = 00000007 00           0         252
# error_counter,err_inj = 00000007 00           0         253
# error_counter,err_inj = 00000007 00           0         254
# error_counter,err_inj = 00000007 00           0         255
# word_count =       10440
```

2c) Transcript: 5 consecutive error injections breaks the decoder

```
# word_count =       10440
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000000 00           0           8
# error_counter,err_inj = 00000000 00           0           9
# error_counter,err_inj = 00000000 00           0          10
# error_counter,err_inj = 00000000 00           0          11
# error_counter,err_inj = 00000001 01           0          12
# error_counter,err_inj = 00000002 01           0          13
# error_counter,err_inj = 00000003 01           1          14
# error_counter,err_inj = 00000004 01           2          15
# error_counter,err_inj = 00000005 01           3          16
# error_counter,err_inj = 00000005 00           4          17
# error_counter,err_inj = 00000005 00           4          18
# error_counter,err_inj = 00000005 00           4          19
# error_counter,err_inj = 00000005 00           4          20
# error_counter,err_inj = 00000005 00           4          21
# error_counter,err_inj = 00000005 00           4          22
# error_counter,err_inj = 00000005 00           4          23
# error_counter,err_inj = 00000005 00           4          24
# error_counter,err_inj = 00000005 00           4          25
# error_counter,err_inj = 00000005 00           4          26
# error_counter,err_inj = 00000005 00           4          27
# error_counter,err_inj = 00000006 01           4          28
# error_counter,err_inj = 00000007 01           4          29
# error_counter,err_inj = 00000008 01           5          30
# error_counter,err_inj = 00000009 01           6          31
# error_counter,err_inj = 0000000a 01           7          32
# error_counter,err_inj = 0000000a 00           8          33
# error_counter,err_inj = 0000000a 00           8          34
# error_counter,err_inj = 0000000a 00           8          35
# error_counter,err_inj = 0000000a 00           8          36
# error_counter,err_inj = 0000000a 00           8          37
# error_counter,err_inj = 0000000a 00           8          38
# error_counter,err_inj = 0000000a 00           8          39
# error_counter,err_inj = 0000000a 00           8          40
# error_counter,err_inj = 0000000a 00           8          41
# error_counter,err_inj = 0000000a 00           8          42
# error_counter,err_inj = 0000000a 00           8          43
# error_counter,err_inj = 0000000b 01           8          44
# error_counter,err_inj = 0000000c 01           8          45
# error_counter,err_inj = 0000000d 01           9          46
# error_counter,err_inj = 0000000e 01          10          47
# error_counter,err_inj = 0000000f 01          11          48
# error_counter,err_inj = 0000000f 00          12          49
# error_counter,err_inj = 0000000f 00          12          50
# error_counter,err_inj = 0000000f 00          12          51
# error_counter,err_inj = 0000000f 00          12          52
# error_counter,err_inj = 0000000f 00          12          53
# error_counter,err_inj = 0000000f 00          12          54
# error_counter,err_inj = 0000000f 00          12          55
# error_counter,err_inj = 0000000f 00          12          56
# error_counter,err_inj = 0000000f 00          12          57
# error_counter,err_inj = 0000000f 00          12          58
# error_counter,err_inj = 0000000f 00          12          59
# error_counter,err_inj = 00000010 01          12          60
# error_counter,err_inj = 00000011 01          12          61
# error_counter,err_inj = 00000012 01          13          62
# error_counter,err_inj = 00000013 01          14          63
# error_counter,err_inj = 00000014 01          15          64
# error_counter,err_inj = 00000014 00          16          65
# error_counter,err_inj = 00000014 00          16          66
# error_counter,err_inj = 00000014 00          16          67
# error_counter,err_inj = 00000014 00          16          68
# error_counter,err_inj = 00000014 00          16          69
# error_counter,err_inj = 00000014 00          16          70
# error_counter,err_inj = 00000014 00          16          71
# error_counter,err_inj = 00000014 00          16          72
# error_counter,err_inj = 00000014 00          16          73
# error_counter,err_inj = 00000014 00          16          74
```

```
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# boo! in = 1, out = 0, w_ct =          97, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =          98, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# boo! in = 1, out = 0, w_ct =         101, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         102, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         103, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         104, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# boo! in = 1, out = 0, w_ct =         145, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         146, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# boo! in = 1, out = 0, w_ct =         149, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         150, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         151, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         152, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
```

```
# error_counter,err_inj = 00000014 00          16          75
# error_counter,err_inj = 00000015 01          16          76
# error_counter,err_inj = 00000016 01          16          77
# error_counter,err_inj = 00000017 01          17          78
# error_counter,err_inj = 00000018 01          18          79
# error_counter,err_inj = 00000019 01          19          80
# error_counter,err_inj = 00000019 00          20          81
# error_counter,err_inj = 00000019 00          20          82
# error_counter,err_inj = 00000019 00          20          83
# error_counter,err_inj = 00000019 00          20          84
# error_counter,err_inj = 00000019 00          20          85
# error_counter,err_inj = 00000019 00          20          86
# error_counter,err_inj = 00000019 00          20          87
# error_counter,err_inj = 00000019 00          20          88
# error_counter,err_inj = 00000019 00          20          89
# error_counter,err_inj = 00000019 00          20          90
# error_counter,err_inj = 00000019 00          20          91
# error_counter,err_inj = 0000001a 01          20          92
# error_counter,err_inj = 0000001b 01          20          93
# error_counter,err_inj = 0000001c 01          21          94
# error_counter,err_inj = 0000001d 01          22          95
# error_counter,err_inj = 0000001e 01          23          96
# error_counter,err_inj = 0000001e 00          24          97
# error_counter,err_inj = 0000001e 00          24          98
# error_counter,err_inj = 0000001e 00          24          99
# error_counter,err_inj = 0000001e 00          24         100
# error_counter,err_inj = 0000001e 00          24         101
# error_counter,err_inj = 0000001e 00          24         102
# error_counter,err_inj = 0000001e 00          24         103
# error_counter,err_inj = 0000001e 00          24         104
# error_counter,err_inj = 0000001e 00          24         105
# error_counter,err_inj = 0000001e 00          24         106
# error_counter,err_inj = 0000001e 00          24         107
# error_counter,err_inj = 0000001f 01          24         108
# error_counter,err_inj = 00000020 01          24         109
# error_counter,err_inj = 00000021 01          25         110
# error_counter,err_inj = 00000022 01          26         111
# error_counter,err_inj = 00000023 01          27         112
# error_counter,err_inj = 00000023 00          28         113
# error_counter,err_inj = 00000023 00          28         114
# error_counter,err_inj = 00000023 00          28         115
# error_counter,err_inj = 00000023 00          28         116
# error_counter,err_inj = 00000023 00          28         117
# error_counter,err_inj = 00000023 00          28         118
# error_counter,err_inj = 00000023 00          28         119
# error_counter,err_inj = 00000023 00          28         120
# error_counter,err_inj = 00000023 00          28         121
# error_counter,err_inj = 00000023 00          28         122
# error_counter,err_inj = 00000023 00          28         123
# error_counter,err_inj = 00000024 01          28         124
# error_counter,err_inj = 00000025 01          28         125
# error_counter,err_inj = 00000026 01          29         126
# error_counter,err_inj = 00000027 01          30         127
# error_counter,err_inj = 00000028 01          31         128
# error_counter,err_inj = 00000028 00          32         129
# error_counter,err_inj = 00000028 00          32         130
# error_counter,err_inj = 00000028 00          32         131
# error_counter,err_inj = 00000028 00          32         132
# error_counter,err_inj = 00000028 00          32         133
# error_counter,err_inj = 00000028 00          32         134
# error_counter,err_inj = 00000028 00          32         135
# error_counter,err_inj = 00000028 00          32         136
# error_counter,err_inj = 00000028 00          32         137
# error_counter,err_inj = 00000028 00          32         138
# error_counter,err_inj = 00000028 00          32         139
# error_counter,err_inj = 00000029 01          32         140
# error_counter,err_inj = 0000002a 01          32         141
# error_counter,err_inj = 0000002b 01          33         142
# error_counter,err_inj = 0000002c 01          34         143
# error_counter,err_inj = 0000002d 01          35         144
# error_counter,err_inj = 0000002d 00          36         145
# error_counter,err_inj = 0000002d 00          36         146
# error_counter,err_inj = 0000002d 00          36         147
# error_counter,err_inj = 0000002d 00          36         148
# error_counter,err_inj = 0000002d 00          36         149
# error_counter,err_inj = 0000002d 00          36         150
# error_counter,err_inj = 0000002d 00          36         151
# error_counter,err_inj = 0000002d 00          36         152
# error_counter,err_inj = 0000002d 00          36         153
# error_counter,err_inj = 0000002d 00          36         154
# error_counter,err_inj = 0000002d 00          36         155
# error_counter,err_inj = 0000002e 01          36         156
# error_counter,err_inj = 0000002f 01          36         157
# error_counter,err_inj = 00000030 01          37         158
# error_counter,err_inj = 00000031 01          38         159
# error_counter,err_inj = 00000032 01          39         160
# error_counter,err_inj = 00000032 00          40         161
# error_counter,err_inj = 00000032 00          40         162
# error_counter,err_inj = 00000032 00          40         163
# error_counter,err_inj = 00000032 00          40         164
# error_counter,err_inj = 00000032 00          40         165
# error_counter,err_inj = 00000032 00          40         166
# error_counter,err_inj = 00000032 00          40         167
```

```
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# boo! in = 1, out = 0, w_ct =         161, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         162, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# boo! in = 1, out = 0, w_ct =         165, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         166, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         167, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         168, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# boo! in = 1, out = 0, w_ct =         177, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         178, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# boo! in = 1, out = 0, w_ct =         181, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         182, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         183, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         184, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# boo! in = 1, out = 0, w_ct =         193, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         194, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# boo! in = 1, out = 0, w_ct =         197, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         198, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         199, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         200, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# boo! in = 1, out = 0, w_ct =         209, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         210, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# boo! in = 1, out = 0, w_ct =         213, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         214, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         215, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         216, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
```

```
# error_counter,err_inj = 00000032 00          40          168
# error_counter,err_inj = 00000032 00          40          169
# error_counter,err_inj = 00000032 00          40          170
# error_counter,err_inj = 00000032 00          40          171
# error_counter,err_inj = 00000033 01          40          172
# error_counter,err_inj = 00000034 01          40          173
# error_counter,err_inj = 00000035 01          41          174
# error_counter,err_inj = 00000036 01          42          175
# error_counter,err_inj = 00000037 01          43          176
# error_counter,err_inj = 00000037 00          44          177
# error_counter,err_inj = 00000037 00          44          178
# error_counter,err_inj = 00000037 00          44          179
# error_counter,err_inj = 00000037 00          44          180
# error_counter,err_inj = 00000037 00          44          181
# error_counter,err_inj = 00000037 00          44          182
# error_counter,err_inj = 00000037 00          44          183
# error_counter,err_inj = 00000037 00          44          184
# error_counter,err_inj = 00000037 00          44          185
# error_counter,err_inj = 00000037 00          44          186
# error_counter,err_inj = 00000037 00          44          187
# error_counter,err_inj = 00000038 01          44          188
# error_counter,err_inj = 00000039 01          44          189
# error_counter,err_inj = 0000003a 01          45          190
# error_counter,err_inj = 0000003b 01          46          191
# error_counter,err_inj = 0000003c 01          47          192
# error_counter,err_inj = 0000003c 00          48          193
# error_counter,err_inj = 0000003c 00          48          194
# error_counter,err_inj = 0000003c 00          48          195
# error_counter,err_inj = 0000003c 00          48          196
# error_counter,err_inj = 0000003c 00          48          197
# error_counter,err_inj = 0000003c 00          48          198
# error_counter,err_inj = 0000003c 00          48          199
# error_counter,err_inj = 0000003c 00          48          200
# error_counter,err_inj = 0000003c 00          48          201
# error_counter,err_inj = 0000003c 00          48          202
# error_counter,err_inj = 0000003c 00          48          203
# error_counter,err_inj = 0000003d 01          48          204
# error_counter,err_inj = 0000003e 01          48          205
# error_counter,err_inj = 0000003f 01          49          206
# error_counter,err_inj = 00000040 01          50          207
# error_counter,err_inj = 00000041 01          51          208
# error_counter,err_inj = 00000041 00          52          209
# error_counter,err_inj = 00000041 00          52          210
# error_counter,err_inj = 00000041 00          52          211
# error_counter,err_inj = 00000041 00          52          212
# error_counter,err_inj = 00000041 00          52          213
# error_counter,err_inj = 00000041 00          52          214
# error_counter,err_inj = 00000041 00          52          215
# error_counter,err_inj = 00000041 00          52          216
# error_counter,err_inj = 00000041 00          52          217
# error_counter,err_inj = 00000041 00          52          218
# error_counter,err_inj = 00000041 00          52          219
# error_counter,err_inj = 00000042 01          52          220
# error_counter,err_inj = 00000043 01          52          221
# error_counter,err_inj = 00000044 01          53          222
# error_counter,err_inj = 00000045 01          54          223
# error_counter,err_inj = 00000046 01          55          224
# error_counter,err_inj = 00000046 00          56          225
# error_counter,err_inj = 00000046 00          56          226
# error_counter,err_inj = 00000046 00          56          227
# error_counter,err_inj = 00000046 00          56          228
# error_counter,err_inj = 00000046 00          56          229
# error_counter,err_inj = 00000046 00          56          230
# error_counter,err_inj = 00000046 00          56          231
# error_counter,err_inj = 00000046 00          56          232
# error_counter,err_inj = 00000046 00          56          233
# error_counter,err_inj = 00000046 00          56          234
# error_counter,err_inj = 00000046 00          56          235
# error_counter,err_inj = 00000047 01          56          236
# error_counter,err_inj = 00000048 01          56          237
# error_counter,err_inj = 00000049 01          57          238
# error_counter,err_inj = 0000004a 01          58          239
# error_counter,err_inj = 0000004b 01          59          240
# error_counter,err_inj = 0000004b 00          60          241
# error_counter,err_inj = 0000004b 00          60          242
# error_counter,err_inj = 0000004b 00          60          243
# error_counter,err_inj = 0000004b 00          60          244
# error_counter,err_inj = 0000004b 00          60          245
# error_counter,err_inj = 0000004b 00          60          246
# error_counter,err_inj = 0000004b 00          60          247
# error_counter,err_inj = 0000004b 00          60          248
# error_counter,err_inj = 0000004b 00          60          249
# error_counter,err_inj = 0000004b 00          60          250
# error_counter,err_inj = 0000004b 00          60          251
# error_counter,err_inj = 0000004c 01          60          252
# error_counter,err_inj = 0000004d 01          60          253
# error_counter,err_inj = 0000004e 01          61          254
# error_counter,err_inj = 0000004f 01          62          255
```

```
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# boo! in = 1, out = 0, w_ct =         225, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         226, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# boo! in = 1, out = 0, w_ct =         229, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         230, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         231, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         232, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          80, OUT: good =         214, bad =
42
```

2d) Transcript: 5 consecutive error injections breaks the decoder

```
# word_count =       10440
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# boo! in = 1, out = 0, w_ct =          21, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# boo! in = 1, out = 0, w_ct =          29, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# boo! in = 0, out = 1, w_ct =          34, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# boo! in = 1, out = 0, w_ct =          38, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000001 10          0         12
# error_counter,err_inj = 00000002 10          0         13
# error_counter,err_inj = 00000003 10          1         14
# error_counter,err_inj = 00000004 10          2         15
# error_counter,err_inj = 00000005 10          3         16
# error_counter,err_inj = 00000005 00          4         17
# error_counter,err_inj = 00000005 00          4         18
# error_counter,err_inj = 00000005 00          4         19
# error_counter,err_inj = 00000005 00          4         20
# error_counter,err_inj = 00000005 00          4         21
# error_counter,err_inj = 00000005 00          4         22
# error_counter,err_inj = 00000005 00          4         23
# error_counter,err_inj = 00000005 00          4         24
# error_counter,err_inj = 00000005 00          4         25
# error_counter,err_inj = 00000005 00          4         26
# error_counter,err_inj = 00000005 00          4         27
# error_counter,err_inj = 00000006 10          4         28
# error_counter,err_inj = 00000007 10          4         29
# error_counter,err_inj = 00000008 10          5         30
# error_counter,err_inj = 00000009 10          6         31
# error_counter,err_inj = 0000000a 10          7         32
# error_counter,err_inj = 0000000a 00          8         33
# error_counter,err_inj = 0000000a 00          8         34
# error_counter,err_inj = 0000000a 00          8         35
# error_counter,err_inj = 0000000a 00          8         36
# error_counter,err_inj = 0000000a 00          8         37
# error_counter,err_inj = 0000000a 00          8         38
# error_counter,err_inj = 0000000a 00          8         39
# error_counter,err_inj = 0000000a 00          8         40
# error_counter,err_inj = 0000000a 00          8         41
# error_counter,err_inj = 0000000a 00          8         42
# error_counter,err_inj = 0000000a 00          8         43
# error_counter,err_inj = 0000000b 10          8         44
# error_counter,err_inj = 0000000c 10          8         45
# error_counter,err_inj = 0000000d 10          9         46
# error_counter,err_inj = 0000000e 10         10         47
# error_counter,err_inj = 0000000f 10         11         48
```

```
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# boo! in = 0, out = 1, w_ct =          69, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# boo! in = 1, out = 0, w_ct =          76, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# boo! in = 1, out = 0, w_ct =         101, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# boo! in = 1, out = 0, w_ct =         108, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# boo! in = 1, out = 0, w_ct =         117, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# boo! in = 1, out = 0, w_ct =         124, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
```

```
# error_counter,err_inj = 0000000f 00          12          49
# error_counter,err_inj = 0000000f 00          12          50
# error_counter,err_inj = 0000000f 00          12          51
# error_counter,err_inj = 0000000f 00          12          52
# error_counter,err_inj = 0000000f 00          12          53
# error_counter,err_inj = 0000000f 00          12          54
# error_counter,err_inj = 0000000f 00          12          55
# error_counter,err_inj = 0000000f 00          12          56
# error_counter,err_inj = 0000000f 00          12          57
# error_counter,err_inj = 0000000f 00          12          58
# error_counter,err_inj = 0000000f 00          12          59
# error_counter,err_inj = 00000010 10          12          60
# error_counter,err_inj = 00000011 10          12          61
# error_counter,err_inj = 00000012 10          13          62
# error_counter,err_inj = 00000013 10          14          63
# error_counter,err_inj = 00000014 10          15          64
# error_counter,err_inj = 00000014 00          16          65
# error_counter,err_inj = 00000014 00          16          66
# error_counter,err_inj = 00000014 00          16          67
# error_counter,err_inj = 00000014 00          16          68
# error_counter,err_inj = 00000014 00          16          69
# error_counter,err_inj = 00000014 00          16          70
# error_counter,err_inj = 00000014 00          16          71
# error_counter,err_inj = 00000014 00          16          72
# error_counter,err_inj = 00000014 00          16          73
# error_counter,err_inj = 00000014 00          16          74
# error_counter,err_inj = 00000014 00          16          75
# error_counter,err_inj = 00000015 10          16          76
# error_counter,err_inj = 00000016 10          16          77
# error_counter,err_inj = 00000017 10          17          78
# error_counter,err_inj = 00000018 10          18          79
# error_counter,err_inj = 00000019 10          19          80
# error_counter,err_inj = 00000019 00          20          81
# error_counter,err_inj = 00000019 00          20          82
# error_counter,err_inj = 00000019 00          20          83
# error_counter,err_inj = 00000019 00          20          84
# error_counter,err_inj = 00000019 00          20          85
# error_counter,err_inj = 00000019 00          20          86
# error_counter,err_inj = 00000019 00          20          87
# error_counter,err_inj = 00000019 00          20          88
# error_counter,err_inj = 00000019 00          20          89
# error_counter,err_inj = 00000019 00          20          90
# error_counter,err_inj = 00000019 00          20          91
# error_counter,err_inj = 0000001a 10          20          92
# error_counter,err_inj = 0000001b 10          20          93
# error_counter,err_inj = 0000001c 10          21          94
# error_counter,err_inj = 0000001d 10          22          95
# error_counter,err_inj = 0000001e 10          23          96
# error_counter,err_inj = 0000001e 00          24          97
# error_counter,err_inj = 0000001e 00          24          98
# error_counter,err_inj = 0000001e 00          24          99
# error_counter,err_inj = 0000001e 00          24         100
# error_counter,err_inj = 0000001e 00          24         101
# error_counter,err_inj = 0000001e 00          24         102
# error_counter,err_inj = 0000001e 00          24         103
# error_counter,err_inj = 0000001e 00          24         104
# error_counter,err_inj = 0000001e 00          24         105
# error_counter,err_inj = 0000001e 00          24         106
# error_counter,err_inj = 0000001e 00          24         107
# error_counter,err_inj = 0000001f 10          24         108
# error_counter,err_inj = 00000020 10          24         109
# error_counter,err_inj = 00000021 10          25         110
# error_counter,err_inj = 00000022 10          26         111
# error_counter,err_inj = 00000023 10          27         112
# error_counter,err_inj = 00000023 00          28         113
# error_counter,err_inj = 00000023 00          28         114
# error_counter,err_inj = 00000023 00          28         115
# error_counter,err_inj = 00000023 00          28         116
# error_counter,err_inj = 00000023 00          28         117
# error_counter,err_inj = 00000023 00          28         118
# error_counter,err_inj = 00000023 00          28         119
# error_counter,err_inj = 00000023 00          28         120
# error_counter,err_inj = 00000023 00          28         121
# error_counter,err_inj = 00000023 00          28         122
# error_counter,err_inj = 00000023 00          28         123
# error_counter,err_inj = 00000024 10          28         124
# error_counter,err_inj = 00000025 10          28         125
# error_counter,err_inj = 00000026 10          29         126
# error_counter,err_inj = 00000027 10          30         127
# error_counter,err_inj = 00000028 10          31         128
# error_counter,err_inj = 00000028 00          32         129
# error_counter,err_inj = 00000028 00          32         130
# error_counter,err_inj = 00000028 00          32         131
# error_counter,err_inj = 00000028 00          32         132
# error_counter,err_inj = 00000028 00          32         133
# error_counter,err_inj = 00000028 00          32         134
# error_counter,err_inj = 00000028 00          32         135
# error_counter,err_inj = 00000028 00          32         136
# error_counter,err_inj = 00000028 00          32         137
# error_counter,err_inj = 00000028 00          32         138
# error_counter,err_inj = 00000028 00          32         139
# error_counter,err_inj = 00000029 10          32         140
# error_counter,err_inj = 0000002a 10          32         141
```

```
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# boo! in = 0, out = 1, w_ct =         133, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# boo! in = 1, out = 0, w_ct =         140, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# boo! in = 1, out = 0, w_ct =         149, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# boo! in = 1, out = 0, w_ct =         156, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# boo! in = 1, out = 0, w_ct =         165, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# boo! in = 1, out = 0, w_ct =         172, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# boo! in = 1, out = 0, w_ct =         181, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# boo! in = 1, out = 0, w_ct =         188, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# boo! in = 1, out = 0, w_ct =         197, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# boo! in = 1, out = 0, w_ct =         204, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# boo! in = 1, out = 0, w_ct =         213, err = 00,
```

```
# error_counter,err_inj = 0000002b 10          33          142
# error_counter,err_inj = 0000002c 10          34          143
# error_counter,err_inj = 0000002d 10          35          144
# error_counter,err_inj = 0000002d 00          36          145
# error_counter,err_inj = 0000002d 00          36          146
# error_counter,err_inj = 0000002d 00          36          147
# error_counter,err_inj = 0000002d 00          36          148
# error_counter,err_inj = 0000002d 00          36          149
# error_counter,err_inj = 0000002d 00          36          150
# error_counter,err_inj = 0000002d 00          36          151
# error_counter,err_inj = 0000002d 00          36          152
# error_counter,err_inj = 0000002d 00          36          153
# error_counter,err_inj = 0000002d 00          36          154
# error_counter,err_inj = 0000002d 00          36          155
# error_counter,err_inj = 0000002e 10          36          156
# error_counter,err_inj = 0000002f 10          36          157
# error_counter,err_inj = 00000030 10          37          158
# error_counter,err_inj = 00000031 10          38          159
# error_counter,err_inj = 00000032 10          39          160
# error_counter,err_inj = 00000032 00          40          161
# error_counter,err_inj = 00000032 00          40          162
# error_counter,err_inj = 00000032 00          40          163
# error_counter,err_inj = 00000032 00          40          164
# error_counter,err_inj = 00000032 00          40          165
# error_counter,err_inj = 00000032 00          40          166
# error_counter,err_inj = 00000032 00          40          167
# error_counter,err_inj = 00000032 00          40          168
# error_counter,err_inj = 00000032 00          40          169
# error_counter,err_inj = 00000032 00          40          170
# error_counter,err_inj = 00000032 00          40          171
# error_counter,err_inj = 00000033 10          40          172
# error_counter,err_inj = 00000034 10          40          173
# error_counter,err_inj = 00000035 10          41          174
# error_counter,err_inj = 00000036 10          42          175
# error_counter,err_inj = 00000037 10          43          176
# error_counter,err_inj = 00000037 00          44          177
# error_counter,err_inj = 00000037 00          44          178
# error_counter,err_inj = 00000037 00          44          179
# error_counter,err_inj = 00000037 00          44          180
# error_counter,err_inj = 00000037 00          44          181
# error_counter,err_inj = 00000037 00          44          182
# error_counter,err_inj = 00000037 00          44          183
# error_counter,err_inj = 00000037 00          44          184
# error_counter,err_inj = 00000037 00          44          185
# error_counter,err_inj = 00000037 00          44          186
# error_counter,err_inj = 00000037 00          44          187
# error_counter,err_inj = 00000038 10          44          188
# error_counter,err_inj = 00000039 10          44          189
# error_counter,err_inj = 0000003a 10          45          190
# error_counter,err_inj = 0000003b 10          46          191
# error_counter,err_inj = 0000003c 10          47          192
# error_counter,err_inj = 0000003c 00          48          193
# error_counter,err_inj = 0000003c 00          48          194
# error_counter,err_inj = 0000003c 00          48          195
# error_counter,err_inj = 0000003c 00          48          196
# error_counter,err_inj = 0000003c 00          48          197
# error_counter,err_inj = 0000003c 00          48          198
# error_counter,err_inj = 0000003c 00          48          199
# error_counter,err_inj = 0000003c 00          48          200
# error_counter,err_inj = 0000003c 00          48          201
# error_counter,err_inj = 0000003c 00          48          202
# error_counter,err_inj = 0000003c 00          48          203
# error_counter,err_inj = 0000003d 10          48          204
# error_counter,err_inj = 0000003e 10          48          205
# error_counter,err_inj = 0000003f 10          49          206
# error_counter,err_inj = 00000040 10          50          207
# error_counter,err_inj = 00000041 10          51          208
# error_counter,err_inj = 00000041 00          52          209
# error_counter,err_inj = 00000041 00          52          210
# error_counter,err_inj = 00000041 00          52          211
# error_counter,err_inj = 00000041 00          52          212
# error_counter,err_inj = 00000041 00          52          213
# error_counter,err_inj = 00000041 00          52          214
# error_counter,err_inj = 00000041 00          52          215
# error_counter,err_inj = 00000041 00          52          216
# error_counter,err_inj = 00000041 00          52          217
# error_counter,err_inj = 00000041 00          52          218
# error_counter,err_inj = 00000041 00          52          219
# error_counter,err_inj = 00000042 10          52          220
# error_counter,err_inj = 00000043 10          52          221
# error_counter,err_inj = 00000044 10          53          222
# error_counter,err_inj = 00000045 10          54          223
# error_counter,err_inj = 00000046 10          55          224
# error_counter,err_inj = 00000046 00          56          225
# error_counter,err_inj = 00000046 00          56          226
# error_counter,err_inj = 00000046 00          56          227
# error_counter,err_inj = 00000046 00          56          228
# error_counter,err_inj = 00000046 00          56          229
# error_counter,err_inj = 00000046 00          56          230
# error_counter,err_inj = 00000046 00          56          231
# error_counter,err_inj = 00000046 00          56          232
# error_counter,err_inj = 00000046 00          56          233
# error_counter,err_inj = 00000046 00          56          234
```

```
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# boo! in = 1, out = 0, w_ct =         220, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# boo! in = 1, out = 0, w_ct =         229, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# boo! in = 1, out = 0, w_ct =         236, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# boo! in = 1, out = 0, w_ct =         245, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# boo! in = 1, out = 0, w_ct =         252, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          80, OUT: good =         230, bad =
26
```

```
# error_counter,err_inj = 00000046 00          56         235
# error_counter,err_inj = 00000047 10          56         236
# error_counter,err_inj = 00000048 10          56         237
# error_counter,err_inj = 00000049 10          57         238
# error_counter,err_inj = 0000004a 10          58         239
# error_counter,err_inj = 0000004b 10          59         240
# error_counter,err_inj = 0000004b 00          60         241
# error_counter,err_inj = 0000004b 00          60         242
# error_counter,err_inj = 0000004b 00          60         243
# error_counter,err_inj = 0000004b 00          60         244
# error_counter,err_inj = 0000004b 00          60         245
# error_counter,err_inj = 0000004b 00          60         246
# error_counter,err_inj = 0000004b 00          60         247
# error_counter,err_inj = 0000004b 00          60         248
# error_counter,err_inj = 0000004b 00          60         249
# error_counter,err_inj = 0000004b 00          60         250
# error_counter,err_inj = 0000004b 00          60         251
# error_counter,err_inj = 0000004c 10          60         252
# error_counter,err_inj = 0000004d 10          60         253
# error_counter,err_inj = 0000004e 10          61         254
# error_counter,err_inj = 0000004f 10          62         255
```

2e) Transcript: 3 consecutive error injections breaks the decoder

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000000 00          0         12
# error_counter,err_inj = 00000000 00          0         13
# error_counter,err_inj = 00000000 00          0         14
# error_counter,err_inj = 00000000 00          0         15
# error_counter,err_inj = 00000000 00          0         16
# error_counter,err_inj = 00000000 00          0         17
# error_counter,err_inj = 00000000 00          0         18
# error_counter,err_inj = 00000000 00          0         19
# error_counter,err_inj = 00000000 00          0         20
# error_counter,err_inj = 00000000 00          0         21
# error_counter,err_inj = 00000000 00          0         22
# error_counter,err_inj = 00000000 00          0         23
# error_counter,err_inj = 00000000 00          0         24
# error_counter,err_inj = 00000000 00          0         25
# error_counter,err_inj = 00000000 00          0         26
# error_counter,err_inj = 00000000 00          0         27
# error_counter,err_inj = 00000000 00          0         28
# error_counter,err_inj = 00000000 00          0         29
# error_counter,err_inj = 00000001 11          0         30
# error_counter,err_inj = 00000002 11          0         31
# error_counter,err_inj = 00000003 11          2         32
# error_counter,err_inj = 00000003 00          4         33
# error_counter,err_inj = 00000003 00          4         34
# error_counter,err_inj = 00000003 00          4         35
# error_counter,err_inj = 00000003 00          4         36
```

```
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# boo! in = 0, out = 1, w_ct =          71, err = 00,
1045000, BAD!
# boo! in = 0, out = 1, w_ct =          72, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# boo! in = 1, out = 0, w_ct =          76, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# boo! in = 1, out = 0, w_ct =         103, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         104, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# boo! in = 1, out = 0, w_ct =         108, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
```

```
# error_counter,err_inj = 00000003 00          4          37
# error_counter,err_inj = 00000003 00          4          38
# error_counter,err_inj = 00000003 00          4          39
# error_counter,err_inj = 00000003 00          4          40
# error_counter,err_inj = 00000003 00          4          41
# error_counter,err_inj = 00000003 00          4          42
# error_counter,err_inj = 00000003 00          4          43
# error_counter,err_inj = 00000003 00          4          44
# error_counter,err_inj = 00000003 00          4          45
# error_counter,err_inj = 00000003 00          4          46
# error_counter,err_inj = 00000003 00          4          47
# error_counter,err_inj = 00000003 00          4          48
# error_counter,err_inj = 00000003 00          4          49
# error_counter,err_inj = 00000003 00          4          50
# error_counter,err_inj = 00000003 00          4          51
# error_counter,err_inj = 00000003 00          4          52
# error_counter,err_inj = 00000003 00          4          53
# error_counter,err_inj = 00000003 00          4          54
# error_counter,err_inj = 00000003 00          4          55
# error_counter,err_inj = 00000003 00          4          56
# error_counter,err_inj = 00000003 00          4          57
# error_counter,err_inj = 00000003 00          4          58
# error_counter,err_inj = 00000003 00          4          59
# error_counter,err_inj = 00000003 00          4          60
# error_counter,err_inj = 00000003 00          4          61
# error_counter,err_inj = 00000004 11          4          62
# error_counter,err_inj = 00000005 11          4          63
# error_counter,err_inj = 00000006 11          6          64
# error_counter,err_inj = 00000006 00          8          65
# error_counter,err_inj = 00000006 00          8          66
# error_counter,err_inj = 00000006 00          8          67
# error_counter,err_inj = 00000006 00          8          68
# error_counter,err_inj = 00000006 00          8          69
# error_counter,err_inj = 00000006 00          8          70
# error_counter,err_inj = 00000006 00          8          71
# error_counter,err_inj = 00000006 00          8          72
# error_counter,err_inj = 00000006 00          8          73
# error_counter,err_inj = 00000006 00          8          74
# error_counter,err_inj = 00000006 00          8          75
# error_counter,err_inj = 00000006 00          8          76
# error_counter,err_inj = 00000006 00          8          77
# error_counter,err_inj = 00000006 00          8          78
# error_counter,err_inj = 00000006 00          8          79
# error_counter,err_inj = 00000006 00          8          80
# error_counter,err_inj = 00000006 00          8          81
# error_counter,err_inj = 00000006 00          8          82
# error_counter,err_inj = 00000006 00          8          83
# error_counter,err_inj = 00000006 00          8          84
# error_counter,err_inj = 00000006 00          8          85
# error_counter,err_inj = 00000006 00          8          86
# error_counter,err_inj = 00000006 00          8          87
# error_counter,err_inj = 00000006 00          8          88
# error_counter,err_inj = 00000006 00          8          89
# error_counter,err_inj = 00000006 00          8          90
# error_counter,err_inj = 00000006 00          8          91
# error_counter,err_inj = 00000006 00          8          92
# error_counter,err_inj = 00000006 00          8          93
# error_counter,err_inj = 00000007 11          8          94
# error_counter,err_inj = 00000008 11          8          95
# error_counter,err_inj = 00000009 11         10          96
# error_counter,err_inj = 00000009 00         12          97
# error_counter,err_inj = 00000009 00         12          98
# error_counter,err_inj = 00000009 00         12          99
# error_counter,err_inj = 00000009 00         12         100
# error_counter,err_inj = 00000009 00         12         101
# error_counter,err_inj = 00000009 00         12         102
# error_counter,err_inj = 00000009 00         12         103
# error_counter,err_inj = 00000009 00         12         104
# error_counter,err_inj = 00000009 00         12         105
# error_counter,err_inj = 00000009 00         12         106
# error_counter,err_inj = 00000009 00         12         107
# error_counter,err_inj = 00000009 00         12         108
# error_counter,err_inj = 00000009 00         12         109
# error_counter,err_inj = 00000009 00         12         110
# error_counter,err_inj = 00000009 00         12         111
# error_counter,err_inj = 00000009 00         12         112
# error_counter,err_inj = 00000009 00         12         113
# error_counter,err_inj = 00000009 00         12         114
# error_counter,err_inj = 00000009 00         12         115
# error_counter,err_inj = 00000009 00         12         116
# error_counter,err_inj = 00000009 00         12         117
# error_counter,err_inj = 00000009 00         12         118
# error_counter,err_inj = 00000009 00         12         119
# error_counter,err_inj = 00000009 00         12         120
# error_counter,err_inj = 00000009 00         12         121
# error_counter,err_inj = 00000009 00         12         122
# error_counter,err_inj = 00000009 00         12         123
# error_counter,err_inj = 00000009 00         12         124
# error_counter,err_inj = 00000009 00         12         125
# error_counter,err_inj = 0000000a 11         12         126
# error_counter,err_inj = 0000000b 11         12         127
# error_counter,err_inj = 0000000c 11         14         128
# error_counter,err_inj = 0000000c 00         16         129
```

```
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# boo! in = 0, out = 1, w_ct =         135, err = 00,
1045000, BAD!
# boo! in = 0, out = 1, w_ct =         136, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# boo! in = 1, out = 0, w_ct =         140, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# boo! in = 1, out = 0, w_ct =         167, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         168, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# boo! in = 1, out = 0, w_ct =         172, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# boo! in = 1, out = 0, w_ct =         199, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         200, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# boo! in = 1, out = 0, w_ct =         204, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
```

```
# error_counter,err_inj = 0000000c 00          16          130
# error_counter,err_inj = 0000000c 00          16          131
# error_counter,err_inj = 0000000c 00          16          132
# error_counter,err_inj = 0000000c 00          16          133
# error_counter,err_inj = 0000000c 00          16          134
# error_counter,err_inj = 0000000c 00          16          135
# error_counter,err_inj = 0000000c 00          16          136
# error_counter,err_inj = 0000000c 00          16          137
# error_counter,err_inj = 0000000c 00          16          138
# error_counter,err_inj = 0000000c 00          16          139
# error_counter,err_inj = 0000000c 00          16          140
# error_counter,err_inj = 0000000c 00          16          141
# error_counter,err_inj = 0000000c 00          16          142
# error_counter,err_inj = 0000000c 00          16          143
# error_counter,err_inj = 0000000c 00          16          144
# error_counter,err_inj = 0000000c 00          16          145
# error_counter,err_inj = 0000000c 00          16          146
# error_counter,err_inj = 0000000c 00          16          147
# error_counter,err_inj = 0000000c 00          16          148
# error_counter,err_inj = 0000000c 00          16          149
# error_counter,err_inj = 0000000c 00          16          150
# error_counter,err_inj = 0000000c 00          16          151
# error_counter,err_inj = 0000000c 00          16          152
# error_counter,err_inj = 0000000c 00          16          153
# error_counter,err_inj = 0000000c 00          16          154
# error_counter,err_inj = 0000000c 00          16          155
# error_counter,err_inj = 0000000c 00          16          156
# error_counter,err_inj = 0000000c 00          16          157
# error_counter,err_inj = 0000000d 11          16          158
# error_counter,err_inj = 0000000e 11          16          159
# error_counter,err_inj = 0000000f 11          18          160
# error_counter,err_inj = 0000000f 00          20          161
# error_counter,err_inj = 0000000f 00          20          162
# error_counter,err_inj = 0000000f 00          20          163
# error_counter,err_inj = 0000000f 00          20          164
# error_counter,err_inj = 0000000f 00          20          165
# error_counter,err_inj = 0000000f 00          20          166
# error_counter,err_inj = 0000000f 00          20          167
# error_counter,err_inj = 0000000f 00          20          168
# error_counter,err_inj = 0000000f 00          20          169
# error_counter,err_inj = 0000000f 00          20          170
# error_counter,err_inj = 0000000f 00          20          171
# error_counter,err_inj = 0000000f 00          20          172
# error_counter,err_inj = 0000000f 00          20          173
# error_counter,err_inj = 0000000f 00          20          174
# error_counter,err_inj = 0000000f 00          20          175
# error_counter,err_inj = 0000000f 00          20          176
# error_counter,err_inj = 0000000f 00          20          177
# error_counter,err_inj = 0000000f 00          20          178
# error_counter,err_inj = 0000000f 00          20          179
# error_counter,err_inj = 0000000f 00          20          180
# error_counter,err_inj = 0000000f 00          20          181
# error_counter,err_inj = 0000000f 00          20          182
# error_counter,err_inj = 0000000f 00          20          183
# error_counter,err_inj = 0000000f 00          20          184
# error_counter,err_inj = 0000000f 00          20          185
# error_counter,err_inj = 0000000f 00          20          186
# error_counter,err_inj = 0000000f 00          20          187
# error_counter,err_inj = 0000000f 00          20          188
# error_counter,err_inj = 0000000f 00          20          189
# error_counter,err_inj = 00000010 11          20          190
# error_counter,err_inj = 00000011 11          20          191
# error_counter,err_inj = 00000012 11          22          192
# error_counter,err_inj = 00000012 00          24          193
# error_counter,err_inj = 00000012 00          24          194
# error_counter,err_inj = 00000012 00          24          195
# error_counter,err_inj = 00000012 00          24          196
# error_counter,err_inj = 00000012 00          24          197
# error_counter,err_inj = 00000012 00          24          198
# error_counter,err_inj = 00000012 00          24          199
# error_counter,err_inj = 00000012 00          24          200
# error_counter,err_inj = 00000012 00          24          201
# error_counter,err_inj = 00000012 00          24          202
# error_counter,err_inj = 00000012 00          24          203
# error_counter,err_inj = 00000012 00          24          204
# error_counter,err_inj = 00000012 00          24          205
# error_counter,err_inj = 00000012 00          24          206
# error_counter,err_inj = 00000012 00          24          207
# error_counter,err_inj = 00000012 00          24          208
# error_counter,err_inj = 00000012 00          24          209
# error_counter,err_inj = 00000012 00          24          210
# error_counter,err_inj = 00000012 00          24          211
# error_counter,err_inj = 00000012 00          24          212
# error_counter,err_inj = 00000012 00          24          213
# error_counter,err_inj = 00000012 00          24          214
# error_counter,err_inj = 00000012 00          24          215
# error_counter,err_inj = 00000012 00          24          216
# error_counter,err_inj = 00000012 00          24          217
# error_counter,err_inj = 00000012 00          24          218
# error_counter,err_inj = 00000012 00          24          219
# error_counter,err_inj = 00000012 00          24          220
# error_counter,err_inj = 00000012 00          24          221
# error_counter,err_inj = 00000013 11          24          222
```

```
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# boo! in = 1, out = 0, w_ct =         231, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         232, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# boo! in = 1, out = 0, w_ct =         236, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          24, OUT: good =         238, bad =
18
```

```
# error_counter,err_inj = 00000014 11          24         223
# error_counter,err_inj = 00000015 11          26         224
# error_counter,err_inj = 00000015 00          28         225
# error_counter,err_inj = 00000015 00          28         226
# error_counter,err_inj = 00000015 00          28         227
# error_counter,err_inj = 00000015 00          28         228
# error_counter,err_inj = 00000015 00          28         229
# error_counter,err_inj = 00000015 00          28         230
# error_counter,err_inj = 00000015 00          28         231
# error_counter,err_inj = 00000015 00          28         232
# error_counter,err_inj = 00000015 00          28         233
# error_counter,err_inj = 00000015 00          28         234
# error_counter,err_inj = 00000015 00          28         235
# error_counter,err_inj = 00000015 00          28         236
# error_counter,err_inj = 00000015 00          28         237
# error_counter,err_inj = 00000015 00          28         238
# error_counter,err_inj = 00000015 00          28         239
# error_counter,err_inj = 00000015 00          28         240
# error_counter,err_inj = 00000015 00          28         241
# error_counter,err_inj = 00000015 00          28         242
# error_counter,err_inj = 00000015 00          28         243
# error_counter,err_inj = 00000015 00          28         244
# error_counter,err_inj = 00000015 00          28         245
# error_counter,err_inj = 00000015 00          28         246
# error_counter,err_inj = 00000015 00          28         247
# error_counter,err_inj = 00000015 00          28         248
# error_counter,err_inj = 00000015 00          28         249
# error_counter,err_inj = 00000015 00          28         250
# error_counter,err_inj = 00000015 00          28         251
# error_counter,err_inj = 00000015 00          28         252
# error_counter,err_inj = 00000015 00          28         253
# error_counter,err_inj = 00000016 11          28         254
# error_counter,err_inj = 00000017 11          28         255
```

Problem:	**# of Errors**	**Pattern of Errors**	**# of Errors Uncorrected**
2a	16	00000000000000010000000000000001…	0
2a1	32	00000001000000001…	0
2a2	32	00000001000000001…	0
2a3	16	00000000000000000000000000000011…	0
2a4	32	00000000000000110000000000000011…	0
2a5	32	00000000000000110000000000000011…	0
2a6	32	00000000000000000000000000001111…	0

2a7	32	00000000000000000000000000000000000000000000000000000000000010101010	0
2a8	16	00000000000000000000000000000011	0
2b, 2b3	18	00010000000000000000000000000000000000000000000000000100000000000001	0
2b1, b2	30	Random bad bits with a random spacing of 0 to 8 bits	0
2b4 &b5	16	Random bad bits with two consecutive injections	0
2b6&7	32	Random bad bits with 4 consecutive injections	12
2b8	8	Random bad bits with 2 consecutive injections with a spacing of 32	0
2c	80	00000000000000000000001010101	42
2d	80	00000000000000000000010101010	26
2e	24	00000000000000000000000000000000000000000000000000000000000111111	18

2A) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000000 00          0         12
# error_counter,err_inj = 00000000 00          0         13
# error_counter,err_inj = 00000000 00          0         14
# error_counter,err_inj = 00000000 00          0         15
# error_counter,err_inj = 00000001 01          0         16
# error_counter,err_inj = 00000001 00          0         17
# error_counter,err_inj = 00000001 00          0         18
# error_counter,err_inj = 00000001 00          0         19
# error_counter,err_inj = 00000001 00          0         20
# error_counter,err_inj = 00000001 00          0         21
# error_counter,err_inj = 00000001 00          0         22
# error_counter,err_inj = 00000001 00          0         23
# error_counter,err_inj = 00000001 00          0         24
# error_counter,err_inj = 00000001 00          0         25
# error_counter,err_inj = 00000001 00          0         26
# error_counter,err_inj = 00000001 00          0         27
# error_counter,err_inj = 00000001 00          0         28
# error_counter,err_inj = 00000001 00          0         29
# error_counter,err_inj = 00000001 00          0         30
# error_counter,err_inj = 00000001 00          0         31
# error_counter,err_inj = 00000002 01          0         32
# error_counter,err_inj = 00000002 00          0         33
# error_counter,err_inj = 00000002 00          0         34
# error_counter,err_inj = 00000002 00          0         35
# error_counter,err_inj = 00000002 00          0         36
# error_counter,err_inj = 00000002 00          0         37
# error_counter,err_inj = 00000002 00          0         38
# error_counter,err_inj = 00000002 00          0         39
# error_counter,err_inj = 00000002 00          0         40
# error_counter,err_inj = 00000002 00          0         41
# error_counter,err_inj = 00000002 00          0         42
# error_counter,err_inj = 00000002 00          0         43
# error_counter,err_inj = 00000002 00          0         44
# error_counter,err_inj = 00000002 00          0         45
# error_counter,err_inj = 00000002 00          0         46
# error_counter,err_inj = 00000002 00          0         47
# error_counter,err_inj = 00000003 01          0         48
# error_counter,err_inj = 00000003 00          0         49
```

```
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
```

```
# error_counter,err_inj = 00000003 00          0          50
# error_counter,err_inj = 00000003 00          0          51
# error_counter,err_inj = 00000003 00          0          52
# error_counter,err_inj = 00000003 00          0          53
# error_counter,err_inj = 00000003 00          0          54
# error_counter,err_inj = 00000003 00          0          55
# error_counter,err_inj = 00000003 00          0          56
# error_counter,err_inj = 00000003 00          0          57
# error_counter,err_inj = 00000003 00          0          58
# error_counter,err_inj = 00000003 00          0          59
# error_counter,err_inj = 00000003 00          0          60
# error_counter,err_inj = 00000003 00          0          61
# error_counter,err_inj = 00000003 00          0          62
# error_counter,err_inj = 00000003 00          0          63
# error_counter,err_inj = 00000004 01          0          64
# error_counter,err_inj = 00000004 00          0          65
# error_counter,err_inj = 00000004 00          0          66
# error_counter,err_inj = 00000004 00          0          67
# error_counter,err_inj = 00000004 00          0          68
# error_counter,err_inj = 00000004 00          0          69
# error_counter,err_inj = 00000004 00          0          70
# error_counter,err_inj = 00000004 00          0          71
# error_counter,err_inj = 00000004 00          0          72
# error_counter,err_inj = 00000004 00          0          73
# error_counter,err_inj = 00000004 00          0          74
# error_counter,err_inj = 00000004 00          0          75
# error_counter,err_inj = 00000004 00          0          76
# error_counter,err_inj = 00000004 00          0          77
# error_counter,err_inj = 00000004 00          0          78
# error_counter,err_inj = 00000004 00          0          79
# error_counter,err_inj = 00000005 01          0          80
# error_counter,err_inj = 00000005 00          0          81
# error_counter,err_inj = 00000005 00          0          82
# error_counter,err_inj = 00000005 00          0          83
# error_counter,err_inj = 00000005 00          0          84
# error_counter,err_inj = 00000005 00          0          85
# error_counter,err_inj = 00000005 00          0          86
# error_counter,err_inj = 00000005 00          0          87
# error_counter,err_inj = 00000005 00          0          88
# error_counter,err_inj = 00000005 00          0          89
# error_counter,err_inj = 00000005 00          0          90
# error_counter,err_inj = 00000005 00          0          91
# error_counter,err_inj = 00000005 00          0          92
# error_counter,err_inj = 00000005 00          0          93
# error_counter,err_inj = 00000005 00          0          94
# error_counter,err_inj = 00000005 00          0          95
# error_counter,err_inj = 00000006 01          0          96
# error_counter,err_inj = 00000006 00          0          97
# error_counter,err_inj = 00000006 00          0          98
# error_counter,err_inj = 00000006 00          0          99
# error_counter,err_inj = 00000006 00          0         100
# error_counter,err_inj = 00000006 00          0         101
# error_counter,err_inj = 00000006 00          0         102
# error_counter,err_inj = 00000006 00          0         103
# error_counter,err_inj = 00000006 00          0         104
# error_counter,err_inj = 00000006 00          0         105
# error_counter,err_inj = 00000006 00          0         106
# error_counter,err_inj = 00000006 00          0         107
# error_counter,err_inj = 00000006 00          0         108
# error_counter,err_inj = 00000006 00          0         109
# error_counter,err_inj = 00000006 00          0         110
# error_counter,err_inj = 00000006 00          0         111
# error_counter,err_inj = 00000007 01          0         112
# error_counter,err_inj = 00000007 00          0         113
# error_counter,err_inj = 00000007 00          0         114
# error_counter,err_inj = 00000007 00          0         115
# error_counter,err_inj = 00000007 00          0         116
# error_counter,err_inj = 00000007 00          0         117
# error_counter,err_inj = 00000007 00          0         118
# error_counter,err_inj = 00000007 00          0         119
# error_counter,err_inj = 00000007 00          0         120
# error_counter,err_inj = 00000007 00          0         121
# error_counter,err_inj = 00000007 00          0         122
# error_counter,err_inj = 00000007 00          0         123
# error_counter,err_inj = 00000007 00          0         124
# error_counter,err_inj = 00000007 00          0         125
# error_counter,err_inj = 00000007 00          0         126
# error_counter,err_inj = 00000007 00          0         127
# error_counter,err_inj = 00000008 01          0         128
# error_counter,err_inj = 00000008 00          0         129
# error_counter,err_inj = 00000008 00          0         130
# error_counter,err_inj = 00000008 00          0         131
# error_counter,err_inj = 00000008 00          0         132
# error_counter,err_inj = 00000008 00          0         133
# error_counter,err_inj = 00000008 00          0         134
# error_counter,err_inj = 00000008 00          0         135
# error_counter,err_inj = 00000008 00          0         136
# error_counter,err_inj = 00000008 00          0         137
# error_counter,err_inj = 00000008 00          0         138
# error_counter,err_inj = 00000008 00          0         139
# error_counter,err_inj = 00000008 00          0         140
# error_counter,err_inj = 00000008 00          0         141
# error_counter,err_inj = 00000008 00          0         142
```

```
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
```

```
# error_counter,err_inj = 00000008 00          0         143
# error_counter,err_inj = 00000009 01          0         144
# error_counter,err_inj = 00000009 00          0         145
# error_counter,err_inj = 00000009 00          0         146
# error_counter,err_inj = 00000009 00          0         147
# error_counter,err_inj = 00000009 00          0         148
# error_counter,err_inj = 00000009 00          0         149
# error_counter,err_inj = 00000009 00          0         150
# error_counter,err_inj = 00000009 00          0         151
# error_counter,err_inj = 00000009 00          0         152
# error_counter,err_inj = 00000009 00          0         153
# error_counter,err_inj = 00000009 00          0         154
# error_counter,err_inj = 00000009 00          0         155
# error_counter,err_inj = 00000009 00          0         156
# error_counter,err_inj = 00000009 00          0         157
# error_counter,err_inj = 00000009 00          0         158
# error_counter,err_inj = 00000009 00          0         159
# error_counter,err_inj = 0000000a 01          0         160
# error_counter,err_inj = 0000000a 00          0         161
# error_counter,err_inj = 0000000a 00          0         162
# error_counter,err_inj = 0000000a 00          0         163
# error_counter,err_inj = 0000000a 00          0         164
# error_counter,err_inj = 0000000a 00          0         165
# error_counter,err_inj = 0000000a 00          0         166
# error_counter,err_inj = 0000000a 00          0         167
# error_counter,err_inj = 0000000a 00          0         168
# error_counter,err_inj = 0000000a 00          0         169
# error_counter,err_inj = 0000000a 00          0         170
# error_counter,err_inj = 0000000a 00          0         171
# error_counter,err_inj = 0000000a 00          0         172
# error_counter,err_inj = 0000000a 00          0         173
# error_counter,err_inj = 0000000a 00          0         174
# error_counter,err_inj = 0000000a 00          0         175
# error_counter,err_inj = 0000000b 01          0         176
# error_counter,err_inj = 0000000b 00          0         177
# error_counter,err_inj = 0000000b 00          0         178
# error_counter,err_inj = 0000000b 00          0         179
# error_counter,err_inj = 0000000b 00          0         180
# error_counter,err_inj = 0000000b 00          0         181
# error_counter,err_inj = 0000000b 00          0         182
# error_counter,err_inj = 0000000b 00          0         183
# error_counter,err_inj = 0000000b 00          0         184
# error_counter,err_inj = 0000000b 00          0         185
# error_counter,err_inj = 0000000b 00          0         186
# error_counter,err_inj = 0000000b 00          0         187
# error_counter,err_inj = 0000000b 00          0         188
# error_counter,err_inj = 0000000b 00          0         189
# error_counter,err_inj = 0000000b 00          0         190
# error_counter,err_inj = 0000000b 00          0         191
# error_counter,err_inj = 0000000c 01          0         192
# error_counter,err_inj = 0000000c 00          0         193
# error_counter,err_inj = 0000000c 00          0         194
# error_counter,err_inj = 0000000c 00          0         195
# error_counter,err_inj = 0000000c 00          0         196
# error_counter,err_inj = 0000000c 00          0         197
# error_counter,err_inj = 0000000c 00          0         198
# error_counter,err_inj = 0000000c 00          0         199
# error_counter,err_inj = 0000000c 00          0         200
# error_counter,err_inj = 0000000c 00          0         201
# error_counter,err_inj = 0000000c 00          0         202
# error_counter,err_inj = 0000000c 00          0         203
# error_counter,err_inj = 0000000c 00          0         204
# error_counter,err_inj = 0000000c 00          0         205
# error_counter,err_inj = 0000000c 00          0         206
# error_counter,err_inj = 0000000c 00          0         207
# error_counter,err_inj = 0000000d 01          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000d 00          0         213
# error_counter,err_inj = 0000000d 00          0         214
# error_counter,err_inj = 0000000d 00          0         215
# error_counter,err_inj = 0000000d 00          0         216
# error_counter,err_inj = 0000000d 00          0         217
# error_counter,err_inj = 0000000d 00          0         218
# error_counter,err_inj = 0000000d 00          0         219
# error_counter,err_inj = 0000000d 00          0         220
# error_counter,err_inj = 0000000d 00          0         221
# error_counter,err_inj = 0000000d 00          0         222
# error_counter,err_inj = 0000000d 00          0         223
# error_counter,err_inj = 0000000e 01          0         224
# error_counter,err_inj = 0000000e 00          0         225
# error_counter,err_inj = 0000000e 00          0         226
# error_counter,err_inj = 0000000e 00          0         227
# error_counter,err_inj = 0000000e 00          0         228
# error_counter,err_inj = 0000000e 00          0         229
# error_counter,err_inj = 0000000e 00          0         230
# error_counter,err_inj = 0000000e 00          0         231
# error_counter,err_inj = 0000000e 00          0         232
# error_counter,err_inj = 0000000e 00          0         233
# error_counter,err_inj = 0000000e 00          0         234
# error_counter,err_inj = 0000000e 00          0         235
```

```
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000e 00           0         236
# error_counter,err_inj = 0000000e 00           0         237
# error_counter,err_inj = 0000000e 00           0         238
# error_counter,err_inj = 0000000e 00           0         239
# error_counter,err_inj = 0000000f 01           0         240
# error_counter,err_inj = 0000000f 00           0         241
# error_counter,err_inj = 0000000f 00           0         242
# error_counter,err_inj = 0000000f 00           0         243
# error_counter,err_inj = 0000000f 00           0         244
# error_counter,err_inj = 0000000f 00           0         245
# error_counter,err_inj = 0000000f 00           0         246
# error_counter,err_inj = 0000000f 00           0         247
# error_counter,err_inj = 0000000f 00           0         248
# error_counter,err_inj = 0000000f 00           0         249
# error_counter,err_inj = 0000000f 00           0         250
# error_counter,err_inj = 0000000f 00           0         251
# error_counter,err_inj = 0000000f 00           0         252
# error_counter,err_inj = 0000000f 00           0         253
# error_counter,err_inj = 0000000f 00           0         254
# error_counter,err_inj = 0000000f 00           0         255
# word_count =       10440
```

2a1) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000001 01           0           8
# error_counter,err_inj = 00000001 00           0           9
# error_counter,err_inj = 00000001 00           0          10
# error_counter,err_inj = 00000001 00           0          11
# error_counter,err_inj = 00000001 00           0          12
# error_counter,err_inj = 00000001 00           0          13
# error_counter,err_inj = 00000001 00           0          14
# error_counter,err_inj = 00000001 00           0          15
# error_counter,err_inj = 00000002 01           0          16
# error_counter,err_inj = 00000002 00           0          17
# error_counter,err_inj = 00000002 00           0          18
# error_counter,err_inj = 00000002 00           0          19
# error_counter,err_inj = 00000002 00           0          20
# error_counter,err_inj = 00000002 00           0          21
# error_counter,err_inj = 00000002 00           0          22
# error_counter,err_inj = 00000002 00           0          23
# error_counter,err_inj = 00000003 01           0          24
# error_counter,err_inj = 00000003 00           0          25
# error_counter,err_inj = 00000003 00           0          26
# error_counter,err_inj = 00000003 00           0          27
# error_counter,err_inj = 00000003 00           0          28
# error_counter,err_inj = 00000003 00           0          29
# error_counter,err_inj = 00000003 00           0          30
# error_counter,err_inj = 00000003 00           0          31
# error_counter,err_inj = 00000004 01           0          32
# error_counter,err_inj = 00000004 00           0          33
# error_counter,err_inj = 00000004 00           0          34
# error_counter,err_inj = 00000004 00           0          35
# error_counter,err_inj = 00000004 00           0          36
# error_counter,err_inj = 00000004 00           0          37
# error_counter,err_inj = 00000004 00           0          38
# error_counter,err_inj = 00000004 00           0          39
# error_counter,err_inj = 00000005 01           0          40
# error_counter,err_inj = 00000005 00           0          41
# error_counter,err_inj = 00000005 00           0          42
# error_counter,err_inj = 00000005 00           0          43
# error_counter,err_inj = 00000005 00           0          44
# error_counter,err_inj = 00000005 00           0          45
# error_counter,err_inj = 00000005 00           0          46
# error_counter,err_inj = 00000005 00           0          47
# error_counter,err_inj = 00000006 01           0          48
# error_counter,err_inj = 00000006 00           0          49
# error_counter,err_inj = 00000006 00           0          50
# error_counter,err_inj = 00000006 00           0          51
# error_counter,err_inj = 00000006 00           0          52
# error_counter,err_inj = 00000006 00           0          53
# error_counter,err_inj = 00000006 00           0          54
# error_counter,err_inj = 00000006 00           0          55
# error_counter,err_inj = 00000007 01           0          56
# error_counter,err_inj = 00000007 00           0          57
# error_counter,err_inj = 00000007 00           0          58
```

```
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
```

```
# error_counter,err_inj = 00000007 00          0          59
# error_counter,err_inj = 00000007 00          0          60
# error_counter,err_inj = 00000007 00          0          61
# error_counter,err_inj = 00000007 00          0          62
# error_counter,err_inj = 00000007 00          0          63
# error_counter,err_inj = 00000008 01          0          64
# error_counter,err_inj = 00000008 00          0          65
# error_counter,err_inj = 00000008 00          0          66
# error_counter,err_inj = 00000008 00          0          67
# error_counter,err_inj = 00000008 00          0          68
# error_counter,err_inj = 00000008 00          0          69
# error_counter,err_inj = 00000008 00          0          70
# error_counter,err_inj = 00000008 00          0          71
# error_counter,err_inj = 00000009 01          0          72
# error_counter,err_inj = 00000009 00          0          73
# error_counter,err_inj = 00000009 00          0          74
# error_counter,err_inj = 00000009 00          0          75
# error_counter,err_inj = 00000009 00          0          76
# error_counter,err_inj = 00000009 00          0          77
# error_counter,err_inj = 00000009 00          0          78
# error_counter,err_inj = 00000009 00          0          79
# error_counter,err_inj = 0000000a 01          0          80
# error_counter,err_inj = 0000000a 00          0          81
# error_counter,err_inj = 0000000a 00          0          82
# error_counter,err_inj = 0000000a 00          0          83
# error_counter,err_inj = 0000000a 00          0          84
# error_counter,err_inj = 0000000a 00          0          85
# error_counter,err_inj = 0000000a 00          0          86
# error_counter,err_inj = 0000000a 00          0          87
# error_counter,err_inj = 0000000b 01          0          88
# error_counter,err_inj = 0000000b 00          0          89
# error_counter,err_inj = 0000000b 00          0          90
# error_counter,err_inj = 0000000b 00          0          91
# error_counter,err_inj = 0000000b 00          0          92
# error_counter,err_inj = 0000000b 00          0          93
# error_counter,err_inj = 0000000b 00          0          94
# error_counter,err_inj = 0000000b 00          0          95
# error_counter,err_inj = 0000000c 01          0          96
# error_counter,err_inj = 0000000c 00          0          97
# error_counter,err_inj = 0000000c 00          0          98
# error_counter,err_inj = 0000000c 00          0          99
# error_counter,err_inj = 0000000c 00          0         100
# error_counter,err_inj = 0000000c 00          0         101
# error_counter,err_inj = 0000000c 00          0         102
# error_counter,err_inj = 0000000c 00          0         103
# error_counter,err_inj = 0000000d 01          0         104
# error_counter,err_inj = 0000000d 00          0         105
# error_counter,err_inj = 0000000d 00          0         106
# error_counter,err_inj = 0000000d 00          0         107
# error_counter,err_inj = 0000000d 00          0         108
# error_counter,err_inj = 0000000d 00          0         109
# error_counter,err_inj = 0000000d 00          0         110
# error_counter,err_inj = 0000000d 00          0         111
# error_counter,err_inj = 0000000e 01          0         112
# error_counter,err_inj = 0000000e 00          0         113
# error_counter,err_inj = 0000000e 00          0         114
# error_counter,err_inj = 0000000e 00          0         115
# error_counter,err_inj = 0000000e 00          0         116
# error_counter,err_inj = 0000000e 00          0         117
# error_counter,err_inj = 0000000e 00          0         118
# error_counter,err_inj = 0000000e 00          0         119
# error_counter,err_inj = 0000000f 01          0         120
# error_counter,err_inj = 0000000f 00          0         121
# error_counter,err_inj = 0000000f 00          0         122
# error_counter,err_inj = 0000000f 00          0         123
# error_counter,err_inj = 0000000f 00          0         124
# error_counter,err_inj = 0000000f 00          0         125
# error_counter,err_inj = 0000000f 00          0         126
# error_counter,err_inj = 0000000f 00          0         127
# error_counter,err_inj = 00000010 01          0         128
# error_counter,err_inj = 00000010 00          0         129
# error_counter,err_inj = 00000010 00          0         130
# error_counter,err_inj = 00000010 00          0         131
# error_counter,err_inj = 00000010 00          0         132
# error_counter,err_inj = 00000010 00          0         133
# error_counter,err_inj = 00000010 00          0         134
# error_counter,err_inj = 00000010 00          0         135
# error_counter,err_inj = 00000011 01          0         136
# error_counter,err_inj = 00000011 00          0         137
# error_counter,err_inj = 00000011 00          0         138
# error_counter,err_inj = 00000011 00          0         139
# error_counter,err_inj = 00000011 00          0         140
# error_counter,err_inj = 00000011 00          0         141
# error_counter,err_inj = 00000011 00          0         142
# error_counter,err_inj = 00000011 00          0         143
# error_counter,err_inj = 00000012 01          0         144
# error_counter,err_inj = 00000012 00          0         145
# error_counter,err_inj = 00000012 00          0         146
# error_counter,err_inj = 00000012 00          0         147
# error_counter,err_inj = 00000012 00          0         148
# error_counter,err_inj = 00000012 00          0         149
# error_counter,err_inj = 00000012 00          0         150
# error_counter,err_inj = 00000012 00          0         151
```

```
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
```

```
# error_counter,err_inj = 00000013 01          0         152
# error_counter,err_inj = 00000013 00          0         153
# error_counter,err_inj = 00000013 00          0         154
# error_counter,err_inj = 00000013 00          0         155
# error_counter,err_inj = 00000013 00          0         156
# error_counter,err_inj = 00000013 00          0         157
# error_counter,err_inj = 00000013 00          0         158
# error_counter,err_inj = 00000013 00          0         159
# error_counter,err_inj = 00000014 01          0         160
# error_counter,err_inj = 00000014 00          0         161
# error_counter,err_inj = 00000014 00          0         162
# error_counter,err_inj = 00000014 00          0         163
# error_counter,err_inj = 00000014 00          0         164
# error_counter,err_inj = 00000014 00          0         165
# error_counter,err_inj = 00000014 00          0         166
# error_counter,err_inj = 00000014 00          0         167
# error_counter,err_inj = 00000015 01          0         168
# error_counter,err_inj = 00000015 00          0         169
# error_counter,err_inj = 00000015 00          0         170
# error_counter,err_inj = 00000015 00          0         171
# error_counter,err_inj = 00000015 00          0         172
# error_counter,err_inj = 00000015 00          0         173
# error_counter,err_inj = 00000015 00          0         174
# error_counter,err_inj = 00000015 00          0         175
# error_counter,err_inj = 00000016 01          0         176
# error_counter,err_inj = 00000016 00          0         177
# error_counter,err_inj = 00000016 00          0         178
# error_counter,err_inj = 00000016 00          0         179
# error_counter,err_inj = 00000016 00          0         180
# error_counter,err_inj = 00000016 00          0         181
# error_counter,err_inj = 00000016 00          0         182
# error_counter,err_inj = 00000016 00          0         183
# error_counter,err_inj = 00000017 01          0         184
# error_counter,err_inj = 00000017 00          0         185
# error_counter,err_inj = 00000017 00          0         186
# error_counter,err_inj = 00000017 00          0         187
# error_counter,err_inj = 00000017 00          0         188
# error_counter,err_inj = 00000017 00          0         189
# error_counter,err_inj = 00000017 00          0         190
# error_counter,err_inj = 00000017 00          0         191
# error_counter,err_inj = 00000018 01          0         192
# error_counter,err_inj = 00000018 00          0         193
# error_counter,err_inj = 00000018 00          0         194
# error_counter,err_inj = 00000018 00          0         195
# error_counter,err_inj = 00000018 00          0         196
# error_counter,err_inj = 00000018 00          0         197
# error_counter,err_inj = 00000018 00          0         198
# error_counter,err_inj = 00000018 00          0         199
# error_counter,err_inj = 00000019 01          0         200
# error_counter,err_inj = 00000019 00          0         201
# error_counter,err_inj = 00000019 00          0         202
# error_counter,err_inj = 00000019 00          0         203
# error_counter,err_inj = 00000019 00          0         204
# error_counter,err_inj = 00000019 00          0         205
# error_counter,err_inj = 00000019 00          0         206
# error_counter,err_inj = 00000019 00          0         207
# error_counter,err_inj = 0000001a 01          0         208
# error_counter,err_inj = 0000001a 00          0         209
# error_counter,err_inj = 0000001a 00          0         210
# error_counter,err_inj = 0000001a 00          0         211
# error_counter,err_inj = 0000001a 00          0         212
# error_counter,err_inj = 0000001a 00          0         213
# error_counter,err_inj = 0000001a 00          0         214
# error_counter,err_inj = 0000001a 00          0         215
# error_counter,err_inj = 0000001b 01          0         216
# error_counter,err_inj = 0000001b 00          0         217
# error_counter,err_inj = 0000001b 00          0         218
# error_counter,err_inj = 0000001b 00          0         219
# error_counter,err_inj = 0000001b 00          0         220
# error_counter,err_inj = 0000001b 00          0         221
# error_counter,err_inj = 0000001b 00          0         222
# error_counter,err_inj = 0000001b 00          0         223
# error_counter,err_inj = 0000001c 01          0         224
# error_counter,err_inj = 0000001c 00          0         225
# error_counter,err_inj = 0000001c 00          0         226
# error_counter,err_inj = 0000001c 00          0         227
# error_counter,err_inj = 0000001c 00          0         228
# error_counter,err_inj = 0000001c 00          0         229
# error_counter,err_inj = 0000001c 00          0         230
# error_counter,err_inj = 0000001c 00          0         231
# error_counter,err_inj = 0000001d 01          0         232
# error_counter,err_inj = 0000001d 00          0         233
# error_counter,err_inj = 0000001d 00          0         234
# error_counter,err_inj = 0000001d 00          0         235
# error_counter,err_inj = 0000001d 00          0         236
# error_counter,err_inj = 0000001d 00          0         237
# error_counter,err_inj = 0000001d 00          0         238
# error_counter,err_inj = 0000001d 00          0         239
# error_counter,err_inj = 0000001e 01          0         240
# error_counter,err_inj = 0000001e 00          0         241
# error_counter,err_inj = 0000001e 00          0         242
# error_counter,err_inj = 0000001e 00          0         243
# error_counter,err_inj = 0000001e 00          0         244
```

```
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001e 00           0         245
# error_counter,err_inj = 0000001e 00           0         246
# error_counter,err_inj = 0000001e 00           0         247
# error_counter,err_inj = 0000001f 01           0         248
# error_counter,err_inj = 0000001f 00           0         249
# error_counter,err_inj = 0000001f 00           0         250
# error_counter,err_inj = 0000001f 00           0         251
# error_counter,err_inj = 0000001f 00           0         252
# error_counter,err_inj = 0000001f 00           0         253
# error_counter,err_inj = 0000001f 00           0         254
# error_counter,err_inj = 0000001f 00           0         255
# word_count =       10440
```

2a2) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000001 10           0           8
# error_counter,err_inj = 00000001 00           0           9
# error_counter,err_inj = 00000001 00           0          10
# error_counter,err_inj = 00000001 00           0          11
# error_counter,err_inj = 00000001 00           0          12
# error_counter,err_inj = 00000001 00           0          13
# error_counter,err_inj = 00000001 00           0          14
# error_counter,err_inj = 00000001 00           0          15
# error_counter,err_inj = 00000002 10           0          16
# error_counter,err_inj = 00000002 00           0          17
# error_counter,err_inj = 00000002 00           0          18
# error_counter,err_inj = 00000002 00           0          19
# error_counter,err_inj = 00000002 00           0          20
# error_counter,err_inj = 00000002 00           0          21
# error_counter,err_inj = 00000002 00           0          22
# error_counter,err_inj = 00000002 00           0          23
# error_counter,err_inj = 00000003 10           0          24
# error_counter,err_inj = 00000003 00           0          25
# error_counter,err_inj = 00000003 00           0          26
# error_counter,err_inj = 00000003 00           0          27
# error_counter,err_inj = 00000003 00           0          28
# error_counter,err_inj = 00000003 00           0          29
# error_counter,err_inj = 00000003 00           0          30
# error_counter,err_inj = 00000003 00           0          31
# error_counter,err_inj = 00000004 10           0          32
# error_counter,err_inj = 00000004 00           0          33
# error_counter,err_inj = 00000004 00           0          34
# error_counter,err_inj = 00000004 00           0          35
# error_counter,err_inj = 00000004 00           0          36
# error_counter,err_inj = 00000004 00           0          37
# error_counter,err_inj = 00000004 00           0          38
# error_counter,err_inj = 00000004 00           0          39
# error_counter,err_inj = 00000005 10           0          40
# error_counter,err_inj = 00000005 00           0          41
# error_counter,err_inj = 00000005 00           0          42
# error_counter,err_inj = 00000005 00           0          43
# error_counter,err_inj = 00000005 00           0          44
# error_counter,err_inj = 00000005 00           0          45
# error_counter,err_inj = 00000005 00           0          46
# error_counter,err_inj = 00000005 00           0          47
# error_counter,err_inj = 00000006 10           0          48
# error_counter,err_inj = 00000006 00           0          49
# error_counter,err_inj = 00000006 00           0          50
# error_counter,err_inj = 00000006 00           0          51
# error_counter,err_inj = 00000006 00           0          52
# error_counter,err_inj = 00000006 00           0          53
# error_counter,err_inj = 00000006 00           0          54
# error_counter,err_inj = 00000006 00           0          55
# error_counter,err_inj = 00000007 10           0          56
# error_counter,err_inj = 00000007 00           0          57
# error_counter,err_inj = 00000007 00           0          58
# error_counter,err_inj = 00000007 00           0          59
# error_counter,err_inj = 00000007 00           0          60
# error_counter,err_inj = 00000007 00           0          61
# error_counter,err_inj = 00000007 00           0          62
# error_counter,err_inj = 00000007 00           0          63
# error_counter,err_inj = 00000008 10           0          64
# error_counter,err_inj = 00000008 00           0          65
# error_counter,err_inj = 00000008 00           0          66
# error_counter,err_inj = 00000008 00           0          67
# error_counter,err_inj = 00000008 00           0          68
# error_counter,err_inj = 00000008 00           0          69
# error_counter,err_inj = 00000008 00           0          70
# error_counter,err_inj = 00000008 00           0          71
# error_counter,err_inj = 00000009 10           0          72
# error_counter,err_inj = 00000009 00           0          73
# error_counter,err_inj = 00000009 00           0          74
```

```
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
```

```
# error_counter,err_inj = 00000009 00          0          75
# error_counter,err_inj = 00000009 00          0          76
# error_counter,err_inj = 00000009 00          0          77
# error_counter,err_inj = 00000009 00          0          78
# error_counter,err_inj = 00000009 00          0          79
# error_counter,err_inj = 0000000a 10          0          80
# error_counter,err_inj = 0000000a 00          0          81
# error_counter,err_inj = 0000000a 00          0          82
# error_counter,err_inj = 0000000a 00          0          83
# error_counter,err_inj = 0000000a 00          0          84
# error_counter,err_inj = 0000000a 00          0          85
# error_counter,err_inj = 0000000a 00          0          86
# error_counter,err_inj = 0000000a 00          0          87
# error_counter,err_inj = 0000000b 10          0          88
# error_counter,err_inj = 0000000b 00          0          89
# error_counter,err_inj = 0000000b 00          0          90
# error_counter,err_inj = 0000000b 00          0          91
# error_counter,err_inj = 0000000b 00          0          92
# error_counter,err_inj = 0000000b 00          0          93
# error_counter,err_inj = 0000000b 00          0          94
# error_counter,err_inj = 0000000b 00          0          95
# error_counter,err_inj = 0000000c 10          0          96
# error_counter,err_inj = 0000000c 00          0          97
# error_counter,err_inj = 0000000c 00          0          98
# error_counter,err_inj = 0000000c 00          0          99
# error_counter,err_inj = 0000000c 00          0         100
# error_counter,err_inj = 0000000c 00          0         101
# error_counter,err_inj = 0000000c 00          0         102
# error_counter,err_inj = 0000000c 00          0         103
# error_counter,err_inj = 0000000d 10          0         104
# error_counter,err_inj = 0000000d 00          0         105
# error_counter,err_inj = 0000000d 00          0         106
# error_counter,err_inj = 0000000d 00          0         107
# error_counter,err_inj = 0000000d 00          0         108
# error_counter,err_inj = 0000000d 00          0         109
# error_counter,err_inj = 0000000d 00          0         110
# error_counter,err_inj = 0000000d 00          0         111
# error_counter,err_inj = 0000000e 10          0         112
# error_counter,err_inj = 0000000e 00          0         113
# error_counter,err_inj = 0000000e 00          0         114
# error_counter,err_inj = 0000000e 00          0         115
# error_counter,err_inj = 0000000e 00          0         116
# error_counter,err_inj = 0000000e 00          0         117
# error_counter,err_inj = 0000000e 00          0         118
# error_counter,err_inj = 0000000e 00          0         119
# error_counter,err_inj = 0000000f 10          0         120
# error_counter,err_inj = 0000000f 00          0         121
# error_counter,err_inj = 0000000f 00          0         122
# error_counter,err_inj = 0000000f 00          0         123
# error_counter,err_inj = 0000000f 00          0         124
# error_counter,err_inj = 0000000f 00          0         125
# error_counter,err_inj = 0000000f 00          0         126
# error_counter,err_inj = 0000000f 00          0         127
# error_counter,err_inj = 00000010 10          0         128
# error_counter,err_inj = 00000010 00          0         129
# error_counter,err_inj = 00000010 00          0         130
# error_counter,err_inj = 00000010 00          0         131
# error_counter,err_inj = 00000010 00          0         132
# error_counter,err_inj = 00000010 00          0         133
# error_counter,err_inj = 00000010 00          0         134
# error_counter,err_inj = 00000010 00          0         135
# error_counter,err_inj = 00000011 10          0         136
# error_counter,err_inj = 00000011 00          0         137
# error_counter,err_inj = 00000011 00          0         138
# error_counter,err_inj = 00000011 00          0         139
# error_counter,err_inj = 00000011 00          0         140
# error_counter,err_inj = 00000011 00          0         141
# error_counter,err_inj = 00000011 00          0         142
# error_counter,err_inj = 00000011 00          0         143
# error_counter,err_inj = 00000012 10          0         144
# error_counter,err_inj = 00000012 00          0         145
# error_counter,err_inj = 00000012 00          0         146
# error_counter,err_inj = 00000012 00          0         147
# error_counter,err_inj = 00000012 00          0         148
# error_counter,err_inj = 00000012 00          0         149
# error_counter,err_inj = 00000012 00          0         150
# error_counter,err_inj = 00000012 00          0         151
# error_counter,err_inj = 00000013 10          0         152
# error_counter,err_inj = 00000013 00          0         153
# error_counter,err_inj = 00000013 00          0         154
# error_counter,err_inj = 00000013 00          0         155
# error_counter,err_inj = 00000013 00          0         156
# error_counter,err_inj = 00000013 00          0         157
# error_counter,err_inj = 00000013 00          0         158
# error_counter,err_inj = 00000013 00          0         159
# error_counter,err_inj = 00000014 10          0         160
# error_counter,err_inj = 00000014 00          0         161
# error_counter,err_inj = 00000014 00          0         162
# error_counter,err_inj = 00000014 00          0         163
# error_counter,err_inj = 00000014 00          0         164
# error_counter,err_inj = 00000014 00          0         165
# error_counter,err_inj = 00000014 00          0         166
# error_counter,err_inj = 00000014 00          0         167
```

```
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000015 10          0         168
# error_counter,err_inj = 00000015 00          0         169
# error_counter,err_inj = 00000015 00          0         170
# error_counter,err_inj = 00000015 00          0         171
# error_counter,err_inj = 00000015 00          0         172
# error_counter,err_inj = 00000015 00          0         173
# error_counter,err_inj = 00000015 00          0         174
# error_counter,err_inj = 00000015 00          0         175
# error_counter,err_inj = 00000016 10          0         176
# error_counter,err_inj = 00000016 00          0         177
# error_counter,err_inj = 00000016 00          0         178
# error_counter,err_inj = 00000016 00          0         179
# error_counter,err_inj = 00000016 00          0         180
# error_counter,err_inj = 00000016 00          0         181
# error_counter,err_inj = 00000016 00          0         182
# error_counter,err_inj = 00000016 00          0         183
# error_counter,err_inj = 00000017 10          0         184
# error_counter,err_inj = 00000017 00          0         185
# error_counter,err_inj = 00000017 00          0         186
# error_counter,err_inj = 00000017 00          0         187
# error_counter,err_inj = 00000017 00          0         188
# error_counter,err_inj = 00000017 00          0         189
# error_counter,err_inj = 00000017 00          0         190
# error_counter,err_inj = 00000017 00          0         191
# error_counter,err_inj = 00000018 10          0         192
# error_counter,err_inj = 00000018 00          0         193
# error_counter,err_inj = 00000018 00          0         194
# error_counter,err_inj = 00000018 00          0         195
# error_counter,err_inj = 00000018 00          0         196
# error_counter,err_inj = 00000018 00          0         197
# error_counter,err_inj = 00000018 00          0         198
# error_counter,err_inj = 00000018 00          0         199
# error_counter,err_inj = 00000019 10          0         200
# error_counter,err_inj = 00000019 00          0         201
# error_counter,err_inj = 00000019 00          0         202
# error_counter,err_inj = 00000019 00          0         203
# error_counter,err_inj = 00000019 00          0         204
# error_counter,err_inj = 00000019 00          0         205
# error_counter,err_inj = 00000019 00          0         206
# error_counter,err_inj = 00000019 00          0         207
# error_counter,err_inj = 0000001a 10          0         208
# error_counter,err_inj = 0000001a 00          0         209
# error_counter,err_inj = 0000001a 00          0         210
# error_counter,err_inj = 0000001a 00          0         211
# error_counter,err_inj = 0000001a 00          0         212
# error_counter,err_inj = 0000001a 00          0         213
# error_counter,err_inj = 0000001a 00          0         214
# error_counter,err_inj = 0000001a 00          0         215
# error_counter,err_inj = 0000001b 10          0         216
# error_counter,err_inj = 0000001b 00          0         217
# error_counter,err_inj = 0000001b 00          0         218
# error_counter,err_inj = 0000001b 00          0         219
# error_counter,err_inj = 0000001b 00          0         220
# error_counter,err_inj = 0000001b 00          0         221
# error_counter,err_inj = 0000001b 00          0         222
# error_counter,err_inj = 0000001b 00          0         223
# error_counter,err_inj = 0000001c 10          0         224
# error_counter,err_inj = 0000001c 00          0         225
# error_counter,err_inj = 0000001c 00          0         226
# error_counter,err_inj = 0000001c 00          0         227
# error_counter,err_inj = 0000001c 00          0         228
# error_counter,err_inj = 0000001c 00          0         229
# error_counter,err_inj = 0000001c 00          0         230
# error_counter,err_inj = 0000001c 00          0         231
# error_counter,err_inj = 0000001d 10          0         232
# error_counter,err_inj = 0000001d 00          0         233
# error_counter,err_inj = 0000001d 00          0         234
# error_counter,err_inj = 0000001d 00          0         235
# error_counter,err_inj = 0000001d 00          0         236
# error_counter,err_inj = 0000001d 00          0         237
# error_counter,err_inj = 0000001d 00          0         238
# error_counter,err_inj = 0000001d 00          0         239
# error_counter,err_inj = 0000001e 10          0         240
# error_counter,err_inj = 0000001e 00          0         241
# error_counter,err_inj = 0000001e 00          0         242
# error_counter,err_inj = 0000001e 00          0         243
# error_counter,err_inj = 0000001e 00          0         244
# error_counter,err_inj = 0000001e 00          0         245
# error_counter,err_inj = 0000001e 00          0         246
# error_counter,err_inj = 0000001e 00          0         247
# error_counter,err_inj = 0000001f 10          0         248
# error_counter,err_inj = 0000001f 00          0         249
# error_counter,err_inj = 0000001f 00          0         250
# error_counter,err_inj = 0000001f 00          0         251
# error_counter,err_inj = 0000001f 00          0         252
# error_counter,err_inj = 0000001f 00          0         253
# error_counter,err_inj = 0000001f 00          0         254
# error_counter,err_inj = 0000001f 00          0         255
```

2a3) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000000 00          0         12
# error_counter,err_inj = 00000000 00          0         13
# error_counter,err_inj = 00000000 00          0         14
# error_counter,err_inj = 00000000 00          0         15
# error_counter,err_inj = 00000001 11          0         16
# error_counter,err_inj = 00000001 00          0         17
# error_counter,err_inj = 00000001 00          0         18
# error_counter,err_inj = 00000001 00          0         19
# error_counter,err_inj = 00000001 00          0         20
# error_counter,err_inj = 00000001 00          0         21
# error_counter,err_inj = 00000001 00          0         22
# error_counter,err_inj = 00000001 00          0         23
# error_counter,err_inj = 00000001 00          0         24
# error_counter,err_inj = 00000001 00          0         25
# error_counter,err_inj = 00000001 00          0         26
# error_counter,err_inj = 00000001 00          0         27
# error_counter,err_inj = 00000001 00          0         28
# error_counter,err_inj = 00000001 00          0         29
# error_counter,err_inj = 00000001 00          0         30
# error_counter,err_inj = 00000001 00          0         31
# error_counter,err_inj = 00000002 11          0         32
# error_counter,err_inj = 00000002 00          0         33
# error_counter,err_inj = 00000002 00          0         34
# error_counter,err_inj = 00000002 00          0         35
# error_counter,err_inj = 00000002 00          0         36
# error_counter,err_inj = 00000002 00          0         37
# error_counter,err_inj = 00000002 00          0         38
# error_counter,err_inj = 00000002 00          0         39
# error_counter,err_inj = 00000002 00          0         40
# error_counter,err_inj = 00000002 00          0         41
# error_counter,err_inj = 00000002 00          0         42
# error_counter,err_inj = 00000002 00          0         43
# error_counter,err_inj = 00000002 00          0         44
# error_counter,err_inj = 00000002 00          0         45
# error_counter,err_inj = 00000002 00          0         46
# error_counter,err_inj = 00000002 00          0         47
# error_counter,err_inj = 00000003 11          0         48
# error_counter,err_inj = 00000003 00          0         49
# error_counter,err_inj = 00000003 00          0         50
# error_counter,err_inj = 00000003 00          0         51
# error_counter,err_inj = 00000003 00          0         52
# error_counter,err_inj = 00000003 00          0         53
# error_counter,err_inj = 00000003 00          0         54
# error_counter,err_inj = 00000003 00          0         55
# error_counter,err_inj = 00000003 00          0         56
# error_counter,err_inj = 00000003 00          0         57
# error_counter,err_inj = 00000003 00          0         58
# error_counter,err_inj = 00000003 00          0         59
# error_counter,err_inj = 00000003 00          0         60
# error_counter,err_inj = 00000003 00          0         61
# error_counter,err_inj = 00000003 00          0         62
# error_counter,err_inj = 00000003 00          0         63
# error_counter,err_inj = 00000004 11          0         64
# error_counter,err_inj = 00000004 00          0         65
# error_counter,err_inj = 00000004 00          0         66
# error_counter,err_inj = 00000004 00          0         67
# error_counter,err_inj = 00000004 00          0         68
# error_counter,err_inj = 00000004 00          0         69
# error_counter,err_inj = 00000004 00          0         70
# error_counter,err_inj = 00000004 00          0         71
# error_counter,err_inj = 00000004 00          0         72
# error_counter,err_inj = 00000004 00          0         73
# error_counter,err_inj = 00000004 00          0         74
# error_counter,err_inj = 00000004 00          0         75
# error_counter,err_inj = 00000004 00          0         76
# error_counter,err_inj = 00000004 00          0         77
# error_counter,err_inj = 00000004 00          0         78
# error_counter,err_inj = 00000004 00          0         79
# error_counter,err_inj = 00000005 11          0         80
# error_counter,err_inj = 00000005 00          0         81
# error_counter,err_inj = 00000005 00          0         82
# error_counter,err_inj = 00000005 00          0         83
# error_counter,err_inj = 00000005 00          0         84
# error_counter,err_inj = 00000005 00          0         85
# error_counter,err_inj = 00000005 00          0         86
# error_counter,err_inj = 00000005 00          0         87
# error_counter,err_inj = 00000005 00          0         88
# error_counter,err_inj = 00000005 00          0         89
# error_counter,err_inj = 00000005 00          0         90
# error_counter,err_inj = 00000005 00          0         91
# error_counter,err_inj = 00000005 00          0         92
```

```
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
```

```
# error_counter,err_inj = 00000005 00          0          93
# error_counter,err_inj = 00000005 00          0          94
# error_counter,err_inj = 00000005 00          0          95
# error_counter,err_inj = 00000006 11          0          96
# error_counter,err_inj = 00000006 00          0          97
# error_counter,err_inj = 00000006 00          0          98
# error_counter,err_inj = 00000006 00          0          99
# error_counter,err_inj = 00000006 00          0         100
# error_counter,err_inj = 00000006 00          0         101
# error_counter,err_inj = 00000006 00          0         102
# error_counter,err_inj = 00000006 00          0         103
# error_counter,err_inj = 00000006 00          0         104
# error_counter,err_inj = 00000006 00          0         105
# error_counter,err_inj = 00000006 00          0         106
# error_counter,err_inj = 00000006 00          0         107
# error_counter,err_inj = 00000006 00          0         108
# error_counter,err_inj = 00000006 00          0         109
# error_counter,err_inj = 00000006 00          0         110
# error_counter,err_inj = 00000006 00          0         111
# error_counter,err_inj = 00000007 11          0         112
# error_counter,err_inj = 00000007 00          0         113
# error_counter,err_inj = 00000007 00          0         114
# error_counter,err_inj = 00000007 00          0         115
# error_counter,err_inj = 00000007 00          0         116
# error_counter,err_inj = 00000007 00          0         117
# error_counter,err_inj = 00000007 00          0         118
# error_counter,err_inj = 00000007 00          0         119
# error_counter,err_inj = 00000007 00          0         120
# error_counter,err_inj = 00000007 00          0         121
# error_counter,err_inj = 00000007 00          0         122
# error_counter,err_inj = 00000007 00          0         123
# error_counter,err_inj = 00000007 00          0         124
# error_counter,err_inj = 00000007 00          0         125
# error_counter,err_inj = 00000007 00          0         126
# error_counter,err_inj = 00000007 00          0         127
# error_counter,err_inj = 00000008 11          0         128
# error_counter,err_inj = 00000008 00          0         129
# error_counter,err_inj = 00000008 00          0         130
# error_counter,err_inj = 00000008 00          0         131
# error_counter,err_inj = 00000008 00          0         132
# error_counter,err_inj = 00000008 00          0         133
# error_counter,err_inj = 00000008 00          0         134
# error_counter,err_inj = 00000008 00          0         135
# error_counter,err_inj = 00000008 00          0         136
# error_counter,err_inj = 00000008 00          0         137
# error_counter,err_inj = 00000008 00          0         138
# error_counter,err_inj = 00000008 00          0         139
# error_counter,err_inj = 00000008 00          0         140
# error_counter,err_inj = 00000008 00          0         141
# error_counter,err_inj = 00000008 00          0         142
# error_counter,err_inj = 00000008 00          0         143
# error_counter,err_inj = 00000009 11          0         144
# error_counter,err_inj = 00000009 00          0         145
# error_counter,err_inj = 00000009 00          0         146
# error_counter,err_inj = 00000009 00          0         147
# error_counter,err_inj = 00000009 00          0         148
# error_counter,err_inj = 00000009 00          0         149
# error_counter,err_inj = 00000009 00          0         150
# error_counter,err_inj = 00000009 00          0         151
# error_counter,err_inj = 00000009 00          0         152
# error_counter,err_inj = 00000009 00          0         153
# error_counter,err_inj = 00000009 00          0         154
# error_counter,err_inj = 00000009 00          0         155
# error_counter,err_inj = 00000009 00          0         156
# error_counter,err_inj = 00000009 00          0         157
# error_counter,err_inj = 00000009 00          0         158
# error_counter,err_inj = 00000009 00          0         159
# error_counter,err_inj = 0000000a 11          0         160
# error_counter,err_inj = 0000000a 00          0         161
# error_counter,err_inj = 0000000a 00          0         162
# error_counter,err_inj = 0000000a 00          0         163
# error_counter,err_inj = 0000000a 00          0         164
# error_counter,err_inj = 0000000a 00          0         165
# error_counter,err_inj = 0000000a 00          0         166
# error_counter,err_inj = 0000000a 00          0         167
# error_counter,err_inj = 0000000a 00          0         168
# error_counter,err_inj = 0000000a 00          0         169
# error_counter,err_inj = 0000000a 00          0         170
# error_counter,err_inj = 0000000a 00          0         171
# error_counter,err_inj = 0000000a 00          0         172
# error_counter,err_inj = 0000000a 00          0         173
# error_counter,err_inj = 0000000a 00          0         174
# error_counter,err_inj = 0000000a 00          0         175
# error_counter,err_inj = 0000000b 11          0         176
# error_counter,err_inj = 0000000b 00          0         177
# error_counter,err_inj = 0000000b 00          0         178
# error_counter,err_inj = 0000000b 00          0         179
# error_counter,err_inj = 0000000b 00          0         180
# error_counter,err_inj = 0000000b 00          0         181
# error_counter,err_inj = 0000000b 00          0         182
# error_counter,err_inj = 0000000b 00          0         183
# error_counter,err_inj = 0000000b 00          0         184
# error_counter,err_inj = 0000000b 00          0         185
```

```
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000b 00          0         186
# error_counter,err_inj = 0000000b 00          0         187
# error_counter,err_inj = 0000000b 00          0         188
# error_counter,err_inj = 0000000b 00          0         189
# error_counter,err_inj = 0000000b 00          0         190
# error_counter,err_inj = 0000000b 00          0         191
# error_counter,err_inj = 0000000c 11          0         192
# error_counter,err_inj = 0000000c 00          0         193
# error_counter,err_inj = 0000000c 00          0         194
# error_counter,err_inj = 0000000c 00          0         195
# error_counter,err_inj = 0000000c 00          0         196
# error_counter,err_inj = 0000000c 00          0         197
# error_counter,err_inj = 0000000c 00          0         198
# error_counter,err_inj = 0000000c 00          0         199
# error_counter,err_inj = 0000000c 00          0         200
# error_counter,err_inj = 0000000c 00          0         201
# error_counter,err_inj = 0000000c 00          0         202
# error_counter,err_inj = 0000000c 00          0         203
# error_counter,err_inj = 0000000c 00          0         204
# error_counter,err_inj = 0000000c 00          0         205
# error_counter,err_inj = 0000000c 00          0         206
# error_counter,err_inj = 0000000c 00          0         207
# error_counter,err_inj = 0000000d 11          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000d 00          0         213
# error_counter,err_inj = 0000000d 00          0         214
# error_counter,err_inj = 0000000d 00          0         215
# error_counter,err_inj = 0000000d 00          0         216
# error_counter,err_inj = 0000000d 00          0         217
# error_counter,err_inj = 0000000d 00          0         218
# error_counter,err_inj = 0000000d 00          0         219
# error_counter,err_inj = 0000000d 00          0         220
# error_counter,err_inj = 0000000d 00          0         221
# error_counter,err_inj = 0000000d 00          0         222
# error_counter,err_inj = 0000000d 00          0         223
# error_counter,err_inj = 0000000e 11          0         224
# error_counter,err_inj = 0000000e 00          0         225
# error_counter,err_inj = 0000000e 00          0         226
# error_counter,err_inj = 0000000e 00          0         227
# error_counter,err_inj = 0000000e 00          0         228
# error_counter,err_inj = 0000000e 00          0         229
# error_counter,err_inj = 0000000e 00          0         230
# error_counter,err_inj = 0000000e 00          0         231
# error_counter,err_inj = 0000000e 00          0         232
# error_counter,err_inj = 0000000e 00          0         233
# error_counter,err_inj = 0000000e 00          0         234
# error_counter,err_inj = 0000000e 00          0         235
# error_counter,err_inj = 0000000e 00          0         236
# error_counter,err_inj = 0000000e 00          0         237
# error_counter,err_inj = 0000000e 00          0         238
# error_counter,err_inj = 0000000e 00          0         239
# error_counter,err_inj = 0000000f 11          0         240
# error_counter,err_inj = 0000000f 00          0         241
# error_counter,err_inj = 0000000f 00          0         242
# error_counter,err_inj = 0000000f 00          0         243
# error_counter,err_inj = 0000000f 00          0         244
# error_counter,err_inj = 0000000f 00          0         245
# error_counter,err_inj = 0000000f 00          0         246
# error_counter,err_inj = 0000000f 00          0         247
# error_counter,err_inj = 0000000f 00          0         248
# error_counter,err_inj = 0000000f 00          0         249
# error_counter,err_inj = 0000000f 00          0         250
# error_counter,err_inj = 0000000f 00          0         251
# error_counter,err_inj = 0000000f 00          0         252
# error_counter,err_inj = 0000000f 00          0         253
# error_counter,err_inj = 0000000f 00          0         254
# error_counter,err_inj = 0000000f 00          0         255
```

2a4) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000001 01          0          15
# error_counter,err_inj = 00000002 01          0          16
```

```
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
```

```
# error_counter,err_inj = 00000002 00          1          17
# error_counter,err_inj = 00000002 00          1          18
# error_counter,err_inj = 00000002 00          1          19
# error_counter,err_inj = 00000002 00          1          20
# error_counter,err_inj = 00000002 00          1          21
# error_counter,err_inj = 00000002 00          1          22
# error_counter,err_inj = 00000002 00          1          23
# error_counter,err_inj = 00000002 00          1          24
# error_counter,err_inj = 00000002 00          1          25
# error_counter,err_inj = 00000002 00          1          26
# error_counter,err_inj = 00000002 00          1          27
# error_counter,err_inj = 00000002 00          1          28
# error_counter,err_inj = 00000002 00          1          29
# error_counter,err_inj = 00000002 00          1          30
# error_counter,err_inj = 00000003 01          1          31
# error_counter,err_inj = 00000004 01          1          32
# error_counter,err_inj = 00000004 00          2          33
# error_counter,err_inj = 00000004 00          2          34
# error_counter,err_inj = 00000004 00          2          35
# error_counter,err_inj = 00000004 00          2          36
# error_counter,err_inj = 00000004 00          2          37
# error_counter,err_inj = 00000004 00          2          38
# error_counter,err_inj = 00000004 00          2          39
# error_counter,err_inj = 00000004 00          2          40
# error_counter,err_inj = 00000004 00          2          41
# error_counter,err_inj = 00000004 00          2          42
# error_counter,err_inj = 00000004 00          2          43
# error_counter,err_inj = 00000004 00          2          44
# error_counter,err_inj = 00000004 00          2          45
# error_counter,err_inj = 00000004 00          2          46
# error_counter,err_inj = 00000005 01          2          47
# error_counter,err_inj = 00000006 01          2          48
# error_counter,err_inj = 00000006 00          3          49
# error_counter,err_inj = 00000006 00          3          50
# error_counter,err_inj = 00000006 00          3          51
# error_counter,err_inj = 00000006 00          3          52
# error_counter,err_inj = 00000006 00          3          53
# error_counter,err_inj = 00000006 00          3          54
# error_counter,err_inj = 00000006 00          3          55
# error_counter,err_inj = 00000006 00          3          56
# error_counter,err_inj = 00000006 00          3          57
# error_counter,err_inj = 00000006 00          3          58
# error_counter,err_inj = 00000006 00          3          59
# error_counter,err_inj = 00000006 00          3          60
# error_counter,err_inj = 00000006 00          3          61
# error_counter,err_inj = 00000006 00          3          62
# error_counter,err_inj = 00000007 01          3          63
# error_counter,err_inj = 00000008 01          3          64
# error_counter,err_inj = 00000008 00          4          65
# error_counter,err_inj = 00000008 00          4          66
# error_counter,err_inj = 00000008 00          4          67
# error_counter,err_inj = 00000008 00          4          68
# error_counter,err_inj = 00000008 00          4          69
# error_counter,err_inj = 00000008 00          4          70
# error_counter,err_inj = 00000008 00          4          71
# error_counter,err_inj = 00000008 00          4          72
# error_counter,err_inj = 00000008 00          4          73
# error_counter,err_inj = 00000008 00          4          74
# error_counter,err_inj = 00000008 00          4          75
# error_counter,err_inj = 00000008 00          4          76
# error_counter,err_inj = 00000008 00          4          77
# error_counter,err_inj = 00000008 00          4          78
# error_counter,err_inj = 00000009 01          4          79
# error_counter,err_inj = 0000000a 01          4          80
# error_counter,err_inj = 0000000a 00          5          81
# error_counter,err_inj = 0000000a 00          5          82
# error_counter,err_inj = 0000000a 00          5          83
# error_counter,err_inj = 0000000a 00          5          84
# error_counter,err_inj = 0000000a 00          5          85
# error_counter,err_inj = 0000000a 00          5          86
# error_counter,err_inj = 0000000a 00          5          87
# error_counter,err_inj = 0000000a 00          5          88
# error_counter,err_inj = 0000000a 00          5          89
# error_counter,err_inj = 0000000a 00          5          90
# error_counter,err_inj = 0000000a 00          5          91
# error_counter,err_inj = 0000000a 00          5          92
# error_counter,err_inj = 0000000a 00          5          93
# error_counter,err_inj = 0000000a 00          5          94
# error_counter,err_inj = 0000000b 01          5          95
# error_counter,err_inj = 0000000c 01          5          96
# error_counter,err_inj = 0000000c 00          6          97
# error_counter,err_inj = 0000000c 00          6          98
# error_counter,err_inj = 0000000c 00          6          99
# error_counter,err_inj = 0000000c 00          6         100
# error_counter,err_inj = 0000000c 00          6         101
# error_counter,err_inj = 0000000c 00          6         102
# error_counter,err_inj = 0000000c 00          6         103
# error_counter,err_inj = 0000000c 00          6         104
# error_counter,err_inj = 0000000c 00          6         105
# error_counter,err_inj = 0000000c 00          6         106
# error_counter,err_inj = 0000000c 00          6         107
# error_counter,err_inj = 0000000c 00          6         108
# error_counter,err_inj = 0000000c 00          6         109
```

```
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
```

```
# error_counter,err_inj = 0000000c 00          6         110
# error_counter,err_inj = 0000000d 01          6         111
# error_counter,err_inj = 0000000e 01          6         112
# error_counter,err_inj = 0000000e 00          7         113
# error_counter,err_inj = 0000000e 00          7         114
# error_counter,err_inj = 0000000e 00          7         115
# error_counter,err_inj = 0000000e 00          7         116
# error_counter,err_inj = 0000000e 00          7         117
# error_counter,err_inj = 0000000e 00          7         118
# error_counter,err_inj = 0000000e 00          7         119
# error_counter,err_inj = 0000000e 00          7         120
# error_counter,err_inj = 0000000e 00          7         121
# error_counter,err_inj = 0000000e 00          7         122
# error_counter,err_inj = 0000000e 00          7         123
# error_counter,err_inj = 0000000e 00          7         124
# error_counter,err_inj = 0000000e 00          7         125
# error_counter,err_inj = 0000000e 00          7         126
# error_counter,err_inj = 0000000f 01          7         127
# error_counter,err_inj = 00000010 01          7         128
# error_counter,err_inj = 00000010 00          8         129
# error_counter,err_inj = 00000010 00          8         130
# error_counter,err_inj = 00000010 00          8         131
# error_counter,err_inj = 00000010 00          8         132
# error_counter,err_inj = 00000010 00          8         133
# error_counter,err_inj = 00000010 00          8         134
# error_counter,err_inj = 00000010 00          8         135
# error_counter,err_inj = 00000010 00          8         136
# error_counter,err_inj = 00000010 00          8         137
# error_counter,err_inj = 00000010 00          8         138
# error_counter,err_inj = 00000010 00          8         139
# error_counter,err_inj = 00000010 00          8         140
# error_counter,err_inj = 00000010 00          8         141
# error_counter,err_inj = 00000010 00          8         142
# error_counter,err_inj = 00000011 01          8         143
# error_counter,err_inj = 00000012 01          8         144
# error_counter,err_inj = 00000012 00          9         145
# error_counter,err_inj = 00000012 00          9         146
# error_counter,err_inj = 00000012 00          9         147
# error_counter,err_inj = 00000012 00          9         148
# error_counter,err_inj = 00000012 00          9         149
# error_counter,err_inj = 00000012 00          9         150
# error_counter,err_inj = 00000012 00          9         151
# error_counter,err_inj = 00000012 00          9         152
# error_counter,err_inj = 00000012 00          9         153
# error_counter,err_inj = 00000012 00          9         154
# error_counter,err_inj = 00000012 00          9         155
# error_counter,err_inj = 00000012 00          9         156
# error_counter,err_inj = 00000012 00          9         157
# error_counter,err_inj = 00000012 00          9         158
# error_counter,err_inj = 00000013 01          9         159
# error_counter,err_inj = 00000014 01          9         160
# error_counter,err_inj = 00000014 00         10         161
# error_counter,err_inj = 00000014 00         10         162
# error_counter,err_inj = 00000014 00         10         163
# error_counter,err_inj = 00000014 00         10         164
# error_counter,err_inj = 00000014 00         10         165
# error_counter,err_inj = 00000014 00         10         166
# error_counter,err_inj = 00000014 00         10         167
# error_counter,err_inj = 00000014 00         10         168
# error_counter,err_inj = 00000014 00         10         169
# error_counter,err_inj = 00000014 00         10         170
# error_counter,err_inj = 00000014 00         10         171
# error_counter,err_inj = 00000014 00         10         172
# error_counter,err_inj = 00000014 00         10         173
# error_counter,err_inj = 00000014 00         10         174
# error_counter,err_inj = 00000015 01         10         175
# error_counter,err_inj = 00000016 01         10         176
# error_counter,err_inj = 00000016 00         11         177
# error_counter,err_inj = 00000016 00         11         178
# error_counter,err_inj = 00000016 00         11         179
# error_counter,err_inj = 00000016 00         11         180
# error_counter,err_inj = 00000016 00         11         181
# error_counter,err_inj = 00000016 00         11         182
# error_counter,err_inj = 00000016 00         11         183
# error_counter,err_inj = 00000016 00         11         184
# error_counter,err_inj = 00000016 00         11         185
# error_counter,err_inj = 00000016 00         11         186
# error_counter,err_inj = 00000016 00         11         187
# error_counter,err_inj = 00000016 00         11         188
# error_counter,err_inj = 00000016 00         11         189
# error_counter,err_inj = 00000016 00         11         190
# error_counter,err_inj = 00000017 01         11         191
# error_counter,err_inj = 00000018 01         11         192
# error_counter,err_inj = 00000018 00         12         193
# error_counter,err_inj = 00000018 00         12         194
# error_counter,err_inj = 00000018 00         12         195
# error_counter,err_inj = 00000018 00         12         196
# error_counter,err_inj = 00000018 00         12         197
# error_counter,err_inj = 00000018 00         12         198
# error_counter,err_inj = 00000018 00         12         199
# error_counter,err_inj = 00000018 00         12         200
# error_counter,err_inj = 00000018 00         12         201
# error_counter,err_inj = 00000018 00         12         202
```

```
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000018 00          12          203
# error_counter,err_inj = 00000018 00          12          204
# error_counter,err_inj = 00000018 00          12          205
# error_counter,err_inj = 00000018 00          12          206
# error_counter,err_inj = 00000019 01          12          207
# error_counter,err_inj = 0000001a 01          12          208
# error_counter,err_inj = 0000001a 00          13          209
# error_counter,err_inj = 0000001a 00          13          210
# error_counter,err_inj = 0000001a 00          13          211
# error_counter,err_inj = 0000001a 00          13          212
# error_counter,err_inj = 0000001a 00          13          213
# error_counter,err_inj = 0000001a 00          13          214
# error_counter,err_inj = 0000001a 00          13          215
# error_counter,err_inj = 0000001a 00          13          216
# error_counter,err_inj = 0000001a 00          13          217
# error_counter,err_inj = 0000001a 00          13          218
# error_counter,err_inj = 0000001a 00          13          219
# error_counter,err_inj = 0000001a 00          13          220
# error_counter,err_inj = 0000001a 00          13          221
# error_counter,err_inj = 0000001a 00          13          222
# error_counter,err_inj = 0000001b 01          13          223
# error_counter,err_inj = 0000001c 01          13          224
# error_counter,err_inj = 0000001c 00          14          225
# error_counter,err_inj = 0000001c 00          14          226
# error_counter,err_inj = 0000001c 00          14          227
# error_counter,err_inj = 0000001c 00          14          228
# error_counter,err_inj = 0000001c 00          14          229
# error_counter,err_inj = 0000001c 00          14          230
# error_counter,err_inj = 0000001c 00          14          231
# error_counter,err_inj = 0000001c 00          14          232
# error_counter,err_inj = 0000001c 00          14          233
# error_counter,err_inj = 0000001c 00          14          234
# error_counter,err_inj = 0000001c 00          14          235
# error_counter,err_inj = 0000001c 00          14          236
# error_counter,err_inj = 0000001c 00          14          237
# error_counter,err_inj = 0000001c 00          14          238
# error_counter,err_inj = 0000001d 01          14          239
# error_counter,err_inj = 0000001e 01          14          240
# error_counter,err_inj = 0000001e 00          15          241
# error_counter,err_inj = 0000001e 00          15          242
# error_counter,err_inj = 0000001e 00          15          243
# error_counter,err_inj = 0000001e 00          15          244
# error_counter,err_inj = 0000001e 00          15          245
# error_counter,err_inj = 0000001e 00          15          246
# error_counter,err_inj = 0000001e 00          15          247
# error_counter,err_inj = 0000001e 00          15          248
# error_counter,err_inj = 0000001e 00          15          249
# error_counter,err_inj = 0000001e 00          15          250
# error_counter,err_inj = 0000001e 00          15          251
# error_counter,err_inj = 0000001e 00          15          252
# error_counter,err_inj = 0000001e 00          15          253
# error_counter,err_inj = 0000001e 00          15          254
# error_counter,err_inj = 0000001f 01          15          255
# word_count =       10440
```

2a5) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000001 10          0          15
# error_counter,err_inj = 00000002 10          0          16
# error_counter,err_inj = 00000002 00          1          17
# error_counter,err_inj = 00000002 00          1          18
# error_counter,err_inj = 00000002 00          1          19
# error_counter,err_inj = 00000002 00          1          20
# error_counter,err_inj = 00000002 00          1          21
# error_counter,err_inj = 00000002 00          1          22
# error_counter,err_inj = 00000002 00          1          23
# error_counter,err_inj = 00000002 00          1          24
# error_counter,err_inj = 00000002 00          1          25
# error_counter,err_inj = 00000002 00          1          26
# error_counter,err_inj = 00000002 00          1          27
# error_counter,err_inj = 00000002 00          1          28
# error_counter,err_inj = 00000002 00          1          29
# error_counter,err_inj = 00000002 00          1          30
# error_counter,err_inj = 00000003 10          1          31
# error_counter,err_inj = 00000004 10          1          32
# error_counter,err_inj = 00000004 00          2          33
```

```
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
```

```
# error_counter,err_inj = 00000004 00          2          34
# error_counter,err_inj = 00000004 00          2          35
# error_counter,err_inj = 00000004 00          2          36
# error_counter,err_inj = 00000004 00          2          37
# error_counter,err_inj = 00000004 00          2          38
# error_counter,err_inj = 00000004 00          2          39
# error_counter,err_inj = 00000004 00          2          40
# error_counter,err_inj = 00000004 00          2          41
# error_counter,err_inj = 00000004 00          2          42
# error_counter,err_inj = 00000004 00          2          43
# error_counter,err_inj = 00000004 00          2          44
# error_counter,err_inj = 00000004 00          2          45
# error_counter,err_inj = 00000004 00          2          46
# error_counter,err_inj = 00000005 10          2          47
# error_counter,err_inj = 00000006 10          2          48
# error_counter,err_inj = 00000006 00          3          49
# error_counter,err_inj = 00000006 00          3          50
# error_counter,err_inj = 00000006 00          3          51
# error_counter,err_inj = 00000006 00          3          52
# error_counter,err_inj = 00000006 00          3          53
# error_counter,err_inj = 00000006 00          3          54
# error_counter,err_inj = 00000006 00          3          55
# error_counter,err_inj = 00000006 00          3          56
# error_counter,err_inj = 00000006 00          3          57
# error_counter,err_inj = 00000006 00          3          58
# error_counter,err_inj = 00000006 00          3          59
# error_counter,err_inj = 00000006 00          3          60
# error_counter,err_inj = 00000006 00          3          61
# error_counter,err_inj = 00000006 00          3          62
# error_counter,err_inj = 00000007 10          3          63
# error_counter,err_inj = 00000008 10          3          64
# error_counter,err_inj = 00000008 00          4          65
# error_counter,err_inj = 00000008 00          4          66
# error_counter,err_inj = 00000008 00          4          67
# error_counter,err_inj = 00000008 00          4          68
# error_counter,err_inj = 00000008 00          4          69
# error_counter,err_inj = 00000008 00          4          70
# error_counter,err_inj = 00000008 00          4          71
# error_counter,err_inj = 00000008 00          4          72
# error_counter,err_inj = 00000008 00          4          73
# error_counter,err_inj = 00000008 00          4          74
# error_counter,err_inj = 00000008 00          4          75
# error_counter,err_inj = 00000008 00          4          76
# error_counter,err_inj = 00000008 00          4          77
# error_counter,err_inj = 00000008 00          4          78
# error_counter,err_inj = 00000009 10          4          79
# error_counter,err_inj = 0000000a 10          4          80
# error_counter,err_inj = 0000000a 00          5          81
# error_counter,err_inj = 0000000a 00          5          82
# error_counter,err_inj = 0000000a 00          5          83
# error_counter,err_inj = 0000000a 00          5          84
# error_counter,err_inj = 0000000a 00          5          85
# error_counter,err_inj = 0000000a 00          5          86
# error_counter,err_inj = 0000000a 00          5          87
# error_counter,err_inj = 0000000a 00          5          88
# error_counter,err_inj = 0000000a 00          5          89
# error_counter,err_inj = 0000000a 00          5          90
# error_counter,err_inj = 0000000a 00          5          91
# error_counter,err_inj = 0000000a 00          5          92
# error_counter,err_inj = 0000000a 00          5          93
# error_counter,err_inj = 0000000a 00          5          94
# error_counter,err_inj = 0000000b 10          5          95
# error_counter,err_inj = 0000000c 10          5          96
# error_counter,err_inj = 0000000c 00          6          97
# error_counter,err_inj = 0000000c 00          6          98
# error_counter,err_inj = 0000000c 00          6          99
# error_counter,err_inj = 0000000c 00          6         100
# error_counter,err_inj = 0000000c 00          6         101
# error_counter,err_inj = 0000000c 00          6         102
# error_counter,err_inj = 0000000c 00          6         103
# error_counter,err_inj = 0000000c 00          6         104
# error_counter,err_inj = 0000000c 00          6         105
# error_counter,err_inj = 0000000c 00          6         106
# error_counter,err_inj = 0000000c 00          6         107
# error_counter,err_inj = 0000000c 00          6         108
# error_counter,err_inj = 0000000c 00          6         109
# error_counter,err_inj = 0000000c 00          6         110
# error_counter,err_inj = 0000000d 10          6         111
# error_counter,err_inj = 0000000e 10          6         112
# error_counter,err_inj = 0000000e 00          7         113
# error_counter,err_inj = 0000000e 00          7         114
# error_counter,err_inj = 0000000e 00          7         115
# error_counter,err_inj = 0000000e 00          7         116
# error_counter,err_inj = 0000000e 00          7         117
# error_counter,err_inj = 0000000e 00          7         118
# error_counter,err_inj = 0000000e 00          7         119
# error_counter,err_inj = 0000000e 00          7         120
# error_counter,err_inj = 0000000e 00          7         121
# error_counter,err_inj = 0000000e 00          7         122
# error_counter,err_inj = 0000000e 00          7         123
# error_counter,err_inj = 0000000e 00          7         124
# error_counter,err_inj = 0000000e 00          7         125
# error_counter,err_inj = 0000000e 00          7         126
```

```
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
```

```
# error_counter,err_inj = 0000000f 10          7         127
# error_counter,err_inj = 00000010 10          7         128
# error_counter,err_inj = 00000010 00          8         129
# error_counter,err_inj = 00000010 00          8         130
# error_counter,err_inj = 00000010 00          8         131
# error_counter,err_inj = 00000010 00          8         132
# error_counter,err_inj = 00000010 00          8         133
# error_counter,err_inj = 00000010 00          8         134
# error_counter,err_inj = 00000010 00          8         135
# error_counter,err_inj = 00000010 00          8         136
# error_counter,err_inj = 00000010 00          8         137
# error_counter,err_inj = 00000010 00          8         138
# error_counter,err_inj = 00000010 00          8         139
# error_counter,err_inj = 00000010 00          8         140
# error_counter,err_inj = 00000010 00          8         141
# error_counter,err_inj = 00000010 00          8         142
# error_counter,err_inj = 00000011 10          8         143
# error_counter,err_inj = 00000012 10          8         144
# error_counter,err_inj = 00000012 00          9         145
# error_counter,err_inj = 00000012 00          9         146
# error_counter,err_inj = 00000012 00          9         147
# error_counter,err_inj = 00000012 00          9         148
# error_counter,err_inj = 00000012 00          9         149
# error_counter,err_inj = 00000012 00          9         150
# error_counter,err_inj = 00000012 00          9         151
# error_counter,err_inj = 00000012 00          9         152
# error_counter,err_inj = 00000012 00          9         153
# error_counter,err_inj = 00000012 00          9         154
# error_counter,err_inj = 00000012 00          9         155
# error_counter,err_inj = 00000012 00          9         156
# error_counter,err_inj = 00000012 00          9         157
# error_counter,err_inj = 00000012 00          9         158
# error_counter,err_inj = 00000013 10          9         159
# error_counter,err_inj = 00000014 10          9         160
# error_counter,err_inj = 00000014 00         10         161
# error_counter,err_inj = 00000014 00         10         162
# error_counter,err_inj = 00000014 00         10         163
# error_counter,err_inj = 00000014 00         10         164
# error_counter,err_inj = 00000014 00         10         165
# error_counter,err_inj = 00000014 00         10         166
# error_counter,err_inj = 00000014 00         10         167
# error_counter,err_inj = 00000014 00         10         168
# error_counter,err_inj = 00000014 00         10         169
# error_counter,err_inj = 00000014 00         10         170
# error_counter,err_inj = 00000014 00         10         171
# error_counter,err_inj = 00000014 00         10         172
# error_counter,err_inj = 00000014 00         10         173
# error_counter,err_inj = 00000014 00         10         174
# error_counter,err_inj = 00000015 10         10         175
# error_counter,err_inj = 00000016 10         10         176
# error_counter,err_inj = 00000016 00         11         177
# error_counter,err_inj = 00000016 00         11         178
# error_counter,err_inj = 00000016 00         11         179
# error_counter,err_inj = 00000016 00         11         180
# error_counter,err_inj = 00000016 00         11         181
# error_counter,err_inj = 00000016 00         11         182
# error_counter,err_inj = 00000016 00         11         183
# error_counter,err_inj = 00000016 00         11         184
# error_counter,err_inj = 00000016 00         11         185
# error_counter,err_inj = 00000016 00         11         186
# error_counter,err_inj = 00000016 00         11         187
# error_counter,err_inj = 00000016 00         11         188
# error_counter,err_inj = 00000016 00         11         189
# error_counter,err_inj = 00000016 00         11         190
# error_counter,err_inj = 00000017 10         11         191
# error_counter,err_inj = 00000018 10         11         192
# error_counter,err_inj = 00000018 00         12         193
# error_counter,err_inj = 00000018 00         12         194
# error_counter,err_inj = 00000018 00         12         195
# error_counter,err_inj = 00000018 00         12         196
# error_counter,err_inj = 00000018 00         12         197
# error_counter,err_inj = 00000018 00         12         198
# error_counter,err_inj = 00000018 00         12         199
# error_counter,err_inj = 00000018 00         12         200
# error_counter,err_inj = 00000018 00         12         201
# error_counter,err_inj = 00000018 00         12         202
# error_counter,err_inj = 00000018 00         12         203
# error_counter,err_inj = 00000018 00         12         204
# error_counter,err_inj = 00000018 00         12         205
# error_counter,err_inj = 00000018 00         12         206
# error_counter,err_inj = 00000019 10         12         207
# error_counter,err_inj = 0000001a 10         12         208
# error_counter,err_inj = 0000001a 00         13         209
# error_counter,err_inj = 0000001a 00         13         210
# error_counter,err_inj = 0000001a 00         13         211
# error_counter,err_inj = 0000001a 00         13         212
# error_counter,err_inj = 0000001a 00         13         213
# error_counter,err_inj = 0000001a 00         13         214
# error_counter,err_inj = 0000001a 00         13         215
# error_counter,err_inj = 0000001a 00         13         216
# error_counter,err_inj = 0000001a 00         13         217
# error_counter,err_inj = 0000001a 00         13         218
# error_counter,err_inj = 0000001a 00         13         219
```

```
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001a 00          13          220
# error_counter,err_inj = 0000001a 00          13          221
# error_counter,err_inj = 0000001a 00          13          222
# error_counter,err_inj = 0000001b 10          13          223
# error_counter,err_inj = 0000001c 10          13          224
# error_counter,err_inj = 0000001c 00          14          225
# error_counter,err_inj = 0000001c 00          14          226
# error_counter,err_inj = 0000001c 00          14          227
# error_counter,err_inj = 0000001c 00          14          228
# error_counter,err_inj = 0000001c 00          14          229
# error_counter,err_inj = 0000001c 00          14          230
# error_counter,err_inj = 0000001c 00          14          231
# error_counter,err_inj = 0000001c 00          14          232
# error_counter,err_inj = 0000001c 00          14          233
# error_counter,err_inj = 0000001c 00          14          234
# error_counter,err_inj = 0000001c 00          14          235
# error_counter,err_inj = 0000001c 00          14          236
# error_counter,err_inj = 0000001c 00          14          237
# error_counter,err_inj = 0000001c 00          14          238
# error_counter,err_inj = 0000001d 10          14          239
# error_counter,err_inj = 0000001e 10          14          240
# error_counter,err_inj = 0000001e 00          15          241
# error_counter,err_inj = 0000001e 00          15          242
# error_counter,err_inj = 0000001e 00          15          243
# error_counter,err_inj = 0000001e 00          15          244
# error_counter,err_inj = 0000001e 00          15          245
# error_counter,err_inj = 0000001e 00          15          246
# error_counter,err_inj = 0000001e 00          15          247
# error_counter,err_inj = 0000001e 00          15          248
# error_counter,err_inj = 0000001e 00          15          249
# error_counter,err_inj = 0000001e 00          15          250
# error_counter,err_inj = 0000001e 00          15          251
# error_counter,err_inj = 0000001e 00          15          252
# error_counter,err_inj = 0000001e 00          15          253
# error_counter,err_inj = 0000001e 00          15          254
# error_counter,err_inj = 0000001f 10          15          255
# word_count =       10440
```

2a6) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000001 01          0          29
# error_counter,err_inj = 00000002 01          0          30
# error_counter,err_inj = 00000003 01          1          31
# error_counter,err_inj = 00000004 01          2          32
# error_counter,err_inj = 00000004 00          3          33
# error_counter,err_inj = 00000004 00          3          34
# error_counter,err_inj = 00000004 00          3          35
# error_counter,err_inj = 00000004 00          3          36
# error_counter,err_inj = 00000004 00          3          37
# error_counter,err_inj = 00000004 00          3          38
# error_counter,err_inj = 00000004 00          3          39
# error_counter,err_inj = 00000004 00          3          40
# error_counter,err_inj = 00000004 00          3          41
# error_counter,err_inj = 00000004 00          3          42
# error_counter,err_inj = 00000004 00          3          43
# error_counter,err_inj = 00000004 00          3          44
# error_counter,err_inj = 00000004 00          3          45
# error_counter,err_inj = 00000004 00          3          46
# error_counter,err_inj = 00000004 00          3          47
# error_counter,err_inj = 00000004 00          3          48
# error_counter,err_inj = 00000004 00          3          49
# error_counter,err_inj = 00000004 00          3          50
```

```
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
```

```
# error_counter,err_inj = 00000004 00          3          51
# error_counter,err_inj = 00000004 00          3          52
# error_counter,err_inj = 00000004 00          3          53
# error_counter,err_inj = 00000004 00          3          54
# error_counter,err_inj = 00000004 00          3          55
# error_counter,err_inj = 00000004 00          3          56
# error_counter,err_inj = 00000004 00          3          57
# error_counter,err_inj = 00000004 00          3          58
# error_counter,err_inj = 00000004 00          3          59
# error_counter,err_inj = 00000004 00          3          60
# error_counter,err_inj = 00000005 01          3          61
# error_counter,err_inj = 00000006 01          3          62
# error_counter,err_inj = 00000007 01          4          63
# error_counter,err_inj = 00000008 01          5          64
# error_counter,err_inj = 00000008 00          6          65
# error_counter,err_inj = 00000008 00          6          66
# error_counter,err_inj = 00000008 00          6          67
# error_counter,err_inj = 00000008 00          6          68
# error_counter,err_inj = 00000008 00          6          69
# error_counter,err_inj = 00000008 00          6          70
# error_counter,err_inj = 00000008 00          6          71
# error_counter,err_inj = 00000008 00          6          72
# error_counter,err_inj = 00000008 00          6          73
# error_counter,err_inj = 00000008 00          6          74
# error_counter,err_inj = 00000008 00          6          75
# error_counter,err_inj = 00000008 00          6          76
# error_counter,err_inj = 00000008 00          6          77
# error_counter,err_inj = 00000008 00          6          78
# error_counter,err_inj = 00000008 00          6          79
# error_counter,err_inj = 00000008 00          6          80
# error_counter,err_inj = 00000008 00          6          81
# error_counter,err_inj = 00000008 00          6          82
# error_counter,err_inj = 00000008 00          6          83
# error_counter,err_inj = 00000008 00          6          84
# error_counter,err_inj = 00000008 00          6          85
# error_counter,err_inj = 00000008 00          6          86
# error_counter,err_inj = 00000008 00          6          87
# error_counter,err_inj = 00000008 00          6          88
# error_counter,err_inj = 00000008 00          6          89
# error_counter,err_inj = 00000008 00          6          90
# error_counter,err_inj = 00000008 00          6          91
# error_counter,err_inj = 00000008 00          6          92
# error_counter,err_inj = 00000009 01          6          93
# error_counter,err_inj = 0000000a 01          6          94
# error_counter,err_inj = 0000000b 01          7          95
# error_counter,err_inj = 0000000c 01          8          96
# error_counter,err_inj = 0000000c 00          9          97
# error_counter,err_inj = 0000000c 00          9          98
# error_counter,err_inj = 0000000c 00          9          99
# error_counter,err_inj = 0000000c 00          9         100
# error_counter,err_inj = 0000000c 00          9         101
# error_counter,err_inj = 0000000c 00          9         102
# error_counter,err_inj = 0000000c 00          9         103
# error_counter,err_inj = 0000000c 00          9         104
# error_counter,err_inj = 0000000c 00          9         105
# error_counter,err_inj = 0000000c 00          9         106
# error_counter,err_inj = 0000000c 00          9         107
# error_counter,err_inj = 0000000c 00          9         108
# error_counter,err_inj = 0000000c 00          9         109
# error_counter,err_inj = 0000000c 00          9         110
# error_counter,err_inj = 0000000c 00          9         111
# error_counter,err_inj = 0000000c 00          9         112
# error_counter,err_inj = 0000000c 00          9         113
# error_counter,err_inj = 0000000c 00          9         114
# error_counter,err_inj = 0000000c 00          9         115
# error_counter,err_inj = 0000000c 00          9         116
# error_counter,err_inj = 0000000c 00          9         117
# error_counter,err_inj = 0000000c 00          9         118
# error_counter,err_inj = 0000000c 00          9         119
# error_counter,err_inj = 0000000c 00          9         120
# error_counter,err_inj = 0000000c 00          9         121
# error_counter,err_inj = 0000000c 00          9         122
# error_counter,err_inj = 0000000c 00          9         123
# error_counter,err_inj = 0000000c 00          9         124
# error_counter,err_inj = 0000000d 01          9         125
# error_counter,err_inj = 0000000e 01          9         126
# error_counter,err_inj = 0000000f 01         10         127
# error_counter,err_inj = 00000010 01         11         128
# error_counter,err_inj = 00000010 00         12         129
# error_counter,err_inj = 00000010 00         12         130
# error_counter,err_inj = 00000010 00         12         131
# error_counter,err_inj = 00000010 00         12         132
# error_counter,err_inj = 00000010 00         12         133
# error_counter,err_inj = 00000010 00         12         134
# error_counter,err_inj = 00000010 00         12         135
# error_counter,err_inj = 00000010 00         12         136
# error_counter,err_inj = 00000010 00         12         137
# error_counter,err_inj = 00000010 00         12         138
# error_counter,err_inj = 00000010 00         12         139
# error_counter,err_inj = 00000010 00         12         140
# error_counter,err_inj = 00000010 00         12         141
# error_counter,err_inj = 00000010 00         12         142
# error_counter,err_inj = 00000010 00         12         143
```

```
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
```

```
# error_counter,err_inj = 00000010 00          12          144
# error_counter,err_inj = 00000010 00          12          145
# error_counter,err_inj = 00000010 00          12          146
# error_counter,err_inj = 00000010 00          12          147
# error_counter,err_inj = 00000010 00          12          148
# error_counter,err_inj = 00000010 00          12          149
# error_counter,err_inj = 00000010 00          12          150
# error_counter,err_inj = 00000010 00          12          151
# error_counter,err_inj = 00000010 00          12          152
# error_counter,err_inj = 00000010 00          12          153
# error_counter,err_inj = 00000010 00          12          154
# error_counter,err_inj = 00000010 00          12          155
# error_counter,err_inj = 00000010 00          12          156
# error_counter,err_inj = 00000011 01          12          157
# error_counter,err_inj = 00000012 01          12          158
# error_counter,err_inj = 00000013 01          13          159
# error_counter,err_inj = 00000014 01          14          160
# error_counter,err_inj = 00000014 00          15          161
# error_counter,err_inj = 00000014 00          15          162
# error_counter,err_inj = 00000014 00          15          163
# error_counter,err_inj = 00000014 00          15          164
# error_counter,err_inj = 00000014 00          15          165
# error_counter,err_inj = 00000014 00          15          166
# error_counter,err_inj = 00000014 00          15          167
# error_counter,err_inj = 00000014 00          15          168
# error_counter,err_inj = 00000014 00          15          169
# error_counter,err_inj = 00000014 00          15          170
# error_counter,err_inj = 00000014 00          15          171
# error_counter,err_inj = 00000014 00          15          172
# error_counter,err_inj = 00000014 00          15          173
# error_counter,err_inj = 00000014 00          15          174
# error_counter,err_inj = 00000014 00          15          175
# error_counter,err_inj = 00000014 00          15          176
# error_counter,err_inj = 00000014 00          15          177
# error_counter,err_inj = 00000014 00          15          178
# error_counter,err_inj = 00000014 00          15          179
# error_counter,err_inj = 00000014 00          15          180
# error_counter,err_inj = 00000014 00          15          181
# error_counter,err_inj = 00000014 00          15          182
# error_counter,err_inj = 00000014 00          15          183
# error_counter,err_inj = 00000014 00          15          184
# error_counter,err_inj = 00000014 00          15          185
# error_counter,err_inj = 00000014 00          15          186
# error_counter,err_inj = 00000014 00          15          187
# error_counter,err_inj = 00000014 00          15          188
# error_counter,err_inj = 00000015 01          15          189
# error_counter,err_inj = 00000016 01          15          190
# error_counter,err_inj = 00000017 01          16          191
# error_counter,err_inj = 00000018 01          17          192
# error_counter,err_inj = 00000018 00          18          193
# error_counter,err_inj = 00000018 00          18          194
# error_counter,err_inj = 00000018 00          18          195
# error_counter,err_inj = 00000018 00          18          196
# error_counter,err_inj = 00000018 00          18          197
# error_counter,err_inj = 00000018 00          18          198
# error_counter,err_inj = 00000018 00          18          199
# error_counter,err_inj = 00000018 00          18          200
# error_counter,err_inj = 00000018 00          18          201
# error_counter,err_inj = 00000018 00          18          202
# error_counter,err_inj = 00000018 00          18          203
# error_counter,err_inj = 00000018 00          18          204
# error_counter,err_inj = 00000018 00          18          205
# error_counter,err_inj = 00000018 00          18          206
# error_counter,err_inj = 00000018 00          18          207
# error_counter,err_inj = 00000018 00          18          208
# error_counter,err_inj = 00000018 00          18          209
# error_counter,err_inj = 00000018 00          18          210
# error_counter,err_inj = 00000018 00          18          211
# error_counter,err_inj = 00000018 00          18          212
# error_counter,err_inj = 00000018 00          18          213
# error_counter,err_inj = 00000018 00          18          214
# error_counter,err_inj = 00000018 00          18          215
# error_counter,err_inj = 00000018 00          18          216
# error_counter,err_inj = 00000018 00          18          217
# error_counter,err_inj = 00000018 00          18          218
# error_counter,err_inj = 00000018 00          18          219
# error_counter,err_inj = 00000018 00          18          220
# error_counter,err_inj = 00000019 01          18          221
# error_counter,err_inj = 0000001a 01          18          222
# error_counter,err_inj = 0000001b 01          19          223
# error_counter,err_inj = 0000001c 01          20          224
# error_counter,err_inj = 0000001c 00          21          225
# error_counter,err_inj = 0000001c 00          21          226
# error_counter,err_inj = 0000001c 00          21          227
# error_counter,err_inj = 0000001c 00          21          228
# error_counter,err_inj = 0000001c 00          21          229
# error_counter,err_inj = 0000001c 00          21          230
# error_counter,err_inj = 0000001c 00          21          231
# error_counter,err_inj = 0000001c 00          21          232
# error_counter,err_inj = 0000001c 00          21          233
# error_counter,err_inj = 0000001c 00          21          234
# error_counter,err_inj = 0000001c 00          21          235
# error_counter,err_inj = 0000001c 00          21          236
```

```
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000001c 00          21         237
# error_counter,err_inj = 0000001c 00          21         238
# error_counter,err_inj = 0000001c 00          21         239
# error_counter,err_inj = 0000001c 00          21         240
# error_counter,err_inj = 0000001c 00          21         241
# error_counter,err_inj = 0000001c 00          21         242
# error_counter,err_inj = 0000001c 00          21         243
# error_counter,err_inj = 0000001c 00          21         244
# error_counter,err_inj = 0000001c 00          21         245
# error_counter,err_inj = 0000001c 00          21         246
# error_counter,err_inj = 0000001c 00          21         247
# error_counter,err_inj = 0000001c 00          21         248
# error_counter,err_inj = 0000001c 00          21         249
# error_counter,err_inj = 0000001c 00          21         250
# error_counter,err_inj = 0000001c 00          21         251
# error_counter,err_inj = 0000001c 00          21         252
# error_counter,err_inj = 0000001d 01          21         253
# error_counter,err_inj = 0000001e 01          21         254
# error_counter,err_inj = 0000001f 01          22         255
```

2a7) Transcript:

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000000 00           0           8
# error_counter,err_inj = 00000000 00           0           9
# error_counter,err_inj = 00000000 00           0          10
# error_counter,err_inj = 00000000 00           0          11
# error_counter,err_inj = 00000000 00           0          12
# error_counter,err_inj = 00000000 00           0          13
# error_counter,err_inj = 00000000 00           0          14
# error_counter,err_inj = 00000000 00           0          15
# error_counter,err_inj = 00000000 00           0          16
# error_counter,err_inj = 00000000 00           0          17
# error_counter,err_inj = 00000000 00           0          18
# error_counter,err_inj = 00000000 00           0          19
# error_counter,err_inj = 00000000 00           0          20
# error_counter,err_inj = 00000000 00           0          21
# error_counter,err_inj = 00000000 00           0          22
# error_counter,err_inj = 00000000 00           0          23
# error_counter,err_inj = 00000000 00           0          24
# error_counter,err_inj = 00000000 00           0          25
# error_counter,err_inj = 00000000 00           0          26
# error_counter,err_inj = 00000000 00           0          27
# error_counter,err_inj = 00000000 00           0          28
# error_counter,err_inj = 00000001 10           0          29
# error_counter,err_inj = 00000002 10           0          30
# error_counter,err_inj = 00000003 10           1          31
# error_counter,err_inj = 00000004 10           2          32
# error_counter,err_inj = 00000004 00           3          33
# error_counter,err_inj = 00000004 00           3          34
# error_counter,err_inj = 00000004 00           3          35
# error_counter,err_inj = 00000004 00           3          36
# error_counter,err_inj = 00000004 00           3          37
# error_counter,err_inj = 00000004 00           3          38
# error_counter,err_inj = 00000004 00           3          39
# error_counter,err_inj = 00000004 00           3          40
# error_counter,err_inj = 00000004 00           3          41
# error_counter,err_inj = 00000004 00           3          42
# error_counter,err_inj = 00000004 00           3          43
# error_counter,err_inj = 00000004 00           3          44
# error_counter,err_inj = 00000004 00           3          45
# error_counter,err_inj = 00000004 00           3          46
# error_counter,err_inj = 00000004 00           3          47
# error_counter,err_inj = 00000004 00           3          48
# error_counter,err_inj = 00000004 00           3          49
# error_counter,err_inj = 00000004 00           3          50
# error_counter,err_inj = 00000004 00           3          51
# error_counter,err_inj = 00000004 00           3          52
# error_counter,err_inj = 00000004 00           3          53
# error_counter,err_inj = 00000004 00           3          54
# error_counter,err_inj = 00000004 00           3          55
# error_counter,err_inj = 00000004 00           3          56
# error_counter,err_inj = 00000004 00           3          57
# error_counter,err_inj = 00000004 00           3          58
# error_counter,err_inj = 00000004 00           3          59
# error_counter,err_inj = 00000004 00           3          60
# error_counter,err_inj = 00000005 10           3          61
# error_counter,err_inj = 00000006 10           3          62
# error_counter,err_inj = 00000007 10           4          63
# error_counter,err_inj = 00000008 10           5          64
# error_counter,err_inj = 00000008 00           6          65
# error_counter,err_inj = 00000008 00           6          66
# error_counter,err_inj = 00000008 00           6          67
```

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
```

```
# error_counter,err_inj = 00000008 00          6          68   # yaa! in = 0, out = 0, w_ct =          68, err = 00
# error_counter,err_inj = 00000008 00          6          69   # yaa! in = 0, out = 0, w_ct =          69, err = 00
# error_counter,err_inj = 00000008 00          6          70   # yaa! in = 0, out = 0, w_ct =          70, err = 00
# error_counter,err_inj = 00000008 00          6          71   # yaa! in = 0, out = 0, w_ct =          71, err = 00
# error_counter,err_inj = 00000008 00          6          72   # yaa! in = 0, out = 0, w_ct =          72, err = 00
# error_counter,err_inj = 00000008 00          6          73   # yaa! in = 1, out = 1, w_ct =          73, err = 00
# error_counter,err_inj = 00000008 00          6          74   # yaa! in = 1, out = 1, w_ct =          74, err = 00
# error_counter,err_inj = 00000008 00          6          75   # yaa! in = 1, out = 1, w_ct =          75, err = 00
# error_counter,err_inj = 00000008 00          6          76   # yaa! in = 1, out = 1, w_ct =          76, err = 00
# error_counter,err_inj = 00000008 00          6          77   # yaa! in = 1, out = 1, w_ct =          77, err = 00
# error_counter,err_inj = 00000008 00          6          78   # yaa! in = 0, out = 0, w_ct =          78, err = 00
# error_counter,err_inj = 00000008 00          6          79   # yaa! in = 1, out = 1, w_ct =          79, err = 00
# error_counter,err_inj = 00000008 00          6          80   # yaa! in = 0, out = 0, w_ct =          80, err = 00
# error_counter,err_inj = 00000008 00          6          81   # yaa! in = 1, out = 1, w_ct =          81, err = 00
# error_counter,err_inj = 00000008 00          6          82   # yaa! in = 0, out = 0, w_ct =          82, err = 00
# error_counter,err_inj = 00000008 00          6          83   # yaa! in = 1, out = 1, w_ct =          83, err = 00
# error_counter,err_inj = 00000008 00          6          84   # yaa! in = 0, out = 0, w_ct =          84, err = 00
# error_counter,err_inj = 00000008 00          6          85   # yaa! in = 1, out = 1, w_ct =          85, err = 00
# error_counter,err_inj = 00000008 00          6          86   # yaa! in = 0, out = 0, w_ct =          86, err = 00
# error_counter,err_inj = 00000008 00          6          87   # yaa! in = 0, out = 0, w_ct =          87, err = 00
# error_counter,err_inj = 00000008 00          6          88   # yaa! in = 0, out = 0, w_ct =          88, err = 00
# error_counter,err_inj = 00000008 00          6          89   # yaa! in = 0, out = 0, w_ct =          89, err = 00
# error_counter,err_inj = 00000008 00          6          90   # yaa! in = 0, out = 0, w_ct =          90, err = 00
# error_counter,err_inj = 00000008 00          6          91   # yaa! in = 0, out = 0, w_ct =          91, err = 00
# error_counter,err_inj = 00000008 00          6          92   # yaa! in = 0, out = 0, w_ct =          92, err = 00
# error_counter,err_inj = 00000009 10          6          93   # yaa! in = 0, out = 0, w_ct =          93, err = 00
# error_counter,err_inj = 0000000a 10          6          94   # yaa! in = 0, out = 0, w_ct =          94, err = 00
# error_counter,err_inj = 0000000b 10          7          95   # yaa! in = 0, out = 0, w_ct =          95, err = 00
# error_counter,err_inj = 0000000c 10          8          96   # yaa! in = 1, out = 1, w_ct =          96, err = 00
# error_counter,err_inj = 0000000c 00          9          97   # yaa! in = 1, out = 1, w_ct =          97, err = 00
# error_counter,err_inj = 0000000c 00          9          98   # yaa! in = 1, out = 1, w_ct =          98, err = 00
# error_counter,err_inj = 0000000c 00          9          99   # yaa! in = 1, out = 1, w_ct =          99, err = 00
# error_counter,err_inj = 0000000c 00          9         100   # yaa! in = 1, out = 1, w_ct =         100, err = 00
# error_counter,err_inj = 0000000c 00          9         101   # yaa! in = 1, out = 1, w_ct =         101, err = 00
# error_counter,err_inj = 0000000c 00          9         102   # yaa! in = 1, out = 1, w_ct =         102, err = 00
# error_counter,err_inj = 0000000c 00          9         103   # yaa! in = 1, out = 1, w_ct =         103, err = 00
# error_counter,err_inj = 0000000c 00          9         104   # yaa! in = 1, out = 1, w_ct =         104, err = 00
# error_counter,err_inj = 0000000c 00          9         105   # yaa! in = 1, out = 1, w_ct =         105, err = 00
# error_counter,err_inj = 0000000c 00          9         106   # yaa! in = 0, out = 0, w_ct =         106, err = 00
# error_counter,err_inj = 0000000c 00          9         107   # yaa! in = 0, out = 0, w_ct =         107, err = 00
# error_counter,err_inj = 0000000c 00          9         108   # yaa! in = 1, out = 1, w_ct =         108, err = 00
# error_counter,err_inj = 0000000c 00          9         109   # yaa! in = 1, out = 1, w_ct =         109, err = 00
# error_counter,err_inj = 0000000c 00          9         110   # yaa! in = 1, out = 1, w_ct =         110, err = 00
# error_counter,err_inj = 0000000c 00          9         111   # yaa! in = 0, out = 0, w_ct =         111, err = 00
# error_counter,err_inj = 0000000c 00          9         112   # yaa! in = 1, out = 1, w_ct =         112, err = 00
# error_counter,err_inj = 0000000c 00          9         113   # yaa! in = 0, out = 0, w_ct =         113, err = 00
# error_counter,err_inj = 0000000c 00          9         114   # yaa! in = 0, out = 0, w_ct =         114, err = 00
# error_counter,err_inj = 0000000c 00          9         115   # yaa! in = 0, out = 0, w_ct =         115, err = 00
# error_counter,err_inj = 0000000c 00          9         116   # yaa! in = 0, out = 0, w_ct =         116, err = 00
# error_counter,err_inj = 0000000c 00          9         117   # yaa! in = 1, out = 1, w_ct =         117, err = 00
# error_counter,err_inj = 0000000c 00          9         118   # yaa! in = 1, out = 1, w_ct =         118, err = 00
# error_counter,err_inj = 0000000c 00          9         119   # yaa! in = 1, out = 1, w_ct =         119, err = 00
# error_counter,err_inj = 0000000c 00          9         120   # yaa! in = 0, out = 0, w_ct =         120, err = 00
# error_counter,err_inj = 0000000c 00          9         121   # yaa! in = 1, out = 1, w_ct =         121, err = 00
# error_counter,err_inj = 0000000c 00          9         122   # yaa! in = 0, out = 0, w_ct =         122, err = 00
# error_counter,err_inj = 0000000c 00          9         123   # yaa! in = 1, out = 1, w_ct =         123, err = 00
# error_counter,err_inj = 0000000c 00          9         124   # yaa! in = 1, out = 1, w_ct =         124, err = 00
# error_counter,err_inj = 0000000d 10          9         125   # yaa! in = 1, out = 1, w_ct =         125, err = 00
# error_counter,err_inj = 0000000e 10          9         126   # yaa! in = 0, out = 0, w_ct =         126, err = 00
# error_counter,err_inj = 0000000f 10         10         127   # yaa! in = 0, out = 0, w_ct =         127, err = 00
# error_counter,err_inj = 00000010 10         11         128   # yaa! in = 0, out = 0, w_ct =         128, err = 00
# error_counter,err_inj = 00000010 00         12         129   # yaa! in = 0, out = 0, w_ct =         129, err = 00
# error_counter,err_inj = 00000010 00         12         130   # yaa! in = 0, out = 0, w_ct =         130, err = 00
# error_counter,err_inj = 00000010 00         12         131   # yaa! in = 0, out = 0, w_ct =         131, err = 00
# error_counter,err_inj = 00000010 00         12         132   # yaa! in = 0, out = 0, w_ct =         132, err = 00
# error_counter,err_inj = 00000010 00         12         133   # yaa! in = 0, out = 0, w_ct =         133, err = 00
# error_counter,err_inj = 00000010 00         12         134   # yaa! in = 0, out = 0, w_ct =         134, err = 00
# error_counter,err_inj = 00000010 00         12         135   # yaa! in = 0, out = 0, w_ct =         135, err = 00
# error_counter,err_inj = 00000010 00         12         136   # yaa! in = 0, out = 0, w_ct =         136, err = 00
# error_counter,err_inj = 00000010 00         12         137   # yaa! in = 1, out = 1, w_ct =         137, err = 00
# error_counter,err_inj = 00000010 00         12         138   # yaa! in = 1, out = 1, w_ct =         138, err = 00
# error_counter,err_inj = 00000010 00         12         139   # yaa! in = 1, out = 1, w_ct =         139, err = 00
# error_counter,err_inj = 00000010 00         12         140   # yaa! in = 1, out = 1, w_ct =         140, err = 00
# error_counter,err_inj = 00000010 00         12         141   # yaa! in = 1, out = 1, w_ct =         141, err = 00
# error_counter,err_inj = 00000010 00         12         142   # yaa! in = 1, out = 1, w_ct =         142, err = 00
# error_counter,err_inj = 00000010 00         12         143   # yaa! in = 1, out = 1, w_ct =         143, err = 00
# error_counter,err_inj = 00000010 00         12         144   # yaa! in = 1, out = 1, w_ct =         144, err = 00
# error_counter,err_inj = 00000010 00         12         145   # yaa! in = 1, out = 1, w_ct =         145, err = 00
# error_counter,err_inj = 00000010 00         12         146   # yaa! in = 1, out = 1, w_ct =         146, err = 00
# error_counter,err_inj = 00000010 00         12         147   # yaa! in = 0, out = 0, w_ct =         147, err = 00
# error_counter,err_inj = 00000010 00         12         148   # yaa! in = 1, out = 1, w_ct =         148, err = 00
# error_counter,err_inj = 00000010 00         12         149   # yaa! in = 1, out = 1, w_ct =         149, err = 00
# error_counter,err_inj = 00000010 00         12         150   # yaa! in = 1, out = 1, w_ct =         150, err = 00
# error_counter,err_inj = 00000010 00         12         151   # yaa! in = 1, out = 1, w_ct =         151, err = 00
# error_counter,err_inj = 00000010 00         12         152   # yaa! in = 1, out = 1, w_ct =         152, err = 00
# error_counter,err_inj = 00000010 00         12         153   # yaa! in = 1, out = 1, w_ct =         153, err = 00
# error_counter,err_inj = 00000010 00         12         154   # yaa! in = 1, out = 1, w_ct =         154, err = 00
# error_counter,err_inj = 00000010 00         12         155   # yaa! in = 1, out = 1, w_ct =         155, err = 00
# error_counter,err_inj = 00000010 00         12         156   # yaa! in = 1, out = 1, w_ct =         156, err = 00
# error_counter,err_inj = 00000011 10         12         157   # yaa! in = 1, out = 1, w_ct =         157, err = 00
# error_counter,err_inj = 00000012 10         12         158   # yaa! in = 1, out = 1, w_ct =         158, err = 00
# error_counter,err_inj = 00000013 10         13         159   # yaa! in = 1, out = 1, w_ct =         159, err = 00
# error_counter,err_inj = 00000014 10         14         160   # yaa! in = 1, out = 1, w_ct =         160, err = 00
```

```
# error_counter,err_inj = 00000014 00          15          161    # yaa! in = 1, out = 1, w_ct =          161, err = 00
# error_counter,err_inj = 00000014 00          15          162    # yaa! in = 1, out = 1, w_ct =          162, err = 00
# error_counter,err_inj = 00000014 00          15          163    # yaa! in = 1, out = 1, w_ct =          163, err = 00
# error_counter,err_inj = 00000014 00          15          164    # yaa! in = 1, out = 1, w_ct =          164, err = 00
# error_counter,err_inj = 00000014 00          15          165    # yaa! in = 1, out = 1, w_ct =          165, err = 00
# error_counter,err_inj = 00000014 00          15          166    # yaa! in = 1, out = 1, w_ct =          166, err = 00
# error_counter,err_inj = 00000014 00          15          167    # yaa! in = 1, out = 1, w_ct =          167, err = 00
# error_counter,err_inj = 00000014 00          15          168    # yaa! in = 1, out = 1, w_ct =          168, err = 00
# error_counter,err_inj = 00000014 00          15          169    # yaa! in = 1, out = 1, w_ct =          169, err = 00
# error_counter,err_inj = 00000014 00          15          170    # yaa! in = 1, out = 1, w_ct =          170, err = 00
# error_counter,err_inj = 00000014 00          15          171    # yaa! in = 1, out = 1, w_ct =          171, err = 00
# error_counter,err_inj = 00000014 00          15          172    # yaa! in = 1, out = 1, w_ct =          172, err = 00
# error_counter,err_inj = 00000014 00          15          173    # yaa! in = 1, out = 1, w_ct =          173, err = 00
# error_counter,err_inj = 00000014 00          15          174    # yaa! in = 1, out = 1, w_ct =          174, err = 00
# error_counter,err_inj = 00000014 00          15          175    # yaa! in = 1, out = 1, w_ct =          175, err = 00
# error_counter,err_inj = 00000014 00          15          176    # yaa! in = 1, out = 1, w_ct =          176, err = 00
# error_counter,err_inj = 00000014 00          15          177    # yaa! in = 1, out = 1, w_ct =          177, err = 00
# error_counter,err_inj = 00000014 00          15          178    # yaa! in = 1, out = 1, w_ct =          178, err = 00
# error_counter,err_inj = 00000014 00          15          179    # yaa! in = 1, out = 1, w_ct =          179, err = 00
# error_counter,err_inj = 00000014 00          15          180    # yaa! in = 1, out = 1, w_ct =          180, err = 00
# error_counter,err_inj = 00000014 00          15          181    # yaa! in = 1, out = 1, w_ct =          181, err = 00
# error_counter,err_inj = 00000014 00          15          182    # yaa! in = 1, out = 1, w_ct =          182, err = 00
# error_counter,err_inj = 00000014 00          15          183    # yaa! in = 1, out = 1, w_ct =          183, err = 00
# error_counter,err_inj = 00000014 00          15          184    # yaa! in = 1, out = 1, w_ct =          184, err = 00
# error_counter,err_inj = 00000014 00          15          185    # yaa! in = 1, out = 1, w_ct =          185, err = 00
# error_counter,err_inj = 00000014 00          15          186    # yaa! in = 1, out = 1, w_ct =          186, err = 00
# error_counter,err_inj = 00000014 00          15          187    # yaa! in = 1, out = 1, w_ct =          187, err = 00
# error_counter,err_inj = 00000014 00          15          188    # yaa! in = 1, out = 1, w_ct =          188, err = 00
# error_counter,err_inj = 00000015 10          15          189    # yaa! in = 1, out = 1, w_ct =          189, err = 00
# error_counter,err_inj = 00000016 10          15          190    # yaa! in = 1, out = 1, w_ct =          190, err = 00
# error_counter,err_inj = 00000017 10          16          191    # yaa! in = 1, out = 1, w_ct =          191, err = 00
# error_counter,err_inj = 00000018 10          17          192    # yaa! in = 1, out = 1, w_ct =          192, err = 00
# error_counter,err_inj = 00000018 00          18          193    # yaa! in = 1, out = 1, w_ct =          193, err = 00
# error_counter,err_inj = 00000018 00          18          194    # yaa! in = 1, out = 1, w_ct =          194, err = 00
# error_counter,err_inj = 00000018 00          18          195    # yaa! in = 1, out = 1, w_ct =          195, err = 00
# error_counter,err_inj = 00000018 00          18          196    # yaa! in = 1, out = 1, w_ct =          196, err = 00
# error_counter,err_inj = 00000018 00          18          197    # yaa! in = 1, out = 1, w_ct =          197, err = 00
# error_counter,err_inj = 00000018 00          18          198    # yaa! in = 1, out = 1, w_ct =          198, err = 00
# error_counter,err_inj = 00000018 00          18          199    # yaa! in = 1, out = 1, w_ct =          199, err = 00
# error_counter,err_inj = 00000018 00          18          200    # yaa! in = 1, out = 1, w_ct =          200, err = 00
# error_counter,err_inj = 00000018 00          18          201    # yaa! in = 1, out = 1, w_ct =          201, err = 00
# error_counter,err_inj = 00000018 00          18          202    # yaa! in = 1, out = 1, w_ct =          202, err = 00
# error_counter,err_inj = 00000018 00          18          203    # yaa! in = 1, out = 1, w_ct =          203, err = 00
# error_counter,err_inj = 00000018 00          18          204    # yaa! in = 1, out = 1, w_ct =          204, err = 00
# error_counter,err_inj = 00000018 00          18          205    # yaa! in = 1, out = 1, w_ct =          205, err = 00
# error_counter,err_inj = 00000018 00          18          206    # yaa! in = 1, out = 1, w_ct =          206, err = 00
# error_counter,err_inj = 00000018 00          18          207    # yaa! in = 1, out = 1, w_ct =          207, err = 00
# error_counter,err_inj = 00000018 00          18          208    # yaa! in = 1, out = 1, w_ct =          208, err = 00
# error_counter,err_inj = 00000018 00          18          209    # yaa! in = 1, out = 1, w_ct =          209, err = 00
# error_counter,err_inj = 00000018 00          18          210    # yaa! in = 1, out = 1, w_ct =          210, err = 00
# error_counter,err_inj = 00000018 00          18          211    # yaa! in = 1, out = 1, w_ct =          211, err = 00
# error_counter,err_inj = 00000018 00          18          212    # yaa! in = 1, out = 1, w_ct =          212, err = 00
# error_counter,err_inj = 00000018 00          18          213    # yaa! in = 1, out = 1, w_ct =          213, err = 00
# error_counter,err_inj = 00000018 00          18          214    # yaa! in = 1, out = 1, w_ct =          214, err = 00
# error_counter,err_inj = 00000018 00          18          215    # yaa! in = 1, out = 1, w_ct =          215, err = 00
# error_counter,err_inj = 00000018 00          18          216    # yaa! in = 1, out = 1, w_ct =          216, err = 00
# error_counter,err_inj = 00000018 00          18          217    # yaa! in = 1, out = 1, w_ct =          217, err = 00
# error_counter,err_inj = 00000018 00          18          218    # yaa! in = 1, out = 1, w_ct =          218, err = 00
# error_counter,err_inj = 00000018 00          18          219    # yaa! in = 1, out = 1, w_ct =          219, err = 00
# error_counter,err_inj = 00000018 00          18          220    # yaa! in = 1, out = 1, w_ct =          220, err = 00
# error_counter,err_inj = 00000019 10          18          221    # yaa! in = 1, out = 1, w_ct =          221, err = 00
# error_counter,err_inj = 0000001a 10          18          222    # yaa! in = 1, out = 1, w_ct =          222, err = 00
# error_counter,err_inj = 0000001b 10          19          223    # yaa! in = 1, out = 1, w_ct =          223, err = 00
# error_counter,err_inj = 0000001c 10          20          224    # yaa! in = 1, out = 1, w_ct =          224, err = 00
# error_counter,err_inj = 0000001c 00          21          225    # yaa! in = 1, out = 1, w_ct =          225, err = 00
# error_counter,err_inj = 0000001c 00          21          226    # yaa! in = 1, out = 1, w_ct =          226, err = 00
# error_counter,err_inj = 0000001c 00          21          227    # yaa! in = 1, out = 1, w_ct =          227, err = 00
# error_counter,err_inj = 0000001c 00          21          228    # yaa! in = 1, out = 1, w_ct =          228, err = 00
# error_counter,err_inj = 0000001c 00          21          229    # yaa! in = 1, out = 1, w_ct =          229, err = 00
# error_counter,err_inj = 0000001c 00          21          230    # yaa! in = 1, out = 1, w_ct =          230, err = 00
# error_counter,err_inj = 0000001c 00          21          231    # yaa! in = 1, out = 1, w_ct =          231, err = 00
# error_counter,err_inj = 0000001c 00          21          232    # yaa! in = 1, out = 1, w_ct =          232, err = 00
# error_counter,err_inj = 0000001c 00          21          233    # yaa! in = 1, out = 1, w_ct =          233, err = 00
# error_counter,err_inj = 0000001c 00          21          234    # yaa! in = 1, out = 1, w_ct =          234, err = 00
# error_counter,err_inj = 0000001c 00          21          235    # yaa! in = 1, out = 1, w_ct =          235, err = 00
# error_counter,err_inj = 0000001c 00          21          236    # yaa! in = 1, out = 1, w_ct =          236, err = 00
# error_counter,err_inj = 0000001c 00          21          237    # yaa! in = 1, out = 1, w_ct =          237, err = 00
# error_counter,err_inj = 0000001c 00          21          238    # yaa! in = 1, out = 1, w_ct =          238, err = 00
# error_counter,err_inj = 0000001c 00          21          239    # yaa! in = 1, out = 1, w_ct =          239, err = 00
# error_counter,err_inj = 0000001c 00          21          240    # yaa! in = 1, out = 1, w_ct =          240, err = 00
# error_counter,err_inj = 0000001c 00          21          241    # yaa! in = 1, out = 1, w_ct =          241, err = 00
# error_counter,err_inj = 0000001c 00          21          242    # yaa! in = 1, out = 1, w_ct =          242, err = 00
# error_counter,err_inj = 0000001c 00          21          243    # yaa! in = 1, out = 1, w_ct =          243, err = 00
# error_counter,err_inj = 0000001c 00          21          244    # yaa! in = 1, out = 1, w_ct =          244, err = 00
# error_counter,err_inj = 0000001c 00          21          245    # yaa! in = 1, out = 1, w_ct =          245, err = 00
# error_counter,err_inj = 0000001c 00          21          246    # yaa! in = 1, out = 1, w_ct =          246, err = 00
# error_counter,err_inj = 0000001c 00          21          247    # yaa! in = 1, out = 1, w_ct =          247, err = 00
# error_counter,err_inj = 0000001c 00          21          248    # yaa! in = 0, out = 0, w_ct =          248, err = 00
# error_counter,err_inj = 0000001c 00          21          249    # yaa! in = 1, out = 1, w_ct =          249, err = 00
# error_counter,err_inj = 0000001c 00          21          250    # yaa! in = 1, out = 1, w_ct =          250, err = 00
# error_counter,err_inj = 0000001c 00          21          251    # yaa! in = 1, out = 1, w_ct =          251, err = 00
# error_counter,err_inj = 0000001c 00          21          252    # yaa! in = 1, out = 1, w_ct =          252, err = 00
# error_counter,err_inj = 0000001d 10          21          253    # yaa! in = 1, out = 1, w_ct =          253, err = 00
```

```
# error_counter,err_inj = 0000001e 10          21         254
# error_counter,err_inj = 0000001f 10          22         255
```

```
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         256, bad =
0
```

2a8) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000000 00          0          29
# error_counter,err_inj = 00000000 00          0          30
# error_counter,err_inj = 00000001 11          0          31
# error_counter,err_inj = 00000002 11          0          32
# error_counter,err_inj = 00000002 00          2          33
# error_counter,err_inj = 00000002 00          2          34
# error_counter,err_inj = 00000002 00          2          35
# error_counter,err_inj = 00000002 00          2          36
# error_counter,err_inj = 00000002 00          2          37
# error_counter,err_inj = 00000002 00          2          38
# error_counter,err_inj = 00000002 00          2          39
# error_counter,err_inj = 00000002 00          2          40
# error_counter,err_inj = 00000002 00          2          41
# error_counter,err_inj = 00000002 00          2          42
# error_counter,err_inj = 00000002 00          2          43
# error_counter,err_inj = 00000002 00          2          44
# error_counter,err_inj = 00000002 00          2          45
# error_counter,err_inj = 00000002 00          2          46
# error_counter,err_inj = 00000002 00          2          47
# error_counter,err_inj = 00000002 00          2          48
# error_counter,err_inj = 00000002 00          2          49
# error_counter,err_inj = 00000002 00          2          50
# error_counter,err_inj = 00000002 00          2          51
# error_counter,err_inj = 00000002 00          2          52
# error_counter,err_inj = 00000002 00          2          53
# error_counter,err_inj = 00000002 00          2          54
# error_counter,err_inj = 00000002 00          2          55
# error_counter,err_inj = 00000002 00          2          56
# error_counter,err_inj = 00000002 00          2          57
# error_counter,err_inj = 00000002 00          2          58
# error_counter,err_inj = 00000002 00          2          59
# error_counter,err_inj = 00000002 00          2          60
# error_counter,err_inj = 00000002 00          2          61
# error_counter,err_inj = 00000002 00          2          62
# error_counter,err_inj = 00000003 11          2          63
# error_counter,err_inj = 00000004 11          2          64
# error_counter,err_inj = 00000004 00          4          65
# error_counter,err_inj = 00000004 00          4          66
# error_counter,err_inj = 00000004 00          4          67
# error_counter,err_inj = 00000004 00          4          68
# error_counter,err_inj = 00000004 00          4          69
# error_counter,err_inj = 00000004 00          4          70
# error_counter,err_inj = 00000004 00          4          71
# error_counter,err_inj = 00000004 00          4          72
# error_counter,err_inj = 00000004 00          4          73
# error_counter,err_inj = 00000004 00          4          74
# error_counter,err_inj = 00000004 00          4          75
# error_counter,err_inj = 00000004 00          4          76
# error_counter,err_inj = 00000004 00          4          77
# error_counter,err_inj = 00000004 00          4          78
# error_counter,err_inj = 00000004 00          4          79
# error_counter,err_inj = 00000004 00          4          80
# error_counter,err_inj = 00000004 00          4          81
# error_counter,err_inj = 00000004 00          4          82
# error_counter,err_inj = 00000004 00          4          83
# error_counter,err_inj = 00000004 00          4          84
```

```
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
```

```
# error_counter,err_inj = 00000004 00          4          85
# error_counter,err_inj = 00000004 00          4          86
# error_counter,err_inj = 00000004 00          4          87
# error_counter,err_inj = 00000004 00          4          88
# error_counter,err_inj = 00000004 00          4          89
# error_counter,err_inj = 00000004 00          4          90
# error_counter,err_inj = 00000004 00          4          91
# error_counter,err_inj = 00000004 00          4          92
# error_counter,err_inj = 00000004 00          4          93
# error_counter,err_inj = 00000004 00          4          94
# error_counter,err_inj = 00000005 11          4          95
# error_counter,err_inj = 00000006 11          4          96
# error_counter,err_inj = 00000006 00          6          97
# error_counter,err_inj = 00000006 00          6          98
# error_counter,err_inj = 00000006 00          6          99
# error_counter,err_inj = 00000006 00          6         100
# error_counter,err_inj = 00000006 00          6         101
# error_counter,err_inj = 00000006 00          6         102
# error_counter,err_inj = 00000006 00          6         103
# error_counter,err_inj = 00000006 00          6         104
# error_counter,err_inj = 00000006 00          6         105
# error_counter,err_inj = 00000006 00          6         106
# error_counter,err_inj = 00000006 00          6         107
# error_counter,err_inj = 00000006 00          6         108
# error_counter,err_inj = 00000006 00          6         109
# error_counter,err_inj = 00000006 00          6         110
# error_counter,err_inj = 00000006 00          6         111
# error_counter,err_inj = 00000006 00          6         112
# error_counter,err_inj = 00000006 00          6         113
# error_counter,err_inj = 00000006 00          6         114
# error_counter,err_inj = 00000006 00          6         115
# error_counter,err_inj = 00000006 00          6         116
# error_counter,err_inj = 00000006 00          6         117
# error_counter,err_inj = 00000006 00          6         118
# error_counter,err_inj = 00000006 00          6         119
# error_counter,err_inj = 00000006 00          6         120
# error_counter,err_inj = 00000006 00          6         121
# error_counter,err_inj = 00000006 00          6         122
# error_counter,err_inj = 00000006 00          6         123
# error_counter,err_inj = 00000006 00          6         124
# error_counter,err_inj = 00000006 00          6         125
# error_counter,err_inj = 00000006 00          6         126
# error_counter,err_inj = 00000007 11          6         127
# error_counter,err_inj = 00000008 11          6         128
# error_counter,err_inj = 00000008 00          8         129
# error_counter,err_inj = 00000008 00          8         130
# error_counter,err_inj = 00000008 00          8         131
# error_counter,err_inj = 00000008 00          8         132
# error_counter,err_inj = 00000008 00          8         133
# error_counter,err_inj = 00000008 00          8         134
# error_counter,err_inj = 00000008 00          8         135
# error_counter,err_inj = 00000008 00          8         136
# error_counter,err_inj = 00000008 00          8         137
# error_counter,err_inj = 00000008 00          8         138
# error_counter,err_inj = 00000008 00          8         139
# error_counter,err_inj = 00000008 00          8         140
# error_counter,err_inj = 00000008 00          8         141
# error_counter,err_inj = 00000008 00          8         142
# error_counter,err_inj = 00000008 00          8         143
# error_counter,err_inj = 00000008 00          8         144
# error_counter,err_inj = 00000008 00          8         145
# error_counter,err_inj = 00000008 00          8         146
# error_counter,err_inj = 00000008 00          8         147
# error_counter,err_inj = 00000008 00          8         148
# error_counter,err_inj = 00000008 00          8         149
# error_counter,err_inj = 00000008 00          8         150
# error_counter,err_inj = 00000008 00          8         151
# error_counter,err_inj = 00000008 00          8         152
# error_counter,err_inj = 00000008 00          8         153
# error_counter,err_inj = 00000008 00          8         154
# error_counter,err_inj = 00000008 00          8         155
# error_counter,err_inj = 00000008 00          8         156
# error_counter,err_inj = 00000008 00          8         157
# error_counter,err_inj = 00000008 00          8         158
# error_counter,err_inj = 00000009 11          8         159
# error_counter,err_inj = 0000000a 11          8         160
# error_counter,err_inj = 0000000a 00         10         161
# error_counter,err_inj = 0000000a 00         10         162
# error_counter,err_inj = 0000000a 00         10         163
# error_counter,err_inj = 0000000a 00         10         164
# error_counter,err_inj = 0000000a 00         10         165
# error_counter,err_inj = 0000000a 00         10         166
# error_counter,err_inj = 0000000a 00         10         167
# error_counter,err_inj = 0000000a 00         10         168
# error_counter,err_inj = 0000000a 00         10         169
# error_counter,err_inj = 0000000a 00         10         170
# error_counter,err_inj = 0000000a 00         10         171
# error_counter,err_inj = 0000000a 00         10         172
# error_counter,err_inj = 0000000a 00         10         173
# error_counter,err_inj = 0000000a 00         10         174
# error_counter,err_inj = 0000000a 00         10         175
# error_counter,err_inj = 0000000a 00         10         176
# error_counter,err_inj = 0000000a 00         10         177
```

```
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000a 00          10          178
# error_counter,err_inj = 0000000a 00          10          179
# error_counter,err_inj = 0000000a 00          10          180
# error_counter,err_inj = 0000000a 00          10          181
# error_counter,err_inj = 0000000a 00          10          182
# error_counter,err_inj = 0000000a 00          10          183
# error_counter,err_inj = 0000000a 00          10          184
# error_counter,err_inj = 0000000a 00          10          185
# error_counter,err_inj = 0000000a 00          10          186
# error_counter,err_inj = 0000000a 00          10          187
# error_counter,err_inj = 0000000a 00          10          188
# error_counter,err_inj = 0000000a 00          10          189
# error_counter,err_inj = 0000000a 00          10          190
# error_counter,err_inj = 0000000b 11          10          191
# error_counter,err_inj = 0000000c 11          10          192
# error_counter,err_inj = 0000000c 00          12          193
# error_counter,err_inj = 0000000c 00          12          194
# error_counter,err_inj = 0000000c 00          12          195
# error_counter,err_inj = 0000000c 00          12          196
# error_counter,err_inj = 0000000c 00          12          197
# error_counter,err_inj = 0000000c 00          12          198
# error_counter,err_inj = 0000000c 00          12          199
# error_counter,err_inj = 0000000c 00          12          200
# error_counter,err_inj = 0000000c 00          12          201
# error_counter,err_inj = 0000000c 00          12          202
# error_counter,err_inj = 0000000c 00          12          203
# error_counter,err_inj = 0000000c 00          12          204
# error_counter,err_inj = 0000000c 00          12          205
# error_counter,err_inj = 0000000c 00          12          206
# error_counter,err_inj = 0000000c 00          12          207
# error_counter,err_inj = 0000000c 00          12          208
# error_counter,err_inj = 0000000c 00          12          209
# error_counter,err_inj = 0000000c 00          12          210
# error_counter,err_inj = 0000000c 00          12          211
# error_counter,err_inj = 0000000c 00          12          212
# error_counter,err_inj = 0000000c 00          12          213
# error_counter,err_inj = 0000000c 00          12          214
# error_counter,err_inj = 0000000c 00          12          215
# error_counter,err_inj = 0000000c 00          12          216
# error_counter,err_inj = 0000000c 00          12          217
# error_counter,err_inj = 0000000c 00          12          218
# error_counter,err_inj = 0000000c 00          12          219
# error_counter,err_inj = 0000000c 00          12          220
# error_counter,err_inj = 0000000c 00          12          221
# error_counter,err_inj = 0000000c 00          12          222
# error_counter,err_inj = 0000000d 11          12          223
# error_counter,err_inj = 0000000e 11          12          224
# error_counter,err_inj = 0000000e 00          14          225
# error_counter,err_inj = 0000000e 00          14          226
# error_counter,err_inj = 0000000e 00          14          227
# error_counter,err_inj = 0000000e 00          14          228
# error_counter,err_inj = 0000000e 00          14          229
# error_counter,err_inj = 0000000e 00          14          230
# error_counter,err_inj = 0000000e 00          14          231
# error_counter,err_inj = 0000000e 00          14          232
# error_counter,err_inj = 0000000e 00          14          233
# error_counter,err_inj = 0000000e 00          14          234
# error_counter,err_inj = 0000000e 00          14          235
# error_counter,err_inj = 0000000e 00          14          236
# error_counter,err_inj = 0000000e 00          14          237
# error_counter,err_inj = 0000000e 00          14          238
# error_counter,err_inj = 0000000e 00          14          239
# error_counter,err_inj = 0000000e 00          14          240
# error_counter,err_inj = 0000000e 00          14          241
# error_counter,err_inj = 0000000e 00          14          242
# error_counter,err_inj = 0000000e 00          14          243
# error_counter,err_inj = 0000000e 00          14          244
# error_counter,err_inj = 0000000e 00          14          245
# error_counter,err_inj = 0000000e 00          14          246
# error_counter,err_inj = 0000000e 00          14          247
# error_counter,err_inj = 0000000e 00          14          248
# error_counter,err_inj = 0000000e 00          14          249
# error_counter,err_inj = 0000000e 00          14          250
# error_counter,err_inj = 0000000e 00          14          251
# error_counter,err_inj = 0000000e 00          14          252
# error_counter,err_inj = 0000000e 00          14          253
# error_counter,err_inj = 0000000e 00          14          254
# error_counter,err_inj = 0000000f 11          14          255
# word_count =       10440
```

2b) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000001 01          0           2
# error_counter,err_inj = 00000001 00          0           3
# error_counter,err_inj = 00000002 00          0           4
# error_counter,err_inj = 00000002 00          0           5
# error_counter,err_inj = 00000002 00          0           6
# error_counter,err_inj = 00000002 00          0           7
# error_counter,err_inj = 00000002 00          0           8
```

```
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
```

```
# error_counter,err_inj = 00000002 00          0           9
# error_counter,err_inj = 00000002 00          0          10
# error_counter,err_inj = 00000002 00          0          11
# error_counter,err_inj = 00000002 00          0          12
# error_counter,err_inj = 00000002 00          0          13
# error_counter,err_inj = 00000002 00          0          14
# error_counter,err_inj = 00000002 00          0          15
# error_counter,err_inj = 00000002 00          0          16
# error_counter,err_inj = 00000002 00          0          17
# error_counter,err_inj = 00000002 00          0          18
# error_counter,err_inj = 00000002 00          0          19
# error_counter,err_inj = 00000002 00          0          20
# error_counter,err_inj = 00000002 00          0          21
# error_counter,err_inj = 00000002 00          0          22
# error_counter,err_inj = 00000002 00          0          23
# error_counter,err_inj = 00000002 00          0          24
# error_counter,err_inj = 00000003 01          0          25
# error_counter,err_inj = 00000003 00          0          26
# error_counter,err_inj = 00000003 00          0          27
# error_counter,err_inj = 00000003 00          0          28
# error_counter,err_inj = 00000003 00          0          29
# error_counter,err_inj = 00000003 00          0          30
# error_counter,err_inj = 00000003 00          0          31
# error_counter,err_inj = 00000003 00          0          32
# error_counter,err_inj = 00000004 01          0          33
# error_counter,err_inj = 00000004 00          0          34
# error_counter,err_inj = 00000004 00          0          35
# error_counter,err_inj = 00000004 00          0          36
# error_counter,err_inj = 00000004 00          0          37
# error_counter,err_inj = 00000004 00          0          38
# error_counter,err_inj = 00000004 00          0          39
# error_counter,err_inj = 00000004 00          0          40
# error_counter,err_inj = 00000004 00          0          41
# error_counter,err_inj = 00000004 00          0          42
# error_counter,err_inj = 00000004 00          0          43
# error_counter,err_inj = 00000004 00          0          44
# error_counter,err_inj = 00000004 00          0          45
# error_counter,err_inj = 00000004 00          0          46
# error_counter,err_inj = 00000004 00          0          47
# error_counter,err_inj = 00000004 00          0          48
# error_counter,err_inj = 00000004 00          0          49
# error_counter,err_inj = 00000004 00          0          50
# error_counter,err_inj = 00000005 11          0          51
# error_counter,err_inj = 00000005 00          0          52
# error_counter,err_inj = 00000005 00          0          53
# error_counter,err_inj = 00000005 00          0          54
# error_counter,err_inj = 00000005 00          0          55
# error_counter,err_inj = 00000005 00          0          56
# error_counter,err_inj = 00000005 00          0          57
# error_counter,err_inj = 00000005 00          0          58
# error_counter,err_inj = 00000005 00          0          59
# error_counter,err_inj = 00000005 00          0          60
# error_counter,err_inj = 00000006 01          0          61
# error_counter,err_inj = 00000006 00          0          62
# error_counter,err_inj = 00000006 00          0          63
# error_counter,err_inj = 00000006 00          0          64
# error_counter,err_inj = 00000006 00          0          65
# error_counter,err_inj = 00000006 00          0          66
# error_counter,err_inj = 00000006 00          0          67
# error_counter,err_inj = 00000006 00          0          68
# error_counter,err_inj = 00000006 00          0          69
# error_counter,err_inj = 00000007 10          0          70
# error_counter,err_inj = 00000007 00          0          71
# error_counter,err_inj = 00000007 00          0          72
# error_counter,err_inj = 00000007 00          0          73
# error_counter,err_inj = 00000007 00          0          74
# error_counter,err_inj = 00000007 00          0          75
# error_counter,err_inj = 00000007 00          0          76
# error_counter,err_inj = 00000007 00          0          77
# error_counter,err_inj = 00000007 00          0          78
# error_counter,err_inj = 00000007 00          0          79
# error_counter,err_inj = 00000007 00          0          80
# error_counter,err_inj = 00000007 00          0          81
# error_counter,err_inj = 00000007 00          0          82
# error_counter,err_inj = 00000007 00          0          83
# error_counter,err_inj = 00000007 00          0          84
# error_counter,err_inj = 00000007 00          0          85
# error_counter,err_inj = 00000007 00          0          86
# error_counter,err_inj = 00000008 11          0          87
# error_counter,err_inj = 00000008 00          0          88
# error_counter,err_inj = 00000008 00          0          89
# error_counter,err_inj = 00000008 00          0          90
# error_counter,err_inj = 00000008 00          0          91
# error_counter,err_inj = 00000008 00          0          92
# error_counter,err_inj = 00000008 00          0          93
# error_counter,err_inj = 00000008 00          0          94
# error_counter,err_inj = 00000008 00          0          95
# error_counter,err_inj = 00000008 00          0          96
# error_counter,err_inj = 00000008 00          0          97
# error_counter,err_inj = 00000008 00          0          98
# error_counter,err_inj = 00000008 00          0          99
# error_counter,err_inj = 00000008 00          0         100
# error_counter,err_inj = 00000008 00          0         101
```

```
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
```

```
# error_counter,err_inj = 00000008 00          0         102
# error_counter,err_inj = 00000008 00          0         103
# error_counter,err_inj = 00000008 00          0         104
# error_counter,err_inj = 00000008 00          0         105
# error_counter,err_inj = 00000008 00          0         106
# error_counter,err_inj = 00000008 00          0         107
# error_counter,err_inj = 00000008 00          0         108
# error_counter,err_inj = 00000008 00          0         109
# error_counter,err_inj = 00000008 00          0         110
# error_counter,err_inj = 00000008 00          0         111
# error_counter,err_inj = 00000008 00          0         112
# error_counter,err_inj = 00000009 01          0         113
# error_counter,err_inj = 00000009 00          0         114
# error_counter,err_inj = 00000009 00          0         115
# error_counter,err_inj = 00000009 00          0         116
# error_counter,err_inj = 00000009 00          0         117
# error_counter,err_inj = 00000009 00          0         118
# error_counter,err_inj = 00000009 00          0         119
# error_counter,err_inj = 00000009 00          0         120
# error_counter,err_inj = 00000009 00          0         121
# error_counter,err_inj = 00000009 00          0         122
# error_counter,err_inj = 00000009 00          0         123
# error_counter,err_inj = 00000009 00          0         124
# error_counter,err_inj = 00000009 00          0         125
# error_counter,err_inj = 00000009 00          0         126
# error_counter,err_inj = 00000009 00          0         127
# error_counter,err_inj = 00000009 00          0         128
# error_counter,err_inj = 00000009 00          0         129
# error_counter,err_inj = 00000009 00          0         130
# error_counter,err_inj = 00000009 00          0         131
# error_counter,err_inj = 00000009 00          0         132
# error_counter,err_inj = 00000009 00          0         133
# error_counter,err_inj = 00000009 00          0         134
# error_counter,err_inj = 0000000a 11          0         135
# error_counter,err_inj = 0000000a 00          0         136
# error_counter,err_inj = 0000000a 00          0         137
# error_counter,err_inj = 0000000a 00          0         138
# error_counter,err_inj = 0000000a 00          0         139
# error_counter,err_inj = 0000000a 00          0         140
# error_counter,err_inj = 0000000a 00          0         141
# error_counter,err_inj = 0000000a 00          0         142
# error_counter,err_inj = 0000000a 00          0         143
# error_counter,err_inj = 0000000a 00          0         144
# error_counter,err_inj = 0000000a 00          0         145
# error_counter,err_inj = 0000000a 00          0         146
# error_counter,err_inj = 0000000a 00          0         147
# error_counter,err_inj = 0000000a 00          0         148
# error_counter,err_inj = 0000000a 00          0         149
# error_counter,err_inj = 0000000a 00          0         150
# error_counter,err_inj = 0000000a 00          0         151
# error_counter,err_inj = 0000000a 00          0         152
# error_counter,err_inj = 0000000a 00          0         153
# error_counter,err_inj = 0000000a 00          0         154
# error_counter,err_inj = 0000000a 00          0         155
# error_counter,err_inj = 0000000a 00          0         156
# error_counter,err_inj = 0000000a 00          0         157
# error_counter,err_inj = 0000000a 00          0         158
# error_counter,err_inj = 0000000b 10          0         159
# error_counter,err_inj = 0000000b 00          0         160
# error_counter,err_inj = 0000000b 00          0         161
# error_counter,err_inj = 0000000b 00          0         162
# error_counter,err_inj = 0000000b 00          0         163
# error_counter,err_inj = 0000000b 00          0         164
# error_counter,err_inj = 0000000b 00          0         165
# error_counter,err_inj = 0000000b 00          0         166
# error_counter,err_inj = 0000000c 10          0         167
# error_counter,err_inj = 0000000c 00          0         168
# error_counter,err_inj = 0000000c 00          0         169
# error_counter,err_inj = 0000000c 00          0         170
# error_counter,err_inj = 0000000c 00          0         171
# error_counter,err_inj = 0000000c 00          0         172
# error_counter,err_inj = 0000000c 00          0         173
# error_counter,err_inj = 0000000c 00          0         174
# error_counter,err_inj = 0000000c 00          0         175
# error_counter,err_inj = 0000000c 00          0         176
# error_counter,err_inj = 0000000c 00          0         177
# error_counter,err_inj = 0000000d 00          0         178
# error_counter,err_inj = 0000000d 00          0         179
# error_counter,err_inj = 0000000d 00          0         180
# error_counter,err_inj = 0000000d 00          0         181
# error_counter,err_inj = 0000000d 00          0         182
# error_counter,err_inj = 0000000d 00          0         183
# error_counter,err_inj = 0000000d 00          0         184
# error_counter,err_inj = 0000000d 00          0         185
# error_counter,err_inj = 0000000d 00          0         186
# error_counter,err_inj = 0000000d 00          0         187
# error_counter,err_inj = 0000000d 00          0         188
# error_counter,err_inj = 0000000d 00          0         189
# error_counter,err_inj = 0000000d 00          0         190
# error_counter,err_inj = 0000000d 00          0         191
# error_counter,err_inj = 0000000d 00          0         192
# error_counter,err_inj = 0000000d 00          0         193
# error_counter,err_inj = 0000000d 00          0         194
```

```
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          18, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000d 00          0         195
# error_counter,err_inj = 0000000d 00          0         196
# error_counter,err_inj = 0000000d 00          0         197
# error_counter,err_inj = 0000000d 00          0         198
# error_counter,err_inj = 0000000d 00          0         199
# error_counter,err_inj = 0000000d 00          0         200
# error_counter,err_inj = 0000000d 00          0         201
# error_counter,err_inj = 0000000d 00          0         202
# error_counter,err_inj = 0000000d 00          0         203
# error_counter,err_inj = 0000000d 00          0         204
# error_counter,err_inj = 0000000d 00          0         205
# error_counter,err_inj = 0000000d 00          0         206
# error_counter,err_inj = 0000000d 00          0         207
# error_counter,err_inj = 0000000d 00          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000e 11          0         213
# error_counter,err_inj = 0000000e 00          0         214
# error_counter,err_inj = 0000000e 00          0         215
# error_counter,err_inj = 0000000e 00          0         216
# error_counter,err_inj = 0000000e 00          0         217
# error_counter,err_inj = 0000000e 00          0         218
# error_counter,err_inj = 0000000f 11          0         219
# error_counter,err_inj = 0000000f 00          0         220
# error_counter,err_inj = 0000000f 00          0         221
# error_counter,err_inj = 00000010 01          0         222
# error_counter,err_inj = 00000010 00          0         223
# error_counter,err_inj = 00000010 00          0         224
# error_counter,err_inj = 00000010 00          0         225
# error_counter,err_inj = 00000010 00          0         226
# error_counter,err_inj = 00000010 00          0         227
# error_counter,err_inj = 00000010 00          0         228
# error_counter,err_inj = 00000010 00          0         229
# error_counter,err_inj = 00000010 00          0         230
# error_counter,err_inj = 00000010 00          0         231
# error_counter,err_inj = 00000010 00          0         232
# error_counter,err_inj = 00000010 00          0         233
# error_counter,err_inj = 00000010 00          0         234
# error_counter,err_inj = 00000010 00          0         235
# error_counter,err_inj = 00000011 10          0         236
# error_counter,err_inj = 00000011 00          0         237
# error_counter,err_inj = 00000011 00          0         238
# error_counter,err_inj = 00000011 00          0         239
# error_counter,err_inj = 00000011 00          0         240
# error_counter,err_inj = 00000011 00          0         241
# error_counter,err_inj = 00000011 00          0         242
# error_counter,err_inj = 00000011 00          0         243
# error_counter,err_inj = 00000011 00          0         244
# error_counter,err_inj = 00000011 00          0         245
# error_counter,err_inj = 00000011 00          0         246
# error_counter,err_inj = 00000011 00          0         247
# error_counter,err_inj = 00000011 00          0         248
# error_counter,err_inj = 00000011 00          0         249
# error_counter,err_inj = 00000011 00          0         250
# error_counter,err_inj = 00000011 00          0         251
# error_counter,err_inj = 00000012 00          0         252
# error_counter,err_inj = 00000012 00          0         253
# error_counter,err_inj = 00000012 00          0         254
# error_counter,err_inj = 00000012 00          0         255
```

2b1, 2b2) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000001 01          0           2
# error_counter,err_inj = 00000001 00          0           3
# error_counter,err_inj = 00000002 00          0           4
# error_counter,err_inj = 00000002 00          0           5
# error_counter,err_inj = 00000003 11          0           6
# error_counter,err_inj = 00000003 00          0           7
# error_counter,err_inj = 00000003 00          0           8
# error_counter,err_inj = 00000003 00          0           9
# error_counter,err_inj = 00000003 00          0          10
# error_counter,err_inj = 00000003 00          0          11
# error_counter,err_inj = 00000003 00          0          12
# error_counter,err_inj = 00000003 00          0          13
# error_counter,err_inj = 00000003 00          0          14
# error_counter,err_inj = 00000003 00          0          15
# error_counter,err_inj = 00000003 00          0          16
# error_counter,err_inj = 00000003 00          0          17
# error_counter,err_inj = 00000003 00          0          18
# error_counter,err_inj = 00000003 00          0          19
# error_counter,err_inj = 00000003 00          0          20
# error_counter,err_inj = 00000003 00          0          21
# error_counter,err_inj = 00000003 00          0          22
# error_counter,err_inj = 00000003 00          0          23
# error_counter,err_inj = 00000003 00          0          24
# error_counter,err_inj = 00000004 01          0          25
```

```
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
```

```
# error_counter,err_inj = 00000004 00          0          26
# error_counter,err_inj = 00000004 00          0          27
# error_counter,err_inj = 00000004 00          0          28
# error_counter,err_inj = 00000004 00          0          29
# error_counter,err_inj = 00000005 00          0          30
# error_counter,err_inj = 00000005 00          0          31
# error_counter,err_inj = 00000005 00          0          32
# error_counter,err_inj = 00000006 01          0          33
# error_counter,err_inj = 00000006 00          0          34
# error_counter,err_inj = 00000007 11          0          35
# error_counter,err_inj = 00000007 00          0          36
# error_counter,err_inj = 00000007 00          0          37
# error_counter,err_inj = 00000007 00          0          38
# error_counter,err_inj = 00000007 00          0          39
# error_counter,err_inj = 00000007 00          0          40
# error_counter,err_inj = 00000007 00          0          41
# error_counter,err_inj = 00000007 00          0          42
# error_counter,err_inj = 00000008 00          0          43
# error_counter,err_inj = 00000008 00          0          44
# error_counter,err_inj = 00000008 00          0          45
# error_counter,err_inj = 00000008 00          0          46
# error_counter,err_inj = 00000008 00          0          47
# error_counter,err_inj = 00000008 00          0          48
# error_counter,err_inj = 00000008 00          0          49
# error_counter,err_inj = 00000008 00          0          50
# error_counter,err_inj = 00000009 11          0          51
# error_counter,err_inj = 00000009 00          0          52
# error_counter,err_inj = 00000009 00          0          53
# error_counter,err_inj = 00000009 00          0          54
# error_counter,err_inj = 00000009 00          0          55
# error_counter,err_inj = 00000009 00          0          56
# error_counter,err_inj = 00000009 00          0          57
# error_counter,err_inj = 00000009 00          0          58
# error_counter,err_inj = 00000009 00          0          59
# error_counter,err_inj = 00000009 00          0          60
# error_counter,err_inj = 0000000a 01          0          61
# error_counter,err_inj = 0000000a 00          0          62
# error_counter,err_inj = 0000000a 00          0          63
# error_counter,err_inj = 0000000a 00          0          64
# error_counter,err_inj = 0000000a 00          0          65
# error_counter,err_inj = 0000000a 00          0          66
# error_counter,err_inj = 0000000a 00          0          67
# error_counter,err_inj = 0000000a 00          0          68
# error_counter,err_inj = 0000000a 00          0          69
# error_counter,err_inj = 0000000b 10          0          70
# error_counter,err_inj = 0000000b 00          0          71
# error_counter,err_inj = 0000000b 00          0          72
# error_counter,err_inj = 0000000b 00          0          73
# error_counter,err_inj = 0000000b 00          0          74
# error_counter,err_inj = 0000000b 00          0          75
# error_counter,err_inj = 0000000b 00          0          76
# error_counter,err_inj = 0000000b 00          0          77
# error_counter,err_inj = 0000000b 00          0          78
# error_counter,err_inj = 0000000b 00          0          79
# error_counter,err_inj = 0000000b 00          0          80
# error_counter,err_inj = 0000000b 00          0          81
# error_counter,err_inj = 0000000c 11          0          82
# error_counter,err_inj = 0000000c 00          0          83
# error_counter,err_inj = 0000000c 00          0          84
# error_counter,err_inj = 0000000c 00          0          85
# error_counter,err_inj = 0000000c 00          0          86
# error_counter,err_inj = 0000000d 11          0          87
# error_counter,err_inj = 0000000d 00          0          88
# error_counter,err_inj = 0000000d 00          0          89
# error_counter,err_inj = 0000000d 00          0          90
# error_counter,err_inj = 0000000d 00          0          91
# error_counter,err_inj = 0000000d 00          0          92
# error_counter,err_inj = 0000000d 00          0          93
# error_counter,err_inj = 0000000d 00          0          94
# error_counter,err_inj = 0000000d 00          0          95
# error_counter,err_inj = 0000000d 00          0          96
# error_counter,err_inj = 0000000d 00          0          97
# error_counter,err_inj = 0000000d 00          0          98
# error_counter,err_inj = 0000000d 00          0          99
# error_counter,err_inj = 0000000d 00          0         100
# error_counter,err_inj = 0000000d 00          0         101
# error_counter,err_inj = 0000000d 00          0         102
# error_counter,err_inj = 0000000d 00          0         103
# error_counter,err_inj = 0000000d 00          0         104
# error_counter,err_inj = 0000000d 00          0         105
# error_counter,err_inj = 0000000d 00          0         106
# error_counter,err_inj = 0000000d 00          0         107
# error_counter,err_inj = 0000000d 00          0         108
# error_counter,err_inj = 0000000d 00          0         109
# error_counter,err_inj = 0000000d 00          0         110
# error_counter,err_inj = 0000000d 00          0         111
# error_counter,err_inj = 0000000d 00          0         112
# error_counter,err_inj = 0000000e 01          0         113
# error_counter,err_inj = 0000000e 00          0         114
# error_counter,err_inj = 0000000e 00          0         115
# error_counter,err_inj = 0000000e 00          0         116
# error_counter,err_inj = 0000000e 00          0         117
# error_counter,err_inj = 0000000e 00          0         118
```

```
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
```

```
# error_counter,err_inj = 0000000e 00          0         119
# error_counter,err_inj = 0000000e 00          0         120
# error_counter,err_inj = 0000000e 00          0         121
# error_counter,err_inj = 0000000e 00          0         122
# error_counter,err_inj = 0000000e 00          0         123
# error_counter,err_inj = 0000000e 00          0         124
# error_counter,err_inj = 0000000f 11          0         125
# error_counter,err_inj = 0000000f 00          0         126
# error_counter,err_inj = 0000000f 00          0         127
# error_counter,err_inj = 0000000f 00          0         128
# error_counter,err_inj = 0000000f 00          0         129
# error_counter,err_inj = 0000000f 00          0         130
# error_counter,err_inj = 0000000f 00          0         131
# error_counter,err_inj = 0000000f 00          0         132
# error_counter,err_inj = 0000000f 00          0         133
# error_counter,err_inj = 0000000f 00          0         134
# error_counter,err_inj = 00000010 11          0         135
# error_counter,err_inj = 00000010 00          0         136
# error_counter,err_inj = 00000010 00          0         137
# error_counter,err_inj = 00000010 00          0         138
# error_counter,err_inj = 00000010 00          0         139
# error_counter,err_inj = 00000010 00          0         140
# error_counter,err_inj = 00000011 01          0         141
# error_counter,err_inj = 00000011 00          0         142
# error_counter,err_inj = 00000011 00          0         143
# error_counter,err_inj = 00000011 00          0         144
# error_counter,err_inj = 00000011 00          0         145
# error_counter,err_inj = 00000011 00          0         146
# error_counter,err_inj = 00000011 00          0         147
# error_counter,err_inj = 00000011 00          0         148
# error_counter,err_inj = 00000011 00          0         149
# error_counter,err_inj = 00000011 00          0         150
# error_counter,err_inj = 00000011 00          0         151
# error_counter,err_inj = 00000011 00          0         152
# error_counter,err_inj = 00000011 00          0         153
# error_counter,err_inj = 00000011 00          0         154
# error_counter,err_inj = 00000011 00          0         155
# error_counter,err_inj = 00000011 00          0         156
# error_counter,err_inj = 00000011 00          0         157
# error_counter,err_inj = 00000011 00          0         158
# error_counter,err_inj = 00000012 10          0         159
# error_counter,err_inj = 00000012 00          0         160
# error_counter,err_inj = 00000012 00          0         161
# error_counter,err_inj = 00000012 00          0         162
# error_counter,err_inj = 00000012 00          0         163
# error_counter,err_inj = 00000012 00          0         164
# error_counter,err_inj = 00000012 00          0         165
# error_counter,err_inj = 00000012 00          0         166
# error_counter,err_inj = 00000013 10          0         167
# error_counter,err_inj = 00000013 00          0         168
# error_counter,err_inj = 00000013 00          0         169
# error_counter,err_inj = 00000013 00          0         170
# error_counter,err_inj = 00000013 00          0         171
# error_counter,err_inj = 00000013 00          0         172
# error_counter,err_inj = 00000013 00          0         173
# error_counter,err_inj = 00000013 00          0         174
# error_counter,err_inj = 00000013 00          0         175
# error_counter,err_inj = 00000013 00          0         176
# error_counter,err_inj = 00000013 00          0         177
# error_counter,err_inj = 00000014 00          0         178
# error_counter,err_inj = 00000014 00          0         179
# error_counter,err_inj = 00000014 00          0         180
# error_counter,err_inj = 00000014 00          0         181
# error_counter,err_inj = 00000014 00          0         182
# error_counter,err_inj = 00000014 00          0         183
# error_counter,err_inj = 00000014 00          0         184
# error_counter,err_inj = 00000014 00          0         185
# error_counter,err_inj = 00000015 11          0         186
# error_counter,err_inj = 00000015 00          0         187
# error_counter,err_inj = 00000015 00          0         188
# error_counter,err_inj = 00000015 00          0         189
# error_counter,err_inj = 00000015 00          0         190
# error_counter,err_inj = 00000015 00          0         191
# error_counter,err_inj = 00000015 00          0         192
# error_counter,err_inj = 00000015 00          0         193
# error_counter,err_inj = 00000015 00          0         194
# error_counter,err_inj = 00000015 00          0         195
# error_counter,err_inj = 00000015 00          0         196
# error_counter,err_inj = 00000015 00          0         197
# error_counter,err_inj = 00000015 00          0         198
# error_counter,err_inj = 00000015 00          0         199
# error_counter,err_inj = 00000015 00          0         200
# error_counter,err_inj = 00000015 00          0         201
# error_counter,err_inj = 00000015 00          0         202
# error_counter,err_inj = 00000015 00          0         203
# error_counter,err_inj = 00000015 00          0         204
# error_counter,err_inj = 00000015 00          0         205
# error_counter,err_inj = 00000015 00          0         206
# error_counter,err_inj = 00000015 00          0         207
# error_counter,err_inj = 00000015 00          0         208
# error_counter,err_inj = 00000015 00          0         209
# error_counter,err_inj = 00000015 00          0         210
# error_counter,err_inj = 00000015 00          0         211
```

```
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          30, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000015 00          0         212
# error_counter,err_inj = 00000016 11          0         213
# error_counter,err_inj = 00000016 00          0         214
# error_counter,err_inj = 00000016 00          0         215
# error_counter,err_inj = 00000016 00          0         216
# error_counter,err_inj = 00000016 00          0         217
# error_counter,err_inj = 00000016 00          0         218
# error_counter,err_inj = 00000017 11          0         219
# error_counter,err_inj = 00000017 00          0         220
# error_counter,err_inj = 00000017 00          0         221
# error_counter,err_inj = 00000018 01          0         222
# error_counter,err_inj = 00000018 00          0         223
# error_counter,err_inj = 00000018 00          0         224
# error_counter,err_inj = 00000018 00          0         225
# error_counter,err_inj = 00000019 11          0         226
# error_counter,err_inj = 00000019 00          0         227
# error_counter,err_inj = 00000019 00          0         228
# error_counter,err_inj = 00000019 00          0         229
# error_counter,err_inj = 00000019 00          0         230
# error_counter,err_inj = 00000019 00          0         231
# error_counter,err_inj = 00000019 00          0         232
# error_counter,err_inj = 00000019 00          0         233
# error_counter,err_inj = 00000019 00          0         234
# error_counter,err_inj = 00000019 00          0         235
# error_counter,err_inj = 0000001a 10          0         236
# error_counter,err_inj = 0000001a 00          0         237
# error_counter,err_inj = 0000001a 00          0         238
# error_counter,err_inj = 0000001a 00          0         239
# error_counter,err_inj = 0000001a 00          0         240
# error_counter,err_inj = 0000001b 10          0         241
# error_counter,err_inj = 0000001b 00          0         242
# error_counter,err_inj = 0000001b 00          0         243
# error_counter,err_inj = 0000001b 00          0         244
# error_counter,err_inj = 0000001b 00          0         245
# error_counter,err_inj = 0000001c 10          0         246
# error_counter,err_inj = 0000001c 00          0         247
# error_counter,err_inj = 0000001c 00          0         248
# error_counter,err_inj = 0000001c 00          0         249
# error_counter,err_inj = 0000001c 00          0         250
# error_counter,err_inj = 0000001c 00          0         251
# error_counter,err_inj = 0000001d 00          0         252
# error_counter,err_inj = 0000001d 00          0         253
# error_counter,err_inj = 0000001d 00          0         254
# error_counter,err_inj = 0000001e 00          0         255
```

2b4 &5) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000001 01          0          16
# error_counter,err_inj = 00000001 00          0          17
# error_counter,err_inj = 00000001 00          0          18
# error_counter,err_inj = 00000001 00          0          19
# error_counter,err_inj = 00000001 00          0          20
# error_counter,err_inj = 00000001 00          0          21
# error_counter,err_inj = 00000001 00          0          22
# error_counter,err_inj = 00000001 00          0          23
# error_counter,err_inj = 00000001 00          0          24
# error_counter,err_inj = 00000001 00          0          25
# error_counter,err_inj = 00000001 00          0          26
# error_counter,err_inj = 00000001 00          0          27
# error_counter,err_inj = 00000001 00          0          28
# error_counter,err_inj = 00000001 00          0          29
# error_counter,err_inj = 00000001 00          0          30
# error_counter,err_inj = 00000001 00          0          31
# error_counter,err_inj = 00000002 10          0          32
# error_counter,err_inj = 00000002 00          0          33
# error_counter,err_inj = 00000002 00          0          34
# error_counter,err_inj = 00000002 00          0          35
# error_counter,err_inj = 00000002 00          0          36
# error_counter,err_inj = 00000002 00          0          37
# error_counter,err_inj = 00000002 00          0          38
# error_counter,err_inj = 00000002 00          0          39
# error_counter,err_inj = 00000002 00          0          40
# error_counter,err_inj = 00000002 00          0          41
# error_counter,err_inj = 00000002 00          0          42
```

```
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
```

```
# error_counter,err_inj = 00000002 00          0          43
# error_counter,err_inj = 00000002 00          0          44
# error_counter,err_inj = 00000002 00          0          45
# error_counter,err_inj = 00000002 00          0          46
# error_counter,err_inj = 00000002 00          0          47
# error_counter,err_inj = 00000003 10          0          48
# error_counter,err_inj = 00000003 00          0          49
# error_counter,err_inj = 00000003 00          0          50
# error_counter,err_inj = 00000003 00          0          51
# error_counter,err_inj = 00000003 00          0          52
# error_counter,err_inj = 00000003 00          0          53
# error_counter,err_inj = 00000003 00          0          54
# error_counter,err_inj = 00000003 00          0          55
# error_counter,err_inj = 00000003 00          0          56
# error_counter,err_inj = 00000003 00          0          57
# error_counter,err_inj = 00000003 00          0          58
# error_counter,err_inj = 00000003 00          0          59
# error_counter,err_inj = 00000003 00          0          60
# error_counter,err_inj = 00000003 00          0          61
# error_counter,err_inj = 00000003 00          0          62
# error_counter,err_inj = 00000003 00          0          63
# error_counter,err_inj = 00000004 00          0          64
# error_counter,err_inj = 00000004 00          0          65
# error_counter,err_inj = 00000004 00          0          66
# error_counter,err_inj = 00000004 00          0          67
# error_counter,err_inj = 00000004 00          0          68
# error_counter,err_inj = 00000004 00          0          69
# error_counter,err_inj = 00000004 00          0          70
# error_counter,err_inj = 00000004 00          0          71
# error_counter,err_inj = 00000004 00          0          72
# error_counter,err_inj = 00000004 00          0          73
# error_counter,err_inj = 00000004 00          0          74
# error_counter,err_inj = 00000004 00          0          75
# error_counter,err_inj = 00000004 00          0          76
# error_counter,err_inj = 00000004 00          0          77
# error_counter,err_inj = 00000004 00          0          78
# error_counter,err_inj = 00000004 00          0          79
# error_counter,err_inj = 00000005 00          0          80
# error_counter,err_inj = 00000005 00          0          81
# error_counter,err_inj = 00000005 00          0          82
# error_counter,err_inj = 00000005 00          0          83
# error_counter,err_inj = 00000005 00          0          84
# error_counter,err_inj = 00000005 00          0          85
# error_counter,err_inj = 00000005 00          0          86
# error_counter,err_inj = 00000005 00          0          87
# error_counter,err_inj = 00000005 00          0          88
# error_counter,err_inj = 00000005 00          0          89
# error_counter,err_inj = 00000005 00          0          90
# error_counter,err_inj = 00000005 00          0          91
# error_counter,err_inj = 00000005 00          0          92
# error_counter,err_inj = 00000005 00          0          93
# error_counter,err_inj = 00000005 00          0          94
# error_counter,err_inj = 00000005 00          0          95
# error_counter,err_inj = 00000006 10          0          96
# error_counter,err_inj = 00000006 00          0          97
# error_counter,err_inj = 00000006 00          0          98
# error_counter,err_inj = 00000006 00          0          99
# error_counter,err_inj = 00000006 00          0         100
# error_counter,err_inj = 00000006 00          0         101
# error_counter,err_inj = 00000006 00          0         102
# error_counter,err_inj = 00000006 00          0         103
# error_counter,err_inj = 00000006 00          0         104
# error_counter,err_inj = 00000006 00          0         105
# error_counter,err_inj = 00000006 00          0         106
# error_counter,err_inj = 00000006 00          0         107
# error_counter,err_inj = 00000006 00          0         108
# error_counter,err_inj = 00000006 00          0         109
# error_counter,err_inj = 00000006 00          0         110
# error_counter,err_inj = 00000006 00          0         111
# error_counter,err_inj = 00000007 10          0         112
# error_counter,err_inj = 00000007 00          0         113
# error_counter,err_inj = 00000007 00          0         114
# error_counter,err_inj = 00000007 00          0         115
# error_counter,err_inj = 00000007 00          0         116
# error_counter,err_inj = 00000007 00          0         117
# error_counter,err_inj = 00000007 00          0         118
# error_counter,err_inj = 00000007 00          0         119
# error_counter,err_inj = 00000007 00          0         120
# error_counter,err_inj = 00000007 00          0         121
# error_counter,err_inj = 00000007 00          0         122
# error_counter,err_inj = 00000007 00          0         123
# error_counter,err_inj = 00000007 00          0         124
# error_counter,err_inj = 00000007 00          0         125
# error_counter,err_inj = 00000007 00          0         126
# error_counter,err_inj = 00000007 00          0         127
# error_counter,err_inj = 00000008 00          0         128
# error_counter,err_inj = 00000008 00          0         129
# error_counter,err_inj = 00000008 00          0         130
# error_counter,err_inj = 00000008 00          0         131
# error_counter,err_inj = 00000008 00          0         132
# error_counter,err_inj = 00000008 00          0         133
# error_counter,err_inj = 00000008 00          0         134
# error_counter,err_inj = 00000008 00          0         135
```

```
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
```

```
# error_counter,err_inj = 00000008 00          0         136
# error_counter,err_inj = 00000008 00          0         137
# error_counter,err_inj = 00000008 00          0         138
# error_counter,err_inj = 00000008 00          0         139
# error_counter,err_inj = 00000008 00          0         140
# error_counter,err_inj = 00000008 00          0         141
# error_counter,err_inj = 00000008 00          0         142
# error_counter,err_inj = 00000008 00          0         143
# error_counter,err_inj = 00000009 00          0         144
# error_counter,err_inj = 00000009 00          0         145
# error_counter,err_inj = 00000009 00          0         146
# error_counter,err_inj = 00000009 00          0         147
# error_counter,err_inj = 00000009 00          0         148
# error_counter,err_inj = 00000009 00          0         149
# error_counter,err_inj = 00000009 00          0         150
# error_counter,err_inj = 00000009 00          0         151
# error_counter,err_inj = 00000009 00          0         152
# error_counter,err_inj = 00000009 00          0         153
# error_counter,err_inj = 00000009 00          0         154
# error_counter,err_inj = 00000009 00          0         155
# error_counter,err_inj = 00000009 00          0         156
# error_counter,err_inj = 00000009 00          0         157
# error_counter,err_inj = 00000009 00          0         158
# error_counter,err_inj = 00000009 00          0         159
# error_counter,err_inj = 0000000a 11          0         160
# error_counter,err_inj = 0000000a 00          0         161
# error_counter,err_inj = 0000000a 00          0         162
# error_counter,err_inj = 0000000a 00          0         163
# error_counter,err_inj = 0000000a 00          0         164
# error_counter,err_inj = 0000000a 00          0         165
# error_counter,err_inj = 0000000a 00          0         166
# error_counter,err_inj = 0000000a 00          0         167
# error_counter,err_inj = 0000000a 00          0         168
# error_counter,err_inj = 0000000a 00          0         169
# error_counter,err_inj = 0000000a 00          0         170
# error_counter,err_inj = 0000000a 00          0         171
# error_counter,err_inj = 0000000a 00          0         172
# error_counter,err_inj = 0000000a 00          0         173
# error_counter,err_inj = 0000000a 00          0         174
# error_counter,err_inj = 0000000a 00          0         175
# error_counter,err_inj = 0000000b 10          0         176
# error_counter,err_inj = 0000000b 00          0         177
# error_counter,err_inj = 0000000b 00          0         178
# error_counter,err_inj = 0000000b 00          0         179
# error_counter,err_inj = 0000000b 00          0         180
# error_counter,err_inj = 0000000b 00          0         181
# error_counter,err_inj = 0000000b 00          0         182
# error_counter,err_inj = 0000000b 00          0         183
# error_counter,err_inj = 0000000b 00          0         184
# error_counter,err_inj = 0000000b 00          0         185
# error_counter,err_inj = 0000000b 00          0         186
# error_counter,err_inj = 0000000b 00          0         187
# error_counter,err_inj = 0000000b 00          0         188
# error_counter,err_inj = 0000000b 00          0         189
# error_counter,err_inj = 0000000b 00          0         190
# error_counter,err_inj = 0000000b 00          0         191
# error_counter,err_inj = 0000000c 01          0         192
# error_counter,err_inj = 0000000c 00          0         193
# error_counter,err_inj = 0000000c 00          0         194
# error_counter,err_inj = 0000000c 00          0         195
# error_counter,err_inj = 0000000c 00          0         196
# error_counter,err_inj = 0000000c 00          0         197
# error_counter,err_inj = 0000000c 00          0         198
# error_counter,err_inj = 0000000c 00          0         199
# error_counter,err_inj = 0000000c 00          0         200
# error_counter,err_inj = 0000000c 00          0         201
# error_counter,err_inj = 0000000c 00          0         202
# error_counter,err_inj = 0000000c 00          0         203
# error_counter,err_inj = 0000000c 00          0         204
# error_counter,err_inj = 0000000c 00          0         205
# error_counter,err_inj = 0000000c 00          0         206
# error_counter,err_inj = 0000000c 00          0         207
# error_counter,err_inj = 0000000d 10          0         208
# error_counter,err_inj = 0000000d 00          0         209
# error_counter,err_inj = 0000000d 00          0         210
# error_counter,err_inj = 0000000d 00          0         211
# error_counter,err_inj = 0000000d 00          0         212
# error_counter,err_inj = 0000000d 00          0         213
# error_counter,err_inj = 0000000d 00          0         214
# error_counter,err_inj = 0000000d 00          0         215
# error_counter,err_inj = 0000000d 00          0         216
# error_counter,err_inj = 0000000d 00          0         217
# error_counter,err_inj = 0000000d 00          0         218
# error_counter,err_inj = 0000000d 00          0         219
# error_counter,err_inj = 0000000d 00          0         220
# error_counter,err_inj = 0000000d 00          0         221
# error_counter,err_inj = 0000000d 00          0         222
# error_counter,err_inj = 0000000d 00          0         223
# error_counter,err_inj = 0000000e 01          0         224
# error_counter,err_inj = 0000000e 00          0         225
# error_counter,err_inj = 0000000e 00          0         226
# error_counter,err_inj = 0000000e 00          0         227
# error_counter,err_inj = 0000000e 00          0         228
```

```
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          16, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 0000000e 00          0         229
# error_counter,err_inj = 0000000e 00          0         230
# error_counter,err_inj = 0000000e 00          0         231
# error_counter,err_inj = 0000000e 00          0         232
# error_counter,err_inj = 0000000e 00          0         233
# error_counter,err_inj = 0000000e 00          0         234
# error_counter,err_inj = 0000000e 00          0         235
# error_counter,err_inj = 0000000e 00          0         236
# error_counter,err_inj = 0000000e 00          0         237
# error_counter,err_inj = 0000000e 00          0         238
# error_counter,err_inj = 0000000e 00          0         239
# error_counter,err_inj = 0000000f 11          0         240
# error_counter,err_inj = 0000000f 00          0         241
# error_counter,err_inj = 0000000f 00          0         242
# error_counter,err_inj = 0000000f 00          0         243
# error_counter,err_inj = 0000000f 00          0         244
# error_counter,err_inj = 0000000f 00          0         245
# error_counter,err_inj = 0000000f 00          0         246
# error_counter,err_inj = 0000000f 00          0         247
# error_counter,err_inj = 0000000f 00          0         248
# error_counter,err_inj = 0000000f 00          0         249
# error_counter,err_inj = 0000000f 00          0         250
# error_counter,err_inj = 0000000f 00          0         251
# error_counter,err_inj = 0000000f 00          0         252
# error_counter,err_inj = 0000000f 00          0         253
# error_counter,err_inj = 0000000f 00          0         254
# error_counter,err_inj = 0000000f 00          0         255
# word_count =       10440
```

2b6 & 7) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000001 00          0          29
# error_counter,err_inj = 00000002 11          0          30
# error_counter,err_inj = 00000003 01          0          31
# error_counter,err_inj = 00000004 11          2          32
# error_counter,err_inj = 00000004 00          3          33
# error_counter,err_inj = 00000004 00          3          34
# error_counter,err_inj = 00000004 00          3          35
# error_counter,err_inj = 00000004 00          3          36
# error_counter,err_inj = 00000004 00          3          37
# error_counter,err_inj = 00000004 00          3          38
# error_counter,err_inj = 00000004 00          3          39
# error_counter,err_inj = 00000004 00          3          40
# error_counter,err_inj = 00000004 00          3          41
# error_counter,err_inj = 00000004 00          3          42
# error_counter,err_inj = 00000004 00          3          43
# error_counter,err_inj = 00000004 00          3          44
# error_counter,err_inj = 00000004 00          3          45
# error_counter,err_inj = 00000004 00          3          46
# error_counter,err_inj = 00000004 00          3          47
# error_counter,err_inj = 00000004 00          3          48
# error_counter,err_inj = 00000004 00          3          49
# error_counter,err_inj = 00000004 00          3          50
# error_counter,err_inj = 00000004 00          3          51
# error_counter,err_inj = 00000004 00          3          52
# error_counter,err_inj = 00000004 00          3          53
# error_counter,err_inj = 00000004 00          3          54
# error_counter,err_inj = 00000004 00          3          55
# error_counter,err_inj = 00000004 00          3          56
# error_counter,err_inj = 00000004 00          3          57
# error_counter,err_inj = 00000004 00          3          58
# error_counter,err_inj = 00000004 00          3          59
```

```
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# boo! in = 0, out = 1, w_ct =          70, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# boo! in = 0, out = 1, w_ct =          72, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =          73, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# boo! in = 0, out = 1, w_ct =         134, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# boo! in = 1, out = 0, w_ct =         141, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
```

```
# error_counter,err_inj = 00000004 00          3          60
# error_counter,err_inj = 00000005 01          3          61
# error_counter,err_inj = 00000006 10          3          62
# error_counter,err_inj = 00000007 01          4          63
# error_counter,err_inj = 00000008 11          5          64
# error_counter,err_inj = 00000008 00          6          65
# error_counter,err_inj = 00000008 00          6          66
# error_counter,err_inj = 00000008 00          6          67
# error_counter,err_inj = 00000008 00          6          68
# error_counter,err_inj = 00000008 00          6          69
# error_counter,err_inj = 00000008 00          6          70
# error_counter,err_inj = 00000008 00          6          71
# error_counter,err_inj = 00000008 00          6          72
# error_counter,err_inj = 00000008 00          6          73
# error_counter,err_inj = 00000008 00          6          74
# error_counter,err_inj = 00000008 00          6          75
# error_counter,err_inj = 00000008 00          6          76
# error_counter,err_inj = 00000008 00          6          77
# error_counter,err_inj = 00000008 00          6          78
# error_counter,err_inj = 00000008 00          6          79
# error_counter,err_inj = 00000008 00          6          80
# error_counter,err_inj = 00000008 00          6          81
# error_counter,err_inj = 00000008 00          6          82
# error_counter,err_inj = 00000008 00          6          83
# error_counter,err_inj = 00000008 00          6          84
# error_counter,err_inj = 00000008 00          6          85
# error_counter,err_inj = 00000008 00          6          86
# error_counter,err_inj = 00000008 00          6          87
# error_counter,err_inj = 00000008 00          6          88
# error_counter,err_inj = 00000008 00          6          89
# error_counter,err_inj = 00000008 00          6          90
# error_counter,err_inj = 00000008 00          6          91
# error_counter,err_inj = 00000008 00          6          92
# error_counter,err_inj = 00000009 01          6          93
# error_counter,err_inj = 0000000a 00          6          94
# error_counter,err_inj = 0000000b 00          7          95
# error_counter,err_inj = 0000000c 11          7          96
# error_counter,err_inj = 0000000c 00          7          97
# error_counter,err_inj = 0000000c 00          7          98
# error_counter,err_inj = 0000000c 00          7          99
# error_counter,err_inj = 0000000c 00          7         100
# error_counter,err_inj = 0000000c 00          7         101
# error_counter,err_inj = 0000000c 00          7         102
# error_counter,err_inj = 0000000c 00          7         103
# error_counter,err_inj = 0000000c 00          7         104
# error_counter,err_inj = 0000000c 00          7         105
# error_counter,err_inj = 0000000c 00          7         106
# error_counter,err_inj = 0000000c 00          7         107
# error_counter,err_inj = 0000000c 00          7         108
# error_counter,err_inj = 0000000c 00          7         109
# error_counter,err_inj = 0000000c 00          7         110
# error_counter,err_inj = 0000000c 00          7         111
# error_counter,err_inj = 0000000c 00          7         112
# error_counter,err_inj = 0000000c 00          7         113
# error_counter,err_inj = 0000000c 00          7         114
# error_counter,err_inj = 0000000c 00          7         115
# error_counter,err_inj = 0000000c 00          7         116
# error_counter,err_inj = 0000000c 00          7         117
# error_counter,err_inj = 0000000c 00          7         118
# error_counter,err_inj = 0000000c 00          7         119
# error_counter,err_inj = 0000000c 00          7         120
# error_counter,err_inj = 0000000c 00          7         121
# error_counter,err_inj = 0000000c 00          7         122
# error_counter,err_inj = 0000000c 00          7         123
# error_counter,err_inj = 0000000c 00          7         124
# error_counter,err_inj = 0000000d 11          7         125
# error_counter,err_inj = 0000000e 10          7         126
# error_counter,err_inj = 0000000f 10          9         127
# error_counter,err_inj = 00000010 01         10         128
# error_counter,err_inj = 00000010 00         11         129
# error_counter,err_inj = 00000010 00         11         130
# error_counter,err_inj = 00000010 00         11         131
# error_counter,err_inj = 00000010 00         11         132
# error_counter,err_inj = 00000010 00         11         133
# error_counter,err_inj = 00000010 00         11         134
# error_counter,err_inj = 00000010 00         11         135
# error_counter,err_inj = 00000010 00         11         136
# error_counter,err_inj = 00000010 00         11         137
# error_counter,err_inj = 00000010 00         11         138
# error_counter,err_inj = 00000010 00         11         139
# error_counter,err_inj = 00000010 00         11         140
# error_counter,err_inj = 00000010 00         11         141
# error_counter,err_inj = 00000010 00         11         142
# error_counter,err_inj = 00000010 00         11         143
# error_counter,err_inj = 00000010 00         11         144
# error_counter,err_inj = 00000010 00         11         145
# error_counter,err_inj = 00000010 00         11         146
# error_counter,err_inj = 00000010 00         11         147
# error_counter,err_inj = 00000010 00         11         148
# error_counter,err_inj = 00000010 00         11         149
# error_counter,err_inj = 00000010 00         11         150
# error_counter,err_inj = 00000010 00         11         151
# error_counter,err_inj = 00000010 00         11         152
```

```
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# boo! in = 1, out = 0, w_ct =         198, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         199, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         200, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# boo! in = 1, out = 0, w_ct =         202, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# boo! in = 1, out = 0, w_ct =         231, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# boo! in = 1, out = 0, w_ct =         233, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         234, err = 00,
```

```
# error_counter,err_inj = 00000010 00          11          153
# error_counter,err_inj = 00000010 00          11          154
# error_counter,err_inj = 00000010 00          11          155
# error_counter,err_inj = 00000010 00          11          156
# error_counter,err_inj = 00000011 10          11          157
# error_counter,err_inj = 00000012 00          11          158
# error_counter,err_inj = 00000013 01          12          159
# error_counter,err_inj = 00000014 00          12          160
# error_counter,err_inj = 00000014 00          13          161
# error_counter,err_inj = 00000014 00          13          162
# error_counter,err_inj = 00000014 00          13          163
# error_counter,err_inj = 00000014 00          13          164
# error_counter,err_inj = 00000014 00          13          165
# error_counter,err_inj = 00000014 00          13          166
# error_counter,err_inj = 00000014 00          13          167
# error_counter,err_inj = 00000014 00          13          168
# error_counter,err_inj = 00000014 00          13          169
# error_counter,err_inj = 00000014 00          13          170
# error_counter,err_inj = 00000014 00          13          171
# error_counter,err_inj = 00000014 00          13          172
# error_counter,err_inj = 00000014 00          13          173
# error_counter,err_inj = 00000014 00          13          174
# error_counter,err_inj = 00000014 00          13          175
# error_counter,err_inj = 00000014 00          13          176
# error_counter,err_inj = 00000014 00          13          177
# error_counter,err_inj = 00000014 00          13          178
# error_counter,err_inj = 00000014 00          13          179
# error_counter,err_inj = 00000014 00          13          180
# error_counter,err_inj = 00000014 00          13          181
# error_counter,err_inj = 00000014 00          13          182
# error_counter,err_inj = 00000014 00          13          183
# error_counter,err_inj = 00000014 00          13          184
# error_counter,err_inj = 00000014 00          13          185
# error_counter,err_inj = 00000014 00          13          186
# error_counter,err_inj = 00000014 00          13          187
# error_counter,err_inj = 00000014 00          13          188
# error_counter,err_inj = 00000015 01          13          189
# error_counter,err_inj = 00000016 01          13          190
# error_counter,err_inj = 00000017 11          14          191
# error_counter,err_inj = 00000018 11          15          192
# error_counter,err_inj = 00000018 00          17          193
# error_counter,err_inj = 00000018 00          17          194
# error_counter,err_inj = 00000018 00          17          195
# error_counter,err_inj = 00000018 00          17          196
# error_counter,err_inj = 00000018 00          17          197
# error_counter,err_inj = 00000018 00          17          198
# error_counter,err_inj = 00000018 00          17          199
# error_counter,err_inj = 00000018 00          17          200
# error_counter,err_inj = 00000018 00          17          201
# error_counter,err_inj = 00000018 00          17          202
# error_counter,err_inj = 00000018 00          17          203
# error_counter,err_inj = 00000018 00          17          204
# error_counter,err_inj = 00000018 00          17          205
# error_counter,err_inj = 00000018 00          17          206
# error_counter,err_inj = 00000018 00          17          207
# error_counter,err_inj = 00000018 00          17          208
# error_counter,err_inj = 00000018 00          17          209
# error_counter,err_inj = 00000018 00          17          210
# error_counter,err_inj = 00000018 00          17          211
# error_counter,err_inj = 00000018 00          17          212
# error_counter,err_inj = 00000018 00          17          213
# error_counter,err_inj = 00000018 00          17          214
# error_counter,err_inj = 00000018 00          17          215
# error_counter,err_inj = 00000018 00          17          216
# error_counter,err_inj = 00000018 00          17          217
# error_counter,err_inj = 00000018 00          17          218
# error_counter,err_inj = 00000018 00          17          219
# error_counter,err_inj = 00000018 00          17          220
# error_counter,err_inj = 00000019 00          17          221
# error_counter,err_inj = 0000001a 11          17          222
# error_counter,err_inj = 0000001b 10          17          223
# error_counter,err_inj = 0000001c 10          19          224
# error_counter,err_inj = 0000001c 00          20          225
# error_counter,err_inj = 0000001c 00          20          226
# error_counter,err_inj = 0000001c 00          20          227
# error_counter,err_inj = 0000001c 00          20          228
# error_counter,err_inj = 0000001c 00          20          229
# error_counter,err_inj = 0000001c 00          20          230
# error_counter,err_inj = 0000001c 00          20          231
# error_counter,err_inj = 0000001c 00          20          232
# error_counter,err_inj = 0000001c 00          20          233
# error_counter,err_inj = 0000001c 00          20          234
# error_counter,err_inj = 0000001c 00          20          235
# error_counter,err_inj = 0000001c 00          20          236
# error_counter,err_inj = 0000001c 00          20          237
# error_counter,err_inj = 0000001c 00          20          238
# error_counter,err_inj = 0000001c 00          20          239
# error_counter,err_inj = 0000001c 00          20          240
# error_counter,err_inj = 0000001c 00          20          241
# error_counter,err_inj = 0000001c 00          20          242
# error_counter,err_inj = 0000001c 00          20          243
# error_counter,err_inj = 0000001c 00          20          244
# error_counter,err_inj = 0000001c 00          20          245
```

```
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          32, OUT: good =         244, bad =
12
```

```
# error_counter,err_inj = 0000001c 00          20         246
# error_counter,err_inj = 0000001c 00          20         247
# error_counter,err_inj = 0000001c 00          20         248
# error_counter,err_inj = 0000001c 00          20         249
# error_counter,err_inj = 0000001c 00          20         250
# error_counter,err_inj = 0000001c 00          20         251
# error_counter,err_inj = 0000001c 00          20         252
# error_counter,err_inj = 0000001d 00          20         253
# error_counter,err_inj = 0000001e 00          20         254
# error_counter,err_inj = 0000001f 01          20         255
# word_count =       10440
```

2b8) Transcript:

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000000 00          0         12
# error_counter,err_inj = 00000000 00          0         13
# error_counter,err_inj = 00000000 00          0         14
# error_counter,err_inj = 00000000 00          0         15
# error_counter,err_inj = 00000000 00          0         16
# error_counter,err_inj = 00000000 00          0         17
# error_counter,err_inj = 00000000 00          0         18
# error_counter,err_inj = 00000000 00          0         19
# error_counter,err_inj = 00000000 00          0         20
# error_counter,err_inj = 00000000 00          0         21
# error_counter,err_inj = 00000000 00          0         22
# error_counter,err_inj = 00000000 00          0         23
# error_counter,err_inj = 00000000 00          0         24
# error_counter,err_inj = 00000000 00          0         25
# error_counter,err_inj = 00000000 00          0         26
# error_counter,err_inj = 00000000 00          0         27
# error_counter,err_inj = 00000000 00          0         28
# error_counter,err_inj = 00000000 00          0         29
# error_counter,err_inj = 00000000 00          0         30
# error_counter,err_inj = 00000000 00          0         31
# error_counter,err_inj = 00000001 10          0         32
# error_counter,err_inj = 00000001 00          0         33
# error_counter,err_inj = 00000001 00          0         34
# error_counter,err_inj = 00000001 00          0         35
# error_counter,err_inj = 00000001 00          0         36
# error_counter,err_inj = 00000001 00          0         37
# error_counter,err_inj = 00000001 00          0         38
# error_counter,err_inj = 00000001 00          0         39
# error_counter,err_inj = 00000001 00          0         40
# error_counter,err_inj = 00000001 00          0         41
# error_counter,err_inj = 00000001 00          0         42
# error_counter,err_inj = 00000001 00          0         43
# error_counter,err_inj = 00000001 00          0         44
# error_counter,err_inj = 00000001 00          0         45
# error_counter,err_inj = 00000001 00          0         46
# error_counter,err_inj = 00000001 00          0         47
# error_counter,err_inj = 00000001 00          0         48
# error_counter,err_inj = 00000001 00          0         49
# error_counter,err_inj = 00000001 00          0         50
# error_counter,err_inj = 00000001 00          0         51
# error_counter,err_inj = 00000001 00          0         52
# error_counter,err_inj = 00000001 00          0         53
# error_counter,err_inj = 00000001 00          0         54
# error_counter,err_inj = 00000001 00          0         55
# error_counter,err_inj = 00000001 00          0         56
# error_counter,err_inj = 00000001 00          0         57
# error_counter,err_inj = 00000001 00          0         58
# error_counter,err_inj = 00000001 00          0         59
# error_counter,err_inj = 00000001 00          0         60
# error_counter,err_inj = 00000001 00          0         61
# error_counter,err_inj = 00000001 00          0         62
# error_counter,err_inj = 00000001 00          0         63
# error_counter,err_inj = 00000002 00          0         64
```

```
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
```

```
# error_counter,err_inj = 00000002 00          0          65
# error_counter,err_inj = 00000002 00          0          66
# error_counter,err_inj = 00000002 00          0          67
# error_counter,err_inj = 00000002 00          0          68
# error_counter,err_inj = 00000002 00          0          69
# error_counter,err_inj = 00000002 00          0          70
# error_counter,err_inj = 00000002 00          0          71
# error_counter,err_inj = 00000002 00          0          72
# error_counter,err_inj = 00000002 00          0          73
# error_counter,err_inj = 00000002 00          0          74
# error_counter,err_inj = 00000002 00          0          75
# error_counter,err_inj = 00000002 00          0          76
# error_counter,err_inj = 00000002 00          0          77
# error_counter,err_inj = 00000002 00          0          78
# error_counter,err_inj = 00000002 00          0          79
# error_counter,err_inj = 00000002 00          0          80
# error_counter,err_inj = 00000002 00          0          81
# error_counter,err_inj = 00000002 00          0          82
# error_counter,err_inj = 00000002 00          0          83
# error_counter,err_inj = 00000002 00          0          84
# error_counter,err_inj = 00000002 00          0          85
# error_counter,err_inj = 00000002 00          0          86
# error_counter,err_inj = 00000002 00          0          87
# error_counter,err_inj = 00000002 00          0          88
# error_counter,err_inj = 00000002 00          0          89
# error_counter,err_inj = 00000002 00          0          90
# error_counter,err_inj = 00000002 00          0          91
# error_counter,err_inj = 00000002 00          0          92
# error_counter,err_inj = 00000002 00          0          93
# error_counter,err_inj = 00000002 00          0          94
# error_counter,err_inj = 00000002 00          0          95
# error_counter,err_inj = 00000003 10          0          96
# error_counter,err_inj = 00000003 00          0          97
# error_counter,err_inj = 00000003 00          0          98
# error_counter,err_inj = 00000003 00          0          99
# error_counter,err_inj = 00000003 00          0         100
# error_counter,err_inj = 00000003 00          0         101
# error_counter,err_inj = 00000003 00          0         102
# error_counter,err_inj = 00000003 00          0         103
# error_counter,err_inj = 00000003 00          0         104
# error_counter,err_inj = 00000003 00          0         105
# error_counter,err_inj = 00000003 00          0         106
# error_counter,err_inj = 00000003 00          0         107
# error_counter,err_inj = 00000003 00          0         108
# error_counter,err_inj = 00000003 00          0         109
# error_counter,err_inj = 00000003 00          0         110
# error_counter,err_inj = 00000003 00          0         111
# error_counter,err_inj = 00000003 00          0         112
# error_counter,err_inj = 00000003 00          0         113
# error_counter,err_inj = 00000003 00          0         114
# error_counter,err_inj = 00000003 00          0         115
# error_counter,err_inj = 00000003 00          0         116
# error_counter,err_inj = 00000003 00          0         117
# error_counter,err_inj = 00000003 00          0         118
# error_counter,err_inj = 00000003 00          0         119
# error_counter,err_inj = 00000003 00          0         120
# error_counter,err_inj = 00000003 00          0         121
# error_counter,err_inj = 00000003 00          0         122
# error_counter,err_inj = 00000003 00          0         123
# error_counter,err_inj = 00000003 00          0         124
# error_counter,err_inj = 00000003 00          0         125
# error_counter,err_inj = 00000003 00          0         126
# error_counter,err_inj = 00000003 00          0         127
# error_counter,err_inj = 00000004 00          0         128
# error_counter,err_inj = 00000004 00          0         129
# error_counter,err_inj = 00000004 00          0         130
# error_counter,err_inj = 00000004 00          0         131
# error_counter,err_inj = 00000004 00          0         132
# error_counter,err_inj = 00000004 00          0         133
# error_counter,err_inj = 00000004 00          0         134
# error_counter,err_inj = 00000004 00          0         135
# error_counter,err_inj = 00000004 00          0         136
# error_counter,err_inj = 00000004 00          0         137
# error_counter,err_inj = 00000004 00          0         138
# error_counter,err_inj = 00000004 00          0         139
# error_counter,err_inj = 00000004 00          0         140
# error_counter,err_inj = 00000004 00          0         141
# error_counter,err_inj = 00000004 00          0         142
# error_counter,err_inj = 00000004 00          0         143
# error_counter,err_inj = 00000004 00          0         144
# error_counter,err_inj = 00000004 00          0         145
# error_counter,err_inj = 00000004 00          0         146
# error_counter,err_inj = 00000004 00          0         147
# error_counter,err_inj = 00000004 00          0         148
# error_counter,err_inj = 00000004 00          0         149
# error_counter,err_inj = 00000004 00          0         150
# error_counter,err_inj = 00000004 00          0         151
# error_counter,err_inj = 00000004 00          0         152
# error_counter,err_inj = 00000004 00          0         153
# error_counter,err_inj = 00000004 00          0         154
# error_counter,err_inj = 00000004 00          0         155
# error_counter,err_inj = 00000004 00          0         156
# error_counter,err_inj = 00000004 00          0         157
```

```
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
```

```
# error_counter,err_inj = 00000004 00          0         158
# error_counter,err_inj = 00000004 00          0         159
# error_counter,err_inj = 00000005 11          0         160
# error_counter,err_inj = 00000005 00          0         161
# error_counter,err_inj = 00000005 00          0         162
# error_counter,err_inj = 00000005 00          0         163
# error_counter,err_inj = 00000005 00          0         164
# error_counter,err_inj = 00000005 00          0         165
# error_counter,err_inj = 00000005 00          0         166
# error_counter,err_inj = 00000005 00          0         167
# error_counter,err_inj = 00000005 00          0         168
# error_counter,err_inj = 00000005 00          0         169
# error_counter,err_inj = 00000005 00          0         170
# error_counter,err_inj = 00000005 00          0         171
# error_counter,err_inj = 00000005 00          0         172
# error_counter,err_inj = 00000005 00          0         173
# error_counter,err_inj = 00000005 00          0         174
# error_counter,err_inj = 00000005 00          0         175
# error_counter,err_inj = 00000005 00          0         176
# error_counter,err_inj = 00000005 00          0         177
# error_counter,err_inj = 00000005 00          0         178
# error_counter,err_inj = 00000005 00          0         179
# error_counter,err_inj = 00000005 00          0         180
# error_counter,err_inj = 00000005 00          0         181
# error_counter,err_inj = 00000005 00          0         182
# error_counter,err_inj = 00000005 00          0         183
# error_counter,err_inj = 00000005 00          0         184
# error_counter,err_inj = 00000005 00          0         185
# error_counter,err_inj = 00000005 00          0         186
# error_counter,err_inj = 00000005 00          0         187
# error_counter,err_inj = 00000005 00          0         188
# error_counter,err_inj = 00000005 00          0         189
# error_counter,err_inj = 00000005 00          0         190
# error_counter,err_inj = 00000005 00          0         191
# error_counter,err_inj = 00000006 01          0         192
# error_counter,err_inj = 00000006 00          0         193
# error_counter,err_inj = 00000006 00          0         194
# error_counter,err_inj = 00000006 00          0         195
# error_counter,err_inj = 00000006 00          0         196
# error_counter,err_inj = 00000006 00          0         197
# error_counter,err_inj = 00000006 00          0         198
# error_counter,err_inj = 00000006 00          0         199
# error_counter,err_inj = 00000006 00          0         200
# error_counter,err_inj = 00000006 00          0         201
# error_counter,err_inj = 00000006 00          0         202
# error_counter,err_inj = 00000006 00          0         203
# error_counter,err_inj = 00000006 00          0         204
# error_counter,err_inj = 00000006 00          0         205
# error_counter,err_inj = 00000006 00          0         206
# error_counter,err_inj = 00000006 00          0         207
# error_counter,err_inj = 00000006 00          0         208
# error_counter,err_inj = 00000006 00          0         209
# error_counter,err_inj = 00000006 00          0         210
# error_counter,err_inj = 00000006 00          0         211
# error_counter,err_inj = 00000006 00          0         212
# error_counter,err_inj = 00000006 00          0         213
# error_counter,err_inj = 00000006 00          0         214
# error_counter,err_inj = 00000006 00          0         215
# error_counter,err_inj = 00000006 00          0         216
# error_counter,err_inj = 00000006 00          0         217
# error_counter,err_inj = 00000006 00          0         218
# error_counter,err_inj = 00000006 00          0         219
# error_counter,err_inj = 00000006 00          0         220
# error_counter,err_inj = 00000006 00          0         221
# error_counter,err_inj = 00000006 00          0         222
# error_counter,err_inj = 00000006 00          0         223
# error_counter,err_inj = 00000007 01          0         224
# error_counter,err_inj = 00000007 00          0         225
# error_counter,err_inj = 00000007 00          0         226
# error_counter,err_inj = 00000007 00          0         227
# error_counter,err_inj = 00000007 00          0         228
# error_counter,err_inj = 00000007 00          0         229
# error_counter,err_inj = 00000007 00          0         230
# error_counter,err_inj = 00000007 00          0         231
# error_counter,err_inj = 00000007 00          0         232
# error_counter,err_inj = 00000007 00          0         233
# error_counter,err_inj = 00000007 00          0         234
# error_counter,err_inj = 00000007 00          0         235
# error_counter,err_inj = 00000007 00          0         236
# error_counter,err_inj = 00000007 00          0         237
# error_counter,err_inj = 00000007 00          0         238
# error_counter,err_inj = 00000007 00          0         239
# error_counter,err_inj = 00000007 00          0         240
# error_counter,err_inj = 00000007 00          0         241
# error_counter,err_inj = 00000007 00          0         242
# error_counter,err_inj = 00000007 00          0         243
# error_counter,err_inj = 00000007 00          0         244
# error_counter,err_inj = 00000007 00          0         245
# error_counter,err_inj = 00000007 00          0         246
# error_counter,err_inj = 00000007 00          0         247
# error_counter,err_inj = 00000007 00          0         248
# error_counter,err_inj = 00000007 00          0         249
# error_counter,err_inj = 00000007 00          0         250
```

```
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =           8, OUT: good =         256, bad =
0
```

```
# error_counter,err_inj = 00000007 00           0         251
# error_counter,err_inj = 00000007 00           0         252
# error_counter,err_inj = 00000007 00           0         253
# error_counter,err_inj = 00000007 00           0         254
# error_counter,err_inj = 00000007 00           0         255
# word_count =       10440
```

2c) Transcript: 5 consecutive error injections breaks the decoder

```
# word_count =       10440
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# yaa! in = 1, out = 1, w_ct =          76, err = 00
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
```

```
# error_counter,err_inj = 00000000 xx           0           0
# error_counter,err_inj = 00000000 00           0           1
# error_counter,err_inj = 00000000 00           0           2
# error_counter,err_inj = 00000000 00           0           3
# error_counter,err_inj = 00000000 00           0           4
# error_counter,err_inj = 00000000 00           0           5
# error_counter,err_inj = 00000000 00           0           6
# error_counter,err_inj = 00000000 00           0           7
# error_counter,err_inj = 00000000 00           0           8
# error_counter,err_inj = 00000000 00           0           9
# error_counter,err_inj = 00000000 00           0          10
# error_counter,err_inj = 00000000 00           0          11
# error_counter,err_inj = 00000001 01           0          12
# error_counter,err_inj = 00000002 01           0          13
# error_counter,err_inj = 00000003 01           1          14
# error_counter,err_inj = 00000004 01           2          15
# error_counter,err_inj = 00000005 01           3          16
# error_counter,err_inj = 00000005 00           4          17
# error_counter,err_inj = 00000005 00           4          18
# error_counter,err_inj = 00000005 00           4          19
# error_counter,err_inj = 00000005 00           4          20
# error_counter,err_inj = 00000005 00           4          21
# error_counter,err_inj = 00000005 00           4          22
# error_counter,err_inj = 00000005 00           4          23
# error_counter,err_inj = 00000005 00           4          24
# error_counter,err_inj = 00000005 00           4          25
# error_counter,err_inj = 00000005 00           4          26
# error_counter,err_inj = 00000005 00           4          27
# error_counter,err_inj = 00000006 01           4          28
# error_counter,err_inj = 00000007 01           4          29
# error_counter,err_inj = 00000008 01           5          30
# error_counter,err_inj = 00000009 01           6          31
# error_counter,err_inj = 0000000a 01           7          32
# error_counter,err_inj = 0000000a 00           8          33
# error_counter,err_inj = 0000000a 00           8          34
# error_counter,err_inj = 0000000a 00           8          35
# error_counter,err_inj = 0000000a 00           8          36
# error_counter,err_inj = 0000000a 00           8          37
# error_counter,err_inj = 0000000a 00           8          38
# error_counter,err_inj = 0000000a 00           8          39
# error_counter,err_inj = 0000000a 00           8          40
# error_counter,err_inj = 0000000a 00           8          41
# error_counter,err_inj = 0000000a 00           8          42
# error_counter,err_inj = 0000000a 00           8          43
# error_counter,err_inj = 0000000b 01           8          44
# error_counter,err_inj = 0000000c 01           8          45
# error_counter,err_inj = 0000000d 01           9          46
# error_counter,err_inj = 0000000e 01          10          47
# error_counter,err_inj = 0000000f 01          11          48
# error_counter,err_inj = 0000000f 00          12          49
# error_counter,err_inj = 0000000f 00          12          50
# error_counter,err_inj = 0000000f 00          12          51
# error_counter,err_inj = 0000000f 00          12          52
# error_counter,err_inj = 0000000f 00          12          53
# error_counter,err_inj = 0000000f 00          12          54
# error_counter,err_inj = 0000000f 00          12          55
# error_counter,err_inj = 0000000f 00          12          56
# error_counter,err_inj = 0000000f 00          12          57
# error_counter,err_inj = 0000000f 00          12          58
# error_counter,err_inj = 0000000f 00          12          59
# error_counter,err_inj = 00000010 01          12          60
# error_counter,err_inj = 00000011 01          12          61
# error_counter,err_inj = 00000012 01          13          62
# error_counter,err_inj = 00000013 01          14          63
# error_counter,err_inj = 00000014 01          15          64
# error_counter,err_inj = 00000014 00          16          65
# error_counter,err_inj = 00000014 00          16          66
# error_counter,err_inj = 00000014 00          16          67
# error_counter,err_inj = 00000014 00          16          68
# error_counter,err_inj = 00000014 00          16          69
# error_counter,err_inj = 00000014 00          16          70
# error_counter,err_inj = 00000014 00          16          71
# error_counter,err_inj = 00000014 00          16          72
# error_counter,err_inj = 00000014 00          16          73
# error_counter,err_inj = 00000014 00          16          74
# error_counter,err_inj = 00000014 00          16          75
# error_counter,err_inj = 00000015 01          16          76
# error_counter,err_inj = 00000016 01          16          77
# error_counter,err_inj = 00000017 01          17          78
# error_counter,err_inj = 00000018 01          18          79
# error_counter,err_inj = 00000019 01          19          80
# error_counter,err_inj = 00000019 00          20          81
```

```
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# boo! in = 1, out = 0, w_ct =          97, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =          98, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# boo! in = 1, out = 0, w_ct =         101, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         102, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         103, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         104, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# yaa! in = 1, out = 1, w_ct =         108, err = 00
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# yaa! in = 1, out = 1, w_ct =         140, err = 00
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# boo! in = 1, out = 0, w_ct =         145, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         146, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# boo! in = 1, out = 0, w_ct =         149, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         150, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         151, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         152, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# boo! in = 1, out = 0, w_ct =         161, err = 00,
```

```
# error_counter,err_inj = 00000019 00          20          82
# error_counter,err_inj = 00000019 00          20          83
# error_counter,err_inj = 00000019 00          20          84
# error_counter,err_inj = 00000019 00          20          85
# error_counter,err_inj = 00000019 00          20          86
# error_counter,err_inj = 00000019 00          20          87
# error_counter,err_inj = 00000019 00          20          88
# error_counter,err_inj = 00000019 00          20          89
# error_counter,err_inj = 00000019 00          20          90
# error_counter,err_inj = 00000019 00          20          91
# error_counter,err_inj = 0000001a 01          20          92
# error_counter,err_inj = 0000001b 01          20          93
# error_counter,err_inj = 0000001c 01          21          94
# error_counter,err_inj = 0000001d 01          22          95
# error_counter,err_inj = 0000001e 01          23          96
# error_counter,err_inj = 0000001e 00          24          97
# error_counter,err_inj = 0000001e 00          24          98
# error_counter,err_inj = 0000001e 00          24          99
# error_counter,err_inj = 0000001e 00          24         100
# error_counter,err_inj = 0000001e 00          24         101
# error_counter,err_inj = 0000001e 00          24         102
# error_counter,err_inj = 0000001e 00          24         103
# error_counter,err_inj = 0000001e 00          24         104
# error_counter,err_inj = 0000001e 00          24         105
# error_counter,err_inj = 0000001e 00          24         106
# error_counter,err_inj = 0000001e 00          24         107
# error_counter,err_inj = 0000001f 01          24         108
# error_counter,err_inj = 00000020 01          24         109
# error_counter,err_inj = 00000021 01          25         110
# error_counter,err_inj = 00000022 01          26         111
# error_counter,err_inj = 00000023 01          27         112
# error_counter,err_inj = 00000023 00          28         113
# error_counter,err_inj = 00000023 00          28         114
# error_counter,err_inj = 00000023 00          28         115
# error_counter,err_inj = 00000023 00          28         116
# error_counter,err_inj = 00000023 00          28         117
# error_counter,err_inj = 00000023 00          28         118
# error_counter,err_inj = 00000023 00          28         119
# error_counter,err_inj = 00000023 00          28         120
# error_counter,err_inj = 00000023 00          28         121
# error_counter,err_inj = 00000023 00          28         122
# error_counter,err_inj = 00000023 00          28         123
# error_counter,err_inj = 00000024 01          28         124
# error_counter,err_inj = 00000025 01          28         125
# error_counter,err_inj = 00000026 01          29         126
# error_counter,err_inj = 00000027 01          30         127
# error_counter,err_inj = 00000028 01          31         128
# error_counter,err_inj = 00000028 00          32         129
# error_counter,err_inj = 00000028 00          32         130
# error_counter,err_inj = 00000028 00          32         131
# error_counter,err_inj = 00000028 00          32         132
# error_counter,err_inj = 00000028 00          32         133
# error_counter,err_inj = 00000028 00          32         134
# error_counter,err_inj = 00000028 00          32         135
# error_counter,err_inj = 00000028 00          32         136
# error_counter,err_inj = 00000028 00          32         137
# error_counter,err_inj = 00000028 00          32         138
# error_counter,err_inj = 00000028 00          32         139
# error_counter,err_inj = 00000029 01          32         140
# error_counter,err_inj = 0000002a 01          32         141
# error_counter,err_inj = 0000002b 01          33         142
# error_counter,err_inj = 0000002c 01          34         143
# error_counter,err_inj = 0000002d 01          35         144
# error_counter,err_inj = 0000002d 00          36         145
# error_counter,err_inj = 0000002d 00          36         146
# error_counter,err_inj = 0000002d 00          36         147
# error_counter,err_inj = 0000002d 00          36         148
# error_counter,err_inj = 0000002d 00          36         149
# error_counter,err_inj = 0000002d 00          36         150
# error_counter,err_inj = 0000002d 00          36         151
# error_counter,err_inj = 0000002d 00          36         152
# error_counter,err_inj = 0000002d 00          36         153
# error_counter,err_inj = 0000002d 00          36         154
# error_counter,err_inj = 0000002d 00          36         155
# error_counter,err_inj = 0000002e 01          36         156
# error_counter,err_inj = 0000002f 01          36         157
# error_counter,err_inj = 00000030 01          37         158
# error_counter,err_inj = 00000031 01          38         159
# error_counter,err_inj = 00000032 01          39         160
# error_counter,err_inj = 00000032 00          40         161
# error_counter,err_inj = 00000032 00          40         162
# error_counter,err_inj = 00000032 00          40         163
# error_counter,err_inj = 00000032 00          40         164
# error_counter,err_inj = 00000032 00          40         165
# error_counter,err_inj = 00000032 00          40         166
# error_counter,err_inj = 00000032 00          40         167
# error_counter,err_inj = 00000032 00          40         168
# error_counter,err_inj = 00000032 00          40         169
# error_counter,err_inj = 00000032 00          40         170
# error_counter,err_inj = 00000032 00          40         171
# error_counter,err_inj = 00000033 01          40         172
# error_counter,err_inj = 00000034 01          40         173
# error_counter,err_inj = 00000035 01          41         174
```

```
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         162, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# boo! in = 1, out = 0, w_ct =         165, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         166, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         167, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         168, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# yaa! in = 1, out = 1, w_ct =         172, err = 00
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# boo! in = 1, out = 0, w_ct =         177, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         178, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# boo! in = 1, out = 0, w_ct =         181, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         182, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         183, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         184, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# boo! in = 1, out = 0, w_ct =         193, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         194, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# boo! in = 1, out = 0, w_ct =         197, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         198, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         199, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         200, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# yaa! in = 1, out = 1, w_ct =         204, err = 00
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# boo! in = 1, out = 0, w_ct =         209, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         210, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# boo! in = 1, out = 0, w_ct =         213, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         214, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         215, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         216, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# boo! in = 1, out = 0, w_ct =         225, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         226, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
```

```
# error_counter,err_inj = 00000036 01          42         175
# error_counter,err_inj = 00000037 01          43         176
# error_counter,err_inj = 00000037 00          44         177
# error_counter,err_inj = 00000037 00          44         178
# error_counter,err_inj = 00000037 00          44         179
# error_counter,err_inj = 00000037 00          44         180
# error_counter,err_inj = 00000037 00          44         181
# error_counter,err_inj = 00000037 00          44         182
# error_counter,err_inj = 00000037 00          44         183
# error_counter,err_inj = 00000037 00          44         184
# error_counter,err_inj = 00000037 00          44         185
# error_counter,err_inj = 00000037 00          44         186
# error_counter,err_inj = 00000037 00          44         187
# error_counter,err_inj = 00000038 01          44         188
# error_counter,err_inj = 00000039 01          44         189
# error_counter,err_inj = 0000003a 01          45         190
# error_counter,err_inj = 0000003b 01          46         191
# error_counter,err_inj = 0000003c 01          47         192
# error_counter,err_inj = 0000003c 00          48         193
# error_counter,err_inj = 0000003c 00          48         194
# error_counter,err_inj = 0000003c 00          48         195
# error_counter,err_inj = 0000003c 00          48         196
# error_counter,err_inj = 0000003c 00          48         197
# error_counter,err_inj = 0000003c 00          48         198
# error_counter,err_inj = 0000003c 00          48         199
# error_counter,err_inj = 0000003c 00          48         200
# error_counter,err_inj = 0000003c 00          48         201
# error_counter,err_inj = 0000003c 00          48         202
# error_counter,err_inj = 0000003c 00          48         203
# error_counter,err_inj = 0000003d 01          48         204
# error_counter,err_inj = 0000003e 01          48         205
# error_counter,err_inj = 0000003f 01          49         206
# error_counter,err_inj = 00000040 01          50         207
# error_counter,err_inj = 00000041 01          51         208
# error_counter,err_inj = 00000041 00          52         209
# error_counter,err_inj = 00000041 00          52         210
# error_counter,err_inj = 00000041 00          52         211
# error_counter,err_inj = 00000041 00          52         212
# error_counter,err_inj = 00000041 00          52         213
# error_counter,err_inj = 00000041 00          52         214
# error_counter,err_inj = 00000041 00          52         215
# error_counter,err_inj = 00000041 00          52         216
# error_counter,err_inj = 00000041 00          52         217
# error_counter,err_inj = 00000041 00          52         218
# error_counter,err_inj = 00000041 00          52         219
# error_counter,err_inj = 00000042 01          52         220
# error_counter,err_inj = 00000043 01          52         221
# error_counter,err_inj = 00000044 01          53         222
# error_counter,err_inj = 00000045 01          54         223
# error_counter,err_inj = 00000046 01          55         224
# error_counter,err_inj = 00000046 00          56         225
# error_counter,err_inj = 00000046 00          56         226
# error_counter,err_inj = 00000046 00          56         227
# error_counter,err_inj = 00000046 00          56         228
# error_counter,err_inj = 00000046 00          56         229
# error_counter,err_inj = 00000046 00          56         230
# error_counter,err_inj = 00000046 00          56         231
# error_counter,err_inj = 00000046 00          56         232
# error_counter,err_inj = 00000046 00          56         233
# error_counter,err_inj = 00000046 00          56         234
# error_counter,err_inj = 00000046 00          56         235
# error_counter,err_inj = 00000047 01          56         236
# error_counter,err_inj = 00000048 01          56         237
# error_counter,err_inj = 00000049 01          57         238
# error_counter,err_inj = 0000004a 01          58         239
# error_counter,err_inj = 0000004b 01          59         240
# error_counter,err_inj = 0000004b 00          60         241
# error_counter,err_inj = 0000004b 00          60         242
# error_counter,err_inj = 0000004b 00          60         243
# error_counter,err_inj = 0000004b 00          60         244
# error_counter,err_inj = 0000004b 00          60         245
# error_counter,err_inj = 0000004b 00          60         246
# error_counter,err_inj = 0000004b 00          60         247
# error_counter,err_inj = 0000004b 00          60         248
# error_counter,err_inj = 0000004b 00          60         249
# error_counter,err_inj = 0000004b 00          60         250
# error_counter,err_inj = 0000004b 00          60         251
# error_counter,err_inj = 0000004c 01          60         252
# error_counter,err_inj = 0000004d 01          60         253
# error_counter,err_inj = 0000004e 01          61         254
# error_counter,err_inj = 0000004f 01          62         255
```

```
# boo! in = 1, out = 0, w_ct =         229, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         230, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         231, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         232, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# yaa! in = 1, out = 1, w_ct =         236, err = 00
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          80, OUT: good =         214, bad =
42
```

2d) Transcript: 5 consecutive error injections breaks the decoder

```
# word_count =       10440
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# boo! in = 1, out = 0, w_ct =          21, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# boo! in = 1, out = 0, w_ct =          29, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# boo! in = 0, out = 1, w_ct =          34, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# boo! in = 1, out = 0, w_ct =          38, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0          0
# error_counter,err_inj = 00000000 00          0          1
# error_counter,err_inj = 00000000 00          0          2
# error_counter,err_inj = 00000000 00          0          3
# error_counter,err_inj = 00000000 00          0          4
# error_counter,err_inj = 00000000 00          0          5
# error_counter,err_inj = 00000000 00          0          6
# error_counter,err_inj = 00000000 00          0          7
# error_counter,err_inj = 00000000 00          0          8
# error_counter,err_inj = 00000000 00          0          9
# error_counter,err_inj = 00000000 00          0         10
# error_counter,err_inj = 00000000 00          0         11
# error_counter,err_inj = 00000001 10          0         12
# error_counter,err_inj = 00000002 10          0         13
# error_counter,err_inj = 00000003 10          1         14
# error_counter,err_inj = 00000004 10          2         15
# error_counter,err_inj = 00000005 10          3         16
# error_counter,err_inj = 00000005 00          4         17
# error_counter,err_inj = 00000005 00          4         18
# error_counter,err_inj = 00000005 00          4         19
# error_counter,err_inj = 00000005 00          4         20
# error_counter,err_inj = 00000005 00          4         21
# error_counter,err_inj = 00000005 00          4         22
# error_counter,err_inj = 00000005 00          4         23
# error_counter,err_inj = 00000005 00          4         24
# error_counter,err_inj = 00000005 00          4         25
# error_counter,err_inj = 00000005 00          4         26
# error_counter,err_inj = 00000005 00          4         27
# error_counter,err_inj = 00000006 10          4         28
# error_counter,err_inj = 00000007 10          4         29
# error_counter,err_inj = 00000008 10          5         30
# error_counter,err_inj = 00000009 10          6         31
# error_counter,err_inj = 0000000a 10          7         32
# error_counter,err_inj = 0000000a 00          8         33
# error_counter,err_inj = 0000000a 00          8         34
# error_counter,err_inj = 0000000a 00          8         35
# error_counter,err_inj = 0000000a 00          8         36
# error_counter,err_inj = 0000000a 00          8         37
# error_counter,err_inj = 0000000a 00          8         38
# error_counter,err_inj = 0000000a 00          8         39
# error_counter,err_inj = 0000000a 00          8         40
# error_counter,err_inj = 0000000a 00          8         41
# error_counter,err_inj = 0000000a 00          8         42
# error_counter,err_inj = 0000000a 00          8         43
# error_counter,err_inj = 0000000b 10          8         44
# error_counter,err_inj = 0000000c 10          8         45
# error_counter,err_inj = 0000000d 10          9         46
# error_counter,err_inj = 0000000e 10         10         47
# error_counter,err_inj = 0000000f 10         11         48
# error_counter,err_inj = 0000000f 00         12         49
# error_counter,err_inj = 0000000f 00         12         50
# error_counter,err_inj = 0000000f 00         12         51
# error_counter,err_inj = 0000000f 00         12         52
# error_counter,err_inj = 0000000f 00         12         53
# error_counter,err_inj = 0000000f 00         12         54
# error_counter,err_inj = 0000000f 00         12         55
```

```
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# boo! in = 0, out = 1, w_ct =          69, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# yaa! in = 0, out = 0, w_ct =          71, err = 00
# yaa! in = 0, out = 0, w_ct =          72, err = 00
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# boo! in = 1, out = 0, w_ct =          76, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# boo! in = 1, out = 0, w_ct =         101, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# yaa! in = 1, out = 1, w_ct =         103, err = 00
# yaa! in = 1, out = 1, w_ct =         104, err = 00
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# boo! in = 1, out = 0, w_ct =         108, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# boo! in = 1, out = 0, w_ct =         117, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# boo! in = 1, out = 0, w_ct =         124, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# boo! in = 0, out = 1, w_ct =         133, err = 00,
1045000, BAD!
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# yaa! in = 0, out = 0, w_ct =         135, err = 00
# yaa! in = 0, out = 0, w_ct =         136, err = 00
```

```
# error_counter,err_inj = 0000000f 00          12          56
# error_counter,err_inj = 0000000f 00          12          57
# error_counter,err_inj = 0000000f 00          12          58
# error_counter,err_inj = 0000000f 00          12          59
# error_counter,err_inj = 00000010 10          12          60
# error_counter,err_inj = 00000011 10          12          61
# error_counter,err_inj = 00000012 10          13          62
# error_counter,err_inj = 00000013 10          14          63
# error_counter,err_inj = 00000014 10          15          64
# error_counter,err_inj = 00000014 00          16          65
# error_counter,err_inj = 00000014 00          16          66
# error_counter,err_inj = 00000014 00          16          67
# error_counter,err_inj = 00000014 00          16          68
# error_counter,err_inj = 00000014 00          16          69
# error_counter,err_inj = 00000014 00          16          70
# error_counter,err_inj = 00000014 00          16          71
# error_counter,err_inj = 00000014 00          16          72
# error_counter,err_inj = 00000014 00          16          73
# error_counter,err_inj = 00000014 00          16          74
# error_counter,err_inj = 00000014 00          16          75
# error_counter,err_inj = 00000015 10          16          76
# error_counter,err_inj = 00000016 10          16          77
# error_counter,err_inj = 00000017 10          17          78
# error_counter,err_inj = 00000018 10          18          79
# error_counter,err_inj = 00000019 10          19          80
# error_counter,err_inj = 00000019 00          20          81
# error_counter,err_inj = 00000019 00          20          82
# error_counter,err_inj = 00000019 00          20          83
# error_counter,err_inj = 00000019 00          20          84
# error_counter,err_inj = 00000019 00          20          85
# error_counter,err_inj = 00000019 00          20          86
# error_counter,err_inj = 00000019 00          20          87
# error_counter,err_inj = 00000019 00          20          88
# error_counter,err_inj = 00000019 00          20          89
# error_counter,err_inj = 00000019 00          20          90
# error_counter,err_inj = 00000019 00          20          91
# error_counter,err_inj = 0000001a 10          20          92
# error_counter,err_inj = 0000001b 10          20          93
# error_counter,err_inj = 0000001c 10          21          94
# error_counter,err_inj = 0000001d 10          22          95
# error_counter,err_inj = 0000001e 10          23          96
# error_counter,err_inj = 0000001e 00          24          97
# error_counter,err_inj = 0000001e 00          24          98
# error_counter,err_inj = 0000001e 00          24          99
# error_counter,err_inj = 0000001e 00          24         100
# error_counter,err_inj = 0000001e 00          24         101
# error_counter,err_inj = 0000001e 00          24         102
# error_counter,err_inj = 0000001e 00          24         103
# error_counter,err_inj = 0000001e 00          24         104
# error_counter,err_inj = 0000001e 00          24         105
# error_counter,err_inj = 0000001e 00          24         106
# error_counter,err_inj = 0000001e 00          24         107
# error_counter,err_inj = 0000001f 10          24         108
# error_counter,err_inj = 00000020 10          24         109
# error_counter,err_inj = 00000021 10          25         110
# error_counter,err_inj = 00000022 10          26         111
# error_counter,err_inj = 00000023 10          27         112
# error_counter,err_inj = 00000023 00          28         113
# error_counter,err_inj = 00000023 00          28         114
# error_counter,err_inj = 00000023 00          28         115
# error_counter,err_inj = 00000023 00          28         116
# error_counter,err_inj = 00000023 00          28         117
# error_counter,err_inj = 00000023 00          28         118
# error_counter,err_inj = 00000023 00          28         119
# error_counter,err_inj = 00000023 00          28         120
# error_counter,err_inj = 00000023 00          28         121
# error_counter,err_inj = 00000023 00          28         122
# error_counter,err_inj = 00000023 00          28         123
# error_counter,err_inj = 00000024 10          28         124
# error_counter,err_inj = 00000025 10          28         125
# error_counter,err_inj = 00000026 10          29         126
# error_counter,err_inj = 00000027 10          30         127
# error_counter,err_inj = 00000028 10          31         128
# error_counter,err_inj = 00000028 00          32         129
# error_counter,err_inj = 00000028 00          32         130
# error_counter,err_inj = 00000028 00          32         131
# error_counter,err_inj = 00000028 00          32         132
# error_counter,err_inj = 00000028 00          32         133
# error_counter,err_inj = 00000028 00          32         134
# error_counter,err_inj = 00000028 00          32         135
# error_counter,err_inj = 00000028 00          32         136
# error_counter,err_inj = 00000028 00          32         137
# error_counter,err_inj = 00000028 00          32         138
# error_counter,err_inj = 00000028 00          32         139
# error_counter,err_inj = 00000029 10          32         140
# error_counter,err_inj = 0000002a 10          32         141
# error_counter,err_inj = 0000002b 10          33         142
# error_counter,err_inj = 0000002c 10          34         143
# error_counter,err_inj = 0000002d 10          35         144
# error_counter,err_inj = 0000002d 00          36         145
# error_counter,err_inj = 0000002d 00          36         146
# error_counter,err_inj = 0000002d 00          36         147
# error_counter,err_inj = 0000002d 00          36         148
```

```
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# boo! in = 1, out = 0, w_ct =         140, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# boo! in = 1, out = 0, w_ct =         149, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# boo! in = 1, out = 0, w_ct =         156, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# boo! in = 1, out = 0, w_ct =         165, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# yaa! in = 1, out = 1, w_ct =         167, err = 00
# yaa! in = 1, out = 1, w_ct =         168, err = 00
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# boo! in = 1, out = 0, w_ct =         172, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# boo! in = 1, out = 0, w_ct =         181, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# boo! in = 1, out = 0, w_ct =         188, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# boo! in = 1, out = 0, w_ct =         197, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# yaa! in = 1, out = 1, w_ct =         199, err = 00
# yaa! in = 1, out = 1, w_ct =         200, err = 00
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# boo! in = 1, out = 0, w_ct =         204, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# boo! in = 1, out = 0, w_ct =         213, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         214, err = 00
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
```

```
# error_counter,err_inj = 0000002d 00          36          149
# error_counter,err_inj = 0000002d 00          36          150
# error_counter,err_inj = 0000002d 00          36          151
# error_counter,err_inj = 0000002d 00          36          152
# error_counter,err_inj = 0000002d 00          36          153
# error_counter,err_inj = 0000002d 00          36          154
# error_counter,err_inj = 0000002d 00          36          155
# error_counter,err_inj = 0000002e 10          36          156
# error_counter,err_inj = 0000002f 10          36          157
# error_counter,err_inj = 00000030 10          37          158
# error_counter,err_inj = 00000031 10          38          159
# error_counter,err_inj = 00000032 10          39          160
# error_counter,err_inj = 00000032 00          40          161
# error_counter,err_inj = 00000032 00          40          162
# error_counter,err_inj = 00000032 00          40          163
# error_counter,err_inj = 00000032 00          40          164
# error_counter,err_inj = 00000032 00          40          165
# error_counter,err_inj = 00000032 00          40          166
# error_counter,err_inj = 00000032 00          40          167
# error_counter,err_inj = 00000032 00          40          168
# error_counter,err_inj = 00000032 00          40          169
# error_counter,err_inj = 00000032 00          40          170
# error_counter,err_inj = 00000032 00          40          171
# error_counter,err_inj = 00000033 10          40          172
# error_counter,err_inj = 00000034 10          40          173
# error_counter,err_inj = 00000035 10          41          174
# error_counter,err_inj = 00000036 10          42          175
# error_counter,err_inj = 00000037 10          43          176
# error_counter,err_inj = 00000037 00          44          177
# error_counter,err_inj = 00000037 00          44          178
# error_counter,err_inj = 00000037 00          44          179
# error_counter,err_inj = 00000037 00          44          180
# error_counter,err_inj = 00000037 00          44          181
# error_counter,err_inj = 00000037 00          44          182
# error_counter,err_inj = 00000037 00          44          183
# error_counter,err_inj = 00000037 00          44          184
# error_counter,err_inj = 00000037 00          44          185
# error_counter,err_inj = 00000037 00          44          186
# error_counter,err_inj = 00000037 00          44          187
# error_counter,err_inj = 00000038 10          44          188
# error_counter,err_inj = 00000039 10          44          189
# error_counter,err_inj = 0000003a 10          45          190
# error_counter,err_inj = 0000003b 10          46          191
# error_counter,err_inj = 0000003c 10          47          192
# error_counter,err_inj = 0000003c 00          48          193
# error_counter,err_inj = 0000003c 00          48          194
# error_counter,err_inj = 0000003c 00          48          195
# error_counter,err_inj = 0000003c 00          48          196
# error_counter,err_inj = 0000003c 00          48          197
# error_counter,err_inj = 0000003c 00          48          198
# error_counter,err_inj = 0000003c 00          48          199
# error_counter,err_inj = 0000003c 00          48          200
# error_counter,err_inj = 0000003c 00          48          201
# error_counter,err_inj = 0000003c 00          48          202
# error_counter,err_inj = 0000003c 00          48          203
# error_counter,err_inj = 0000003d 10          48          204
# error_counter,err_inj = 0000003e 10          48          205
# error_counter,err_inj = 0000003f 10          49          206
# error_counter,err_inj = 00000040 10          50          207
# error_counter,err_inj = 00000041 10          51          208
# error_counter,err_inj = 00000041 00          52          209
# error_counter,err_inj = 00000041 00          52          210
# error_counter,err_inj = 00000041 00          52          211
# error_counter,err_inj = 00000041 00          52          212
# error_counter,err_inj = 00000041 00          52          213
# error_counter,err_inj = 00000041 00          52          214
# error_counter,err_inj = 00000041 00          52          215
# error_counter,err_inj = 00000041 00          52          216
# error_counter,err_inj = 00000041 00          52          217
# error_counter,err_inj = 00000041 00          52          218
# error_counter,err_inj = 00000041 00          52          219
# error_counter,err_inj = 00000042 10          52          220
# error_counter,err_inj = 00000043 10          52          221
# error_counter,err_inj = 00000044 10          53          222
# error_counter,err_inj = 00000045 10          54          223
# error_counter,err_inj = 00000046 10          55          224
# error_counter,err_inj = 00000046 00          56          225
# error_counter,err_inj = 00000046 00          56          226
# error_counter,err_inj = 00000046 00          56          227
# error_counter,err_inj = 00000046 00          56          228
# error_counter,err_inj = 00000046 00          56          229
# error_counter,err_inj = 00000046 00          56          230
# error_counter,err_inj = 00000046 00          56          231
# error_counter,err_inj = 00000046 00          56          232
# error_counter,err_inj = 00000046 00          56          233
# error_counter,err_inj = 00000046 00          56          234
# error_counter,err_inj = 00000046 00          56          235
# error_counter,err_inj = 00000047 10          56          236
# error_counter,err_inj = 00000048 10          56          237
# error_counter,err_inj = 00000049 10          57          238
# error_counter,err_inj = 0000004a 10          58          239
# error_counter,err_inj = 0000004b 10          59          240
# error_counter,err_inj = 0000004b 00          60          241
```

```
# boo! in = 1, out = 0, w_ct =         220, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# boo! in = 1, out = 0, w_ct =         229, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# yaa! in = 1, out = 1, w_ct =         231, err = 00
# yaa! in = 1, out = 1, w_ct =         232, err = 00
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# boo! in = 1, out = 0, w_ct =         236, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# boo! in = 1, out = 0, w_ct =         245, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# boo! in = 1, out = 0, w_ct =         252, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          80, OUT: good =         230, bad =
26
```

```
# error_counter,err_inj = 0000004b 00          60          242
# error_counter,err_inj = 0000004b 00          60          243
# error_counter,err_inj = 0000004b 00          60          244
# error_counter,err_inj = 0000004b 00          60          245
# error_counter,err_inj = 0000004b 00          60          246
# error_counter,err_inj = 0000004b 00          60          247
# error_counter,err_inj = 0000004b 00          60          248
# error_counter,err_inj = 0000004b 00          60          249
# error_counter,err_inj = 0000004b 00          60          250
# error_counter,err_inj = 0000004b 00          60          251
# error_counter,err_inj = 0000004c 10          60          252
# error_counter,err_inj = 0000004d 10          60          253
# error_counter,err_inj = 0000004e 10          61          254
# error_counter,err_inj = 0000004f 10          62          255
```

2e) Transcript: 3 consecutive error injections breaks the decoder

```
# yaa! in = 0, out = 0, w_ct =           0, err = 00
# yaa! in = 0, out = 0, w_ct =           1, err = 00
# yaa! in = 0, out = 0, w_ct =           2, err = 00
# yaa! in = 0, out = 0, w_ct =           3, err = 00
# yaa! in = 0, out = 0, w_ct =           4, err = 00
# yaa! in = 0, out = 0, w_ct =           5, err = 00
# yaa! in = 0, out = 0, w_ct =           6, err = 00
# yaa! in = 0, out = 0, w_ct =           7, err = 00
# yaa! in = 0, out = 0, w_ct =           8, err = 00
# yaa! in = 0, out = 0, w_ct =           9, err = 00
# yaa! in = 0, out = 0, w_ct =          10, err = 00
# yaa! in = 1, out = 1, w_ct =          11, err = 00
# yaa! in = 0, out = 0, w_ct =          12, err = 00
# yaa! in = 0, out = 0, w_ct =          13, err = 00
# yaa! in = 1, out = 1, w_ct =          14, err = 00
# yaa! in = 1, out = 1, w_ct =          15, err = 00
# yaa! in = 0, out = 0, w_ct =          16, err = 00
# yaa! in = 0, out = 0, w_ct =          17, err = 00
# yaa! in = 0, out = 0, w_ct =          18, err = 00
# yaa! in = 1, out = 1, w_ct =          19, err = 00
# yaa! in = 1, out = 1, w_ct =          20, err = 00
# yaa! in = 1, out = 1, w_ct =          21, err = 00
# yaa! in = 0, out = 0, w_ct =          22, err = 00
# yaa! in = 0, out = 0, w_ct =          23, err = 00
# yaa! in = 0, out = 0, w_ct =          24, err = 00
# yaa! in = 0, out = 0, w_ct =          25, err = 00
# yaa! in = 1, out = 1, w_ct =          26, err = 00
# yaa! in = 1, out = 1, w_ct =          27, err = 00
# yaa! in = 1, out = 1, w_ct =          28, err = 00
# yaa! in = 1, out = 1, w_ct =          29, err = 00
# yaa! in = 0, out = 0, w_ct =          30, err = 00
# yaa! in = 0, out = 0, w_ct =          31, err = 00
# yaa! in = 0, out = 0, w_ct =          32, err = 00
# yaa! in = 0, out = 0, w_ct =          33, err = 00
# yaa! in = 0, out = 0, w_ct =          34, err = 00
# yaa! in = 1, out = 1, w_ct =          35, err = 00
# yaa! in = 1, out = 1, w_ct =          36, err = 00
# yaa! in = 1, out = 1, w_ct =          37, err = 00
# yaa! in = 1, out = 1, w_ct =          38, err = 00
# yaa! in = 1, out = 1, w_ct =          39, err = 00
# yaa! in = 0, out = 0, w_ct =          40, err = 00
# yaa! in = 1, out = 1, w_ct =          41, err = 00
# yaa! in = 0, out = 0, w_ct =          42, err = 00
# yaa! in = 1, out = 1, w_ct =          43, err = 00
```

```
# error_counter,err_inj = 00000000 xx          0           0
# error_counter,err_inj = 00000000 00          0           1
# error_counter,err_inj = 00000000 00          0           2
# error_counter,err_inj = 00000000 00          0           3
# error_counter,err_inj = 00000000 00          0           4
# error_counter,err_inj = 00000000 00          0           5
# error_counter,err_inj = 00000000 00          0           6
# error_counter,err_inj = 00000000 00          0           7
# error_counter,err_inj = 00000000 00          0           8
# error_counter,err_inj = 00000000 00          0           9
# error_counter,err_inj = 00000000 00          0          10
# error_counter,err_inj = 00000000 00          0          11
# error_counter,err_inj = 00000000 00          0          12
# error_counter,err_inj = 00000000 00          0          13
# error_counter,err_inj = 00000000 00          0          14
# error_counter,err_inj = 00000000 00          0          15
# error_counter,err_inj = 00000000 00          0          16
# error_counter,err_inj = 00000000 00          0          17
# error_counter,err_inj = 00000000 00          0          18
# error_counter,err_inj = 00000000 00          0          19
# error_counter,err_inj = 00000000 00          0          20
# error_counter,err_inj = 00000000 00          0          21
# error_counter,err_inj = 00000000 00          0          22
# error_counter,err_inj = 00000000 00          0          23
# error_counter,err_inj = 00000000 00          0          24
# error_counter,err_inj = 00000000 00          0          25
# error_counter,err_inj = 00000000 00          0          26
# error_counter,err_inj = 00000000 00          0          27
# error_counter,err_inj = 00000000 00          0          28
# error_counter,err_inj = 00000000 00          0          29
# error_counter,err_inj = 00000001 11          0          30
# error_counter,err_inj = 00000002 11          0          31
# error_counter,err_inj = 00000003 11          2          32
# error_counter,err_inj = 00000003 00          4          33
# error_counter,err_inj = 00000003 00          4          34
# error_counter,err_inj = 00000003 00          4          35
# error_counter,err_inj = 00000003 00          4          36
# error_counter,err_inj = 00000003 00          4          37
# error_counter,err_inj = 00000003 00          4          38
# error_counter,err_inj = 00000003 00          4          39
# error_counter,err_inj = 00000003 00          4          40
# error_counter,err_inj = 00000003 00          4          41
# error_counter,err_inj = 00000003 00          4          42
# error_counter,err_inj = 00000003 00          4          43
```

```
# yaa! in = 0, out = 0, w_ct =          44, err = 00
# yaa! in = 1, out = 1, w_ct =          45, err = 00
# yaa! in = 0, out = 0, w_ct =          46, err = 00
# yaa! in = 1, out = 1, w_ct =          47, err = 00
# yaa! in = 0, out = 0, w_ct =          48, err = 00
# yaa! in = 1, out = 1, w_ct =          49, err = 00
# yaa! in = 0, out = 0, w_ct =          50, err = 00
# yaa! in = 0, out = 0, w_ct =          51, err = 00
# yaa! in = 1, out = 1, w_ct =          52, err = 00
# yaa! in = 1, out = 1, w_ct =          53, err = 00
# yaa! in = 0, out = 0, w_ct =          54, err = 00
# yaa! in = 0, out = 0, w_ct =          55, err = 00
# yaa! in = 0, out = 0, w_ct =          56, err = 00
# yaa! in = 1, out = 1, w_ct =          57, err = 00
# yaa! in = 1, out = 1, w_ct =          58, err = 00
# yaa! in = 1, out = 1, w_ct =          59, err = 00
# yaa! in = 0, out = 0, w_ct =          60, err = 00
# yaa! in = 0, out = 0, w_ct =          61, err = 00
# yaa! in = 0, out = 0, w_ct =          62, err = 00
# yaa! in = 0, out = 0, w_ct =          63, err = 00
# yaa! in = 1, out = 1, w_ct =          64, err = 00
# yaa! in = 1, out = 1, w_ct =          65, err = 00
# yaa! in = 1, out = 1, w_ct =          66, err = 00
# yaa! in = 1, out = 1, w_ct =          67, err = 00
# yaa! in = 0, out = 0, w_ct =          68, err = 00
# yaa! in = 0, out = 0, w_ct =          69, err = 00
# yaa! in = 0, out = 0, w_ct =          70, err = 00
# boo! in = 0, out = 1, w_ct =          71, err = 00,
1045000, BAD!
# boo! in = 0, out = 1, w_ct =          72, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          73, err = 00
# yaa! in = 1, out = 1, w_ct =          74, err = 00
# yaa! in = 1, out = 1, w_ct =          75, err = 00
# boo! in = 1, out = 0, w_ct =          76, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =          77, err = 00
# yaa! in = 0, out = 0, w_ct =          78, err = 00
# yaa! in = 1, out = 1, w_ct =          79, err = 00
# yaa! in = 0, out = 0, w_ct =          80, err = 00
# yaa! in = 1, out = 1, w_ct =          81, err = 00
# yaa! in = 0, out = 0, w_ct =          82, err = 00
# yaa! in = 1, out = 1, w_ct =          83, err = 00
# yaa! in = 0, out = 0, w_ct =          84, err = 00
# yaa! in = 1, out = 1, w_ct =          85, err = 00
# yaa! in = 0, out = 0, w_ct =          86, err = 00
# yaa! in = 0, out = 0, w_ct =          87, err = 00
# yaa! in = 0, out = 0, w_ct =          88, err = 00
# yaa! in = 0, out = 0, w_ct =          89, err = 00
# yaa! in = 0, out = 0, w_ct =          90, err = 00
# yaa! in = 0, out = 0, w_ct =          91, err = 00
# yaa! in = 0, out = 0, w_ct =          92, err = 00
# yaa! in = 0, out = 0, w_ct =          93, err = 00
# yaa! in = 0, out = 0, w_ct =          94, err = 00
# yaa! in = 0, out = 0, w_ct =          95, err = 00
# yaa! in = 1, out = 1, w_ct =          96, err = 00
# yaa! in = 1, out = 1, w_ct =          97, err = 00
# yaa! in = 1, out = 1, w_ct =          98, err = 00
# yaa! in = 1, out = 1, w_ct =          99, err = 00
# yaa! in = 1, out = 1, w_ct =         100, err = 00
# yaa! in = 1, out = 1, w_ct =         101, err = 00
# yaa! in = 1, out = 1, w_ct =         102, err = 00
# boo! in = 1, out = 0, w_ct =         103, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         104, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         105, err = 00
# yaa! in = 0, out = 0, w_ct =         106, err = 00
# yaa! in = 0, out = 0, w_ct =         107, err = 00
# boo! in = 1, out = 0, w_ct =         108, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         109, err = 00
# yaa! in = 1, out = 1, w_ct =         110, err = 00
# yaa! in = 0, out = 0, w_ct =         111, err = 00
# yaa! in = 1, out = 1, w_ct =         112, err = 00
# yaa! in = 0, out = 0, w_ct =         113, err = 00
# yaa! in = 0, out = 0, w_ct =         114, err = 00
# yaa! in = 0, out = 0, w_ct =         115, err = 00
# yaa! in = 0, out = 0, w_ct =         116, err = 00
# yaa! in = 1, out = 1, w_ct =         117, err = 00
# yaa! in = 1, out = 1, w_ct =         118, err = 00
# yaa! in = 1, out = 1, w_ct =         119, err = 00
# yaa! in = 0, out = 0, w_ct =         120, err = 00
# yaa! in = 1, out = 1, w_ct =         121, err = 00
# yaa! in = 0, out = 0, w_ct =         122, err = 00
# yaa! in = 1, out = 1, w_ct =         123, err = 00
# yaa! in = 1, out = 1, w_ct =         124, err = 00
# yaa! in = 1, out = 1, w_ct =         125, err = 00
# yaa! in = 0, out = 0, w_ct =         126, err = 00
# yaa! in = 0, out = 0, w_ct =         127, err = 00
# yaa! in = 0, out = 0, w_ct =         128, err = 00
# yaa! in = 0, out = 0, w_ct =         129, err = 00
# yaa! in = 0, out = 0, w_ct =         130, err = 00
```

```
# error_counter,err_inj = 00000003 00          4          44
# error_counter,err_inj = 00000003 00          4          45
# error_counter,err_inj = 00000003 00          4          46
# error_counter,err_inj = 00000003 00          4          47
# error_counter,err_inj = 00000003 00          4          48
# error_counter,err_inj = 00000003 00          4          49
# error_counter,err_inj = 00000003 00          4          50
# error_counter,err_inj = 00000003 00          4          51
# error_counter,err_inj = 00000003 00          4          52
# error_counter,err_inj = 00000003 00          4          53
# error_counter,err_inj = 00000003 00          4          54
# error_counter,err_inj = 00000003 00          4          55
# error_counter,err_inj = 00000003 00          4          56
# error_counter,err_inj = 00000003 00          4          57
# error_counter,err_inj = 00000003 00          4          58
# error_counter,err_inj = 00000003 00          4          59
# error_counter,err_inj = 00000003 00          4          60
# error_counter,err_inj = 00000003 00          4          61
# error_counter,err_inj = 00000004 11          4          62
# error_counter,err_inj = 00000005 11          4          63
# error_counter,err_inj = 00000006 11          6          64
# error_counter,err_inj = 00000006 00          8          65
# error_counter,err_inj = 00000006 00          8          66
# error_counter,err_inj = 00000006 00          8          67
# error_counter,err_inj = 00000006 00          8          68
# error_counter,err_inj = 00000006 00          8          69
# error_counter,err_inj = 00000006 00          8          70
# error_counter,err_inj = 00000006 00          8          71
# error_counter,err_inj = 00000006 00          8          72
# error_counter,err_inj = 00000006 00          8          73
# error_counter,err_inj = 00000006 00          8          74
# error_counter,err_inj = 00000006 00          8          75
# error_counter,err_inj = 00000006 00          8          76
# error_counter,err_inj = 00000006 00          8          77
# error_counter,err_inj = 00000006 00          8          78
# error_counter,err_inj = 00000006 00          8          79
# error_counter,err_inj = 00000006 00          8          80
# error_counter,err_inj = 00000006 00          8          81
# error_counter,err_inj = 00000006 00          8          82
# error_counter,err_inj = 00000006 00          8          83
# error_counter,err_inj = 00000006 00          8          84
# error_counter,err_inj = 00000006 00          8          85
# error_counter,err_inj = 00000006 00          8          86
# error_counter,err_inj = 00000006 00          8          87
# error_counter,err_inj = 00000006 00          8          88
# error_counter,err_inj = 00000006 00          8          89
# error_counter,err_inj = 00000006 00          8          90
# error_counter,err_inj = 00000006 00          8          91
# error_counter,err_inj = 00000006 00          8          92
# error_counter,err_inj = 00000006 00          8          93
# error_counter,err_inj = 00000007 11          8          94
# error_counter,err_inj = 00000008 11          8          95
# error_counter,err_inj = 00000009 11         10          96
# error_counter,err_inj = 00000009 00         12          97
# error_counter,err_inj = 00000009 00         12          98
# error_counter,err_inj = 00000009 00         12          99
# error_counter,err_inj = 00000009 00         12         100
# error_counter,err_inj = 00000009 00         12         101
# error_counter,err_inj = 00000009 00         12         102
# error_counter,err_inj = 00000009 00         12         103
# error_counter,err_inj = 00000009 00         12         104
# error_counter,err_inj = 00000009 00         12         105
# error_counter,err_inj = 00000009 00         12         106
# error_counter,err_inj = 00000009 00         12         107
# error_counter,err_inj = 00000009 00         12         108
# error_counter,err_inj = 00000009 00         12         109
# error_counter,err_inj = 00000009 00         12         110
# error_counter,err_inj = 00000009 00         12         111
# error_counter,err_inj = 00000009 00         12         112
# error_counter,err_inj = 00000009 00         12         113
# error_counter,err_inj = 00000009 00         12         114
# error_counter,err_inj = 00000009 00         12         115
# error_counter,err_inj = 00000009 00         12         116
# error_counter,err_inj = 00000009 00         12         117
# error_counter,err_inj = 00000009 00         12         118
# error_counter,err_inj = 00000009 00         12         119
# error_counter,err_inj = 00000009 00         12         120
# error_counter,err_inj = 00000009 00         12         121
# error_counter,err_inj = 00000009 00         12         122
# error_counter,err_inj = 00000009 00         12         123
# error_counter,err_inj = 00000009 00         12         124
# error_counter,err_inj = 00000009 00         12         125
# error_counter,err_inj = 0000000a 11         12         126
# error_counter,err_inj = 0000000b 11         12         127
# error_counter,err_inj = 0000000c 11         14         128
# error_counter,err_inj = 0000000c 00         16         129
# error_counter,err_inj = 0000000c 00         16         130
# error_counter,err_inj = 0000000c 00         16         131
# error_counter,err_inj = 0000000c 00         16         132
# error_counter,err_inj = 0000000c 00         16         133
# error_counter,err_inj = 0000000c 00         16         134
# error_counter,err_inj = 0000000c 00         16         135
# error_counter,err_inj = 0000000c 00         16         136
```

```
# yaa! in = 0, out = 0, w_ct =         131, err = 00
# yaa! in = 0, out = 0, w_ct =         132, err = 00
# yaa! in = 0, out = 0, w_ct =         133, err = 00
# yaa! in = 0, out = 0, w_ct =         134, err = 00
# boo! in = 0, out = 1, w_ct =         135, err = 00,
1045000, BAD!
# boo! in = 0, out = 1, w_ct =         136, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         137, err = 00
# yaa! in = 1, out = 1, w_ct =         138, err = 00
# yaa! in = 1, out = 1, w_ct =         139, err = 00
# boo! in = 1, out = 0, w_ct =         140, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         141, err = 00
# yaa! in = 1, out = 1, w_ct =         142, err = 00
# yaa! in = 1, out = 1, w_ct =         143, err = 00
# yaa! in = 1, out = 1, w_ct =         144, err = 00
# yaa! in = 1, out = 1, w_ct =         145, err = 00
# yaa! in = 1, out = 1, w_ct =         146, err = 00
# yaa! in = 0, out = 0, w_ct =         147, err = 00
# yaa! in = 1, out = 1, w_ct =         148, err = 00
# yaa! in = 1, out = 1, w_ct =         149, err = 00
# yaa! in = 1, out = 1, w_ct =         150, err = 00
# yaa! in = 1, out = 1, w_ct =         151, err = 00
# yaa! in = 1, out = 1, w_ct =         152, err = 00
# yaa! in = 1, out = 1, w_ct =         153, err = 00
# yaa! in = 1, out = 1, w_ct =         154, err = 00
# yaa! in = 1, out = 1, w_ct =         155, err = 00
# yaa! in = 1, out = 1, w_ct =         156, err = 00
# yaa! in = 1, out = 1, w_ct =         157, err = 00
# yaa! in = 1, out = 1, w_ct =         158, err = 00
# yaa! in = 1, out = 1, w_ct =         159, err = 00
# yaa! in = 1, out = 1, w_ct =         160, err = 00
# yaa! in = 1, out = 1, w_ct =         161, err = 00
# yaa! in = 1, out = 1, w_ct =         162, err = 00
# yaa! in = 1, out = 1, w_ct =         163, err = 00
# yaa! in = 1, out = 1, w_ct =         164, err = 00
# yaa! in = 1, out = 1, w_ct =         165, err = 00
# yaa! in = 1, out = 1, w_ct =         166, err = 00
# boo! in = 1, out = 0, w_ct =         167, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         168, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         169, err = 00
# yaa! in = 1, out = 1, w_ct =         170, err = 00
# yaa! in = 1, out = 1, w_ct =         171, err = 00
# boo! in = 1, out = 0, w_ct =         172, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         173, err = 00
# yaa! in = 1, out = 1, w_ct =         174, err = 00
# yaa! in = 1, out = 1, w_ct =         175, err = 00
# yaa! in = 1, out = 1, w_ct =         176, err = 00
# yaa! in = 1, out = 1, w_ct =         177, err = 00
# yaa! in = 1, out = 1, w_ct =         178, err = 00
# yaa! in = 1, out = 1, w_ct =         179, err = 00
# yaa! in = 1, out = 1, w_ct =         180, err = 00
# yaa! in = 1, out = 1, w_ct =         181, err = 00
# yaa! in = 1, out = 1, w_ct =         182, err = 00
# yaa! in = 1, out = 1, w_ct =         183, err = 00
# yaa! in = 1, out = 1, w_ct =         184, err = 00
# yaa! in = 1, out = 1, w_ct =         185, err = 00
# yaa! in = 1, out = 1, w_ct =         186, err = 00
# yaa! in = 1, out = 1, w_ct =         187, err = 00
# yaa! in = 1, out = 1, w_ct =         188, err = 00
# yaa! in = 1, out = 1, w_ct =         189, err = 00
# yaa! in = 1, out = 1, w_ct =         190, err = 00
# yaa! in = 1, out = 1, w_ct =         191, err = 00
# yaa! in = 1, out = 1, w_ct =         192, err = 00
# yaa! in = 1, out = 1, w_ct =         193, err = 00
# yaa! in = 1, out = 1, w_ct =         194, err = 00
# yaa! in = 1, out = 1, w_ct =         195, err = 00
# yaa! in = 1, out = 1, w_ct =         196, err = 00
# yaa! in = 1, out = 1, w_ct =         197, err = 00
# yaa! in = 1, out = 1, w_ct =         198, err = 00
# boo! in = 1, out = 0, w_ct =         199, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         200, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         201, err = 00
# yaa! in = 1, out = 1, w_ct =         202, err = 00
# yaa! in = 1, out = 1, w_ct =         203, err = 00
# boo! in = 1, out = 0, w_ct =         204, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         205, err = 00
# yaa! in = 1, out = 1, w_ct =         206, err = 00
# yaa! in = 1, out = 1, w_ct =         207, err = 00
# yaa! in = 1, out = 1, w_ct =         208, err = 00
# yaa! in = 1, out = 1, w_ct =         209, err = 00
# yaa! in = 1, out = 1, w_ct =         210, err = 00
# yaa! in = 1, out = 1, w_ct =         211, err = 00
# yaa! in = 1, out = 1, w_ct =         212, err = 00
# yaa! in = 1, out = 1, w_ct =         213, err = 00
# yaa! in = 1, out = 1, w_ct =         214, err = 00
```

```
# error_counter,err_inj = 0000000c 00          16          137
# error_counter,err_inj = 0000000c 00          16          138
# error_counter,err_inj = 0000000c 00          16          139
# error_counter,err_inj = 0000000c 00          16          140
# error_counter,err_inj = 0000000c 00          16          141
# error_counter,err_inj = 0000000c 00          16          142
# error_counter,err_inj = 0000000c 00          16          143
# error_counter,err_inj = 0000000c 00          16          144
# error_counter,err_inj = 0000000c 00          16          145
# error_counter,err_inj = 0000000c 00          16          146
# error_counter,err_inj = 0000000c 00          16          147
# error_counter,err_inj = 0000000c 00          16          148
# error_counter,err_inj = 0000000c 00          16          149
# error_counter,err_inj = 0000000c 00          16          150
# error_counter,err_inj = 0000000c 00          16          151
# error_counter,err_inj = 0000000c 00          16          152
# error_counter,err_inj = 0000000c 00          16          153
# error_counter,err_inj = 0000000c 00          16          154
# error_counter,err_inj = 0000000c 00          16          155
# error_counter,err_inj = 0000000c 00          16          156
# error_counter,err_inj = 0000000c 00          16          157
# error_counter,err_inj = 0000000d 11          16          158
# error_counter,err_inj = 0000000e 11          16          159
# error_counter,err_inj = 0000000f 11          18          160
# error_counter,err_inj = 0000000f 00          20          161
# error_counter,err_inj = 0000000f 00          20          162
# error_counter,err_inj = 0000000f 00          20          163
# error_counter,err_inj = 0000000f 00          20          164
# error_counter,err_inj = 0000000f 00          20          165
# error_counter,err_inj = 0000000f 00          20          166
# error_counter,err_inj = 0000000f 00          20          167
# error_counter,err_inj = 0000000f 00          20          168
# error_counter,err_inj = 0000000f 00          20          169
# error_counter,err_inj = 0000000f 00          20          170
# error_counter,err_inj = 0000000f 00          20          171
# error_counter,err_inj = 0000000f 00          20          172
# error_counter,err_inj = 0000000f 00          20          173
# error_counter,err_inj = 0000000f 00          20          174
# error_counter,err_inj = 0000000f 00          20          175
# error_counter,err_inj = 0000000f 00          20          176
# error_counter,err_inj = 0000000f 00          20          177
# error_counter,err_inj = 0000000f 00          20          178
# error_counter,err_inj = 0000000f 00          20          179
# error_counter,err_inj = 0000000f 00          20          180
# error_counter,err_inj = 0000000f 00          20          181
# error_counter,err_inj = 0000000f 00          20          182
# error_counter,err_inj = 0000000f 00          20          183
# error_counter,err_inj = 0000000f 00          20          184
# error_counter,err_inj = 0000000f 00          20          185
# error_counter,err_inj = 0000000f 00          20          186
# error_counter,err_inj = 0000000f 00          20          187
# error_counter,err_inj = 0000000f 00          20          188
# error_counter,err_inj = 0000000f 00          20          189
# error_counter,err_inj = 00000010 11          20          190
# error_counter,err_inj = 00000011 11          20          191
# error_counter,err_inj = 00000012 11          22          192
# error_counter,err_inj = 00000012 00          24          193
# error_counter,err_inj = 00000012 00          24          194
# error_counter,err_inj = 00000012 00          24          195
# error_counter,err_inj = 00000012 00          24          196
# error_counter,err_inj = 00000012 00          24          197
# error_counter,err_inj = 00000012 00          24          198
# error_counter,err_inj = 00000012 00          24          199
# error_counter,err_inj = 00000012 00          24          200
# error_counter,err_inj = 00000012 00          24          201
# error_counter,err_inj = 00000012 00          24          202
# error_counter,err_inj = 00000012 00          24          203
# error_counter,err_inj = 00000012 00          24          204
# error_counter,err_inj = 00000012 00          24          205
# error_counter,err_inj = 00000012 00          24          206
# error_counter,err_inj = 00000012 00          24          207
# error_counter,err_inj = 00000012 00          24          208
# error_counter,err_inj = 00000012 00          24          209
# error_counter,err_inj = 00000012 00          24          210
# error_counter,err_inj = 00000012 00          24          211
# error_counter,err_inj = 00000012 00          24          212
# error_counter,err_inj = 00000012 00          24          213
# error_counter,err_inj = 00000012 00          24          214
# error_counter,err_inj = 00000012 00          24          215
# error_counter,err_inj = 00000012 00          24          216
# error_counter,err_inj = 00000012 00          24          217
# error_counter,err_inj = 00000012 00          24          218
# error_counter,err_inj = 00000012 00          24          219
# error_counter,err_inj = 00000012 00          24          220
# error_counter,err_inj = 00000012 00          24          221
# error_counter,err_inj = 00000013 11          24          222
# error_counter,err_inj = 00000014 11          24          223
# error_counter,err_inj = 00000015 11          26          224
# error_counter,err_inj = 00000015 00          28          225
# error_counter,err_inj = 00000015 00          28          226
# error_counter,err_inj = 00000015 00          28          227
# error_counter,err_inj = 00000015 00          28          228
# error_counter,err_inj = 00000015 00          28          229
```

```
# yaa! in = 1, out = 1, w_ct =         215, err = 00
# yaa! in = 1, out = 1, w_ct =         216, err = 00
# yaa! in = 1, out = 1, w_ct =         217, err = 00
# yaa! in = 1, out = 1, w_ct =         218, err = 00
# yaa! in = 1, out = 1, w_ct =         219, err = 00
# yaa! in = 1, out = 1, w_ct =         220, err = 00
# yaa! in = 1, out = 1, w_ct =         221, err = 00
# yaa! in = 1, out = 1, w_ct =         222, err = 00
# yaa! in = 1, out = 1, w_ct =         223, err = 00
# yaa! in = 1, out = 1, w_ct =         224, err = 00
# yaa! in = 1, out = 1, w_ct =         225, err = 00
# yaa! in = 1, out = 1, w_ct =         226, err = 00
# yaa! in = 1, out = 1, w_ct =         227, err = 00
# yaa! in = 1, out = 1, w_ct =         228, err = 00
# yaa! in = 1, out = 1, w_ct =         229, err = 00
# yaa! in = 1, out = 1, w_ct =         230, err = 00
# boo! in = 1, out = 0, w_ct =         231, err = 00,
1045000, BAD!
# boo! in = 1, out = 0, w_ct =         232, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         233, err = 00
# yaa! in = 1, out = 1, w_ct =         234, err = 00
# yaa! in = 1, out = 1, w_ct =         235, err = 00
# boo! in = 1, out = 0, w_ct =         236, err = 00,
1045000, BAD!
# yaa! in = 1, out = 1, w_ct =         237, err = 00
# yaa! in = 1, out = 1, w_ct =         238, err = 00
# yaa! in = 1, out = 1, w_ct =         239, err = 00
# yaa! in = 1, out = 1, w_ct =         240, err = 00
# yaa! in = 1, out = 1, w_ct =         241, err = 00
# yaa! in = 1, out = 1, w_ct =         242, err = 00
# yaa! in = 1, out = 1, w_ct =         243, err = 00
# yaa! in = 1, out = 1, w_ct =         244, err = 00
# yaa! in = 1, out = 1, w_ct =         245, err = 00
# yaa! in = 1, out = 1, w_ct =         246, err = 00
# yaa! in = 1, out = 1, w_ct =         247, err = 00
# yaa! in = 0, out = 0, w_ct =         248, err = 00
# yaa! in = 1, out = 1, w_ct =         249, err = 00
# yaa! in = 1, out = 1, w_ct =         250, err = 00
# yaa! in = 1, out = 1, w_ct =         251, err = 00
# yaa! in = 1, out = 1, w_ct =         252, err = 00
# yaa! in = 1, out = 1, w_ct =         253, err = 00
# yaa! in = 1, out = 1, w_ct =         254, err = 00
# yaa! in = 1, out = 1, w_ct =         255, err = 00
# corrupted_bits =          24, OUT: good =         238, bad =
18
```

```
# error_counter,err_inj = 00000015 00          28          230
# error_counter,err_inj = 00000015 00          28          231
# error_counter,err_inj = 00000015 00          28          232
# error_counter,err_inj = 00000015 00          28          233
# error_counter,err_inj = 00000015 00          28          234
# error_counter,err_inj = 00000015 00          28          235
# error_counter,err_inj = 00000015 00          28          236
# error_counter,err_inj = 00000015 00          28          237
# error_counter,err_inj = 00000015 00          28          238
# error_counter,err_inj = 00000015 00          28          239
# error_counter,err_inj = 00000015 00          28          240
# error_counter,err_inj = 00000015 00          28          241
# error_counter,err_inj = 00000015 00          28          242
# error_counter,err_inj = 00000015 00          28          243
# error_counter,err_inj = 00000015 00          28          244
# error_counter,err_inj = 00000015 00          28          245
# error_counter,err_inj = 00000015 00          28          246
# error_counter,err_inj = 00000015 00          28          247
# error_counter,err_inj = 00000015 00          28          248
# error_counter,err_inj = 00000015 00          28          249
# error_counter,err_inj = 00000015 00          28          250
# error_counter,err_inj = 00000015 00          28          251
# error_counter,err_inj = 00000015 00          28          252
# error_counter,err_inj = 00000015 00          28          253
# error_counter,err_inj = 00000016 11          28          254
# error_counter,err_inj = 00000017 11          28          255
```